

mielke-geometrodynamics: formula evidence

1 row shows a REFINEMENT below, not the OCR reading: 1 verified against the census. Each is marked [refined] beside its identifier. The model is unchanged — the original reading is still what ‘latex’ holds; this report chose the refinement for it. Rows: mielke-geometrodynamics\_FO0431

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

| Identifier                     | Page | Conf.   | LaTeX source                                                                                                                                      | Rendered                                                                                                                  | Image                                                                                                                                                              |
|--------------------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mielke-geometrodynamics_F00001 | 7    | ■ 1.000 | A                                                                                                                                                 | $A$                                                                                                                       | potential 1-form $A$ via electron interference.                                                                                                                    |
| mielke-geometrodynamics_F00002 | 7    | ■ 0.980 | $\operatorname{SL}(5, \mathbb{R})$                                                                                                                | $\mathrm{SL}(5, \mathbb{R})$                                                                                              | model with a primordial $\mathrm{SL}(5, \mathbb{R})$ gauge group is presented. After a tiny                                                                        |
| mielke-geometrodynamics_F00003 | 33   | ■ 0.985 | $\mathbb{R}$                                                                                                                                      | $\mathbb{R}$                                                                                                              | tiable functions $f$ to the space $\mathbb{R}$ of real numbers, defines the <i>tangent vector</i> $e(m)$ or                                                        |
| mielke-geometrodynamics_F00004 | 17   | ■ 1.000 | $\beta$                                                                                                                                           | $\beta$                                                                                                                   | and Fermi’s theory of weak interactions, being the cause for the $\beta$ -decay of certain                                                                         |
| mielke-geometrodynamics_F00005 | 17   | ■ 0.965 | $\mathrm{W}^{\pm}$                                                                                                                                | $\mathrm{W}^{\pm}$                                                                                                        | empirical evidence by the detection of the mediating $\mathrm{W}^{\pm}$ and $\mathrm{Z}^0$ bosons as the                                                           |
| mielke-geometrodynamics_F00006 | 17   | ■ 0.965 | $\mathrm{Z}^0$                                                                                                                                    | $\mathrm{Z}^0$                                                                                                            | empirical evidence by the detection of the mediating $\mathrm{W}^{\pm}$ and $\mathrm{Z}^0$ bosons as the                                                           |
| mielke-geometrodynamics_F00007 | 19   | ■ 1.000 | $1 \times 10^{-12}$                                                                                                                               | $1 \times 10^{-12}$                                                                                                       | mass is measured nowadays with a precision better than $1 \times 10^{-12}$ ; this accuracy                                                                         |
| mielke-geometrodynamics_F00008 | 20   | ■ 1.000 | $\partial_j$                                                                                                                                      | $\partial_j$                                                                                                              | to the electromagnetic fields if the partial derivatives $\partial_j$ in the Dirac equation are                                                                    |
| mielke-geometrodynamics_F00009 | 21   | ■ 1.000 | $D_j := \partial_j + i A_j$                                                                                                                       | $D_j := \partial_j + i A_j$                                                                                               | replaced by covariant derivatives $D_j := \partial_j + i A_j$ . In the latter concept, a “minimal”                                                                 |
| mielke-geometrodynamics_F00010 | 21   | ■ 1.000 | $A_j$                                                                                                                                             | $A_j$                                                                                                                     | substitution of the electromagnetic vector potential $A_j$ is taken into consideration.                                                                            |
| mielke-geometrodynamics_F00011 | 21   | ■ 1.000 | $\psi \rightarrow e^{i\theta} \psi$                                                                                                               | $\psi \rightarrow e^{i\theta} \psi$                                                                                       | remains invariant under the <i>local</i> transformation $\psi \rightarrow e^{i\theta} \psi$ of the phase of the                                                    |
| mielke-geometrodynamics_F00012 | 21   | ■ 1.000 | $\psi$                                                                                                                                            | $\psi$                                                                                                                    | wave function $\psi$ and the simultaneous transformation $A_j \rightarrow A_j - \partial_j \theta$ of the four-                                                    |
| mielke-geometrodynamics_F00013 | 21   | ■ 1.000 | $A_j \rightarrow A_j - \partial_j \theta$                                                                                                         | $A_j \rightarrow A_j - \partial_j \theta$                                                                                 | wave function $\psi$ and the simultaneous transformation $A_j \rightarrow A_j - \partial_j \theta$ of the four-                                                    |
| mielke-geometrodynamics_F00014 | 22   | ■ 0.526 | $n > 4$                                                                                                                                           | $n > 4$                                                                                                                   | ( $n > 4$ ) spaces and was, to a certain extent, anticipated by the pioneering works of                                                                            |
| mielke-geometrodynamics_F00015 | 23   | ■ 0.343 | ${}^{+}$                                                                                                                                          | $+$                                                                                                                       | an SU(3) nonet that consists of the massive spin-2 <sup>+</sup> mesons $f$ , $f'$ , $A_2$ , $K^*(1420)$ .                                                          |
| mielke-geometrodynamics_F00016 | 23   | ■ 0.343 | $A_2, K^*(1420)$                                                                                                                                  | $A_2, K^*(1420)$                                                                                                          | an SU(3) nonet that consists of the massive spin-2 <sup>+</sup> mesons $f$ , $f'$ , $A_2$ , $K^*(1420)$ .                                                          |
| mielke-geometrodynamics_F00017 | 23   | ■ 0.987 | $\operatorname{SL}(2, \mathbb{C})$                                                                                                                | $\mathrm{SL}(2, \mathbb{C})$                                                                                              | $\mathrm{SL}(2, \mathbb{C})$ , respectively. If we apply these findings to <i>strong gravity</i> , then it is obvious                                              |
| mielke-geometrodynamics_F00018 | 23   | ■ 0.697 | $\operatorname{SU}(3)$                                                                                                                            | $\mathrm{SU}(3)$                                                                                                          | to tie together the internal symmetries, such as the SU(3) one, with the spacetime                                                                                 |
| mielke-geometrodynamics_F00019 | 23   | ■ 0.997 | $\operatorname{SL}(2, \mathbb{C}) \subset \operatorname{SL}(2, \mathbb{C}) \otimes \operatorname{SU}(3) \subset \operatorname{SL}(6, \mathbb{C})$ | $\mathrm{SL}(2, \mathbb{C}) \subset \mathrm{SL}(2, \mathbb{C}) \otimes \mathrm{SU}(3) \subset \mathrm{SL}(6, \mathbb{C})$ | chain $\mathrm{SL}(2, \mathbb{C}) \subset \mathrm{SL}(2, \mathbb{C}) \otimes \mathrm{SU}(3) \subset \mathrm{SL}(6, \mathbb{C})$ . If these symmetries are regarded |
| mielke-geometrodynamics_F00020 | 24   | ■ 0.834 | $\operatorname{SL}(6, \mathbb{C})$                                                                                                                | $\mathrm{SL}(6, \mathbb{C})$                                                                                              | as local ones, we get an $\mathrm{SL}(6, \mathbb{C})$ gauge theory of strong interaction (SALAM 1973)                                                              |
| mielke-geometrodynamics_F00021 | 24   | ■ 1.000 | $\ell := \ell^* / \sqrt{\kappa}$                                                                                                                  | $\ell := \ell^* / \sqrt{\kappa}$                                                                                          | a further length $\ell := \ell^* / \sqrt{\kappa}$ is introduced into particle physics that corresponds to                                                          |

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| mielke-geometrodynamics_F00022 | 24   | ■ 0.995 | $\mathrm{G}_{\mathrm{S}}=\mathrm{G}_{\mathrm{N}} / \kappa$                                                                                          | $G_S = G_N/\kappa$                                | the scaled “Newtonian” coupling constant $\overline{G_S = G_N/\kappa}$ of strong gravity. SALAM &                        |
| mielke-geometrodynamics_F00023 | 25   | ■ 1.000 | $\sqrt{s}=540 \mathrm{GeV}$                                                                                                                         | $\sqrt{s} = 540\mathrm{GeV}$                      | associated missing energy at $\sqrt{s} = 540 \mathrm{GeV}$ . Phys Lett B 122:103–116                                     |
| mielke-geometrodynamics_F00024 | 30   | ■ 1.000 | $\mathrm{x}: \mathrm{M} \rightarrow \mathbb{R}^{\mathrm{n}}$                                                                                        | $x : M \rightarrow \mathbb{R}^n$                  | achieve this, we consider injective (reversible) mappings $\overline{x : M \rightarrow \mathbb{R}^n}$ , whose range      |
| mielke-geometrodynamics_F00025 | 30   | ■ 1.000 | $\mathrm{E}^{\mathrm{n}}$                                                                                                                           | $E^n$                                             | extends to an open subset of the usual Euclidean space $E^n$ . Such a map together with                                  |
| mielke-geometrodynamics_F00026 | 30   | ■ 0.993 | $\mathrm{U}_{\mathrm{i}}$                                                                                                                           | $U_i$                                             | its domain $U_i$ is called an n-dimensional chart. By the projection $\pi_i$ onto the single                             |
| mielke-geometrodynamics_F00027 | 30   | ■ 0.993 | $\pi_{\mathrm{i}}$                                                                                                                                  | $\pi_i$                                           | its domain $U_i$ is called an n-dimensional chart. By the projection $\pi_i$ onto the single                             |
| mielke-geometrodynamics_F00028 | 30   | ■ 1.000 | $\mathbb{R}^{\mathrm{n}}$                                                                                                                           | $\mathbb{R}^n$                                    | axes of the coordinate system of $\mathbb{R}^n$ , the <i>local coordinates</i> $x_i = \pi_i \circ x, i = 1, \dots, n$    |
| mielke-geometrodynamics_F00029 | 30   | ■ 1.000 | $\mathrm{x}_{\mathrm{i}}=\pi_{\mathrm{i}} \circ \mathrm{x}, \mathrm{i}=1, \ldots, \mathrm{n}$                                                       | $x_i = \pi_i \circ x, i = 1, \dots, n$            | axes of the coordinate system of $\mathbb{R}^n$ , the <i>local coordinates</i> $x_i = \pi_i \circ x, i = 1, \dots, n$    |
| mielke-geometrodynamics_F00030 | 30   | ■ 1.000 | $\mathrm{S}^{\mathrm{n}}$                                                                                                                           | $S^n$                                             | sphere $S^n$ ). However, it is possible to construct a collection of charts of identical                                 |
| mielke-geometrodynamics_F00031 | 31   | ■ 0.996 | $\mathrm{U}_{\mathrm{i}} \subset \mathrm{M}$                                                                                                        | $U_i \subset M$                                   | dimensions so that the set-theoretic union of their domains $U_i \subset M$ cover M com-                                 |
| mielke-geometrodynamics_F00032 | 31   | ■ 0.999 | $\mathscr{D}(\mathrm{M})$                                                                                                                           | $\mathscr{D}(M)$                                  | form an infinite-dimensional group with respect to composition, i.e., the group $\mathscr{D}(M)$                         |
| mielke-geometrodynamics_F00033 | 31   | ■ 1.000 | $\left(\mathrm{U}_{\mathrm{i}}, \mathrm{x}_{\mathrm{i}}\right)$                                                                                     | $(U_i, x_i)$                                      | A collection $(U_i, x_i)$ of compatible charts is called a $C^\infty$ -atlas of M. (The common                           |
| mielke-geometrodynamics_F00034 | 31   | ■ 1.000 | $\mathrm{C}^{\infty}$                                                                                                                               | $C^\infty$                                        | A collection $(U_i, x_i)$ of compatible charts is called a $C^\infty$ -atlas of M. (The common                           |
| mielke-geometrodynamics_F00035 | 31   | ■ 0.997 | $\mathrm{C}^{\mathrm{r}}$                                                                                                                           | $C^r$                                             | of class $C^r$ ). This atlas again can be extended unequivocally to a <i>complete</i> one, which                         |
| mielke-geometrodynamics_F00036 | 32   | ■ 1.000 | $\mathrm{E}^{\mathrm{n}}(\mathrm{n}+1) / 2$                                                                                                         | $E^{n(n+1)/2}$                                    | locally isometric embeddings of analytic Riemannian manifolds into $E^{n(n+1)/2}$ are                                    |
| mielke-geometrodynamics_F00037 | 32   | ■ 0.936 | $N=n(3 \ n+11) / 2$                                                                                                                                 | $N = n(3n + 11)/2$                                | dimension $\overline{N = n(3n + 11)/2}$ was proven by NASH (1956); for a noncompact one,                                 |
| mielke-geometrodynamics_F00038 | 32   | ■ 1.000 | $N=n(n+1)(3 \ n+11) / 2$                                                                                                                            | $N = n(n + 1)(3n + 11)/2$                         | the extravagant dimension $\overline{N = n(n + 1)(3n + 11)/2}$ results; cf. CHEN (2000) for a                            |
| mielke-geometrodynamics_F00039 | 32   | ■ 1.000 | $\mathbb{E}^4$                                                                                                                                      | $\mathbb{E}^4$                                    | surface of a rotating Kerr–Newman black hole into $\mathbb{E}^4$ .                                                       |
| mielke-geometrodynamics_F00040 | 32   | ■ 0.997 | $\mathrm{T}_{\mathrm{m}}(\mathrm{M})$                                                                                                               | $T_m(M)$                                          | <i>space</i> $T_m(M)$ attached to a point $m \in M$ . Contrary to the given illustration (see                            |
| mielke-geometrodynamics_F00041 | 32   | ■ 0.997 | $\mathrm{m} \in \mathrm{M}$                                                                                                                         | $m \in M$                                         | <i>space</i> $T_m(M)$ attached to a point $m \in M$ . Contrary to the given illustration (see                            |
| mielke-geometrodynamics_F00042 | 32   | ■ 0.573 | $\mathrm{s}(\mathrm{t})$                                                                                                                            | $s(t)$                                            | to achieve this, a curve $s(t)$ on the manifold is chosen that passes through a point                                    |
| mielke-geometrodynamics_F00043 | 33   | ■ 0.985 | $\mathrm{e}(\mathrm{m})$                                                                                                                            | $e(m)$                                            | tiabile functions $f$ to the space $\mathbb{R}$ of real numbers, defines the <i>tangent vector</i> $e(m)$ or             |
| mielke-geometrodynamics_F00044 | 33   | ■ 0.999 | $\mathrm{e}_{\alpha}(\mathrm{m}), \alpha=1, \ldots, \mathrm{n}$                                                                                     | $e_\alpha(m), \alpha = 1, \dots, n$               | independent vectors $\overline{e_\alpha(m), \alpha = 1, \dots, n}$ . It follows from (2.1.2) that the partial            |
| mielke-geometrodynamics_F00045 | 33   | ■ 1.000 | $\partial_{\mathrm{i}}:=\partial / \partial \mathrm{x}^{\mathrm{i}}$                                                                                | $\partial_i := \partial/\partial x^i$             | derivates $\partial_i := \partial/\partial x^i$ constitute a local basis for $T_m(M)$ with respect to a <i>holonomic</i> |
| mielke-geometrodynamics_F00046 | 33   | ■ 1.000 | $\left[\mathrm{e}_{\mathrm{i}}(\mathrm{m}), \mathrm{e}_{\mathrm{j}}(\mathrm{m})\right]=\left[\partial_{\mathrm{i}}, \partial_{\mathrm{j}}\right]=0$ | $[e_i(m), e_j(m)] = [\partial_i, \partial_j] = 0$ | coordinate system. By definition, the relation $[e_i(m), e_j(m)] = [\partial_i, \partial_j] = 0$ is satis-               |
| mielke-geometrodynamics_F00047 | 33   | ■ 1.000 | $\mathrm{e}_{\alpha}^{\mathrm{i}}(\mathrm{m})$                                                                                                      | $e_\alpha^i(m)$                                   | The occurring nonsingular matrices $e_\alpha^i(m)$ depend on the point in question and                                   |
| mielke-geometrodynamics_F00048 | 33   | ■ 0.628 | $\mathrm{T}(\mathrm{M})$                                                                                                                            | $T(M)$                                            | In addition to the tangent bundle $T(M)$ , a so-called <i>dual</i> bundle $T^*(M) = \bigcup_{m \in M} T_m^*(M)$          |

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| mielke-geometrodynamics_F00049 | 33   | 0.628 | $\mathrm{T}^{\ast}(\mathrm{M})=\bigcup_{\mathrm{m}\in\mathrm{M}}\mathrm{T}_{\mathrm{m}}^{\ast}(\mathrm{M})$ | $T^*(M) = \bigcup_{m \in M} T_m^*(M)$                      | In addition to the tangent bundle $T(M)$ , a so-called <i>dual</i> bundle $T^*(M) = \bigcup_{m \in M} T_m^*(M)$                                                                                   |
| mielke-geometrodynamics_F00050 | 33   | 1.000 | $\mathrm{T}_{\mathrm{m}}^{\ast}(\mathrm{M})$                                                                | $T_m^*(M)$                                                 | consisting of all possible cotangent spaces $T_m^*(M)$ on the manifold can be constructed.                                                                                                        |
| mielke-geometrodynamics_F00051 | 33   | 1.000 | $\vartheta^{\alpha}, \alpha=1, \ldots, n$                                                                   | $\vartheta^\alpha, \alpha = 1, \dots, n$                   | Let <span style="border: 1px solid red;"><math>\vartheta^\alpha, \alpha = 1, \dots, n</math></span> , be a basis of $T_m^*(M)$ . Then the “duality” <sup>3</sup> requires that $\vartheta^\alpha$ |
| mielke-geometrodynamics_F00052 | 33   | 1.000 | $\vartheta^{\alpha}$                                                                                        | $\vartheta^\alpha$                                         | Let $\vartheta^\alpha, \alpha = 1, \dots, n$ , be a basis of $T_m^*(M)$ . Then the “duality” <sup>3</sup> requires that $\vartheta^\alpha$                                                        |
| mielke-geometrodynamics_F00053 | 33   | 0.999 | $\mathrm{d}x^i$                                                                                             | $dx^i$                                                     | by $dx^i$ . In terms of the rigid basis of $E^n$ , the 1-forms                                                                                                                                    |
| mielke-geometrodynamics_F00054 | 34   | 0.998 | $\mathrm{d}\vartheta^{\alpha}$                                                                              | $d\vartheta^\alpha$                                        | derivative $d\vartheta^\alpha$ , in contrast to that of $dx^j$ , is generally different from zero. Following                                                                                      |
| mielke-geometrodynamics_F00055 | 34   | 0.998 | $\mathrm{d}x^j$                                                                                             | $dx^j$                                                     | derivative $d\vartheta^\alpha$ , in contrast to that of $dx^j$ , is generally different from zero. Following                                                                                      |
| mielke-geometrodynamics_F00056 | 34   | 0.599 | $\mathrm{E}_{\cdot j}^{\alpha}$                                                                             | $E_{\cdot j}^\alpha$                                       | (2.2.1), $E_{\cdot j}^\alpha$ can be understood as the dual tetrad field “reciprocal” to $e_\beta^i$ (cf. (2.1.3)).                                                                               |
| mielke-geometrodynamics_F00057 | 34   | 0.599 | $\mathrm{e}_{\beta}^i$                                                                                      | $e_\beta^i$                                                | (2.2.1), $E_{\cdot j}^\alpha$ can be understood as the dual tetrad field “reciprocal” to $e_\beta^i$ (cf. (2.1.3)).                                                                               |
| mielke-geometrodynamics_F00058 | 34   | 0.979 | $\otimes_{\mathrm{s}, \mathrm{a}}$                                                                          | $\otimes_{s,a}$                                            | Completely symmetric or antisymmetric products are denoted by $\otimes_{s,a}$ or, even more                                                                                                       |
| mielke-geometrodynamics_F00059 | 34   | 1.000 | $\vee$                                                                                                      | $\vee$                                                     | commonly, by the symbols $\vee$ and $\wedge$ , respectively. Note that a tensor is a geometric                                                                                                    |
| mielke-geometrodynamics_F00060 | 34   | 1.000 | $\wedge$                                                                                                    | $\wedge$                                                   | commonly, by the symbols $\vee$ and $\wedge$ , respectively. Note that a tensor is a geometric                                                                                                    |
| mielke-geometrodynamics_F00061 | 34   | 0.962 | $\mathrm{T}_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(\mathrm{m})$                                | $T_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(m)$ | The quantities $T_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(m)$ are called its p covariant and q contravariant components,                                                              |
| mielke-geometrodynamics_F00062 | 34   | 0.919 | $\mathrm{T}_{\mathrm{p}}^{\mathrm{q}}(\mathrm{M})$                                                          | $T_p^q(M)$                                                 | and $T_p^q(M)$ may be regarded as a bundle associated with $L(M)$ having the manifold                                                                                                             |
| mielke-geometrodynamics_F00063 | 34   | 0.919 | $\mathrm{L}(\mathrm{M})$                                                                                    | $L(M)$                                                     | and $T_p^q(M)$ may be regarded as a bundle associated with $L(M)$ having the manifold                                                                                                             |
| mielke-geometrodynamics_F00064 | 34   | 0.708 | $\mathrm{D}^{(\mathrm{p}, \mathrm{q})}$                                                                     | $D^{(p,q)}$                                                | $M$ as a base, the tensor representation $D^{(p,q)}$ of $GL(n, \mathbb{R})$ , i.e., $\rho^{(p,q)}(GL(n, \mathbb{R}))$ as a                                                                        |
| mielke-geometrodynamics_F00065 | 34   | 0.708 | $\rho^{(\mathrm{p}, \mathrm{q})}(\mathrm{GL}(\mathrm{n}, \mathbb{R}))$                                      | $\rho^{(p,q)}(GL(n, \mathbb{R}))$                          | $M$ as a base, the tensor representation $D^{(p,q)}$ of $GL(n, \mathbb{R})$ , i.e., $\rho^{(p,q)}(GL(n, \mathbb{R}))$ as a                                                                        |
| mielke-geometrodynamics_F00066 | 34   | 0.907 | $\otimes^{\mathrm{p}} \mathbb{R}^n \otimes^{\mathrm{q}} \mathbb{R}^n$                                       | $\otimes^p \mathbb{R}^n \otimes^q \mathbb{R}^n$            | structure group and the (Cartesian) product $\otimes^p \mathbb{R}^n \otimes^q \mathbb{R}^n$ as a typical fiber.                                                                                   |
| mielke-geometrodynamics_F00067 | 34   | 1.000 | $\mathrm{g}_{\mu \nu}$                                                                                      | $g_{\mu\nu}$                                               | the scale of the manifold. Due to the postulated symmetry of the components $g_{\mu \nu}$ ,                                                                                                       |
| mielke-geometrodynamics_F00068 | 35   | 0.651 | $(\mathrm{p}, 0)$                                                                                           | $(p, 0)$                                                   | $(p, 0)$ . Usually, $\wedge := \otimes_a$ is an abbreviation for the antisymmetrized tensor product,                                                                                              |
| mielke-geometrodynamics_F00069 | 35   | 0.651 | $\wedge := \otimes_{\mathrm{a}}$                                                                            | $\wedge := \otimes_a$                                      | $(p, 0)$ . Usually, $\wedge := \otimes_a$ is an abbreviation for the antisymmetrized tensor product,                                                                                              |
| mielke-geometrodynamics_F00070 | 35   | 1.000 | (1939, 1957)                                                                                                | (1939, 1957)                                               | <span style="border: 1px solid red;">[1939, 1957]</span> of the representations of the transformation group in flat spacetime, i.e.,                                                              |
| mielke-geometrodynamics_F00071 | 36   | 1.000 | $\mathscr{H}$                                                                                               | $\mathcal{H}$                                              | rays in a Hilbert space $\mathcal{H}$ , which is to be transformed according to an irreducible                                                                                                    |
| mielke-geometrodynamics_F00072 | 36   | 0.997 | $\varphi$                                                                                                   | $\varphi$                                                  | assumed that $\varphi$ at a point $m$ is equal to the value $\varphi(m)$ in a complex vector space                                                                                                |
| mielke-geometrodynamics_F00073 | 36   | 0.997 | $\varphi(\mathrm{m})$                                                                                       | $\varphi(m)$                                               | assumed that $\varphi$ at a point $m$ is equal to the value $\varphi(m)$ in a complex vector space                                                                                                |
| mielke-geometrodynamics_F00074 | 36   | 1.000 | $\mathrm{V}_{\mathrm{m}}$                                                                                   | $V_m$                                                      | $V_m$ . Additionally, it can be postulated that a vector space of this kind is attached                                                                                                           |
| mielke-geometrodynamics_F00075 | 36   | 0.941 | $\pi$                                                                                                       | $\pi$                                                      | <b>Definition</b> A fiber bundle $(F, M, \pi)$ consists of two $C^\infty$ -differentiable manifolds $F$                                                                                           |

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| mielke-geotrodynamics_F00076 | 36   | 0.362 | $\mathrm{F}_{\mathrm{m}} := -\pi^{-1}(\mathrm{m})$                                             | $F_m := -\pi^{-1}(m)$                       | and $\pi$ the <i>projection</i> . The closed submanifold $F_m := \pi^{-1}(m)$ will be called a <i>typical</i> |
| mielke-geotrodynamics_F00077 | 37   | 0.980 | $\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}_{\mathrm{m}}$                       | $\pi^{-1}(U) \approx U \times F_m$          | point $m \in M$ has a neighborhood $U$ such that $\pi^{-1}(U) \approx U \times F_m$ is isomorphic to the      |
| mielke-geotrodynamics_F00078 | 37   | 0.686 | $\left(\mathrm{U} \times \mathrm{F}_{\mathrm{m}}, \pi_u, \pi_u\right)$                         | $(U \times F_m, U, \pi_u)$                  | product bundle $(U \times F_m, U, \pi_u)$ . The mapping                                                       |
| mielke-geotrodynamics_F00079 | 37   | 0.637 | $\mathrm{P}(\mathrm{M}, \mathrm{G}, \pi, \delta)$                                              | $P(M, G, \pi, \delta)$                      | group, then the collection $P(M, G, \pi, \delta)$ is called a <i>principal fiber bundle</i> if                |
| mielke-geotrodynamics_F00080 | 37   | 0.790 | $(\mathrm{P}, \mathrm{M}, \pi)$                                                                | $(P, M, \pi)$                               | (i) $(P, M, \pi)$ is a fiber bundle with typical fiber $G$ ;                                                  |
| mielke-geotrodynamics_F00081 | 37   | 0.866 | $\mathrm{P}, \mathrm{G}, \delta$                                                               | $P, G, \delta$                              | (ii) $(P, G, \delta)$ is a $G$ -manifold (whereby $G$ is acting on $P$ from the right);                       |
| mielke-geotrodynamics_F00082 | 37   | 0.662 | $\mathrm{U} \subset$                                                                           | $U \subset$                                 | (iii) $P$ is <i>locally</i> trivial, which means that every point of $M$ has a neighborhood $U \subset$       |
| mielke-geotrodynamics_F00083 | 37   | 0.941 | $\iota: \mathrm{U} \times \mathrm{G} \rightarrow \pi^{-1}(\mathrm{U})$                         | $\iota: U \times G \rightarrow \pi^{-1}(U)$ | $M$ together with an isomorphism $\iota: U \times G \rightarrow \pi^{-1}(U)$ for which the following          |
| mielke-geotrodynamics_F00084 | 37   | 0.857 | $G$                                                                                            | $G$                                         | A manifold is called a <i>G-manifold</i> $(P, G, \delta)$ if $G$ acts on it as a free transformation          |
| mielke-geotrodynamics_F00085 | 37   | 0.970 | $\mathrm{p}_2 = g\mathrm{p}_1$                                                                 | $p_2 = gp_1$                                | there always existed a $g$ that would provide a relation $p_2 = g p_1$ for given $p_1$                        |
| mielke-geotrodynamics_F00086 | 37   | 0.970 | $\mathrm{p}_1$                                                                                 | $p_1$                                       | there always existed a $g$ that would provide a relation $p_2 = g p_1$ for given $p_1$                        |
| mielke-geotrodynamics_F00087 | 37   | 0.666 | $\mathrm{p}_2$                                                                                 | $p_2$                                       | and $p_2$ , then $G$ considered as a manifold would be isomorphic to the base $M$ .)                          |
| mielke-geotrodynamics_F00088 | 38   | 0.985 | $\pi, \delta$                                                                                  | $\pi, \delta$                               | this purpose, the fiber bundles $V(M, F; G, P)$ associated with $P(M, G, \pi, \delta)$ are                    |
| mielke-geotrodynamics_F00089 | 38   | 0.999 | $\mathrm{P} \times \mathrm{F}$                                                                 | $P \times F$                                | i.e., on the product space $P \times F$ , an action from the right-hand side is defined by                    |
| mielke-geotrodynamics_F00090 | 38   | 0.678 | $\mathrm{V} = \mathrm{P} \times_{\mathrm{G}} \mathrm{F}$                                       | $V = P \times_G F$                          | by $V = P \times_G F$ . The isomorphism $\pi^{-1}(U) \approx U \times G$ concerning the original domain       |
| mielke-geotrodynamics_F00091 | 38   | 0.678 | $\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$                                    | $\pi^{-1}(U) \approx U \times G$            | by $V = P \times_G F$ . The isomorphism $\pi^{-1}(U) \approx U \times G$ concerning the original domain       |
| mielke-geotrodynamics_F00092 | 38   | 0.307 | $\pi_v^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}$                                  | $\pi_v^{-1}(U) \approx U \times F$          | induce the isomorphism $\pi_v^{-1}(U) \approx U \times F$ in the associated bundle. Provided that             |
| mielke-geotrodynamics_F00093 | 38   | 0.907 | $\pi_v^{-1}(\mathrm{U})$                                                                       | $\pi_v^{-1}(U)$                             | $\pi_v^{-1}(U)$ is an open submanifold of $V$ , then $V$ can be equipped with a differentiable                |
| mielke-geotrodynamics_F00094 | 38   | 0.998 | $\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$                             | $\rho: G \rightarrow GL(N, \mathbb{C})$     | occurs, but its linear representation $\rho: G \rightarrow GL(N, \mathbb{C})$ , i.e., the Lie homomorphism    |
| mielke-geotrodynamics_F00095 | 38   | 0.852 | $\mathrm{N} \times \mathrm{N}$                                                                 | $N \times N$                                | of $G$ into the general linear group of complex $N \times N$ matrices. If their representation                |
| mielke-geotrodynamics_F00096 | 38   | 0.780 | $\mathbb{C}^{\mathrm{N}}$                                                                      | $\mathbb{C}^N$                              | space, the $N$ -dimensional vector space $\mathbb{C}^N$ over the field of complex numbers, is                 |
| mielke-geotrodynamics_F00097 | 38   | 1.000 | $\mathrm{T}_{\mathbb{C}}^*(\mathrm{M})$                                                        | $T_{\mathbb{C}}^*(M)$                       | with the complexified <sup>4</sup> cotangent bundle $T_{\mathbb{C}}^*(M)$ , further vector bundles will come  |
| mielke-geotrodynamics_F00098 | 38   | 1.000 | $\rho$                                                                                         | $\rho$                                      | $\rho$ , i.e., the cross section                                                                              |
| mielke-geotrodynamics_F00099 | 38   | 0.986 | $\mathrm{N} \geq 1$                                                                            | $N \geq 1$                                  | For $N \geq 1$ , these fields form an infinite-dimensional vector space over $\mathbb{C}^N$ . In              |
| mielke-geotrodynamics_F00100 | 38   | 1.000 | $\mathrm{b} = \left(\mathrm{b}_{\mathrm{A}}\right): \mathrm{U} \subset \mathrm{M} \rightarrow$ | $b = (b_A): U \subset M \rightarrow$        | (2.4.3), the bundle coordinates relative to the local basis fields $b = (b_A): U \subset M \rightarrow$       |
| mielke-geotrodynamics_F00101 | 38   | 1.000 | $\mathrm{V}^{\rho}$                                                                            | $V^{\rho}$                                  | $V^{\rho}$ are denoted by                                                                                     |

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| mielke-geometrodynamics_F00102 | 39   | ■ 1.000 | $\rho: \mathrm{G} \rightarrow \operatorname{GL}(\mathrm{N}, \mathbb{C})$                       | $\rho: G \rightarrow GL(N, \mathbb{C})$            | Let the Lie homomorphism $\rho: G \rightarrow GL(N, \mathbb{C})$ be explicitly given by the nonsin-                                                                                              |
| mielke-geometrodynamics_F00103 | 39   | ■ 0.964 | $\rho_{\mathrm{A}}^{\mathrm{B}}(\mathrm{g}(\xi))$                                              | $\rho_{\mathrm{A}}^{\mathrm{B}}(\mathrm{g}(\xi))$  | gular complex $N \times N$ matrices $\rho_{\mathrm{A}}^{\mathrm{B}}(\mathrm{g}(\xi))$ , where $\xi^j (j = 1, \dots, \dim G)$ prescribes a                                                        |
| mielke-geometrodynamics_F00104 | 39   | ■ 0.964 | $\xi^j (\mathrm{j}=1, \ldots, \operatorname{dim} \mathrm{G})$                                  | $\xi^j (j = 1, \dots, \dim G)$                     | gular complex $N \times N$ matrices $\rho_{\mathrm{A}}^{\mathrm{B}}(\mathrm{g}(\xi))$ , where <span style="border: 1px solid red;"><math>\xi^j (j = 1, \dots, \dim G)</math></span> prescribes a |
| mielke-geometrodynamics_F00105 | 39   | ■ 1.000 | $\rho=\mathrm{id}$                                                                             | $\rho = \mathrm{id}$                               | identical representation, i.e., for $\rho = \mathrm{id}$ , the infinitesimal generators $I_j$ form a basis                                                                                       |
| mielke-geometrodynamics_F00106 | 39   | ■ 1.000 | $\mathrm{I}_{\mathrm{j}}$                                                                      | $I_j$                                              | identical representation, i.e., for $\rho = \mathrm{id}$ , the infinitesimal generators $I_j$ form a basis                                                                                       |
| mielke-geometrodynamics_F00107 | 39   | ■ 0.938 | $\mathfrak{g}$                                                                                 | $\mathfrak{g}$                                     | of the Lie algebra $\mathfrak{g}$ of $G$ . As a representation of $I_j$ , the infinitesimal operators inherit                                                                                    |
| mielke-geometrodynamics_F00108 | 39   | ■ 1.000 | $\mathrm{c}_{ij}^{\mathrm{k}}$                                                                 | $c_{ij}^{\mathrm{k}}$                              | These algebraic relations are determined by the structure constants $c_{ij}^{\mathrm{k}}$ of the Lie                                                                                             |
| mielke-geometrodynamics_F00109 | 39   | ■ 0.960 | $\mathrm{V}^{\mathrm{Ad}}$                                                                     | $V^{\mathrm{Ad}}$                                  | vector bundle $V^{\mathrm{Ad}}$ respecting the adjoint representation of $G$ as a structure group.                                                                                               |
| mielke-geometrodynamics_F00110 | 39   | ■ 0.983 | $\mathrm{s}(\mathrm{t}) \subset \mathrm{M}$                                                    | $s(t) \subset M$                                   | in a differentiable manner along a given curve $s(t) \subset M$ in the base manifold. In order                                                                                                   |
| mielke-geometrodynamics_F00111 | 39   | ■ 1.000 | $\mathrm{T}_{\mathrm{p}}^*(\mathrm{P})$                                                        | $T_p^*(P)$                                         | cotangent bundle $T_p^*(P)$ at the point $p \in P$ (see KN I, Chap. 1). Then decompose it                                                                                                        |
| mielke-geometrodynamics_F00112 | 39   | ■ 1.000 | $\mathrm{p} \in \mathrm{P}$                                                                    | $p \in P$                                          | cotangent bundle $T_p^*(P)$ at the point $p \in P$ (see KN I, Chap. 1). Then decompose it                                                                                                        |
| mielke-geometrodynamics_F00113 | 40   | ■ 1.000 | $\mathrm{T}_{\mathrm{p}}^*(\mathrm{P})=\mathrm{H}_{\mathrm{p}} \oplus \mathrm{V}_{\mathrm{p}}$ | $T_p^*(P) = H_p \oplus V_p$                        | into the direct sum of horizontal and the vertical subspaces, i.e., $T_p^*(P) = H_p \oplus V_p$ .                                                                                                |
| mielke-geometrodynamics_F00114 | 40   | ■ 1.000 | $\mathrm{F}_{\mathrm{m}}(\mathrm{M})$                                                          | $F_{\mathrm{m}}(M)$                                | i.e., in the cotangent space of the bundle $F_{\mathrm{m}}(M)$ of typical fibers on $M$ .                                                                                                        |
| mielke-geometrodynamics_F00115 | 40   | ■ 0.999 | $\omega(\mathrm{e}) \in \mathscr{C}$                                                           | $\omega(e) \in \mathcal{C}$                        | This will be achieved by the assertion of a 1-form $\omega(e) \in \mathcal{C}$ with values in the                                                                                                |
| mielke-geometrodynamics_F00116 | 40   | ■ 1.000 | $\mathrm{e}_{\mathrm{v}}=\mathrm{pdg} \in \mathrm{V}_{\mathrm{p}}$                             | $e_v = \mathrm{pdg} \in V_p$                       | (ii) For vertical elements <span style="border: 1px solid red;"><math>e_v = \mathrm{pdg} \in V_p</math></span> , it will be mapped to the left-invariant                                         |
| mielke-geometrodynamics_F00117 | 40   | ■ 1.000 | $\bigwedge^{\mathrm{p}} \mathrm{T}_{\mathbb{C}}^*(\mathrm{M}) \otimes V^{\rho}$                | $\bigwedge^p T_{\mathbb{C}}^*(M) \otimes V^{\rho}$ | connection in an associated vector bundle $\bigwedge^p T_{\mathbb{C}}^*(M) \otimes V^{\rho}$ is defined via the <i>covari-</i>                                                                   |
| mielke-geometrodynamics_F00118 | 40   | ■ 1.000 | $\mathrm{T}_{\mathbb{C}}^*(\mathrm{M}) \otimes V^{\rho}$                                       | $T_{\mathbb{C}}^*(M) \otimes V^{\rho}$             | onto those of the product bundle $T_{\mathbb{C}}^*(M) \otimes V^{\rho}$ . This operator will be defined in such                                                                                  |
| mielke-geometrodynamics_F00119 | 41   | ■ 1.000 | $V^{\rho}$                                                                                     | $V^{\rho}$                                         | expressed as a linear transformation of the selfsame basis of $V^{\rho}$ , i.e.,                                                                                                                 |
| mielke-geometrodynamics_F00120 | 41   | ■ 0.856 | $\mathrm{b}$                                                                                   | $b$                                                | The image of $\mathrm{D}b$ will be referred to as the <i>(G-)covariant exterior derivative</i> of $b$ .                                                                                          |
| mielke-geometrodynamics_F00121 | 41   | ■ 0.641 | $\mathrm{b}_{\mathrm{A}}, \mathrm{A}=1, \ldots, \mathrm{N}$                                    | $b_A, A = 1, \dots, N$                             | restriction to a neighborhood $U$ of $M$ . Let <span style="border: 1px solid red;"><math>b_A, A = 1, \dots, N</math></span> be a local basis for the                                            |
| mielke-geometrodynamics_F00122 | 41   | ■ 0.996 | $\mathrm{V}_{\mathbb{U}}^{\rho}$                                                               | $V_{\mathbb{U}}^{\rho}$                            | cross section $V_{\mathbb{U}}^{\rho}$ restricted to $U$ . With respect to this local system of reference, the                                                                                    |
| mielke-geometrodynamics_F00123 | 41   | ■ 0.998 | $\mathrm{b}_{\mathrm{A}}$                                                                      | $b_A$                                              | resulting effect of $\mathrm{D}$ on $b_A$ can be expanded as follows:                                                                                                                            |
| mielke-geometrodynamics_F00124 | 41   | ■ 0.998 | $\left[\omega_{\mathrm{A}}^{\mathrm{B}}\right]$                                                | $[\omega_A^{\mathrm{B}}]$                          | matrix $[\omega_A^{\mathrm{B}}]$ acquires values in $T_{\mathbb{C}}^*(M)$ and will consequently be called a <i>connection</i>                                                                    |
| mielke-geometrodynamics_F00125 | 41   | ■ 1.000 | $\Gamma_{\alpha}^j(\mathrm{m})$                                                                | $\Gamma_{\alpha}^j(\mathrm{m})$                    | The coefficients $\Gamma_{\alpha}^j(\mathrm{m})$ will turn out to be generalized gauge potentials as they                                                                                        |
| mielke-geometrodynamics_F00126 | 42   | ■ 0.997 | $\mathrm{d} \mathrm{d} \equiv 0$                                                               | $\mathrm{d} \mathrm{d} \equiv 0$                   | For differential forms in general, the identity $\mathrm{d} \mathrm{d} \equiv 0$ holds, which may be regarded as                                                                                 |
| mielke-geometrodynamics_F00127 | 42   | ■ 1.000 | $\partial \partial \equiv 0$                                                                   | $\partial \partial \equiv 0$                       | a counterpart to the homological relation $\partial \partial \equiv 0$ . Figuratively speaking, this means                                                                                       |
| mielke-geometrodynamics_F00128 | 42   | ■ 0.905 | $\Omega$                                                                                       | $\Omega$                                           | that the curvature 2-form $\Omega$ satisfies the <i>second structure equation</i> of É. CARTAN:                                                                                                  |

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| mielke-geometrodynamics_F00129 | 42   | ■ 1.000 | $\mathrm{G}=\mathrm{U}(1)$                                                                                           | $G = U(1)$                                      | which can be formulated in a fiber bundle with abelian structure group $G = U(1)$ ,                                     |
| mielke-geometrodynamics_F00130 | 43   | ■ 1.000 | $\circlearrowleft=\partial \mathrm{U} \subset \mathrm{M}$                                                            | $\circlearrowleft = \partial U \subset M$       | along an <i>infinitesimal</i> closed curve $\circlearrowleft = \partial U \subset M$ . The very concept of a connection |
| mielke-geometrodynamics_F00131 | 43   | ■ 1.000 | $\partial \mathrm{U}$                                                                                                | $\partial U$                                    | closed curve $\partial U$ . Figure 2.2 shows this for a parallel displacement of a pair of tangent                      |
| mielke-geometrodynamics_F00132 | 43   | ■ 1.000 | $\mathrm{e}(\mathrm{m}) \in \mathrm{T}_{\mathrm{m}}(\mathrm{M})$                                                     | $e(m) \in T_m(M)$                               | $T_m(M)$ onto itself. These are generated by displacements of $\mathfrak{e}(m) \in T_m(M)$ along                        |
| mielke-geometrodynamics_F00133 | 44   | ■ 0.996 | $\mathrm{G}(\mathrm{p})$                                                                                             | $G(p)$                                          | (i) $G(p)$ is equivariant, i.e.,                                                                                        |
| mielke-geometrodynamics_F00134 | 44   | ■ 0.897 | $\mathrm{F}_{\mathrm{m}}=\pi^{-1}(\mathrm{~m})$                                                                      | $F_m = \pi^{-1}(m)$                             | (ii) $G(p)$ preserves each fiber $F_m = \pi^{-1}(m)$ , i.e., acts trivially on the base space,                          |
| mielke-geometrodynamics_F00135 | 44   | ■ 1.000 | $\mathscr{G}_{\mathrm{P}}$                                                                                           | $\mathcal{G}_P$                                 | position, they constitute an infinite-dimensional group, the group $\mathcal{G}_P$ of <i>local</i>                      |
| mielke-geometrodynamics_F00136 | 45   | ■ 0.513 | $\exp : \mathfrak{g} \rightarrow \mathrm{G}$                                                                         | $\exp : \mathfrak{g} \rightarrow G$             | The exponential mapping $\exp : \mathfrak{g} \rightarrow G$ from the Lie algebra $\mathfrak{g}$ into the structure      |
| mielke-geometrodynamics_F00137 | 45   | ■ 0.614 | $P \times_{\mathrm{Ad}} \mathrm{G}$                                                                                  | $P \times_{\mathrm{Ad}} G$                      | group $G$ of $P$ induces a corresponding mapping within $P \times_{\mathrm{Ad}} G$ . As a result,                       |
| mielke-geometrodynamics_F00138 | 45   | ■ 1.000 | $\theta^k(m) \in C^{\infty}(M)$                                                                                      | $\theta^k(m) \in C^\infty(M)$                   | Here $\mathfrak{\theta}^k(m) \in C^\infty(M)$ denote real functions on the base space. Gauge trans-                     |
| mielke-geometrodynamics_F00139 | 45   | ■ 0.984 | $\mathrm{V}^{\rho}\left(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P}\right)$ | $V^\rho(M, \mathbb{C}^N, GL(N, \mathbb{C}), P)$ | formations within an associate vector bundle $V^\rho(M, \mathbb{C}^N, GL(N, \mathbb{C}), P)$ , i.e.,                    |
| mielke-geometrodynamics_F00140 | 45   | ■ 1.000 | $\mathscr{G}_{\mathrm{V}}$                                                                                           | $\mathcal{G}_V$                                 | elements of $\mathcal{G}_V$ , are represented by the same expression to the extent that $L^k$                           |
| mielke-geometrodynamics_F00141 | 45   | ■ 1.000 | $\mathrm{L}^k$                                                                                                       | $L^k$                                           | elements of $\mathcal{G}_V$ , are represented by the same expression to the extent that $L^k$                           |
| mielke-geometrodynamics_F00142 | 45   | ■ 0.482 | $\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$                                                   | $\rho : G \rightarrow GL(N, \mathbb{C})$        | $\bar{\rho} : G \rightarrow GL(N, \mathbb{C})$ .                                                                        |
| mielke-geometrodynamics_F00143 | 45   | ■ 1.000 | $\bar{\varphi}$                                                                                                      | $\bar{\varphi}$                                 | It has to be emphasized that the bundle coordinates $\bar{\varphi}$ of the vector bundle constructed                    |
| mielke-geometrodynamics_F00144 | 45   | ■ 1.000 | $\bar{\rho}$                                                                                                         | $\bar{\rho}$                                    | on the conjugate (“Dirac adjoint”) representation $\bar{\rho}$ transforms according to                                  |
| mielke-geometrodynamics_F00145 | 46   | ■ 0.798 | $\mathrm{V}\left(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P}\right)$        | $V(M, \mathbb{C}^N, GL(N, \mathbb{C}), P)$      | Let $V(M, \mathbb{C}^N, GL(N, \mathbb{C}), P)$ be a complex vector bundle that is associated with                       |
| mielke-geometrodynamics_F00146 | 46   | ■ 0.438 | $\gamma_k$                                                                                                           | $\gamma_k$                                      | 2k-forms $\gamma_k$ on $P$ whose inverse images after the projection on the base space $M$ read                         |
| mielke-geometrodynamics_F00147 | 46   | ■ 1.000 | $k \leq n / 2$                                                                                                       | $k \leq n/2$                                    | The number of curvature 2-forms $\Omega$ in the exterior product is $k \leq n/2$ . From the                             |
| mielke-geometrodynamics_F00148 | 46   | ■ 1.000 | $\omega$                                                                                                             | $\omega$                                        | of the latter is not determined by the particular choice of the connection $\omega$ , but depends                       |
| mielke-geometrodynamics_F00149 | 46   | ■ 0.959 | $k^{\text{th}}$                                                                                                      | $k^{\text{th}}$                                 | precisely (see: KN II, Theorem 3.1), it can be stated that the abstractly defined $k^{\text{th}}$                       |
| mielke-geometrodynamics_F00150 | 46   | ■ 0.570 | $c_k(\mathrm{V})$                                                                                                    | $c_k(V)$                                        | Chern class $c_k(V)$ of a complex vector bundle $V$ is represented by the closed 2k-form                                |
| mielke-geometrodynamics_F00151 | 46   | ■ 0.546 | $V^{\mathbb{R}}$                                                                                                     | $V^{\mathbb{R}}$                                | In order to obtain the characteristic classes of a real vector bundle $V^{\mathbb{R}}$ over $M$ with                    |
| mielke-geometrodynamics_F00152 | 46   | ■ 0.983 | $\mathbb{R}^{\mathrm{N}}$                                                                                            | $\mathbb{R}^N$                                  | the typical fiber $\mathbb{R}^N$ , it will be enlarged to a complex vector bundle $V$ with the typical                  |
| mielke-geometrodynamics_F00153 | 46   | ■ 0.753 | $\mathbb{C}^{\mathrm{N}}$                                                                                            | $\mathbb{C}^N$                                  | fiber $\mathbb{C}^N$ . The latter arises from the complexification of each fiber of $V^{\mathbb{R}}$ . Then the         |
| mielke-geometrodynamics_F00154 | 46   | ■ 0.753 | $\mathrm{V}^{\mathbb{R}}$                                                                                            | $V^{\mathbb{R}}$                                | fiber $\mathbb{C}^N$ . The latter arises from the complexification of each fiber of $V^{\mathbb{R}}$ . Then the         |
| mielke-geometrodynamics_F00155 | 47   | ■ 1.000 | $s=0$                                                                                                                | $s = 0$                                         | the Euclidean signature $s = 0$ , we find locally, cf. DANIEL & VIALLET (1980),                                         |

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|--------------------------------|------|---------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mielke-geometrodynamics_F00156 | 47   | ■ 1.000 | $\mathrm{P}\left(\mathrm{S}^4, \mathrm{SU}(\mathrm{f}), \pi, \delta\right)$ | $\mathrm{P}\left(\mathrm{S}^4, \mathrm{SU}(\mathrm{f}), \pi, \delta\right)$ | It is typical for the principal fiber bundles $\mathrm{P}(\mathrm{S}^4, \mathrm{SU}(\mathrm{f}), \pi, \delta)$ , which are to be found in                                                    |
| mielke-geometrodynamics_F00157 | 47   | ■ 1.000 | $\operatorname{det} \mathrm{G}=1$                                           | $\det \mathrm{G}=1$                                                         | for which $\det \mathrm{G}=1$ holds. Inasmuch as the trace of the Lie-algebra-valued curvature                                                                                               |
| mielke-geometrodynamics_F00158 | 47   | ■ 0.565 | $\operatorname{Tr}(\Omega \wedge \Omega)$                                   | $\mathrm{Tr}(\Omega \wedge \Omega)$                                         | 2-form $\Omega$ vanishes in such instances, the gauge-invariant 4-form $\mathrm{Tr}(\Omega \wedge \Omega)$ suffices                                                                          |
| mielke-geometrodynamics_F00159 | 47   | ■ 0.892 | $\stackrel{*}{\pi}\left(\gamma_2\right)$                                    | $\stackrel{*}{\pi}\left(\gamma_2\right)$                                    | The integration of the second Chern class $\stackrel{*}{\pi}\left(\gamma_2\right)$ over the base space yields a                                                                              |
| mielke-geometrodynamics_F00160 | 47   | ■ 0.992 | $\mathrm{G}=\mathrm{SU}(\mathrm{f})$                                        | $\mathrm{G}=\mathrm{SU}(\mathrm{f})$                                        | index”; see JACKIW (1977). Concerning bundles with structure group $\mathrm{G}=\mathrm{SU}(\mathrm{f})$ ,                                                                                    |
| mielke-geometrodynamics_F00161 | 47   | ■ 0.827 | $\mathrm{SU}(2)$                                                            | $\mathrm{SU}(2)$                                                            | et al. (1975) is correct, since in their paper, the isospin group $\mathrm{SU}(2)$ has been enlarged                                                                                         |
| mielke-geometrodynamics_F00162 | 47   | ■ 0.749 | $\widetilde{\mathrm{SO}}(4) \approx \mathrm{SU}(2) \otimes \mathrm{SU}(2)$  | $\widetilde{\mathrm{SO}}(4) \approx \mathrm{SU}(2) \otimes \mathrm{SU}(2)$  | to the real structure group <span style="border: 1px solid red;"><math>\widetilde{\mathrm{SO}}(4) \approx \mathrm{SU}(2) \otimes \mathrm{SU}(2)</math></span> , due to a special isomorphism |
| mielke-geometrodynamics_F00163 | 47   | ■ 0.995 | $\gamma_2$                                                                  | $\gamma_2$                                                                  | of the closed form $\gamma_2$ onto $\mathrm{M}$ is even an exact one. This is a property that it shares                                                                                      |
| mielke-geometrodynamics_F00164 | 48   | ■ 1.000 | $\mathrm{F}_{\alpha \beta}^{\mathrm{j}}$                                    | $\mathrm{F}_{\alpha \beta}^{\mathrm{j}}$                                    | that the field strengths $\mathrm{F}_{\alpha \beta}^{\mathrm{j}}$ (components of the curvature form) have to vanish faster                                                                   |
| mielke-geometrodynamics_F00165 | 48   | ■ 1.000 | $ \mathrm{x} ^{-2}$                                                         | $ \mathrm{x} ^{-2}$                                                         | than $ \mathrm{x} ^{-2}$ at infinity, since otherwise, the integral (2.8.13) would not exist. In order to                                                                                    |
| mielke-geometrodynamics_F00166 | 49   | ■ 1.000 | $ \mathrm{x}  \rightarrow \infty$                                           | $ \mathrm{x}  \rightarrow \infty$                                           | should hold for $ \mathrm{x}  \rightarrow \infty$ . Here $\stackrel{\infty}{\omega}$ denotes a Maurer–Cartan connection, i.e., a left-                                                       |
| mielke-geometrodynamics_F00167 | 49   | ■ 1.000 | $\stackrel{\infty}{\omega}$                                                 | $\stackrel{\infty}{\omega}$                                                 | should hold for $ \mathrm{x}  \rightarrow \infty$ . Here $\stackrel{\infty}{\omega}$ denotes a Maurer–Cartan connection, i.e., a left-                                                       |
| mielke-geometrodynamics_F00168 | 49   | ■ 0.978 | $\partial \mathrm{M}$                                                       | $\partial \mathrm{M}$                                                       | If the boundary $\partial \mathrm{M}$ is chosen in such a way that only a pure gauge connection $\stackrel{\infty}{\omega}$                                                                  |
| mielke-geometrodynamics_F00169 | 49   | ■ 0.999 | $\mathrm{G}=\mathrm{SU}(2)$                                                 | $\mathrm{G}=\mathrm{SU}(2)$                                                 | in full compliance with (2.8.15); cf. JACKIW (1980). If $\mathrm{G}=\mathrm{SU}(2)$ serves as a struc-                                                                                       |
| mielke-geometrodynamics_F00170 | 49   | ■ 1.000 | $\mathrm{S}^3$                                                              | $\mathrm{S}^3$                                                              | $\mathrm{S}^3$ topologically, then it can be shown that $\mathrm{c}_2(\mathrm{M})$ determines the “mapping degree”                                                                           |
| mielke-geometrodynamics_F00171 | 49   | ■ 1.000 | $\mathrm{c}_2(\mathrm{M})$                                                  | $\mathrm{c}_2(\mathrm{M})$                                                  | $\mathrm{S}^3$ topologically, then it can be shown that $\mathrm{c}_2(\mathrm{M})$ determines the “mapping degree”                                                                           |
| mielke-geometrodynamics_F00172 | 49   | ■ 0.954 | $\mathrm{S}^3 \approx \mathrm{SU}(2)$                                       | $\mathrm{S}^3 \approx \mathrm{SU}(2)$                                       | or the <i>winding number</i> <sup>6</sup> of the mapping of $\mathrm{S}^3$ in $\mathrm{S}^3 \approx \mathrm{SU}(2)$ . These considerations                                                   |
| mielke-geometrodynamics_F00173 | 49   | ■ 0.995 | $\mathrm{n}^{\text{th}}$                                                    | $\mathrm{n}^{\text{th}}$                                                    | <sup>6</sup> Speaking more generally, the $\mathrm{n}^{\text{th}}$ -homotopy group of the $\mathrm{n}$ -dimensional sphere $\mathrm{S}^{\mathrm{n}}$ is given by the                         |
| mielke-geometrodynamics_F00174 | 49   | ■ 1.000 | $\mathbb{Z}$                                                                | $\mathbb{Z}$                                                                | group $\mathbb{Z}$ of integers, i.e., by $\pi_n\left(\mathrm{S}^n\right)=\mathbb{Z}$ . As such, it determines the winding number of the mapping                                              |
| mielke-geometrodynamics_F00175 | 49   | ■ 1.000 | $\pi_n\left(\mathrm{S}^n\right)=\mathbb{Z}$                                 | $\pi_n\left(\mathrm{S}^n\right)=\mathbb{Z}$                                 | group $\mathbb{Z}$ of integers, i.e., by $\pi_n\left(\mathrm{S}^n\right)=\mathbb{Z}$ . As such, it determines the winding number of the mapping                                              |
| mielke-geometrodynamics_F00176 | 49   | ■ 1.000 | $\mathrm{S}^{\mathrm{n}} \rightarrow \mathrm{S}^{\mathrm{n}}$               | $\mathrm{S}^{\mathrm{n}} \rightarrow \mathrm{S}^{\mathrm{n}}$               | <span style="border: 1px solid red;"><math>\mathrm{S}^{\mathrm{n}} \rightarrow \mathrm{S}^{\mathrm{n}}</math></span> .                                                                       |
| mielke-geometrodynamics_F00177 | 52   | ■ 1.000 | $\delta \varphi$                                                            | $\delta \varphi$                                                            | The operator $\delta \varphi$ of <i>small variations</i> $\varphi+\delta \varphi$ of the fields $\varphi$ is satisfying the Leibniz                                                          |
| mielke-geometrodynamics_F00178 | 52   | ■ 1.000 | $\varphi+\delta \varphi$                                                    | $\varphi+\delta \varphi$                                                    | The operator $\delta \varphi$ of <i>small variations</i> $\varphi+\delta \varphi$ of the fields $\varphi$ is satisfying the Leibniz                                                          |
| mielke-geometrodynamics_F00179 | 53   | ■ 0.937 | $(n-1)$                                                                     | $(n-1)$                                                                     | be the $(n-1)$ -form of the <i>field momenta</i> which are <i>canonically conjugated</i> to $d \varphi$ . It                                                                                 |
| mielke-geometrodynamics_F00180 | 53   | ■ 0.937 | $d \varphi$                                                                 | $d \varphi$                                                                 | be the $(n-1)$ -form of the <i>field momenta</i> which are <i>canonically conjugated</i> to $d \varphi$ . It                                                                                 |
| mielke-geometrodynamics_F00181 | 53   | ■ 1.000 | $\delta$                                                                    | $\delta$                                                                    | after $\delta$ has been interchanged with the integration. The last term is an exact form.                                                                                                   |
| mielke-geometrodynamics_F00182 | 53   | ■ 1.000 | $s$                                                                         | $s$                                                                         | where $s$ is the signature of the pseudo-Riemannian manifold.                                                                                                                                |

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| mielke-geometrodynamics_F00183 | 53   | ■ 1.000 | $d \, d \, \text{\texttt{\textbackslash}equiv} \, 0$                                                                                                                                                                                                                                                                                                            | $dd \equiv 0$                                                                  | where the last term vanishes identically, due to $dd \equiv 0$ . Following the physically                               |
| mielke-geometrodynamics_F00184 | 53   | ■ 0.913 | $\text{\texttt{\textbackslash}mathrm{x}}^{\text{\texttt{\textbackslash}i}}$                                                                                                                                                                                                                                                                                     | $x^i$                                                                          | quently does not depend upon the coordinates $x^i$ explicitly, we get the well-known                                    |
| mielke-geometrodynamics_F00185 | 54   | ■ 1.000 | $\text{\texttt{\textbackslash}varphi} \, \text{\texttt{\textbackslash}rightarrow} \, \text{\texttt{\textbackslash}psi} := \text{\texttt{\textbackslash}left}(\text{\texttt{\textbackslash}varphi}, \, \text{\texttt{\textbackslash}Theta}^{\text{\texttt{\textbackslash}1}}) := d \, \text{\texttt{\textbackslash}varphi} \text{\texttt{\textbackslash}right})$ | $\varphi \rightarrow \psi := \left( \varphi, \Theta^{(1)} := d\varphi \right)$ | via the substitution $\varphi \rightarrow \psi := (\varphi, \Theta^{(1)} := d\varphi)$ into a first order system of the |
| mielke-geometrodynamics_F00186 | 54   | ■ 1.000 | $\text{\texttt{\textbackslash}mathscr{D}}(M)$                                                                                                                                                                                                                                                                                                                   | $\mathscr{D}(M)$                                                               | respect to the group $\mathscr{D}(M)$ of general coordinate-transformations of the spacetime                            |
| mielke-geometrodynamics_F00187 | 54   | ■ 1.000 | $\text{\texttt{\textbackslash}mathscr{G}}$                                                                                                                                                                                                                                                                                                                      | $\mathscr{G}$                                                                  | labels, but as well with respect to the “internal covariance group” $\mathscr{G}$ of gauge trans-                       |
| mielke-geometrodynamics_F00188 | 54   | ■ 0.989 | $L_{\text{\texttt{\textbackslash}text} \, \text{\texttt{\textbackslash}mat} \, \text{\texttt{\textbackslash}}} \}$                                                                                                                                                                                                                                              | $L_{\text{mat}}$                                                               | The invariance of $L_{\text{mat}}$ with respect to $\mathscr{D}(M)$ is already guaranteed by the use                    |
| mielke-geometrodynamics_F00189 | 55   | ■ 1.000 | $\text{\texttt{\textbackslash}bar{\text{\texttt{\textbackslash}varphi}}} \, \text{\texttt{\textbackslash}mathscr{0}} \, \text{\texttt{\textbackslash}varphi}$                                                                                                                                                                                                   | $\overline{\varphi} \mathscr{O} \varphi$                                       | depend upon $\overline{\varphi} \mathscr{O} \varphi$ , where $\mathscr{O}$ denotes a gauge-covariant operator           |
| mielke-geometrodynamics_F00190 | 55   | ■ 1.000 | $\text{\texttt{\textbackslash}mathscr{0}}$                                                                                                                                                                                                                                                                                                                      | $\mathscr{O}$                                                                  | depend upon $\overline{\varphi} \mathscr{O} \varphi$ , where $\mathscr{O}$ denotes a gauge-covariant operator           |
| mielke-geometrodynamics_F00191 | 55   | ■ 0.996 | $\text{\texttt{\textbackslash}omega}^{\text{\texttt{\textbackslash}prime}}$                                                                                                                                                                                                                                                                                     | $\omega'$                                                                      | by means of a 1-form $\omega'$ in such a way that the term $GdG^{-1}$ resulting from gauge                              |
| mielke-geometrodynamics_F00192 | 55   | ■ 0.996 | $\text{\texttt{\textbackslash}mathrm{GdG}}^{\text{\texttt{\textbackslash}{-1}}}$                                                                                                                                                                                                                                                                                | $GdG^{-1}$                                                                     | by means of a 1-form $\omega'$ in such a way that the term $GdG^{-1}$ resulting from gauge                              |
| mielke-geometrodynamics_F00193 | 55   | ■ 1.000 | $\text{\texttt{\textbackslash}mathrm{G}} \, \text{\texttt{\textbackslash}in} \, \text{\texttt{\textbackslash}mathscr{G}}_{\text{\texttt{\textbackslash}mathrm{V}}}$                                                                                                                                                                                             | $G \in \mathscr{G}_V$                                                          | concerning the “actively” applied gauge transformation $G \in \mathscr{G}_V$ . For the derivation                       |
| mielke-geometrodynamics_F00194 | 55   | ■ 1.000 | $\{ \, \}^{\text{\texttt{\textbackslash}mathrm{G}}} \, \text{\texttt{\textbackslash}omega}$                                                                                                                                                                                                                                                                     | ${}^G\omega$                                                                   | matrix ${}^G\omega$ of the connection relative to a basis ${}^G\text{b}$ , which is locally “rotated” by                |
| mielke-geometrodynamics_F00195 | 55   | ■ 1.000 | $\{ \, \}^{\text{\texttt{\textbackslash}mathrm{G}}} \, \text{\texttt{\textbackslash}mathrm{b}}$                                                                                                                                                                                                                                                                 | ${}^G\text{b}$                                                                 | matrix ${}^G\omega$ of the connection relative to a basis ${}^G\text{b}$ , which is locally “rotated” by                |
| mielke-geometrodynamics_F00196 | 55   | ■ 1.000 | $\text{\texttt{\textbackslash}mathrm{G}}^{\text{\texttt{\textbackslash}{-1}}}$                                                                                                                                                                                                                                                                                  | $G^{-1}$                                                                       | If $G$ ist substituted by $G^{-1}$ exactly the relation (3.2.3) comes into existence, which                             |
| mielke-geometrodynamics_F00197 | 56   | ■ 0.749 | $\{ \, \}_{\text{\texttt{\textbackslash}sigma}_{\text{\texttt{\textbackslash}alpha}}}^{\text{\texttt{\textbackslash}{*}}} \, \text{\texttt{\textbackslash}omega}$                                                                                                                                                                                               | ${}_{\sigma_\alpha}^* \omega$                                                  | image or pull-back $\sigma_\alpha^* \omega$ due to the defining properties of a connection (see: KN I,                  |
| mielke-geometrodynamics_F00198 | 56   | ■ 0.818 | $G^{\text{\texttt{\textbackslash}{-1}}} \, \text{\texttt{\textbackslash}omega}$                                                                                                                                                                                                                                                                                 | $G^{-1}\omega$                                                                 | The transformed connection $G^{-1}\omega$ deviates from the original one solely by a gauge-                             |
| mielke-geometrodynamics_F00199 | 56   | ■ 1.000 | $\text{\texttt{\textbackslash}bar{\text{\texttt{\textbackslash}omega}}}$                                                                                                                                                                                                                                                                                        | $\overline{\omega}$                                                            | For a so-called “copy” $\overline{\omega}$ of a connection $\omega$ , this expression is to be taken as different       |
| mielke-geometrodynamics_F00200 | 56   | ■ 1.000 | $\text{\texttt{\textbackslash}bar{\text{\texttt{\textbackslash}Omega}}} = \text{\texttt{\textbackslash}mathrm{G}}^{\text{\texttt{\textbackslash}{-1}}} \, \text{\texttt{\textbackslash}Omega} \, \text{\texttt{\textbackslash}mathrm{G}}$                                                                                                                       | $\overline{\Omega} = G^{-1}\Omega G$                                           | from zero, although - according to the definition - its curvature $\overline{\Omega} = G^{-1}\Omega G$ is               |
| mielke-geometrodynamics_F00201 | 56   | ■ 1.000 | $d^{\text{\texttt{\textbackslash}{*}}} \text{\texttt{\textbackslash}omega} \, \text{\texttt{\textbackslash}simeq} \, 0$                                                                                                                                                                                                                                         | $d^*\omega \simeq 0$                                                           | transversal or Coulomb condition $d^*\omega \simeq 0$ does not fix uniquely (large) gauge trans-                        |
| mielke-geometrodynamics_F00202 | 56   | ■ 1.000 | $A = A_{\text{\texttt{\textbackslash}i}} \, d \, x^{\text{\texttt{\textbackslash}i}}$                                                                                                                                                                                                                                                                           | $A = A_i dx^i$                                                                 | tions of Maxwell, the electromagnetic four-potential $A = A_i dx^i$ , a one-form, is the                                |
| mielke-geometrodynamics_F00203 | 56   | ■ 1.000 | $F$                                                                                                                                                                                                                                                                                                                                                             | $F$                                                                            | fundamental variable, whereas Faraday’s field strength $F$ is a derived concept and                                     |
| mielke-geometrodynamics_F00204 | 57   | ■ 0.992 | $d \, d \, \text{\texttt{\textbackslash}Phi} \, \text{\texttt{\textbackslash}equiv} \, 0$                                                                                                                                                                                                                                                                       | $dd\Phi \equiv 0$                                                              | On the other hand, $dd\Phi \equiv 0$ for any exterior p-form $\Phi$ , due to the Poincaré lemma.                        |
| mielke-geometrodynamics_F00205 | 57   | ■ 0.992 | $\text{\texttt{\textbackslash}Phi}$                                                                                                                                                                                                                                                                                                                             | $\Phi$                                                                         | On the other hand, $dd\Phi \equiv 0$ for any exterior p-form $\Phi$ , due to the Poincaré lemma.                        |
| mielke-geometrodynamics_F00206 | 57   | ■ 1.000 | $d \, F \, \text{\texttt{\textbackslash}equiv} \, 0$                                                                                                                                                                                                                                                                                                            | $dF \equiv 0$                                                                  | Thus if $dF \equiv 0$ , there exists locally a one-form $A$ such that $F = dA$ . However, the                           |
| mielke-geometrodynamics_F00207 | 57   | ■ 1.000 | $F = d \, A$                                                                                                                                                                                                                                                                                                                                                    | $F = dA$                                                                       | Thus if $dF \equiv 0$ , there exists locally a one-form $A$ such that $F = dA$ . However, the                           |
| mielke-geometrodynamics_F00208 | 57   | ■ 0.981 | $U(1)$                                                                                                                                                                                                                                                                                                                                                          | $U(1)$                                                                         | local $U(1)$ gauge transformation                                                                                       |
| mielke-geometrodynamics_F00209 | 57   | ■ 1.000 | $D \, \text{\texttt{\textbackslash}rightarrow} \, \text{\texttt{\textbackslash}mathscr{D}} := D + i \, e \, A$                                                                                                                                                                                                                                                  | $D \rightarrow \mathscr{D} := D + ieA$                                         | Although, $A$ is needed for the standard minimal coupling $D \rightarrow \mathscr{D} := D + ieA$ to                     |

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| mielke-geometrodynamics_F00210 | 57   | ■ 1.000 | L                                                                                      | $L$                                                                               | For a Lagrangian formulation, let us depart from the four-form $L$ which consist of                                                                                         |
| mielke-geometrodynamics_F00211 | 57   | ■ 0.980 | \Psi                                                                                   | $\Psi$                                                                            | the gauge field part and $L_{\text{mat}}$ governing the matter field $\Psi$ and its minimal coupling                                                                        |
| mielke-geometrodynamics_F00212 | 57   | ■ 1.000 | S=\int L                                                                               | $S = \int L$                                                                      | usually be assumed to be of first order in the fields. Stationarity of the action $S = \int L$                                                                              |
| mielke-geometrodynamics_F00213 | 57   | ■ 1.000 | H                                                                                      | $H$                                                                               | The excitation $H$ is the field momentum conjugated to $A$ , i.e.                                                                                                           |
| mielke-geometrodynamics_F00214 | 58   | ■ 1.000 | d d H \equiv 0                                                                         | $ddH \equiv 0$                                                                    | $ddH \equiv 0$ , the field equations imposes ‘on shell’ that the electric current $j$ is con-                                                                               |
| mielke-geometrodynamics_F00215 | 58   | ■ 1.000 | j                                                                                      | $j$                                                                               | $ddH \equiv 0$ , the field equations imposes ‘on shell’ that the electric current $j$ is con-                                                                               |
| mielke-geometrodynamics_F00216 | 58   | ■ 1.000 | C:=A \wedge F=A \wedge d A                                                             | $C := A \wedge F = A \wedge dA$                                                   | where $C := A \wedge F = A \wedge dA$ is the abelian Chern–Simons term, known to violate                                                                                    |
| mielke-geometrodynamics_F00217 | 58   | ■ 1.000 | P                                                                                      | $P$                                                                               | parity $P$ .                                                                                                                                                                |
| mielke-geometrodynamics_F00218 | 58   | ■ 0.885 | \stackrel{\theta}{L}=-\left(\cos \theta F \wedge * F+\sin \theta F \wedge F\right) / 2 | $\stackrel{\theta}{L} = -(\cos \theta F \wedge * F + \sin \theta F \wedge F) / 2$ | $\stackrel{\theta}{L} = -(\cos \theta F \wedge * F + \sin \theta F \wedge F) / 2$ , where $\theta$ is the ‘vacuum’ angle of duality rotations or                            |
| mielke-geometrodynamics_F00219 | 58   | ■ 0.885 | \theta                                                                                 | $\theta$                                                                          | $\stackrel{\theta}{L} = -(\cos \theta F \wedge * F + \sin \theta F \wedge F) / 2$ , where $\theta$ is the ‘vacuum’ angle of duality rotations or                            |
| mielke-geometrodynamics_F00220 | 59   | ■ 0.950 | *                                                                                      | $*$                                                                               | The Hodge dual $*$ depends on the metric $g$ and on the orientation of the manifold, as                                                                                     |
| mielke-geometrodynamics_F00221 | 59   | ■ 0.950 | g                                                                                      | $g$                                                                               | The Hodge dual $*$ depends on the metric $g$ and on the orientation of the manifold, as                                                                                     |
| mielke-geometrodynamics_F00222 | 59   | ■ 1.000 | \eta                                                                                   | $\eta$                                                                            | where $\eta$ is the standard volume four-form. The Pontryagin four-form can be written                                                                                      |
| mielke-geometrodynamics_F00223 | 59   | ■ 1.000 | \partial \mathfrak{D} / \partial t                                                     | $\partial \mathfrak{D} / \partial t$                                              | <sup>3</sup> The displacement current $\partial \mathfrak{D} / \partial t$ , where $\mathfrak{D} = \varepsilon \mathbf{E}$ , was anticipated 1839 by James Mac Cullagh, cf. |
| mielke-geometrodynamics_F00224 | 59   | ■ 1.000 | \mathfrak{D}=\varepsilon \mathbf{E}                                                    | $\mathfrak{D} = \varepsilon \mathbf{E}$                                           | <sup>3</sup> The displacement current $\partial \mathfrak{D} / \partial t$ , where $\mathfrak{D} = \varepsilon \mathbf{E}$ , was anticipated 1839 by James Mac Cullagh, cf. |
| mielke-geometrodynamics_F00225 | 60   | ■ 1.000 | \vartheta_{[\alpha} \wedge \Sigma_{\beta]}=0                                           | $\vartheta_{[\alpha} \wedge \Sigma_{\beta]} = 0$                                  | and turns out to be symmetric, i.e., $\vartheta_{[\alpha} \wedge \Sigma_{\beta]} = 0$ . For non-vanishing charge current                                                    |
| mielke-geometrodynamics_F00226 | 60   | ■ 1.000 | B=\frac{1}{2} B_{i j} d x^i \wedge d x^j                                               | $B = \frac{1}{2} B_{ij} dx^i \wedge dx^j$                                         | two-form $B = \frac{1}{2} B_{ij} dx^i \wedge dx^j$ are varied <i>independently</i> , slightly reminiscent of the                                                            |
| mielke-geometrodynamics_F00227 | 60   | ■ 1.000 | B                                                                                      | $B$                                                                               | Independent variations with respect to $A$ and $B$ lead to $dB \cong 0$ and to the <i>constraint</i> of                                                                     |
| mielke-geometrodynamics_F00228 | 60   | ■ 1.000 | d B \cong 0                                                                            | $dB \cong 0$                                                                      | Independent variations with respect to $A$ and $B$ lead to $dB \cong 0$ and to the <i>constraint</i> of                                                                     |
| mielke-geometrodynamics_F00229 | 60   | ■ 1.000 | F:=d A \cong 0                                                                         | $F := dA \cong 0$                                                                 | vanishing field strength $F := dA \cong 0$ . This implies, however, that such a primordial                                                                                  |
| mielke-geometrodynamics_F00230 | 60   | ■ 1.000 | d C                                                                                    | $dC$                                                                              | (3.3.22) only by a boundary term $dC$ derived from a <i>Chern–Simons</i> (CS) three-form                                                                                    |
| mielke-geometrodynamics_F00231 | 60   | ■ 1.000 | H=d B                                                                                  | $H = dB$                                                                          | excitation $H = dB$ , the Kalb–Ramond <i>axion</i> three-form.                                                                                                              |
| mielke-geometrodynamics_F00232 | 61   | ■ 1.000 | \delta d C / \delta A=d F \equiv 0                                                     | $\delta dC / \delta A = dF \equiv 0$                                              | can be recovered via the variation of the Pontryagin term, e.g., $\delta dC / \delta A = dF \equiv 0$                                                                       |
| mielke-geometrodynamics_F00233 | 61   | ■ 0.998 | \psi=\psi_{i} d x^i                                                                    | $\psi = \psi_i dx^i$                                                              | where $\psi = \psi_i dx^i$ is a one-form or a covector; cf. LUCCHESI et al. (1993). In a                                                                                    |
| mielke-geometrodynamics_F00234 | 61   | ■ 1.000 | s \psi=D \phi                                                                          | $s\psi = D\phi$                                                                   | BRST quantization, due to the zero mode $s\psi = D\phi$ with $s\phi = 0$ , the “topological”                                                                                |
| mielke-geometrodynamics_F00235 | 61   | ■ 1.000 | s \phi=0                                                                               | $s\phi = 0$                                                                       | BRST quantization, due to the zero mode $s\psi = D\phi$ with $s\phi = 0$ , the “topological”                                                                                |
| mielke-geometrodynamics_F00236 | 61   | ■ 1.000 | B F                                                                                    | $BF$                                                                              | to topological field theory is the $BF$ formalism which, in the case of gravity, was                                                                                        |

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| mielke-geometrodynamics_F00237 | 61   | ■ 1.000 | $F=\{ \ }^{*} B$                               | $F = {}^*B$                                    | the definition of the field strength (3.3.24), but as a bonus, from the relation $F = {}^*B$                                           |
| mielke-geometrodynamics_F00238 | 61   | ■ 0.998 | $j:=\delta L_{\text {matter }} / \delta A$     | $j := \delta L_{\text{matter}} / \delta A$     | where $j := \delta L_{\text{matter}} / \delta A$ is the matter current.                                                                |
| mielke-geometrodynamics_F00239 | 61   | ■ 1.000 | $d j \cong 0$                                  | $dj \cong 0$                                   | which “on shell,” is conserved classically, i.e., $dj \cong 0$ . In general, this includes                                             |
| mielke-geometrodynamics_F00240 | 61   | ■ 0.999 | $d A$                                          | $dA$                                           | <i>Pauli-type terms</i> generated by the variation of the Lagrangian with respect to $dA$ ; cf.                                        |
| mielke-geometrodynamics_F00241 | 62   | ■ 1.000 | $V(B)$                                         | $V(B)$                                         | The quadratic term in (3.3.28) can be generalized to a four-form potential $V(B)$ .                                                    |
| mielke-geometrodynamics_F00242 | 62   | ■ 1.000 | $d B+j \cong 0$                                | $dB + j \cong 0$                               | Then the field equation $dB + j \cong 0$ together with the relation $F \cong \partial V / \partial B$ for                              |
| mielke-geometrodynamics_F00243 | 62   | ■ 1.000 | $F \cong \partial V / \partial B$              | $F \cong \partial V / \partial B$              | Then the field equation $dB + j \cong 0$ together with the relation $F \cong \partial V / \partial B$ for                              |
| mielke-geometrodynamics_F00244 | 62   | ■ 1.000 | $-B \wedge F+B \wedge B \wedge B / 3$          | $-B \wedge F + B \wedge B \wedge B / 3$        | systems with a cubic term $-B \wedge F + B \wedge B \wedge B / 3$ are directly related to Chern–Simons                                 |
| mielke-geometrodynamics_F00245 | 62   | ■ 0.999 | $C:=A \wedge F-A \wedge A \wedge A / 3$        | $C := A \wedge F - A \wedge A \wedge A / 3$    | theories departing from the noninvariant three-form $C := A \wedge F - A \wedge A \wedge A / 3$ ; cf.                                  |
| mielke-geometrodynamics_F00246 | 62   | ■ 1.000 | $F=-B \wedge B$                                | $F = -B \wedge B$                              | $F = -B \wedge B$ together with the Bianchi identity $DB = DF \equiv 0$ correspond, in 3D                                              |
| mielke-geometrodynamics_F00247 | 62   | ■ 1.000 | $D B=D F \equiv 0$                             | $DB = DF \equiv 0$                             | $F = -B \wedge B$ together with the Bianchi identity $DB = DF \equiv 0$ correspond, in 3D                                              |
| mielke-geometrodynamics_F00248 | 62   | ■ 0.999 | $\Gamma_{\alpha}^j(\mathrm{m})$                | $\Gamma_{\alpha}^j(\mathrm{m})$                | Speaking a priori, the connection $\omega$ and the corresponding gauge potentials $\Gamma_{\alpha}^j(\mathrm{m})$                      |
| mielke-geometrodynamics_F00249 | 62   | ■ 1.000 | $d \omega$                                     | $d\omega$                                      | only of its first derivative $d\omega$ . The only <i>gauge-covariant</i> object of this kind that meets                                |
| mielke-geometrodynamics_F00250 | 62   | ■ 0.668 | $\mathscr{F}$                                  | $\mathscr{F}$                                  | 4-forms into 0-forms, i.e., scalar functions. In quantized gauge theories, $\mathscr{F}$ and ${}^*\mathscr{F}$                         |
| mielke-geometrodynamics_F00251 | 62   | ■ 0.668 | $\{ \ }^{*} \mathscr{F}$                       | ${}^*\mathscr{F}$                              | 4-forms into 0-forms, i.e., scalar functions. In quantized gauge theories, $\mathscr{F}$ and ${}^*\mathscr{F}$                         |
| mielke-geometrodynamics_F00252 | 63   | ■ 0.731 | $\Pi^{\Omega}$                                 | $\Pi^{\Omega}$                                 | The 2-form $\Pi^{\Omega}$ of the <i>field momenta</i> being <i>canonically conjugated</i> to $\Omega$ are formally                     |
| mielke-geometrodynamics_F00253 | 63   | ■ 0.894 | $n-1$                                          | $n - 1$                                        | $(n - 1)$ -form $\tau$ of the matter current as a source. The latter is canonically conjugate                                          |
| mielke-geometrodynamics_F00254 | 63   | ■ 0.894 | $\tau$                                         | $\tau$                                         | $(n - 1)$ -form $\tau$ of the matter current as a source. The latter is canonically conjugate                                          |
| mielke-geometrodynamics_F00255 | 64   | ■ 1.000 | $\delta L / \delta \omega$                     | $\delta L / \delta \omega$                     | the variation $\delta L / \delta \omega$ of the total system. They are complemented by the Bianchi                                     |
| mielke-geometrodynamics_F00256 | 64   | ■ 1.000 | (1978, 1980)                                   | (1978, 1980)                                   | it has been shown by HORNDESKI (1978, 1980) that a field theory of Yang–Mills                                                          |
| mielke-geometrodynamics_F00257 | 65   | ■ 1.000 | $\mathrm{H}^3$                                 | $\mathrm{H}^3$                                 | hypersurface $\mathrm{H}^3$ or its boundary $\partial \mathrm{H}^3$ , respectively. Note that $\iota$ involves not only                |
| mielke-geometrodynamics_F00258 | 65   | ■ 1.000 | $\partial \mathrm{H}^3$                        | $\partial \mathrm{H}^3$                        | hypersurface $\mathrm{H}^3$ or its boundary $\partial \mathrm{H}^3$ , respectively. Note that $\iota$ involves not only                |
| mielke-geometrodynamics_F00259 | 65   | ■ 1.000 | $\iota$                                        | $\iota$                                        | hypersurface $\mathrm{H}^3$ or its boundary $\partial \mathrm{H}^3$ , respectively. Note that $\iota$ involves not only                |
| mielke-geometrodynamics_F00260 | 65   | ■ 0.879 | $\mathrm{SU}(\mathrm{N})$                      | $\mathrm{SU}(\mathrm{N})$                      | In more generalized versions, these theories make use of $\mathrm{SU}(\mathrm{N})$ as a structure group,                               |
| mielke-geometrodynamics_F00261 | 66   | ■ 0.971 | $L(\mathscr{F}, {}^*\mathscr{F})$              | $L(\mathscr{F}, {}^*\mathscr{F})$              | $L(\mathscr{F}, {}^*\mathscr{F})$ is of course a mere empirical one and can be answered only by means of                               |
| mielke-geometrodynamics_F00262 | 66   | ■ 0.435 | $\mathrm{I}^j=\mathrm{i} \sigma^j$             | $\mathbb{I}^j = i\sigma^j$                     | with the infinitesimal generators $\mathbb{I}^j = i\sigma^j$ , thus obtaining the group $\mathrm{SU}(2) \times \mathrm{SU}(2) \approx$ |
| mielke-geometrodynamics_F00263 | 66   | ■ 0.435 | $\mathrm{SU}(2) \times \mathrm{SU}(2) \approx$ | $\mathrm{SU}(2) \times \mathrm{SU}(2) \approx$ | with the infinitesimal generators $\mathbb{I}^j = i\sigma^j$ , thus obtaining the group $\mathrm{SU}(2) \times \mathrm{SU}(2) \approx$ |

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| mielke-geometrodynamics_F00264 | 66   | ■ 0.765 | $\mathrm{SO}(4)$                                 | $\mathrm{SO}(4)$                                   | $\mathrm{SO}(4)$ by proceeding from the array of matrices                                                                                |
| mielke-geometrodynamics_F00265 | 67   | ■ 0.998 | $\mathrm{h}=\mathrm{h}( \mathrm{x} )$            | $h = h( x )$                                       | i.e., $h = h( x )$ in the spherically symmetric case. A similar nonlinear equation occurs                                                |
| mielke-geometrodynamics_F00266 | 67   | ■ 0.499 | $\mathrm{M}^4$                                   | $M^4$                                              | cf. JACKIW et al. (1977). In a flat space $M^4$ with Euclidean signature $s = 0$ , this is a                                             |
| mielke-geometrodynamics_F00267 | 67   | ■ 0.999 | $\mathrm{k}=1$                                   | $k = 1$                                            | in the case of $k = 1$ ('T HOOFT 1977). The configuration as given above depends                                                         |
| mielke-geometrodynamics_F00268 | 67   | ■ 0.512 | $5\mathrm{k}+4$                                  | $5k + 4$                                           | on $5k + 4$ parameters. From studies of related objects in algebraic topology it is                                                      |
| mielke-geometrodynamics_F00269 | 67   | ■ 1.000 | $\mathbb{R}^4\cup\{\infty\}\approx\mathrm{S}^4$  | $\mathbb{R}^4 \cup \{\infty\} \approx S^4$         | of the Yang–Mills equation in a Euclidean space $\mathbb{R}^4 \cup \{\infty\} \approx S^4$ , which is confor-                            |
| mielke-geometrodynamics_F00270 | 68   | ■ 1.000 | $ \text{-}\infty\rangle$                         | $ \text{-}\infty\rangle$                           | transition amplitude between the asymptotic states $ \text{-}\infty\rangle$ and $\langle\infty $ . This ampli-                           |
| mielke-geometrodynamics_F00271 | 68   | ■ 1.000 | $\langle\infty $                                 | $\langle\infty $                                   | transition amplitude between the asymptotic states $ \text{-}\infty\rangle$ and $\langle\infty $ . This ampli-                           |
| mielke-geometrodynamics_F00272 | 68   | ■ 1.000 | $\mathrm{M}_{\omega}$                            | $\mathcal{M}_{\omega}$                             | configuration space $\mathcal{M}_{\omega}$ of all inequivalent gauge fields, i.e., of all possible connec-                               |
| mielke-geometrodynamics_F00273 | 68   | ■ 0.991 | $\mathrm{M}_{\omega}=\mathrm{C}/\mathcal{G}_p$   | $\mathcal{M}_{\omega} = \mathcal{C}/\mathcal{G}_p$ | $\mathcal{M}_{\omega} = \mathcal{C}/\mathcal{G}_p$ has to be taken as the configuration space proper. Under not too stringent            |
| mielke-geometrodynamics_F00274 | 68   | ■ 0.999 | $\mu[\omega]$                                    | $\mu[\omega]$                                      | can be shown that the “measure” $\mu[\omega]$ for the functional integration, as given by the                                            |
| mielke-geometrodynamics_F00275 | 69   | ■ 0.976 | $S$                                              | $S$                                                | regime of gauge fields, the so-called <i>S-duality</i> .                                                                                 |
| mielke-geometrodynamics_F00276 | 69   | ■ 1.000 | $\operatorname{Spin}^c(4)$                       | $\mathrm{Spin}^c(4)$                               | 2006) coupled to spinors $\psi$ regarded as a lift from the frame bundle to $\mathrm{Spin}^c(4)$ .                                       |
| mielke-geometrodynamics_F00277 | 69   | ■ 0.988 | $\psi=\psi_L+\psi_R=P_{-}\psi+P_{+}\psi$         | $\psi = \psi_L + \psi_R = P_{-}\psi + P_{+}\psi$   | cf. JOST et al. (1996), where here the decomposition $\psi = \psi_L + \psi_R = P_{-}\psi + P_{+}\psi$                                    |
| mielke-geometrodynamics_F00278 | 70   | ■ 1.000 | $\bar{\psi}$                                     | $\bar{\psi}$                                       | The variation with respect to $\bar{\psi}$ and $A$ leads to the <i>convective</i> -type spinor equation                                  |
| mielke-geometrodynamics_F00279 | 70   | ■ 1.000 | $R$                                              | $R$                                                | (3.6.1). For a nonnegative scalar curvature $R$ in the Weizenböck formula, all solutions                                                 |
| mielke-geometrodynamics_F00280 | 70   | ■ 1.000 | $\psi=0$                                         | $\psi = 0$                                         | satisfy $\psi \equiv 0$ and then are $U(1)$ instantons.                                                                                  |
| mielke-geometrodynamics_F00281 | 70   | ■ 0.996 | $F^{\pm}$                                        | $F^{\pm}$                                          | $F^{\pm}$ resembles (MIELKE 1984) the modified (double) duality ansatz                                                                   |
| mielke-geometrodynamics_F00282 | 71   | ■ 1.000 | $\mathrm{A}_{\alpha}^{\mathrm{j}}$               | $A_{\alpha}^j$                                     | gauge potentials $A_{\alpha}^j$ in the Lagrangian n-form. However, there is only one massless                                            |
| mielke-geometrodynamics_F00283 | 71   | ■ 1.000 | $\varphi^j$                                      | $\varphi^j$                                        | fields $\varphi^j$ with f components but also the vector fields $\Theta^{(1)j}$ will be coupled to the                                   |
| mielke-geometrodynamics_F00284 | 71   | ■ 1.000 | $\Theta^{(1)j}$                                  | $\Theta^{(1)j}$                                    | fields $\varphi^j$ with f components but also the vector fields $\Theta^{(1)j}$ will be coupled to the                                   |
| mielke-geometrodynamics_F00285 | 71   | ■ 0.902 | $\phi^{(0)}$                                     | $\phi^{(0)}$                                       | the 0-form $\phi^{(0)}$ and the 1-form $\Theta^{(1)}$ with values in the adjoint representation of G.                                    |
| mielke-geometrodynamics_F00286 | 71   | ■ 0.902 | $\Theta^{(1)}$                                   | $\Theta^{(1)}$                                     | the 0-form $\phi^{(0)}$ and the 1-form $\Theta^{(1)}$ with values in the adjoint representation of G.                                    |
| mielke-geometrodynamics_F00287 | 71   | ■ 1.000 | $\mathrm{A}_{\alpha}^{\mathrm{j}}(\mathrm{m})$   | $A_{\alpha}^j(m)$                                  | gies to the Higgs model. The coupling of the gauge potentials $A_{\alpha}^j(m)$ to the Higgs                                             |
| mielke-geometrodynamics_F00288 | 71   | ■ 1.000 | $\delta L_{\mathrm{H}}/\delta\bar{\phi}^{(0)}$   | $\delta L_H/\delta\bar{\phi}^{(0)}$                | the G-equivalence principle. From the variations and $\delta L_H/\delta\bar{\phi}^{(0)}$ and $\delta L_H/\delta\bar{\Theta}^{(1)}$ , the |
| mielke-geometrodynamics_F00289 | 71   | ■ 1.000 | $\delta L_{\mathrm{H}}/\delta\bar{\Theta}^{(1)}$ | $\delta L_H/\delta\bar{\Theta}^{(1)}$              | the G-equivalence principle. From the variations and $\delta L_H/\delta\bar{\phi}^{(0)}$ and $\delta L_H/\delta\bar{\Theta}^{(1)}$ , the |
| mielke-geometrodynamics_F00290 | 72   | ■ 1.000 | $\mu$                                            | $\mu$                                              | It should be noted that the parameter $\mu$ in (3.7.5) complies formally with an imaginary                                               |

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|--------------------------------|------|---------|------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| mielke-geometrodynamics_F00291 | 72   | ■ 1.000 | $\varphi=0$                                                      | $\varphi = 0$                                                    | mass of the Klein–Gordon-type field. However, the trivial field configuration $\varphi = 0$                                          |
| mielke-geometrodynamics_F00292 | 72   | ■ 1.000 | $ \varphi $                                                      | $ \varphi $                                                      | This is a consequence of the <i>nonlinear term</i> in (3.7.1), which is quartic in $\varphi$ .                                       |
| mielke-geometrodynamics_F00293 | 72   | ■ 1.000 | $\lambda \varphi$                                                | $\lambda \varphi$                                                | quantum-mechanical state $\lambda \varphi$ , for which the vacuum expectation value                                                  |
| mielke-geometrodynamics_F00294 | 72   | ■ 1.000 | $\varphi_{\mathrm{H}}:=\varphi-\mu/\sqrt{\lambda}$               | $\varphi_{\mathrm{H}}:=\varphi-\mu/\sqrt{\lambda}$               | $\varphi_{\mathrm{H}}:=\varphi-\mu/\sqrt{\lambda}$ along with the gauge fields $A_{\alpha}^j$ acquire a real, i.e., physical, mass   |
| mielke-geometrodynamics_F00295 | 72   | ■ 1.000 | $m_{\mathrm{H}}=\sqrt{2}\mu$                                     | $m_{\mathrm{H}}=\sqrt{2}\mu$                                     | that is determined by the curvature of the potential at the bottom, i.e., by $m_{\mathrm{H}}=\sqrt{2}\mu$ ,                          |
| mielke-geometrodynamics_F00296 | 72   | ■ 0.882 | $G/H$                                                            | $G/H$                                                            | field $\varphi$ such that its range is an orbit $G/H$ of $G$ in $V^{\rho}$ with respect to a subgroup $H$ ,                          |
| mielke-geometrodynamics_F00297 | 72   | ■ 1.000 | $v_0\in V^{\rho}$                                                | $v_0\in V^{\rho}$                                                | Let $v_0\in V^{\rho}$ be a fixed vector for which the cross section $\mathring{\varphi}_{\min}:=\sigma(v_0)\in C^{\infty}(V^{\rho})$ |
| mielke-geometrodynamics_F00298 | 72   | ■ 1.000 | $\mathring{\varphi}_{\min}:=\sigma(v_0)\in C^{\infty}(V^{\rho})$ | $\mathring{\varphi}_{\min}:=\sigma(v_0)\in C^{\infty}(V^{\rho})$ | Let $v_0\in V^{\rho}$ be a fixed vector for which the cross section $\mathring{\varphi}_{\min}:=\sigma(v_0)\in C^{\infty}(V^{\rho})$ |
| mielke-geometrodynamics_F00299 | 72   | ■ 0.683 | $ag\in G$                                                        | $ag\in G$                                                        | to find a $g\in G$ that connects any two vectors $v$ and $v_0$ from $V^{\rho}$ via the relation                                      |
| mielke-geometrodynamics_F00300 | 72   | ■ 0.683 | $v_0$                                                            | $v_0$                                                            | to find a $g\in G$ that connects any two vectors $v$ and $v_0$ from $V^{\rho}$ via the relation                                      |
| mielke-geometrodynamics_F00301 | 72   | ■ 1.000 | $v=\rho(g)\cdot v_0$                                             | $v=\rho(g)\cdot v_0$                                             | $v=\rho(g)\cdot v_0$ . An invariant subgroup of $G$ , the so-called isotropy group                                                   |
| mielke-geometrodynamics_F00302 | 73   | ■ 0.996 | $\mathring{\varphi}_{\min}=\sigma(v_0)$                          | $\mathring{\varphi}_{\min}=\sigma(v_0)$                          | can be <i>uniquely</i> adjoined to the fixed $\mathring{\varphi}_{\min}=\sigma(v_0)$ . Then                                          |
| mielke-geometrodynamics_F00303 | 73   | ■ 1.000 | $P(G,M,\pi,\delta)$                                              | $P(G,M,\pi,\delta)$                                              | is a subbundle of $P(G,M,\pi,\delta)$ over the same base space; its structure group is $H$                                           |
| mielke-geometrodynamics_F00304 | 73   | ■ 0.702 | $\mathfrak{h}$                                                   | $\mathfrak{h}$                                                   | <b>Proposition</b> A connection $\omega$ in $P$ is reducible to an $\mathfrak{h}$ -valued connection $\omega_{\mathrm{H}}$ in $Q$ if |
| mielke-geometrodynamics_F00305 | 73   | ■ 0.702 | $\omega_{\mathrm{H}}$                                            | $\omega_{\mathrm{H}}$                                            | <b>Proposition</b> A connection $\omega$ in $P$ is reducible to an $\mathfrak{h}$ -valued connection $\omega_{\mathrm{H}}$ in $Q$ if |
| mielke-geometrodynamics_F00306 | 73   | ■ 0.702 | $Q$                                                              | $Q$                                                              | <b>Proposition</b> A connection $\omega$ in $P$ is reducible to an $\mathfrak{h}$ -valued connection $\omega_{\mathrm{H}}$ in $Q$ if |
| mielke-geometrodynamics_F00307 | 73   | ■ 1.000 | $\omega_0$                                                       | $\omega_0$                                                       | theories of gravity that will be developed later on. Let $\omega_0$ be a timelike component                                          |
| mielke-geometrodynamics_F00308 | 74   | ■ 1.000 | $\phi$                                                           | $\phi$                                                           | of the connection can be identified with the Higgs field $\phi$ without self-interaction:                                            |
| mielke-geometrodynamics_F00309 | 74   | ■ 1.000 | $\lambda\rightarrow 0$                                           | $\lambda\rightarrow 0$                                           | limit $\lambda\rightarrow 0$ may be obtained by identifying $\omega_0$ in the interaction-free Yang–Mills                            |
| mielke-geometrodynamics_F00310 | 74   | ■ 1.000 | $\vec{\sigma}\vec{\omega}$                                       | $\vec{\sigma}\vec{\omega}$                                       | provides the static Yang–Mills potential $\vec{\sigma}\vec{\omega}$ of the monopole. After this identifica-                          |
| mielke-geometrodynamics_F00311 | 74   | ■ 0.503 | (1981,1983)                                                      | (1981,1983)                                                      | (1977), TCHRAKIAN (1981, 1983), PRASAD & ROSSI (1981a, b). The topological                                                           |
| mielke-geometrodynamics_F00312 | 75   | ■ 1.000 | $\sigma$                                                         | $\sigma$                                                         | In this context, it is worthwhile to mention that in the nonlinear $\sigma$ -model, a variant                                        |
| mielke-geometrodynamics_F00313 | 75   | ■ 1.000 | $\mathbb{R}^2\times S^2$                                         | $\mathbb{R}^2\times S^2$                                         | connection in a nontrivial $U(1)$ -bundle over $\mathbb{R}^2\times S^2$                                                              |
| mielke-geometrodynamics_F00314 | 75   | ■ 0.657 | $S^4$                                                            | $S^4$                                                            | classification of $SU(2)$ -bundles over $S^4$ according to the                                                                       |
| mielke-geometrodynamics_F00315 | 78   | ■ 1.000 | $N=2$                                                            | $N=2$                                                            | $N=2$ supersymmetric Yang–Mills theory. Nuclear Phys B 426(1):19–52                                                                  |
| mielke-geometrodynamics_F00316 | 79   | ■ 1.000 | $\ell^*$                                                         | $\ell^*$                                                         | diminutive coupling constant given by the Planck length $\ell^*$ . It may even provide the                                           |
| mielke-geometrodynamics_F00317 | 80   | ■ 0.998 | (1980,1981)                                                      | (1980,1981)                                                      | those theories of gravity, was worked out in more detail by HEHL (1980, 1981) using                                                  |

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| mielke-geometrodynamics_F00318 | 81   | ■ 0.951 | $\mathrm{A}(\mathrm{n}, \mathbb{R})$                                              | $A(n, \mathbb{R})$                                                       | and to start off deductively from the affine group $A(n, \mathbb{R})$ as structure group. As a                                               |
| mielke-geometrodynamics_F00319 | 81   | ■ 1.000 | $\mathrm{A}^{\mathrm{n}}$                                                         | $A^n$                                                                    | Euclidean space $E^n$ , regarded as an affine space $A^n$ .                                                                                  |
| mielke-geometrodynamics_F00320 | 81   | ■ 0.999 | $\mathrm{a}=\left(\mathrm{a}^{\alpha}\right) \in \mathbb{R}^{\mathrm{n}}$         | $a = (a^\alpha) \in \mathbb{R}^n$                                        | Let $\mathbf{a} = (a^\alpha) \in \mathbb{R}^n$ be a row vector. According to its definition, the action of an                                |
| mielke-geometrodynamics_F00321 | 81   | ■ 0.992 | $\mathrm{x} \in \mathrm{A}^{\mathrm{n}}$                                          | $x \in A^n$                                                              | element of $A(n, \mathbb{R})$ on a vector $\mathbf{x} \in A^n$ can expressed as the composition of a general                                 |
| mielke-geometrodynamics_F00322 | 81   | ■ 1.000 | $\tilde{\mathrm{x}}:=\mathrm{binom}{\mathrm{x}}{1} \in \mathbb{R}^{\mathrm{n}+1}$ | $\tilde{x} := \begin{pmatrix} x \\ 1 \end{pmatrix} \in \mathbb{R}^{n+1}$ | In the following, it will be convenient to introduce a vector $\tilde{x} := \begin{pmatrix} x \\ 1 \end{pmatrix} \in \mathbb{R}^{n+1}$ with  |
| mielke-geometrodynamics_F00323 | 81   | ■ 1.000 | $\mathrm{n}+1$                                                                    | $n + 1$                                                                  | $n + 1$ components and to replace the group action (4.1.1) by the abstract transfor-                                                         |
| mielke-geometrodynamics_F00324 | 81   | ■ 0.998 | $\tilde{\mathrm{x}}$                                                              | $\tilde{x}$                                                              | mations of $\tilde{x}$ via the matrices                                                                                                      |
| mielke-geometrodynamics_F00325 | 81   | ■ 0.890 | $\mathrm{GL}(\mathrm{n}+1, \mathbb{R})$                                           | $GL(n + 1, \mathbb{R})$                                                  | This faithful representation of $A(n, \mathbb{R})$ by a subgroup of $GL(n + 1, \mathbb{R})$ shows clearly                                    |
| mielke-geometrodynamics_F00326 | 81   | ■ 0.623 | $\mathrm{a} \in \mathbb{R}^{\mathrm{n}}$                                          | $a \in \mathbb{R}^n$                                                     | of the translations $a \in \mathbb{R}^n$ and the linear transformations $g \in GL(n, \mathbb{R})$ . The rule for                             |
| mielke-geometrodynamics_F00327 | 81   | ■ 0.623 | $\mathrm{g} \in \mathrm{GL}(\mathrm{n}, \mathbb{R})$                              | $g \in GL(n, \mathbb{R})$                                                | of the translations $a \in \mathbb{R}^n$ and the linear transformations $\mathbf{g} \in GL(n, \mathbb{R})$ . The rule for                    |
| mielke-geometrodynamics_F00328 | 81   | ■ 1.000 | $0(1, \mathrm{n}-1) \subset \mathrm{GL}(\mathrm{n}, \mathbb{R})$                  | $0(1, n - 1) \subset GL(n, \mathbb{R})$                                  | $0(1, n - 1) \subset GL(n, \mathbb{R})$ . The invariance of the (flat) metric ground form of the                                             |
| mielke-geometrodynamics_F00329 | 81   | ■ 1.000 | $\mathfrak{a}(\mathrm{n}, \mathbb{R})$                                            | $\mathfrak{a}(n, \mathbb{R})$                                            | $\mathfrak{a}(n, \mathbb{R})$ of the affine group, the analogous decomposition                                                               |
| mielke-geometrodynamics_F00330 | 82   | ■ 0.999 | $L_{\alpha \beta} \in \mathfrak{gl}(\mathrm{n}, \mathbb{R})$                      | $L_{\alpha\beta} \in \mathfrak{gl}(n, \mathbb{R})$                       | In these formulas, $L_{\alpha\beta} \in \mathfrak{gl}(n, \mathbb{R})$ and $P_\alpha \in \mathbb{R}^n$ denote the infinitesimal generators of |
| mielke-geometrodynamics_F00331 | 82   | ■ 0.999 | $P_{\alpha} \in \mathbb{R}^{\mathrm{n}}$                                          | $P_\alpha \in \mathbb{R}^n$                                              | In these formulas, $L_{\alpha\beta} \in \mathfrak{gl}(n, \mathbb{R})$ and $P_\alpha \in \mathbb{R}^n$ denote the infinitesimal generators of |
| mielke-geometrodynamics_F00332 | 82   | ■ 1.000 | $\mathrm{GL}(\mathrm{n}, \mathbb{R})$                                             | $GL(n, \mathbb{R})$                                                      | for the tangent space. The general linear group $GL(n, \mathbb{R})$ acts on this basis as follows:                                           |
| mielke-geometrodynamics_F00333 | 82   | ■ 0.833 | $L_{\mathrm{m}}(\mathrm{M})$                                                      | $L_m(M)$                                                                 | It thus generates a space $L_m(M)$ of <i>linear frames of reference</i> at the point m. Let                                                  |
| mielke-geometrodynamics_F00334 | 82   | ■ 0.993 | $\mathrm{U} \subset \mathrm{M}$                                                   | $U \subset M$                                                            | remember that in a neighborhood $U \subset M$ of a point $m \in M$ with coordinates $x^i$ ,                                                  |
| mielke-geometrodynamics_F00335 | 82   | ■ 0.994 | $\mathrm{e}_{\mathrm{a}}(\mathrm{m})$                                             | $e_a(m)$                                                                 | a basis vector $e_a(m)$ of $T_m(M)$ may always be written in the way given above.                                                            |
| mielke-geometrodynamics_F00336 | 82   | ■ 0.852 | $\mathbb{R}^{\mathrm{n}^2}$                                                       | $\mathbb{R}^{n^2}$                                                       | an open neighborhood of $\mathbb{R}^{n^2}$ . Since $e_\alpha^j(m)$ represents a nonsingular $n \times n$ matrix,                             |
| mielke-geometrodynamics_F00337 | 82   | ■ 0.852 | $\mathrm{e}_{\alpha}^j(\mathrm{m})$                                               | $e_\alpha^j(m)$                                                          | an open neighborhood of $\mathbb{R}^{n^2}$ . Since $e_\alpha^j(m)$ represents a nonsingular $n \times n$ matrix,                             |
| mielke-geometrodynamics_F00338 | 82   | ■ 0.852 | $n \times n$                                                                      | $n \times n$                                                             | an open neighborhood of $\mathbb{R}^{n^2}$ . Since $e_\alpha^j(m)$ represents a nonsingular $n \times n$ matrix,                             |
| mielke-geometrodynamics_F00339 | 82   | ■ 0.987 | $\left(\mathrm{x}^i, \mathrm{e}_{\alpha}^{-j}\right)$                             | $(x^i, e_\alpha^{-j})$                                                   | the collection $(x^i, e_\alpha^j)$ may be regarded as a choice of coordinates of the Cartesian                                               |
| mielke-geometrodynamics_F00340 | 82   | ■ 0.840 | $\mathrm{U} \times \mathrm{GL}(\mathrm{n}, \mathbb{R})$                           | $U \times GL(n, \mathbb{R})$                                             | product $\mathbf{U} \times GL(n, \mathbb{R})$ . These coordinates arise from the original domain $\pi^{-1}(U)$ , for                         |
| mielke-geometrodynamics_F00341 | 82   | ■ 0.840 | $\pi^{-1}(\mathrm{U})$                                                            | $\pi^{-1}(U)$                                                            | product $U \times GL(n, \mathbb{R})$ . These coordinates arise from the original domain $\pi^{-1}(U)$ , for                                  |
| mielke-geometrodynamics_F00342 | 83   | ■ 0.858 | $\operatorname{GL}(\mathrm{n})$                                                   | $GL(n)$                                                                  | which the projection $\pi$ of the principal fiber bundle with the structure group $GL(n,$                                                    |
| mielke-geometrodynamics_F00343 | 83   | ■ 0.899 | $\vartheta$                                                                       | $\vartheta$                                                              | of linear frames. This is the case if an $\mathbb{R}^n$ -valued 1-form $\vartheta$ exists that not only is                                   |
| mielke-geometrodynamics_F00344 | 83   | ■ 0.825 | $\mathrm{n}=\operatorname{dim} \mathrm{M}$                                        | $n = \dim M$                                                             | this close contact and due to the fact that $\vartheta$ has rank $n = \dim M$ , it is natural to call                                        |

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| mielke-geometrodynamics_F00345 | 83   | ■ 0.394 | $\mathbb{R}^n$                                                                                                         | $\mathbb{R}^n$                                         | associated with $L(M)$ . Its standard fiber is $\mathbb{R}^n$ , and thus $T(M)$ has the same dimension                                                |
| mielke-geometrodynamics_F00346 | 83   | ■ 0.569 | $\mathrm{A}(\mathrm{M}) / \mathbb{R}^n$                                                                                | $A(M)/\mathbb{R}^n$                                    | The possibility of identifying $L(M)$ with the quotient bundle $A(M)/\mathbb{R}^n$ results from                                                       |
| mielke-geometrodynamics_F00347 | 84   | ■ 0.999 | $\mathrm{e}_{\alpha}(\mathrm{m})$                                                                                      | $e_{\alpha}(m)$                                        | the basis vectors $e_{\alpha}(m)$ in the tangent space $T_m(M)$ at the point $m \in M$ appears to                                                     |
| mielke-geometrodynamics_F00348 | 84   | ■ 0.593 | $\mathrm{T}=\mathbb{R}^n$                                                                                              | $T = \mathbb{R}^n$                                     | group $T = \mathbb{R}^n$ of translations can be regarded as the quotient space $A(n, \mathbb{R})/GL(n, \mathbb{R})$ ,                                 |
| mielke-geometrodynamics_F00349 | 84   | ■ 0.593 | $\mathrm{A}(\mathrm{n}, \mathbb{R}), \mathbb{R} / \mathrm{GL}(\mathrm{n}, \mathbb{R})$                                 | $A(n, \mathbb{R})/GL(n, \mathbb{R})$                   | group $T = \mathbb{R}^n$ of translations can be regarded as the quotient space $A(n, \mathbb{R})/GL(n, \mathbb{R})$ ,                                 |
| mielke-geometrodynamics_F00350 | 84   | ■ 1.000 | $\mathrm{T}=\mathrm{P} / \mathrm{L}$                                                                                   | $T = P/L$                                              | realized by translations acting on the coset space $T = P/L$ , which is isomorphic to the                                                             |
| mielke-geometrodynamics_F00351 | 84   | ■ 0.550 | $\widetilde{\omega}$                                                                                                   | $\widetilde{\omega}$                                   | connection. Let $\widetilde{\omega}$ be the 1-form of such a <i>generalized affine connection</i> , and let                                           |
| mielke-geometrodynamics_F00352 | 84   | ■ 0.999 | $\chi: \mathrm{L}(\mathrm{M}) \rightarrow 0_{\mathrm{m}} \times \mathrm{L}(\mathrm{M}) \subset \mathrm{A}(\mathrm{M})$ | $\chi : L(M) \rightarrow 0_m \times L(M) \subset A(M)$ | $\chi : L(M) \rightarrow 0_m \times L(M) \subset A(M)$ be the natural injection of linear frames into the                                             |
| mielke-geometrodynamics_F00353 | 85   | ■ 0.448 | $\chi^* \widetilde{\omega}$                                                                                            | $\chi^* \widetilde{\omega}$                            | image $\chi^* \widetilde{\omega}$ of the connection $\widetilde{\omega}$ is an $\mathfrak{a}(n, \mathbb{R})$ -valued 1-form in $L(M)$ that decomposes |
| mielke-geometrodynamics_F00354 | 85   | ■ 0.507 | $\omega^L$                                                                                                             | $\omega^L$                                             | tation (4.1.2) of $A(n, \mathbb{R})$ . In the above formula, $\omega^L$ denotes a $\mathfrak{gl}(n, \mathbb{R})$ -valued 1-form                       |
| mielke-geometrodynamics_F00355 | 85   | ■ 0.507 | $\mathfrak{gl}(n, \mathbb{R})$                                                                                         | $\mathfrak{gl}(n, \mathbb{R})$                         | tation (4.1.2) of $A(n, \mathbb{R})$ . In the above formula, $\omega^L$ denotes a $\mathfrak{gl}(n, \mathbb{R})$ -valued 1-form                       |
| mielke-geometrodynamics_F00356 | 85   | ■ 0.912 | $\omega^T$                                                                                                             | $\omega^T$                                             | in $M$ , while $\omega^T$ is an $\mathbb{R}^n$ -valued 1-form, i.e., a tensorial form of the representation                                           |
| mielke-geometrodynamics_F00357 | 85   | ■ 0.990 | $\mathrm{H} \subset \mathrm{G}$                                                                                        | $H \subset G$                                          | structure with respect to a subgroup $H \subset G$ .                                                                                                  |
| mielke-geometrodynamics_F00358 | 85   | ■ 0.944 | $\widetilde{\Omega}$                                                                                                   | $\widetilde{\Omega}$                                   | The curvature $\widetilde{\Omega}$ in $A(M)$ can be derived from $\widetilde{\omega}$ by an analogous application of                                  |
| mielke-geometrodynamics_F00359 | 85   | ■ 1.000 | $[\omega^T, \omega^T]$                                                                                                 | $[\omega^T, \omega^T]$                                 | The commutator $[\omega^T, \omega^T]$ occurring in the calculation                                                                                    |
| mielke-geometrodynamics_F00360 | 85   | ■ 0.989 | $\omega^T, \omega^L$                                                                                                   | $\omega^T, \omega^L$                                   | which is induced by $\omega^L$ . It is of importance to realize that the commutator $[\omega^T, \omega^L]$                                            |
| mielke-geometrodynamics_F00361 | 86   | ■ 1.000 | $\mathcal{A}(n, \mathbb{R})$                                                                                           | $\mathcal{A}(n, \mathbb{R})$                           | of local translations are subgroups of $\mathcal{A}(n, \mathbb{R})$ . Taking the cross section in the asso-                                           |
| mielke-geometrodynamics_F00362 | 86   | ■ 0.996 | $C^{\infty}$                                                                                                           | $C^{\infty}$                                           | ciated bundle is abbreviated by $C^{\infty}$ , and $\mathrm{Ad}$ denotes the adjoint representation with                                              |
| mielke-geometrodynamics_F00363 | 86   | ■ 0.998 | $GL(n, \mathbb{R})$                                                                                                    | $GL(n, \mathbb{R})$                                    | respect to $GL(n, \mathbb{R})$ . Due to its construction, the group of local translations $\mathcal{T}(n, \mathbb{R})$                                |
| mielke-geometrodynamics_F00364 | 86   | ■ 0.998 | $\mathcal{T}(n, \mathbb{R})$                                                                                           | $\mathcal{T}(n, \mathbb{R})$                           | respect to $GL(n, \mathbb{R})$ . Due to its construction, the group of local translations $\mathcal{T}(n, \mathbb{R})$                                |
| mielke-geometrodynamics_F00365 | 86   | ■ 1.000 | $\mathrm{Diff}(n, \mathbb{R})$                                                                                         | $\mathrm{Diff}(n, \mathbb{R})$                         | is <i>locally</i> isomorphic to the group of active diffeomorphisms $\mathrm{Diff}(n, \mathbb{R})$ of the man-                                        |
| mielke-geometrodynamics_F00366 | 86   | ■ 0.905 | $A(n, \mathbb{R})_{\mathrm{H}}$                                                                                        | $(n, \mathbb{R})$                                      | $\mathrm{Diff}(n, \mathbb{R})$ contains the $(n + n^2)$ -dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of <i>holonomic</i> affine                 |
| mielke-geometrodynamics_F00367 | 86   | ■ 0.905 | $(n + n^2)$                                                                                                            | $(n + n^2)$                                            | $\mathrm{Diff}(n, \mathbb{R})$ contains the $(n + n^2)$ -dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of <i>holonomic</i> affine                 |
| mielke-geometrodynamics_F00368 | 86   | ■ 0.905 | $A(n, \mathbb{R})_{\mathrm{H}}$                                                                                        | $A(n, \mathbb{R})_{\mathrm{H}}$                        | $\mathrm{Diff}(n, \mathbb{R})$ contains the $(n + n^2)$ -dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of <i>holonomic</i> affine                 |
| mielke-geometrodynamics_F00369 | 86   | ■ 0.999 | $P_i = \partial_i := \partial / \partial x^i$                                                                          | $P_i = \partial_i := \partial / \partial x^i$          | transformations as a subgroup, generated by the vector fields $P_i = \partial_i := \partial / \partial x^i$ and                                       |
| mielke-geometrodynamics_F00370 | 86   | ■ 1.000 | $L^i_j = x^i \partial_j$                                                                                               | $L^i_j = x^i \partial_j$                               | $L^i_j = x^i \partial_j$ . Note that differentiable coordinate transformations, which leave exterior                                                  |

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| mielke-geometrodynamics_F00371 | 86   | ■ 1.000 | $\chi$                                                           | $\chi$                                                       | the natural injection $\chi$ , this formula decomposes into                                                                       |
| mielke-geometrodynamics_F00372 | 87   | ■ 1.000 | $\mathrm{V}^{\mathrm{id}}$                                       | $V^{\mathrm{id}}$                                            | in $V^{\mathrm{id}}$ consists of an orbit $\mathbb{R}^n$ of $A(n, \mathbb{R})$ . The isotropy group of the zero section           |
| mielke-geometrodynamics_F00373 | 87   | ■ 0.928 | $\stackrel{\circ}{\varphi} = \sigma(\mathrm{GL}(n, \mathbb{R}))$ | $\overset{\circ}{\varphi} = \sigma(0_m)$                     | $\overset{\circ}{\varphi} = \sigma(0_m)$ is $H = \mathrm{GL}(n, \mathbb{R})$ , i.e., the general linear group. Then the subbundle |
| mielke-geometrodynamics_F00374 | 87   | ■ 0.928 | $H = \mathrm{GL}(n, \mathbb{R})$                                 | $H = \mathrm{GL}(n, \mathbb{R})$                             | $\overset{\circ}{\varphi} = \sigma(0_m)$ is $H = \mathrm{GL}(n, \mathbb{R})$ , i.e., the general linear group. Then the subbundle |
| mielke-geometrodynamics_F00375 | 87   | ■ 0.863 | $\widetilde{\vartheta}$                                          | $\widetilde{\vartheta}$                                      | bution $\widetilde{\vartheta}$ transforms under $\mathcal{A}_p$ like                                                              |
| mielke-geometrodynamics_F00376 | 87   | ■ 0.863 | $\mathcal{A}_p$                                                  | $\mathcal{A}_p$                                              | bution $\widetilde{\vartheta}$ transforms under $\mathcal{A}_p$ like                                                              |
| mielke-geometrodynamics_F00377 | 88   | ■ 0.968 | $D = (\chi \widetilde{D})$                                       | $D = (\chi \widetilde{D})$                                   | to $D = (\chi \widetilde{D})$ , it follows that the torsion 2-form transforms like                                                |
| mielke-geometrodynamics_F00378 | 88   | ■ 0.999 | $\mathcal{G}_p$                                                  | $\mathcal{G}_p$                                              | i.e., as a vector with respect to the subgroup $\mathcal{G}_p$ of linear gauge transformations. As                                |
| mielke-geometrodynamics_F00379 | 89   | ■ 0.993 | $\widetilde{D} = D$                                              | $\widetilde{D} = D$                                          | $\widetilde{D} = D$ it remains valid) the affine connection to the linear $\mathfrak{gl}(n, \mathbb{R})$ -valued connection       |
| mielke-geometrodynamics_F00380 | 90   | ■ 1.000 | $\mathcal{C} \subset M$                                          | $\mathcal{C} \subset M$                                      | to a smooth curve $\mathcal{C} \subset M$ that is parametrized by $s(t)$ . In order to satisfy                                    |
| mielke-geometrodynamics_F00381 | 90   | ■ 1.000 | $F(m, \vartheta) : M \times T^*(M) \rightarrow \mathbb{R}^+$     | $F(m, \vartheta) : M \times T^*(M) \rightarrow \mathbb{R}^+$ | where $F(m, \vartheta) : M \times T^*(M) \rightarrow \mathbb{R}^+$ is a smooth, <i>positive</i> , and <i>homogeneous</i> funda-   |
| mielke-geometrodynamics_F00382 | 91   | ■ 1.000 | $\mathcal{C}$                                                    | $\mathcal{C}$                                                | ance of (4.3.4) versus admissible reparametrizations of the curve $\mathcal{C}$ . Generally, these                                |
| mielke-geometrodynamics_F00383 | 91   | ■ 1.000 | $(2, 0)$                                                         | $(2, 0)$                                                     | Subsequently, a covariant symmetric tensor field of degree (2, 0) is imprinted onto                                               |
| mielke-geometrodynamics_F00384 | 91   | ■ 1.000 | $g_{ij} \in C^\infty(M)$                                         | $g_{ij} \in C^\infty(M)$                                     | Indeed, the functions $g_{ij} \in C^\infty(M)$ describe “concerning the chosen, arbitrary coor-                                   |
| mielke-geometrodynamics_F00385 | 92   | ■ 0.689 | $e_\alpha^i$                                                     | $e_\alpha^i$                                                 | 1-forms $\vartheta$ admitted. Since the $e_\alpha^i$ have been considered as local bundle coordinates                             |
| mielke-geometrodynamics_F00386 | 92   | ■ 1.000 | $\mathfrak{D}(s, n - s)$                                         | $\mathfrak{D}(s, n - s)$                                     | $\mathfrak{D}(s, n - s)$ -valued connection $\omega^g$ , the constraint                                                           |
| mielke-geometrodynamics_F00387 | 92   | ■ 1.000 | $\omega^g$                                                       | $\omega^g$                                                   | $\mathfrak{D}(s, n - s)$ -valued connection $\omega^g$ , the constraint                                                           |
| mielke-geometrodynamics_F00388 | 92   | ■ 0.532 | $\phi^{(\circ)}$                                                 | $\phi^{(\circ)}$                                             | is a necessary and sufficient condition. For tensor-valued 0-forms $\phi^{(\circ)}$ , the exterior                                |
| mielke-geometrodynamics_F00389 | 92   | ■ 0.651 | $\Gamma_i^{\alpha\beta}$                                         | $\Gamma_i^{\alpha\beta}$                                     | where the $\Gamma_i^{\alpha\beta}$ denote the (local) <i>Ricci rotation coefficients</i> . Due to the fact that                   |
| mielke-geometrodynamics_F00390 | 92   | ■ 1.000 | $L_{\alpha\beta}$                                                | $L_{\alpha\beta}$                                            | the infinitesimal generators $L_{\alpha\beta}$ become skew after the restriction to the orthogonal                                |
| mielke-geometrodynamics_F00391 | 92   | ■ 1.000 | $\omega^{\alpha\beta} = -\omega^{\beta\alpha}$                   | $\omega^{\alpha\beta} = -\omega^{\beta\alpha}$               | subgroup, the constraint (4.3.14) analogously yields $\omega^{\alpha\beta} = -\omega^{\beta\alpha}$ , i.e., the skew-             |
| mielke-geometrodynamics_F00392 | 92   | ■ 0.988 | $m_o \in M$                                                      | $m_o \in M$                                                  | neighborhood of a point $m_o \in M$ depends entirely on Minkowski’s metric tensor $o_{ij}$                                        |
| mielke-geometrodynamics_F00393 | 92   | ■ 0.988 | $o_{ij}$                                                         | $o_{ij}$                                                     | neighborhood of a point $m_o \in M$ depends entirely on Minkowski’s metric tensor $o_{ij}$                                        |
| mielke-geometrodynamics_F00394 | 93   | ■ 1.000 | $H = O(s, n - s)$                                                | $H = O(s, n - s)$                                            | gauge theory has only the orthogonal “isotropy group” $H = O(s, n - s)$ as resid-                                                 |
| mielke-geometrodynamics_F00395 | 93   | ■ 1.000 | $\mathfrak{o}(s, n - s)$                                         | $\mathfrak{o}(s, n - s)$                                     | the reduction of the linear connection $\omega^L$ to an $\mathfrak{o}(s, n - s)$ -valued connection $\omega^g$                    |
| mielke-geometrodynamics_F00396 | 93   | ■ 0.999 | $L^g(M)$                                                         | $L^g(M)$                                                     | in $L^g(M)$ . This consistency does not, as is implied by HANSON & REGGE (1979),                                                  |

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| mielke-geometrodynamics_F00397 | 93   | ■ 1.000 | $\mathrm{D} \vartheta = 0$                                                 | $D\vartheta = 0$                                                           | demand the vanishing of the torsion, i.e., $D\vartheta = 0$ . In a Lagrangian formulation of                                                    |
| mielke-geometrodynamics_F00398 | 93   | ■ 0.995 | $\Theta$                                                                   | $\Theta$                                                                   | Higgs field $\vartheta$ , i.e., the canonical <i>orthogonal</i> 1-form, <sup>4</sup> and the resulting torsion $\Theta$                         |
| mielke-geometrodynamics_F00399 | 93   | ■ 0.913 | $\vartheta^{\mathrm{g}}$                                                   | $\vartheta^{\mathrm{g}}$                                                   | <sup>4</sup> As for the nomenclature, the 1-form $\vartheta$ , which is restricted by (4.3.10), will not be denoted by $\vartheta^{\mathrm{g}}$ |
| mielke-geometrodynamics_F00400 | 93   | ■ 1.000 | $\Omega^{\mathrm{g}}$                                                      | $\Omega^{\mathrm{g}}$                                                      | and thus differs from $\omega^{\mathrm{g}}$ and $\Omega^{\mathrm{g}}$ , since we prefer to stick to (4.3.10) or (4.3.9), respectively, as       |
| mielke-geometrodynamics_F00401 | 94   | ■ 0.997 | $\omega^{\{\}}$                                                            | $\omega^{\{\}}$                                                            | Here $\omega^{\{\}}$ denotes the 1-form of the torsion-free Christoffel-type connection for which                                               |
| mielke-geometrodynamics_F00402 | 94   | ■ 0.890 | $K$                                                                        | $K$                                                                        | between the 2-form $\Theta$ and the 1-form $K$ of the so-called <i>contortion</i> , having values                                               |
| mielke-geometrodynamics_F00403 | 94   | ■ 1.000 | $\mathrm{U}_{\mathrm{n}}$                                                  | $\mathrm{U}_{\mathrm{n}}$                                                  | denoted by $\mathrm{U}_{\mathrm{n}}$ , in contrast to a purely Riemannian manifold, which is symbolized by $\mathrm{V}_{\mathrm{n}}$ .          |
| mielke-geometrodynamics_F00404 | 94   | ■ 1.000 | $\mathrm{V}_{\mathrm{n}}$                                                  | $\mathrm{V}_{\mathrm{n}}$                                                  | denoted by $\mathrm{U}_{\mathrm{n}}$ , in contrast to a purely Riemannian manifold, which is symbolized by $\mathrm{V}_{\mathrm{n}}$ .          |
| mielke-geometrodynamics_F00405 | 95   | ■ 0.456 | $R_{\mathrm{i} \mathrm{j} \mathrm{c}}^{\{\dots \mathrm{d}\}}$              | $R_{ijc}^{\{\dots d\}}$                                                    | In this formula, $R_{ijc}^{\{\dots d\}}$ are the mixed components of the Riemannian curvature tensor,                                           |
| mielke-geometrodynamics_F00406 | 95   | ■ 0.999 | $R_{\mathrm{i} \mathrm{j} \mathrm{k} \mathrm{l}}^{\{\}}$                   | $R_{ijkl}^{\{\}}$                                                          | holonomic components $R_{ijcl}^{\{\}}$ have the following symmetries with respect to an inter-                                                  |
| mielke-geometrodynamics_F00407 | 95   | ■ 0.666 | $S \ S \ B$                                                                | $SSB$                                                                      | cation of a Higgs–Kibble-type mechanism of $SSB$ that resulted in the construction of                                                           |
| mielke-geometrodynamics_F00408 | 95   | ■ 1.000 | $\mathrm{O}(1, \mathrm{n}-1)$                                              | $\mathrm{O}(1, \mathrm{n}-1)$                                              | $\mathrm{O}(1, \mathrm{n}-1)$ as the “residual” symmetry of the theory via the general linear group                                             |
| mielke-geometrodynamics_F00409 | 95   | ■ 0.778 | $\operatorname{GL}(\mathrm{n}, \mathbb{R})$                                | $\mathrm{GL}(\mathrm{n}, \mathbb{R})$                                      | $\mathrm{GL}(\mathrm{n}, \mathbb{R})$ .                                                                                                         |
| mielke-geometrodynamics_F00410 | 96   | ■ 1.000 | $\mathrm{R}_{\alpha \beta}^{\mathrm{cd}}(\mathrm{m})$                      | $\mathrm{R}_{\alpha\beta}^{\mathrm{cd}}(\mathrm{m})$                       | spacetime-dependent “structure constants” $\mathrm{R}_{\alpha\beta}^{\mathrm{cd}}(\mathrm{m})$ occur in the generic case. (The                  |
| mielke-geometrodynamics_F00411 | 96   | ■ 1.000 | $\vartheta, \Theta, \Omega^{\mathrm{g}}$                                   | $\vartheta, \Theta, \Omega^{\mathrm{g}}$                                   | However, contrasting with $\vartheta, \Theta, \Omega^{\mathrm{g}}$ (see (4.2.20), (4.2.14), and (4.2.13)), the                                  |
| mielke-geometrodynamics_F00412 | 97   | ■ 1.000 | $L_{\mathrm{g}}$                                                           | $L_{\mathrm{g}}$                                                           | only via of the forms $\Theta$ and $\Omega^{\mathrm{g}}$ . The variation of $L_{\mathrm{g}}$ , regarded as a total differential,                |
| mielke-geometrodynamics_F00413 | 97   | ■ 1.000 | $t$                                                                        | $t$                                                                        | defines the <i>translational</i> and ( <i>Lorentz</i> -)rotational canonical gauge field momenta $\Pi^{\mathrm{T}}$                             |
| mielke-geometrodynamics_F00414 | 97   | ■ 1.000 | $\Pi^{\mathrm{T}}$                                                         | $\Pi^{\mathrm{T}}$                                                         | defines the <i>translational</i> and ( <i>Lorentz</i> -)rotational canonical gauge field momenta $\Pi^{\mathrm{T}}$                             |
| mielke-geometrodynamics_F00415 | 97   | ■ 0.999 | $\Pi^{\mathrm{L}}$                                                         | $\Pi^{\mathrm{L}}$                                                         | and $\Pi^{\mathrm{L}}$ . Due to the special role of $\vartheta$ , there occurs additionally the (n-1)-form $E$ ,                                |
| mielke-geometrodynamics_F00416 | 97   | ■ 0.999 | $E$                                                                        | $E$                                                                        | and $\Pi^{\mathrm{L}}$ . Due to the special role of $\vartheta$ , there occurs additionally the (n-1)-form $E$ ,                                |
| mielke-geometrodynamics_F00417 | 97   | ■ 0.998 | $\Pi_{\square}^{\mathrm{T}}$                                               | $\Pi_{\square}^{\mathrm{T}}$                                               | where the “contortional” 1-form $\Pi_{\square}^{\mathrm{T}}$ , having values in the Lie algebra of the                                          |
| mielke-geometrodynamics_F00418 | 97   | ■ 0.931 | $\Pi^{\mathrm{T}} := \left[ \Pi_{\square}^{\mathrm{T}}, \vartheta \right]$ | $\Pi^{\mathrm{T}} := \left[ \Pi_{\square}^{\mathrm{T}}, \vartheta \right]$ | $\Pi^{\mathrm{T}} := [\Pi_{\square}^{\mathrm{T}}, \vartheta]$ . This is in complete analogy to the relation (4.3.24) for the contor-            |
| mielke-geometrodynamics_F00419 | 97   | ■ 1.000 | $\delta \vartheta$                                                         | $\delta \vartheta$                                                         | The application of Hamilton’s principle to the independent variations of $S$ for $\delta \vartheta$                                             |
| mielke-geometrodynamics_F00420 | 97   | ■ 1.000 | $\delta \omega^{\mathrm{g}}$                                               | $\delta \omega^{\mathrm{g}}$                                               | and $\delta \omega^{\mathrm{g}}$ leads to the <i>Euler–Lagrange</i> equations                                                                   |
| mielke-geometrodynamics_F00421 | 98   | ■ 0.844 | $\Sigma$                                                                   | $\Sigma$                                                                   | fied with the 3-form $\Sigma$ , the energy–momentum current, and the 3-form $\tau$ of the spin                                                  |
| mielke-geometrodynamics_F00422 | 98   | ■ 0.999 | $\lambda^{\varphi} = 0$                                                    | $\lambda^{\varphi} = 0$                                                    | If the Euler–Lagrange equations $\lambda^{\varphi} = 0$ of the matter fields are satisfied, it will                                             |
| mielke-geometrodynamics_F00423 | 98   | ■ 1.000 | $\tau_{\mathrm{s}}$                                                        | $\tau_{\mathrm{s}}$                                                        | turn out that $\Sigma$ and $\tau_{\mathrm{s}}$ are the field momenta of matter that are <i>canonically conjugate</i>                            |

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| mielke-geometrodynamics_F00424 | 98   | ■ 1.000 | \Upsilon                                                                                              | $\Upsilon$                                                                                    | hypermomentum current $\Upsilon$ of the matter fields in such a generalization (HEHL &                                                                                                  |
| mielke-geometrodynamics_F00425 | 100  | ■ 1.000 | \Gamma^{\alpha\beta}=-\Gamma^{\beta\alpha}                                                            | $\Gamma^{\alpha\beta} = -\Gamma^{\beta\alpha}$                                                | symmetric connection 1-form $\Gamma^{\alpha\beta} = -\Gamma^{\beta\alpha}$ is defined by $\delta L = \delta \Gamma^{\alpha\beta} \wedge (\partial L / \partial \Gamma^{\alpha\beta})$ . |
| mielke-geometrodynamics_F00426 | 100  | ■ 1.000 | \delta L=\delta\Gamma^{\alpha\beta}\wedge\left(\partial L/\partial\Gamma^{\alpha\beta}\right)         | $\delta L = \delta \Gamma^{\alpha\beta} \wedge (\partial L / \partial \Gamma^{\alpha\beta})$  | symmetric connection 1-form $\Gamma^{\alpha\beta} = -\Gamma^{\beta\alpha}$ is defined by $\delta L = \delta \Gamma^{\alpha\beta} \wedge (\partial L / \partial \Gamma^{\alpha\beta})$ . |
| mielke-geometrodynamics_F00427 | 100  | ■ 0.629 | S=\int_M L_{\text{mat}}                                                                               | $S = \int_M L_{\text{mat}}$                                                                   | The action $S = \int_M L_{\text{mat}}$ for the matter Lagrangian is, by construction, invari-                                                                                           |
| mielke-geometrodynamics_F00428 | 100  | ■ 1.000 | \operatorname{Diff}(M)                                                                                | $\text{Diff}(M)$                                                                              | ant under the group $\text{Diff}(M)$ of coordinate transformations and local frame rota-                                                                                                |
| mielke-geometrodynamics_F00429 | 100  | ■ 0.924 | \mathscr{T}\subset\operatorname{Diff}(M)                                                              | $\mathcal{T} \subset \text{Diff}(M)$                                                          | one-parameter group of <i>local</i> translations $\mathcal{T} \subset \text{Diff}(M)$ , we employ the $SO(1, 3)$ -                                                                      |
| mielke-geometrodynamics_F00430 | 100  | ■ 0.924 | S O(1,3)                                                                                              | $SO(1, 3)$                                                                                    | one-parameter group of <i>local</i> translations $\mathcal{T} \subset \text{Diff}(M)$ , we employ the $SO(1, 3)$ -                                                                      |
| mielke-geometrodynamics_F00431 | 100  | ■ 0.590 | \mathrm{\xi}_{\lfloor D+\xi\rfloor}                                                                   | $\mathbb{L}_\xi := \xi \rfloor D + D\xi \rfloor$                                              | covariant Lie derivative $\mathbb{L}_\xi := \xi \rfloor D + D\xi \rfloor$ on $M$ with respect to an arbitrary vector                                                                    |
| mielke-geometrodynamics_F00432 | 100  | ■ 0.590 | M                                                                                                     | $M$                                                                                           | covariant Lie derivative $\mathbb{L}_\xi := \xi \rfloor D + D\xi \rfloor$ on $M$ with respect to an arbitrary vector                                                                    |
| mielke-geometrodynamics_F00433 | 100  | ■ 1.000 | \xi                                                                                                   | $\xi$                                                                                         | field $\xi$ . Since $Dg_{\alpha\beta} = 0$ , we obtain                                                                                                                                  |
| mielke-geometrodynamics_F00434 | 100  | ■ 1.000 | D g_{\alpha\beta}=0                                                                                   | $Dg_{\alpha\beta} = 0$                                                                        | field $\xi$ . Since $Dg_{\alpha\beta} = 0$ , we obtain                                                                                                                                  |
| mielke-geometrodynamics_F00435 | 101  | ■ 0.779 | \mathrm{L}_{\xi}L=D\xi jL                                                                             | $L_\xi L = D\xi jL$                                                                           | $\mathbb{L}_\xi L = D\xi \rfloor L$ . Comparing the boundary term, we can read off the identity                                                                                         |
| mielke-geometrodynamics_F00436 | 101  | ■ 1.000 | \xi\rightarrow e_\alpha                                                                               | $\xi \rightarrow e_\alpha$                                                                    | If we replace $\xi \rightarrow e_\alpha$ by the basis frame, then (4.5.4) yields directly the explicit                                                                                  |
| mielke-geometrodynamics_F00437 | 101  | ■ 0.995 | \left.F_{\alpha}^{\text{Max}}\right =\left(e_{\alpha}\right)\wedge j                                  | $F_\alpha^{\text{Max}} = (e_\alpha \rfloor F) \wedge j$                                       | calculus $F_\alpha^{\text{Max}} = (e_\alpha \rfloor F) \wedge j$ .                                                                                                                      |
| mielke-geometrodynamics_F00438 | 101  | ■ 1.000 | \simeq                                                                                                | $\simeq$                                                                                      | no field equations are invoked. <i>Weak</i> identities, which are denoted by $\simeq$ , hold only                                                                                       |
| mielke-geometrodynamics_F00439 | 101  | ■ 0.999 | \delta L/\delta\Psi=0                                                                                 | $\delta L / \delta \Psi = 0$                                                                  | if the matter field equation $\delta L / \delta \Psi = 0$ is satisfied.                                                                                                                 |
| mielke-geometrodynamics_F00440 | 101  | ■ 0.972 | \delta D\Psi=D\delta\Psi+\delta\Gamma_{\alpha}^{\beta}\wedge I^{\alpha}_{\beta}\Psi                   | $\delta D\Psi = D\delta\Psi + \delta \Gamma_{\alpha}^{\beta} \wedge I^{\alpha}_{\beta} \Psi$  | Since $\delta D\Psi = D\delta\Psi + \delta \Gamma_{\alpha}^{\beta} \wedge I^{\alpha}_{\beta} \Psi$ , this is equivalent to                                                              |
| mielke-geometrodynamics_F00441 | 102  | ■ 1.000 | \varepsilon_{\alpha}^{\beta}(x):=\Lambda_{\alpha}^{\beta}(x)-\delta_{\alpha}^{\beta}                  | $\varepsilon_{\alpha}^{\beta}(x) := \Lambda_{\alpha}^{\beta}(x) - \delta_{\alpha}^{\beta}$    | Under an infinitesimal Lorentz rotation $\varepsilon_{\alpha}^{\beta}(x) := \Lambda_{\alpha}^{\beta}(x) - \delta_{\alpha}^{\beta}$ of the frames, we                                    |
| mielke-geometrodynamics_F00445 | 129  | ■ 1.000 | \Gamma_{\alpha}^{\beta}                                                                               | $\Gamma_{\alpha}^{\beta}$                                                                     | contains the linear connection $\Gamma_{\alpha}^{\beta}$ , which is combined with the complex translational                                                                             |
| mielke-geometrodynamics_F00449 | 102  | ■ 0.986 | \overparen{D}                                                                                         | $\widehat{D}$                                                                                 | $\widehat{D}$ with respect to the <i>transposed</i> connection $\widehat{\Gamma}_{\alpha}^{\beta} := \Gamma_{\alpha}^{\beta} + e_{\alpha} \rfloor T^{\beta}$ .                          |
| mielke-geometrodynamics_F00450 | 102  | ■ 0.986 | \left.\overparen{\Gamma}_{\alpha}^{\beta}\right :=\Gamma_{\alpha}^{\beta}+e_{\alpha}\rfloor T^{\beta} | $\widehat{\Gamma}_{\alpha}^{\beta} := \Gamma_{\alpha}^{\beta} + e_{\alpha} \rfloor T^{\beta}$ | $\widehat{D}$ with respect to the <i>transposed</i> connection $\widehat{\Gamma}_{\alpha}^{\beta} := \Gamma_{\alpha}^{\beta} + e_{\alpha} \rfloor T^{\beta}$ .                          |
| mielke-geometrodynamics_F00451 | 102  | ■ 0.999 | \tau_{\alpha\beta}=-\tau_{\beta\alpha}                                                                | $\tau_{\alpha\beta} = -\tau_{\beta\alpha}$                                                    | identity, consists of an intrinsic or <i>spin</i> part $\tau_{\alpha\beta} = -\tau_{\beta\alpha}$ and an orbital part $x_{[\alpha} \wedge \Sigma_{\beta]}$ ,                            |
| mielke-geometrodynamics_F00452 | 102  | ■ 0.999 | x_{[\alpha}\wedge\Sigma_{\beta]}                                                                      | $x_{[\alpha} \wedge \Sigma_{\beta]}$                                                          | identity, consists of an intrinsic or <i>spin</i> part $\tau_{\alpha\beta} = -\tau_{\beta\alpha}$ and an orbital part $x_{[\alpha} \wedge \Sigma_{\beta]}$ ,                            |
| mielke-geometrodynamics_F00453 | 102  | ■ 1.000 | x^i                                                                                                   | $x^i$                                                                                         | coordinates $x^i$ with $x^{\alpha} = \delta_i^{\alpha} x^i$ .                                                                                                                           |
| mielke-geometrodynamics_F00454 | 102  | ■ 1.000 | x^{\alpha}=\delta_i^{\alpha}x^i                                                                       | $x^{\alpha} = \delta_i^{\alpha} x^i$                                                          | coordinates $x^i$ with $x^{\alpha} = \delta_i^{\alpha} x^i$ .                                                                                                                           |
| mielke-geometrodynamics_F00455 | 103  | ■ 1.000 | \xi=\partial/\partial t                                                                               | $\xi = \partial / \partial t$                                                                 | solution, if $\xi = \partial / \partial t$ or $\xi = \partial / \partial \varphi$ , respectively, is chosen. Here no “factor of two”                                                    |

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|--------------------------------|------|---------|------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| mielke-geometrodynamics_F00456 | 103  | ■ 1.000 | $\xi=\partial / \partial \varphi$                                      | $\xi = \partial / \partial \varphi$                                    | solution, if $\xi = \partial / \partial t$ or $\xi = \partial / \partial \varphi$ , respectively, is chosen. Here no “factor of two”                |
| mielke-geometrodynamics_F00457 | 104  | ■ 1.000 | $\pm (\Lambda / 3) r^3$                                                | $\pm (\Lambda / 3) r^3$                                                | complex each contribute <i>one-half</i> of the mass, so that the divergent terms $\pm (\Lambda / 3) r^3$                                            |
| mielke-geometrodynamics_F00458 | 104  | ■ 1.000 | $\Lambda$                                                              | $\Lambda$                                                              | proportional to the cosmological constant $\Lambda$ exactly cancel each other.                                                                      |
| mielke-geometrodynamics_F00459 | 104  | ■ 0.539 | $\Gamma^{(T) \alpha}$                                                  | $\Gamma^{(T) \alpha}$                                                  | includes the true translational potential $\Gamma^{(T) \alpha}$ and the $GL(n, \mathbb{R})$ -gauge connection                                       |
| mielke-geometrodynamics_F00460 | 104  | ■ 1.000 | $\Gamma_{\alpha}^{(L) \beta}$                                          | $\Gamma_{\alpha}^{(L) \beta}$                                          | $\Gamma_{\alpha}^{(L) \beta}$ , as can be deduced from the Möbius-type five-dimensional representation of                                           |
| mielke-geometrodynamics_F00461 | 104  | ■ 1.000 | $A(n, \mathbb{R}) := \mathbb{R}^n \otimes GL(n, \mathbb{R})$           | $A(n, \mathbb{R}) := \mathbb{R}^n \otimes GL(n, \mathbb{R})$           | the affine gauge group $A(n, \mathbb{R}) := \mathbb{R}^n \otimes GL(n, \mathbb{R})$ .                                                               |
| mielke-geometrodynamics_F00462 | 104  | ■ 0.559 | $\widetilde{\xi}$                                                      | $\widetilde{\xi}$                                                      | view: the condition for a parallel transport of an affine vector $\widetilde{\xi}$ around a small closed                                            |
| mielke-geometrodynamics_F00463 | 104  | ■ 0.993 | $\widetilde{\Gamma}$                                                   | $\widetilde{\Gamma}$                                                   | loop by means of $\widetilde{\Gamma}$ reads                                                                                                         |
| mielke-geometrodynamics_F00464 | 104  | ■ 0.272 | $\mathbb{L}_y := y \rfloor D + Dy$                                     | $\mathbb{L}_y := y \rfloor D + Dy$                                     | where $\mathbb{L}_y := y \rfloor D + Dy \rfloor$ denotes the gauge-covariant Lie derivative and $R^{(T) \alpha}$ the                                |
| mielke-geometrodynamics_F00465 | 104  | ■ 0.272 | $R^{(T) \alpha}$                                                       | $R^{(T) \alpha}$                                                       | where $\mathbb{L}_y := y \rfloor D + Dy \rfloor$ denotes the gauge-covariant Lie derivative and $R^{(T) \alpha}$ the                                |
| mielke-geometrodynamics_F00466 | 104  | ■ 1.000 | $y$                                                                    | $y$                                                                    | loop parametrized by $y$ yields                                                                                                                     |
| mielke-geometrodynamics_F00467 | 105  | ■ 1.000 | $A = A_i^J \lambda_J dx^i$                                             | $A = A_i^J \lambda_J dx^i$                                             | where $A = A_i^J \lambda_J dx^i$ is a Yang–Mills-type connection and $P$ is the principal value                                                     |
| mielke-geometrodynamics_F00468 | 105  | ■ 0.996 | $\gamma$                                                               | $\gamma$                                                               | of the integral. If the loop $\gamma$ lies in a field-free region, i.e., one with $U(1)$ field                                                      |
| mielke-geometrodynamics_F00469 | 105  | ■ 1.000 | $F := dA = 0$                                                          | $F := dA = 0$                                                          | strength $F := dA = 0$ or Yang–Mills curvature $F = dA + A \wedge A = 0$ , but encloses                                                             |
| mielke-geometrodynamics_F00470 | 105  | ■ 1.000 | $F = dA + A \wedge A = 0$                                              | $F = dA + A \wedge A = 0$                                              | strength $F := dA = 0$ or Yang–Mills curvature $F = dA + A \wedge A = 0$ , but encloses                                                             |
| mielke-geometrodynamics_F00471 | 105  | ■ 1.000 | $F \neq 0$                                                             | $F \neq 0$                                                             | a “confined” region with nontrivial “magnetic” flux $F \neq 0$ , the potential $A$ can still                                                        |
| mielke-geometrodynamics_F00472 | 105  | ■ 1.000 | $A(4, \mathbb{R})$                                                     | $A(4, \mathbb{R})$                                                     | arising from a gauge theory of the affine structure group $A(4, \mathbb{R})$ .                                                                      |
| mielke-geometrodynamics_F00473 | 105  | ■ 0.924 | $GL(4, \mathbb{R})$                                                    | $GL(4, \mathbb{R})$                                                    | For a so-called manifold $\Psi$ carrying no $GL(4, \mathbb{R})$ -excitations, i.e., no spin, no                                                     |
| mielke-geometrodynamics_F00474 | 105  | ■ 1.000 | $\Gamma^{(T) \alpha} \stackrel{*}{=} 0$                                | $\Gamma^{(T) \alpha} \stackrel{*}{=} 0$                                | of the cone there is $\Gamma^{(T) \alpha} \stackrel{*}{=} 0$ locally.                                                                               |
| mielke-geometrodynamics_F00475 | 105  | ■ 0.651 | (1993, 1994)                                                           | (1993, 1994)                                                           | time manifold; cf. ANANDAN (1993, 1994). Because the coframe is nondegenerate                                                                       |
| mielke-geometrodynamics_F00476 | 105  | ■ 1.000 | $P_{\alpha}$                                                           | $P_{\alpha}$                                                           | Since the physical dimension of $P_{\alpha}$ is $2\pi \hbar / \ell$ , the gravitational analogue of Dirac’s                                         |
| mielke-geometrodynamics_F00477 | 105  | ■ 1.000 | $2 \pi \hbar / \ell$                                                   | $2\pi \hbar / \ell$                                                    | Since the physical dimension of $P_{\alpha}$ is $2\pi \hbar / \ell$ , the gravitational analogue of Dirac’s                                         |
| mielke-geometrodynamics_F00478 | 105  | ■ 1.000 | $\mathcal{M}_n = n M_{\text{Planck}}$                                  | $\mathcal{M}_n = n M_{\text{Planck}}$                                  | i.e., the mass would turn out to be a multiple $\mathcal{M}_n = n M_{\text{Planck}}$ of the (huge) Planck                                           |
| mielke-geometrodynamics_F00479 | 109  | ■ 0.485 | $\Gamma^{\alpha \beta} = \Gamma^{\{j \alpha \beta} - K^{\alpha \beta}$ | $\Gamma^{\alpha \beta} = \Gamma^{\{j \alpha \beta} - K^{\alpha \beta}$ | The Riemann–Cartan (RC) connection $\Gamma^{\alpha \beta} = \Gamma^{\{j \alpha \beta} - K^{\alpha \beta}$ , a one-form, can be                      |
| mielke-geometrodynamics_F00480 | 109  | ■ 0.469 | $\Gamma^{\{\alpha \beta\}}$                                            | $\Gamma^{\{\alpha \beta\}}$                                            | split into the unique Levi-Civita connection $\Gamma^{\{j \alpha \beta}$ of Riemannian geometry and                                                 |
| mielke-geometrodynamics_F00481 | 109  | ■ 1.000 | $K_{\alpha \beta} = -K_{\beta \alpha}$                                 | $K_{\alpha \beta} = -K_{\beta \alpha}$                                 | a <i>contortion</i> one-form $K_{\alpha \beta} = -K_{\beta \alpha}$ , a tensor-valued one-form $K = i K^{\alpha \beta} \sigma_{\alpha \beta} / 4$ , |
| mielke-geometrodynamics_F00482 | 109  | ■ 1.000 | $K = i K^{\alpha \beta} \sigma_{\alpha \beta} / 4$                     | $K = i K^{\alpha \beta} \sigma_{\alpha \beta} / 4$                     | a <i>contortion</i> one-form $K_{\alpha \beta} = -K_{\beta \alpha}$ , a tensor-valued one-form $K = i K^{\alpha \beta} \sigma_{\alpha \beta} / 4$ , |

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| mielke-geometrodynamics_F00483 | 109  | ■ 1.000 | $T^{\alpha}=K^{\alpha}{}_{\beta}\wedge\vartheta^{\beta}$                                                         | $T^{\alpha}=K^{\alpha}{}_{\beta}\wedge\vartheta^{\beta}$                                                         | implicitly related to torsion via $T^{\alpha}=K^{\alpha}{}_{\beta}\wedge\vartheta^{\beta}$ . Recently, it has been stressed by                                       |
| mielke-geometrodynamics_F00484 | 109  | ■ 1.000 | $\Gamma=\Gamma^{\{\}}-K$                                                                                         | $\Gamma=\Gamma^{\{\}}-K$                                                                                         | WEINBERG (2005) that this RC connection $\Gamma=\Gamma^{\{\}}-K$ is just a <i>deformation</i> (or                                                                    |
| mielke-geometrodynamics_F00485 | 109  | ■ 1.000 | $\Gamma^{\{\}}$                                                                                                  | $\Gamma^{\{\}}$                                                                                                  | field redefinition) of the Christoffel connection $\Gamma^{\{\}}$ by the contortion, at least from                                                                   |
| mielke-geometrodynamics_F00486 | 110  | ■ 1.000 | $D>2$                                                                                                            | $D>2$                                                                                                            | renormalizable in $D>2$ without Chern–Simons (CS) terms (ALVES et al. 1999),                                                                                         |
| mielke-geometrodynamics_F00487 | 110  | ■ 1.000 | $j_5$                                                                                                            | $j_5$                                                                                                            | equation relates torsion to spin, i.e., to the axial current $j_5$ in the case of Dirac fields,                                                                      |
| mielke-geometrodynamics_F00488 | 110  | ■ 0.972 | $\bar{\psi}:=\psi^{\dagger}\gamma_0$                                                                             | $\bar{\psi}:=\psi^{\dagger}\gamma_0$                                                                             | $\bar{\psi}:=\psi^{\dagger}\gamma_0$ denotes the Dirac adjoint and $D\psi:=d\psi+\Gamma\wedge\psi$ is the exterior covari-                                           |
| mielke-geometrodynamics_F00489 | 110  | ■ 0.972 | $D\psi:=d\psi+\Gamma\wedge\psi$                                                                                  | $D\psi:=d\psi+\Gamma\wedge\psi$                                                                                  | $\bar{\psi}:=\psi^{\dagger}\gamma_0$ denotes the Dirac adjoint and $D\psi:=d\psi+\Gamma\wedge\psi$ is the exterior covari-                                           |
| mielke-geometrodynamics_F00490 | 110  | ■ 1.000 | $\gamma:=\gamma_{\alpha}\vartheta^{\alpha}$                                                                      | $\gamma:=\gamma_{\alpha}\vartheta^{\alpha}$                                                                      | where $\gamma:=\gamma_{\alpha}\vartheta^{\alpha}$ is the Clifford-algebra-valued coframe satisfying $D\gamma=[\gamma,K]$                                             |
| mielke-geometrodynamics_F00491 | 110  | ■ 1.000 | $D\gamma=[\gamma,K]$                                                                                             | $D\gamma=[\gamma,K]$                                                                                             | where $\gamma:=\gamma_{\alpha}\vartheta^{\alpha}$ is the Clifford-algebra-valued coframe satisfying $D\gamma=[\gamma,K]$                                             |
| mielke-geometrodynamics_F00492 | 110  | ■ 1.000 | $=\gamma_{\alpha}T^{\alpha}$                                                                                     | $=\gamma_{\alpha}T^{\alpha}$                                                                                     | $=\gamma_{\alpha}T^{\alpha}$ , and $T^{\alpha}:=D\vartheta^{\alpha}$ is the torsion two-form.                                                                        |
| mielke-geometrodynamics_F00493 | 110  | ■ 1.000 | $T^{\alpha}:=D\vartheta^{\alpha}$                                                                                | $T^{\alpha}:=D\vartheta^{\alpha}$                                                                                | $=\gamma_{\alpha}T^{\alpha}$ , and $T^{\alpha}:=D\vartheta^{\alpha}$ is the torsion two-form.                                                                        |
| mielke-geometrodynamics_F00494 | 110  | ■ 1.000 | $L_D=\bar{L}_D=L_D^{\dagger}$                                                                                    | $L_D=\bar{L}_D=L_D^{\dagger}$                                                                                    | Since $L_D=\bar{L}_D=L_D^{\dagger}$ even in an unholonomic frame, the minimal coupling pro-                                                                          |
| mielke-geometrodynamics_F00495 | 111  | ■ 1.000 | $\tau_{\alpha\beta\gamma}=\tau_{[\alpha\beta\gamma]}$                                                            | $\tau_{\alpha\beta\gamma}=\tau_{[\alpha\beta\gamma]}$                                                            | with <i>totally antisymmetric</i> components $\tau_{\alpha\beta\gamma}=\tau_{[\alpha\beta\gamma]}$ . Eventually, torsion merely                                      |
| mielke-geometrodynamics_F00496 | 111  | ■ 0.975 | $\mu_{\alpha}=\vartheta_{\alpha}\wedge*j_5/4$                                                                    | $\mu_{\alpha}=\vartheta_{\alpha}\wedge*j_5/4$                                                                    | couples to the <i>spin–energy potential</i> $\mu_{\alpha}=\vartheta_{\alpha}\wedge*j_5/4$ , i.e., to a two-form propor-                                              |
| mielke-geometrodynamics_F00497 | 111  | ■ 0.792 | $\mathrm{GR}_{\parallel}$                                                                                        | $\mathrm{GR}_{\parallel}$                                                                                        | Transitions from EC theory or GR to $\mathrm{GR}_{\parallel}$ can be <i>generated</i> via the Chern–Simons-                                                          |
| mielke-geometrodynamics_F00498 | 111  | ■ 0.701 | $C_{\mathrm{TT}^*}:=\vartheta^{\alpha}\wedge*T_{\alpha}$                                                         | $C_{\mathrm{TT}^*}:=\vartheta^{\alpha}\wedge*T_{\alpha}$                                                         | type term $C_{\mathrm{TT}^*}:=\vartheta^{\alpha}\wedge*T_{\alpha}$ , involving the Hodge dual of the torsion. On the material                                        |
| mielke-geometrodynamics_F00499 | 111  | ■ 1.000 | $L_D\rightarrow L_D+dU$                                                                                          | $L_D\rightarrow L_D+dU$                                                                                          | side, a related change of the Dirac Lagrangian can be induced via $L_D\rightarrow L_D+dU$ ,                                                                          |
| mielke-geometrodynamics_F00500 | 111  | ■ 1.000 | $U=\vartheta^{\alpha}\wedge\mu_{\alpha}$                                                                         | $U=\vartheta^{\alpha}\wedge\mu_{\alpha}$                                                                         | where $U=\vartheta^{\alpha}\wedge\mu_{\alpha}$ features as a superpotential. Then the corresponding boundary                                                         |
| mielke-geometrodynamics_F00501 | 111  | ■ 1.000 | $\hat{\tau}_{\alpha\beta}$                                                                                       | $\hat{\tau}_{\alpha\beta}$                                                                                       | sions (R1) and (R2) of MIELKE et al. (1989), such that the relocalized spin $\hat{\tau}_{\alpha\beta}$                                                               |
| mielke-geometrodynamics_F00502 | 111  | ■ 1.000 | $U=\vartheta^{\alpha}\wedge\vartheta_{\alpha}\wedge*j_5/4$                                                       | $U=\vartheta^{\alpha}\wedge\vartheta_{\alpha}\wedge*j_5/4$                                                       | vanishes. Observe that for Dirac spinors, $U=\vartheta^{\alpha}\wedge\vartheta_{\alpha}\wedge*j_5/4$ is trivial “on shell.”                                          |
| mielke-geometrodynamics_F00503 | 111  | ■ 1.000 | $\kappa=8\pi G_{\mathrm{N}}$                                                                                     | $\kappa=8\pi G_{\mathrm{N}}$                                                                                     | and the orthogonal tetrads, whereas $\kappa=8\pi G_{\mathrm{N}}$ is the gravitational constant in natural                                                            |
| mielke-geometrodynamics_F00504 | 112  | ■ 1.000 | $\Sigma_{\alpha}$                                                                                                | $\Sigma_{\alpha}$                                                                                                | coupled to the canonical energy–momentum current $\Sigma_{\alpha}$ of matter, is obtained by                                                                         |
| mielke-geometrodynamics_F00505 | 112  | ■ 1.000 | $L_{\mathrm{EC}}$                                                                                                | $L_{\mathrm{EC}}$                                                                                                | varying the matter coupled Lagrangian $L_{\mathrm{EC}}$ with respect to the coframe $\vartheta^{\alpha}$ .                                                           |
| mielke-geometrodynamics_F00506 | 112  | ■ 0.973 | $\widehat{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+e_{\alpha}\rfloor T^{\beta}$                     | $\widehat{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+e_{\alpha}\rfloor T^{\beta}$                     | with respect to the transposed connection $\widehat{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+e_{\alpha}\rfloor T^{\beta}$ . Observe that the            |
| mielke-geometrodynamics_F00507 | 113  | ■ 1.000 | $\sigma:=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}=\frac{i}{2}\gamma\wedge\gamma$ | $\sigma:=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}=\frac{i}{2}\gamma\wedge\gamma$ | involving the unit-curvature two-form $\sigma:=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}=\frac{i}{2}\gamma\wedge\gamma$ , the equiva- |
| mielke-geometrodynamics_F00508 | 113  | ■ 1.000 | $\Theta=0$                                                                                                       | $\Theta=0$                                                                                                       | for vanishing torsion, i.e., for $\Theta=0$ in Einstein’s standard GR.                                                                                               |
| mielke-geometrodynamics_F00509 | 113  | ■ 1.000 | $\Gamma^{\alpha\beta}$                                                                                           | $\Gamma^{\alpha\beta}$                                                                                           | By varying (5.3.1) with respect to the linear connection $\Gamma^{\alpha\beta}$ , we obtain the second                                                               |

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| mielke-geometrodynamics_F00510 | 113  | 0.839 | $\tilde{\mathrm{T}}_{\mu\nu}$                                                                                                          | $\tilde{T}_{\mu\nu}$                                                                              | $\tilde{T}_{\mu\nu}$ contains additional terms depending quadratically on the spin angular momen-                                                                 |
| mielke-geometrodynamics_F00511 | 113  | 1.000 | $\tau_{\mathrm{s}}=\mathrm{o}$                                                                                                         | $\tau_{\mathrm{s}} = \mathrm{o}$                                                                  | tum of matter; cf. HEHL et al. (1976). In the instance that $\tau_{\mathrm{s}} = \mathrm{o}$ , the EC equation                                                    |
| mielke-geometrodynamics_F00512 | 113  | 1.000 | $\ell^{*4}$                                                                                                                            | $\ell^{*4}$                                                                                       | terms in (5.3.12) are proportional to $\ell^{*4}$ and could thus, with a modified Planck                                                                          |
| mielke-geometrodynamics_F00513 | 114  | 1.000 | $\ell^{*}=10^{-33} \mathrm{~cm}$                                                                                                       | $\ell^{*} = 10^{-33} \mathrm{~cm}$                                                                | length $\ell^{*} = 10^{-33} \mathrm{~cm}$ , play a role on the macroscopic scale only in cosmology; cf.,                                                          |
| mielke-geometrodynamics_F00514 | 114  | 1.000 | $10^{80}$                                                                                                                              | $10^{80}$                                                                                         | be threatened by a final singularity, since the gravitational collapse of $10^{80}$ baryons                                                                       |
| mielke-geometrodynamics_F00515 | 114  | 0.528 | $R^{\{\backslash\lambda\alpha\backslash\beta\}}$                                                                                       | $R^{\{\lambda\alpha\beta\}}$                                                                      | into the Riemannian curvature $R^{[\alpha\beta]}$ plus contortion pieces.                                                                                         |
| mielke-geometrodynamics_F00516 | 114  | 1.000 | $\mathrm{T}_{\alpha}^{(i)}$                                                                                                            | $^{(i)}T_{\alpha}$                                                                                | allelism as a consequence. Here $^{(i)}T_{\alpha}$ are the three irreducible pieces of the torsion.                                                               |
| mielke-geometrodynamics_F00517 | 114  | 1.000 | $\mathscr{A}$                                                                                                                          | $\mathscr{A}$                                                                                     | When only this axial torsion $\mathscr{A}$ enters algebraically, the EC Lagrangian                                                                                |
| mielke-geometrodynamics_F00518 | 114  | 1.000 | $L_{\mathrm{HE}}$                                                                                                                      | $L_{\mathrm{HE}}$                                                                                 | generalizes <sup>1</sup> the metric Hilbert–Einstein Lagrangian $L_{\mathrm{HE}}$ to an RC spacetime with                                                         |
| mielke-geometrodynamics_F00519 | 114  | 0.647 | $T^{\alpha}\wedge^{*}\left(\mathrm{T}_{\alpha}^{(1)}-\frac{1}{2}\mathrm{T}_{\alpha}^{(2)}+\frac{1}{2}\mathrm{T}_{\alpha}^{(3)}\right)$ | $T^{\alpha}\wedge^{*}\left(^{(1)}T_{\alpha}-2^{(2)}T_{\alpha}-\frac{1}{2}^{(3)}T_{\alpha}\right)$ | dure, since the particular combination $T^{\alpha}\wedge^{*}\left(^{(1)}T_{\alpha}-2^{(2)}T_{\alpha}-\frac{1}{2}^{(3)}T_{\alpha}\right)$ of irreducible pieces is |
| mielke-geometrodynamics_F00520 | 114  | 0.558 | $dC_{\mathrm{TT}}^{*}$                                                                                                                 | $dC_{\mathrm{TT}}^{*}$                                                                            | the space of gravity theories, the term $dC_{\mathrm{TT}}^{*}$ interrelates GR with its teleparallelism equivalent                                                |
| mielke-geometrodynamics_F00521 | 115  | 1.000 | $G_{\alpha}^{\{\backslash\backslash\}}=G_{\alpha}^{\{\backslash\beta\}}\eta_{\beta}$                                                   | $G_{\alpha}^{[\backslash]}=G_{\alpha}^{\beta}\eta_{\beta}$                                        | can be decomposed into the Einstein three-form $G_{\alpha}^{[\backslash]}=G_{\alpha}^{\beta}\eta_{\beta}$ with respect to the                                     |
| mielke-geometrodynamics_F00522 | 115  | 1.000 | $\ell=\sqrt{\kappa}$                                                                                                                   | $\ell = \sqrt{\kappa}$                                                                            | coupled via the “bare” fundamental length $\ell=\sqrt{\kappa}$ . Then “on shell,” EC theory                                                                       |
| mielke-geometrodynamics_F00523 | 115  | 1.000 | $k\rightarrow\infty$                                                                                                                   | $k\rightarrow\infty$                                                                              | bounded in the ultraviolet limit $k\rightarrow\infty$ . Quite generally, the renormalization flow is                                                              |
| mielke-geometrodynamics_F00524 | 115  | 1.000 | $\Gamma_k$                                                                                                                             | $\Gamma_k$                                                                                        | for the effective action $\Gamma_k$ , i.e.,                                                                                                                       |
| mielke-geometrodynamics_F00525 | 116  | 0.999 | $k\partial_k$                                                                                                                          | $k\partial_k$                                                                                     | where $k\partial_k$ is a scale derivative returning the Euler degree of homogeneous functionals.                                                                  |
| mielke-geometrodynamics_F00526 | 116  | 1.000 | $k$                                                                                                                                    | $k$                                                                                               | where $k$ is the renormalization group (RG) scale in momentum space and $\Lambda$ the                                                                             |
| mielke-geometrodynamics_F00527 | 116  | 1.000 | $\rho_{\Lambda}$                                                                                                                       | $\rho_{\Lambda}$                                                                                  | cosmological constant related to dark energy (DE) of density $\rho_{\Lambda}$ ; see also BJORKEN                                                                  |
| mielke-geometrodynamics_F00528 | 116  | 1.000 | $d_{\mathrm{N}}$                                                                                                                       | $d_{\mathrm{N}}$                                                                                  | where $d_{\mathrm{N}}$ is the anomalous dimension of the running Newton coupling $g_{\mathrm{N}}$ .                                                               |
| mielke-geometrodynamics_F00529 | 116  | 1.000 | $g_{\mathrm{N}}$                                                                                                                       | $g_{\mathrm{N}}$                                                                                  | where $d_{\mathrm{N}}$ is the anomalous dimension of the running Newton coupling $g_{\mathrm{N}}$ .                                                               |
| mielke-geometrodynamics_F00530 | 116  | 1.000 | $g_{\mathrm{N}*}$                                                                                                                      | $g_{\mathrm{N}*}$                                                                                 | stants (5.4.3) run into some nontrivial fixed points $g_{\mathrm{N}*}$ and $\lambda_{*}$ , depending on the                                                       |
| mielke-geometrodynamics_F00531 | 116  | 1.000 | $\lambda_{*}$                                                                                                                          | $\lambda_{*}$                                                                                     | stants (5.4.3) run into some nontrivial fixed points $g_{\mathrm{N}*}$ and $\lambda_{*}$ , depending on the                                                       |
| mielke-geometrodynamics_F00532 | 116  | 1.000 | $R^n$                                                                                                                                  | $R^n$                                                                                             | high-order polynomials $R^n$ of the Ricci scalar, similarly as in the classically bifur-                                                                          |
| mielke-geometrodynamics_F00533 | 116  | 1.000 | $f(R)$                                                                                                                                 | $f(R)$                                                                                            | cating $f(R)$ models (SCHUNCK et al. 2005), but then the issue of physical ghosts or                                                                              |
| mielke-geometrodynamics_F00534 | 117  | 1.000 | $\kappa$                                                                                                                               | $\kappa$                                                                                          | also scales with the gravitational constant $\kappa$ , but inversely compared to the Hilbert–                                                                     |
| mielke-geometrodynamics_F00535 | 117  | 1.000 | $1/k^4$                                                                                                                                | $1/k^4$                                                                                           | bilinears is mandatory, including the $1/k^4$ dependence in the functional integral.                                                                              |
| mielke-geometrodynamics_F00536 | 117  | 1.000 | $f_{*}^2$                                                                                                                              | $f_{*}^2$                                                                                         | and the corresponding dimensionless renormalized running coupling $f_{*}^2$ becomes                                                                               |

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| mielke-geometrodynamics_F00537 | 117  | ■ 1.000 | $R_{\alpha\beta} \equiv 0$                                                 | $R_{\alpha\beta} \equiv 0$                                                 | One way to avoid such anomalies is to employ curvature constraints like $R_{\alpha\beta} \equiv 0$ ,                               |
| mielke-geometrodynamics_F00538 | 117  | ■ 1.000 | $SL(5, \mathbb{R})$                                                        | $SL(5, \mathbb{R})$                                                        | from a “minimalist” $SL(5, \mathbb{R})$ gauge model that includes only a “bare” Pontryagin-                                        |
| mielke-geometrodynamics_F00539 | 118  | ■ 1.000 | $\Delta c / c \simeq 10^{-17}$                                             | $\Delta c / c \simeq 10^{-17}$                                             | Lorentz-invariant, nowadays measured at the unprecedented $\Delta c / c \simeq 10^{-17}$ level                                     |
| mielke-geometrodynamics_F00540 | 118  | ■ 1.000 | $10^{-13}$                                                                 | $10^{-13}$                                                                 | et al. 2012) at the $10^{-13}$ level. Moreover, the upper bound for a violation of the                                             |
| mielke-geometrodynamics_F00541 | 118  | ■ 1.000 | $10^{-8}$                                                                  | $10^{-8}$                                                                  | quarks, is about $10^{-8}$ ; cf. HUGHES (1993).                                                                                    |
| mielke-geometrodynamics_F00542 | 118  | ■ 0.979 | $\bar{\nu}_e$                                                              | $\bar{\nu}_e$                                                              | Assuming that not only the dominant delayed antineutrinos $\bar{\nu}_e$ but at least one prompt                                    |
| mielke-geometrodynamics_F00543 | 118  | ■ 0.813 | $\nu_e$                                                                    | $\nu_e$                                                                    | neutrino $\nu_e$ was in the pulse, one could infer that the equivalence principle is valid                                         |
| mielke-geometrodynamics_F00544 | 118  | ■ 1.000 | $10^{-6}$                                                                  | $10^{-6}$                                                                  | (GUZZO et al. 2002) for electron–neutrinos $\nu_e$ up to $10^{-6}$ .                                                               |
| mielke-geometrodynamics_F00545 | 118  | ■ 0.996 | $CPT$                                                                      | $CPT$                                                                      | Moreover, a test of the $CPT$ theorem for the neutral kaon system indirectly                                                       |
| mielke-geometrodynamics_F00546 | 118  | ■ 1.000 | $K^0$                                                                      | $K^0$                                                                      | yielded an upper bound for the relative difference of the <i>inertial masses</i> of the $K^0$                                      |
| mielke-geometrodynamics_F00547 | 118  | ■ 1.000 | $\bar{K}^0$                                                                | $\bar{K}^0$                                                                | meson and its antiparticle $\bar{K}^0$ of less than $10^{-18}$ , when employing (MURAYAMA                                          |
| mielke-geometrodynamics_F00548 | 118  | ■ 1.000 | $10^{-18}$                                                                 | $10^{-18}$                                                                 | meson and its antiparticle $\bar{K}^0$ of less than $10^{-18}$ , when employing (MURAYAMA                                          |
| mielke-geometrodynamics_F00549 | 120  | ■ 0.761 | $(5, \mathbb{R})$                                                          | $(5, \mathbb{R})$                                                          | Mielke EW (2012) Einstein-Weyl gravity from a topological gauge $SL(5, \mathbb{R})$ invariant action. Adv                          |
| mielke-geometrodynamics_F00550 | 122  | ■ 1.000 | (1986, 1991)                                                               | (1986, 1991)                                                               | On the other hand, Ashtekar’s reformulation ASHTEKAR (1986, 1991) of general                                                       |
| mielke-geometrodynamics_F00551 | 122  | ■ 0.996 | $dC_{\mathrm{TT}}$                                                         | $dC_{\mathrm{TT}}$                                                         | ary term $dC_{\mathrm{TT}}$ constructed from a <i>translational Chern–Simons term</i> (CS) multiplied                              |
| mielke-geometrodynamics_F00552 | 122  | ■ 0.987 | $(\mathrm{GR}_{\parallel})$                                                | $(\mathrm{GR}_{\parallel})$                                                | as was already suggested by EINSTEIN (1928), a theory with <i>teleparallelism</i> ( $\mathrm{GR}_{\parallel}$ ).                   |
| mielke-geometrodynamics_F00553 | 123  | ■ 1.000 | $\ell := \sqrt{8\pi\hbar G_{\mathrm{N}}/c^3} \simeq 10^{-33} \mathrm{~cm}$ | $\ell := \sqrt{8\pi\hbar G_{\mathrm{N}}/c^3} \simeq 10^{-33} \mathrm{~cm}$ | scale $\ell := \sqrt{8\pi\hbar G_{\mathrm{N}}/c^3} \simeq 10^{-33} \mathrm{~cm}$ , a topologically rich spectrum of dislocations   |
| mielke-geometrodynamics_F00554 | 123  | ■ 1.000 | $q$                                                                        | $q$                                                                        | In the one-dimensional harmonic oscillator model (MIELKE 1998) with $q$ as gen-                                                    |
| mielke-geometrodynamics_F00555 | 123  | ■ 0.994 | $\mathcal{C} = q^2/2$                                                      | $\mathcal{C} = q^2/2$                                                      | term derived from the <i>Chern–Simons</i> (CS)-type term $\mathcal{C} = q^2/2$ as <i>generating func-</i>                          |
| mielke-geometrodynamics_F00556 | 123  | ■ 0.999 | $\mathcal{S} = \exp(-\mathcal{C}) = \exp(-q^2/2)$                          | $\mathcal{S} = \exp(-\mathcal{C}) = \exp(-q^2/2)$                          | <i>tion</i> . After quantization, the corresponding operator $\mathcal{S} = \exp(-\mathcal{C}) = \exp(-q^2/2)$                     |
| mielke-geometrodynamics_F00557 | 123  | ■ 0.999 | $\Psi = N \exp(i\theta \int \mathcal{C})$                                  | $\Psi = N \exp\left(i\theta \int \mathcal{C}\right)$                       | shown that $\Psi = N \exp(i\theta \int \mathcal{C})$ is, up to an overall factor, the unique solution of the                       |
| mielke-geometrodynamics_F00558 | 123  | ■ 1.000 | $P := \mathbb{R}^4 \otimes SO(1, 3)$                                       | $P := \mathbb{R}^4 \otimes SO(1, 3)$                                       | Since the Poincaré group $P := \mathbb{R}^4 \ltimes SO(1, 3)$ is the semidirect product of trans-                                  |
| mielke-geometrodynamics_F00559 | 124  | ■ 1.000 | $\mathcal{A} := \ast(\vartheta_{\alpha} \wedge T^{\alpha}) =$              | $\mathcal{A} := \ast(\vartheta_{\alpha} \wedge T^{\alpha}) =$              | They are <i>translational</i> and Lorentz-rotational CS terms, where $\mathcal{A} := \ast(\vartheta_{\alpha} \wedge T^{\alpha}) =$ |
| mielke-geometrodynamics_F00560 | 124  | ■ 1.000 | $\mathcal{A}_i dx^i$                                                       | $\mathcal{A}_i dx^i$                                                       | $\mathcal{A}_i dx^i$ is the <i>axial torsion</i> one-form. In the first <i>translational</i> CS three-form, there                  |
| mielke-geometrodynamics_F00561 | 124  | ■ 1.000 | $\ell$                                                                     | $\ell$                                                                     | necessarily occurs a fundamental length $\ell$ for dimensional reasons.                                                            |
| mielke-geometrodynamics_F00562 | 124  | ■ 0.907 | $SO(1, 4)$                                                                 | $SO(1, 4)$                                                                 | the de Sitter group $SO(1, 4)$ or $SO(2, 3)$ as a subgroup, both CS three-forms are                                                |
| mielke-geometrodynamics_F00563 | 124  | ■ 0.907 | $SO(2, 3)$                                                                 | $SO(2, 3)$                                                                 | the de Sitter group $SO(1, 4)$ or $SO(2, 3)$ as a subgroup, both CS three-forms are                                                |

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| mielke-geometrodynamics_F00564 | 124  | ■ 1.000 | $g_{\{\alpha \beta\}}$                                                                                                                                                                                                                                    | $g_{\alpha\beta}$                                                                                                                                                                                                   | free Pontryagin form, in the NY term (6.2.4), a metric $g_{\alpha\beta}$ is needed to raise and                                                                                                                                           |
| mielke-geometrodynamics_F00565 | 124  | ■ 1.000 | $T_{\{\alpha\}}=g_{\{\alpha \beta\}} T^{\{\beta\}}$                                                                                                                                                                                                       | $T_{\alpha} = g_{\alpha\beta} T^{\beta}$                                                                                                                                                                            | lower the indices, for instance in $T_{\alpha} = g_{\alpha\beta} T^{\beta}$ .                                                                                                                                                             |
| mielke-geometrodynamics_F00566 | 124  | ■ 1.000 | $\vartheta_{\{\alpha\}}$                                                                                                                                                                                                                                  | $\vartheta_{\alpha}$                                                                                                                                                                                                | $\vartheta_{\alpha}$ , on the first Bianchi identity                                                                                                                                                                                      |
| mielke-geometrodynamics_F00567 | 125  | ■ 1.000 | $R_{\{\alpha \beta\}}^{\{\ \ \}}$                                                                                                                                                                                                                         | $R_{\alpha\beta}^{\{\ \}}$                                                                                                                                                                                          | of general relativity (GR), where $R_{\alpha\beta}^{\{\ \}}$ denotes the Riemannian curvature with respect                                                                                                                                |
| mielke-geometrodynamics_F00568 | 125  | ■ 1.000 | $\Gamma_{\{\alpha \beta\}}^{\{\ \ \}}$                                                                                                                                                                                                                    | $\Gamma_{\alpha\beta}^{\{\ \}}$                                                                                                                                                                                     | to the Levi-Civita connection $\Gamma_{\alpha\beta}^{\{\ \}}$ .                                                                                                                                                                           |
| mielke-geometrodynamics_F00569 | 125  | ■ 1.000 | $K_{\{\alpha \beta\}}$                                                                                                                                                                                                                                    | $K_{\alpha\beta}$                                                                                                                                                                                                   | is dual to the contortion one-form $K_{\alpha\beta}$ , which features in the decomposition $\Gamma_{\alpha\beta} =$                                                                                                                       |
| mielke-geometrodynamics_F00570 | 125  | ■ 1.000 | $\Gamma_{\{\alpha \beta\}}=$                                                                                                                                                                                                                              | $\Gamma_{\alpha\beta} =$                                                                                                                                                                                            | is dual to the contortion one-form $K_{\alpha\beta}$ , which features in the decomposition $\Gamma_{\alpha\beta} =$                                                                                                                       |
| mielke-geometrodynamics_F00571 | 125  | ■ 0.985 | $\left.\left.\left.\left.\Gamma_{\{\beta \alpha\}}=\Gamma_{\{\alpha \beta\}}-K_{\{\alpha \beta\}}=\Gamma_{\{\alpha \beta\}}+e_{\alpha\rfloor}T_{\beta}+(e_{\alpha\rfloor}e_{\beta\rfloor}T_{\gamma})\wedge\vartheta^{\gamma}\right.\right.\right.\right.$ | $-\Gamma_{\beta\alpha} = \Gamma_{\alpha\beta}^{\{\ \}} - K_{\alpha\beta} = \Gamma_{\alpha\beta}^{\{\ \}} + e_{\alpha\rfloor} T_{\beta} + (e_{\alpha\rfloor} e_{\beta\rfloor} T_{\gamma}) \wedge \vartheta^{\gamma}$ | $-\Gamma_{\beta\alpha} = \Gamma_{\alpha\beta}^{\{\ \}} - K_{\alpha\beta} = \Gamma_{\alpha\beta}^{\{\ \}} + e_{\alpha\rfloor} T_{\beta} + (e_{\alpha\rfloor} e_{\beta\rfloor} T_{\gamma}) \wedge \vartheta^{\gamma}$ of the RC connection. |
| mielke-geometrodynamics_F00572 | 125  | ■ 0.971 | $C_{\mathrm{TT}^*}:=\vartheta^{\alpha}\wedge T_{\alpha}/2\ell^2$                                                                                                                                                                                          | $C_{\mathrm{TT}^*} := \vartheta^{\alpha} \wedge {}^*T_{\alpha}/2\ell^2$                                                                                                                                             | where $C_{\mathrm{TT}^*} := \vartheta^{\alpha} \wedge {}^*T_{\alpha}/2\ell^2$ is a dual CS term, proper <i>teleparallelism</i> (GR <sub>  </sub> ) spec-                                                                                  |
| mielke-geometrodynamics_F00573 | 125  | ■ 1.000 | $R^{\{\alpha \beta\}}=0$                                                                                                                                                                                                                                  | $R^{\alpha\beta} = 0$                                                                                                                                                                                               | the proper constraint $R^{\alpha\beta} = 0$ has to be imposed by subtracting $R^{\alpha\beta} \wedge \lambda_{\alpha\beta}$ from                                                                                                          |
| mielke-geometrodynamics_F00574 | 125  | ■ 1.000 | $R^{\{\alpha \beta\}}\wedge\lambda_{\{\alpha \beta\}}$                                                                                                                                                                                                    | $R^{\alpha\beta} \wedge \lambda_{\alpha\beta}$                                                                                                                                                                      | the proper constraint $R^{\alpha\beta} = 0$ has to be imposed by subtracting $R^{\alpha\beta} \wedge \lambda_{\alpha\beta}$ from                                                                                                          |
| mielke-geometrodynamics_F00575 | 125  | ■ 1.000 | $\lambda_{\{\alpha \beta\}}=-\lambda_{\{\beta \alpha\}}$                                                                                                                                                                                                  | $\lambda_{\alpha\beta} = -\lambda_{\beta\alpha}$                                                                                                                                                                    | (6.2.9), where the 2-form $\lambda_{\alpha\beta} = -\lambda_{\beta\alpha}$ is a Lagrange multiplier, see HEHL et al.                                                                                                                      |
| mielke-geometrodynamics_F00576 | 126  | ■ 1.000 | $\vartheta^{\{\alpha\}}, \Gamma^{\{\alpha \beta\}}$                                                                                                                                                                                                       | $\vartheta^{\alpha}, \Gamma^{\alpha\beta}$                                                                                                                                                                          | By varying this Lagrangian independently with respect to $\vartheta^{\alpha}$ , $\Gamma^{\alpha\beta}$ , and the mul-                                                                                                                     |
| mielke-geometrodynamics_F00577 | 126  | ■ 1.000 | $\lambda_{\{\alpha \beta\}}$                                                                                                                                                                                                                              | $\lambda_{\alpha\beta}$                                                                                                                                                                                             | tiplier $\lambda_{\alpha\beta}$ , we obtain as field equations                                                                                                                                                                            |
| mielke-geometrodynamics_F00578 | 126  | ■ 0.856 | $L_{\{\ \}}$                                                                                                                                                                                                                                              | $L_{  }$                                                                                                                                                                                                            | first field Eq. (6.2.13) is the same as that of the unconstrained Lagrangian $L_{  }$ . The                                                                                                                                               |
| mielke-geometrodynamics_F00579 | 126  | ■ 1.000 | $\lambda_{\{\alpha \beta\}}=\left(\gamma/\ell^2\right)\eta_{\{\alpha \beta\}}$                                                                                                                                                                            | $\lambda_{\alpha\beta} = (\gamma/\ell^2) \eta_{\alpha\beta}$                                                                                                                                                        | <i>spinless</i> matter, a possible solution is $\lambda_{\alpha\beta} = (\gamma/\ell^2) \eta_{\alpha\beta}$ , as will be shown below.                                                                                                     |
| mielke-geometrodynamics_F00581 | 127  | ■ 1.000 | $\sigma=\pm\mathrm{i}\,\gamma$                                                                                                                                                                                                                            | $\sigma = \pm i\gamma$                                                                                                                                                                                              | of BAEKLER et al. (1982); if $\sigma = \pm i\gamma$ is chosen for one of the free parameters there.                                                                                                                                       |
| mielke-geometrodynamics_F00582 | 127  | ■ 1.000 | $\theta_{\mathrm{T}}=\pm\mathrm{i}$                                                                                                                                                                                                                       | $\theta_{\mathrm{T}} = \pm i$                                                                                                                                                                                       | version (6.2.9) of Einstein’s GR for the choice $\theta_{\mathrm{T}} = \pm i$ into a <i>chiral gauge theory</i>                                                                                                                           |
| mielke-geometrodynamics_F00583 | 127  | ■ 0.984 | $\stackrel{(\pm)}{T}^{\alpha}:=\frac{1}{2}\left(T^{\alpha}\pm\mathrm{i}^*T^{\alpha}\right)$                                                                                                                                                               | $\stackrel{(\pm)}{T}^{\alpha} := \frac{1}{2} (T^{\alpha} \pm i^* T^{\alpha})$                                                                                                                                       | where $\stackrel{(\pm)}{T}^{\alpha} := \frac{1}{2} (T^{\alpha} \pm i^* T^{\alpha})$ denotes the self-dual or anti-self-dual torsion. The                                                                                                  |
| mielke-geometrodynamics_F00584 | 128  | ■ 1.000 | $\lambda_{\{\alpha \beta\}}^{(\tau)}$                                                                                                                                                                                                                     | $^{(\tau)}\lambda_{\alpha\beta}$                                                                                                                                                                                    | et al. 1989) an additional piece $^{(\tau)}\lambda_{\alpha\beta}$ in the ansatz for the Lagrangian multiplier,                                                                                                                            |
| mielke-geometrodynamics_F00585 | 128  | ■ 1.000 | $D^{\{\tau\}}\lambda_{\{\alpha \beta\}}=\tau_{\{\alpha \beta\}}$                                                                                                                                                                                          | $D^{(\tau)}\lambda_{\alpha\beta} = \tau_{\alpha\beta}$                                                                                                                                                              | provided it satisfies $D^{(\tau)}\lambda_{\alpha\beta} = \tau_{\alpha\beta}$ .                                                                                                                                                            |
| mielke-geometrodynamics_F00586 | 128  | ■ 0.584 | $\Pi^{\{\alpha\}}$                                                                                                                                                                                                                                        | $\stackrel{(\pm)}{\Pi}^{\alpha}$                                                                                                                                                                                    | The complex translational field momenta $\stackrel{(\pm)}{\Pi}^{\alpha}$ replace the role of the torsion if we                                                                                                                            |
| mielke-geometrodynamics_F00587 | 128  | ■ 0.731 | $\delta L_{  }/\delta\vartheta^{\alpha}$                                                                                                                                                                                                                  | $\delta \stackrel{(\pm)}{L}_{  } / \delta \vartheta^{\alpha}$                                                                                                                                                       | allow us to rederive the field Eq. (6.2.24) via $\delta \stackrel{(\pm)}{L}_{  } / \delta \vartheta^{\alpha}$ from a variational principle.                                                                                               |
| mielke-geometrodynamics_F00588 | 128  | ■ 1.000 | $R_{\{\alpha \beta\}}=0$                                                                                                                                                                                                                                  | $R_{\alpha\beta} = 0$                                                                                                                                                                                               | Cartan spacetime with $R_{\alpha\beta} = 0$ and in view of the first Noether identity (6.8.11),                                                                                                                                           |

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| mielke-geometrodynamics_F00589 | 128  | 0.999 | $\stackrel{\circ}{\Pi}_{\alpha}$                                                                           | $\stackrel{(\pm)}{\Pi}_{\alpha}$                                                                           | terms of the momenta $\stackrel{(\pm)}{\Pi}_{\alpha}$ , it would be convenient to have a <i>relocalized</i> energy–momentum current            |
| mielke-geometrodynamics_F00590 | 128  | 0.121 | $\stackrel{\circ}{\Sigma}_{\alpha}$                                                                        | $\stackrel{(\pm)}{\Sigma}_{\alpha}$                                                                        | $\stackrel{(\pm)}{\Sigma}_{\alpha}$ for which $\stackrel{(\pm)(\pm)}{D} \stackrel{(\pm)}{\Sigma}_{\alpha} \cong 0$ holds.                      |
| mielke-geometrodynamics_F00591 | 128  | 0.121 | $\stackrel{\circ}{D}$                                                                                      | $\stackrel{(\pm)(\pm)}{D}$                                                                                 | $\stackrel{(\pm)}{\Sigma}_{\alpha}$ for which $\stackrel{(\pm)(\pm)}{D} \stackrel{(\pm)}{\Sigma}_{\alpha} \cong 0$ holds.                      |
| mielke-geometrodynamics_F00592 | 129  | 0.821 | $\stackrel{\circ}{A}_{\alpha}{}^{\beta}$                                                                   | $\stackrel{(\mp)}{A}_{\alpha}{}^{\beta}$                                                                   | valued 1-form $\stackrel{(\mp)}{A}_{\alpha}{}^{\beta}$ deforms our original Riemann–Cartan connection $\Gamma_{\alpha}{}^{\beta}$ similarly    |
| mielke-geometrodynamics_F00593 | 129  | 0.946 | $\underline{\Gamma}_{\parallel}^{\alpha\beta} \rightarrow 0$                                               | $\Gamma_{\parallel}^{\alpha\beta} \rightarrow 0$                                                           | (1973), THIRRING (1978), HEHL et al. (1980). In the teleparallelism limit $\underline{\Gamma}_{\parallel}^{\alpha\beta} \rightarrow 0$ ,       |
| mielke-geometrodynamics_F00594 | 129  | 1.000 | $H_{\alpha}$                                                                                               | $H_{\alpha}$                                                                                               | the “superpotential” $H_{\alpha}$ is the <i>Møller</i> or <i>Freud complex</i> (FREUD 1939)                                                    |
| mielke-geometrodynamics_F00595 | 129  | 0.700 | $\overparen{D}\eta_{\alpha} := D\eta_{\alpha} -$                                                           | $\widehat{D}\eta_{\alpha} := D\eta_{\alpha} -$                                                             | <sup>3</sup> The <i>transposed</i> connection, which has the property (MIELKE et al. 1989) that $\widehat{D}\eta_{\alpha} := D\eta_{\alpha} -$ |
| mielke-geometrodynamics_F00596 | 129  | 0.521 | $(e_{\alpha} \downarrow T^{\beta}) \wedge \eta_{\beta} \equiv 0$                                           | $(e_{\alpha} \downarrow T^{\beta}) \wedge \eta_{\beta} \equiv 0$                                           | $(e_{\alpha} \downarrow T^{\beta}) \wedge \eta_{\beta} \equiv 0$ , may be regarded as a special real version of our Sen-type connection, for   |
| mielke-geometrodynamics_F00597 | 129  | 0.593 | $\stackrel{\circ}{D}\eta_{\alpha} = \pm(i/2)\ell^2\eta_{\alpha\beta} \wedge \stackrel{(\pm)}{\Pi}^{\beta}$ | $\stackrel{(\pm)}{D}\eta_{\alpha} = \pm(i/2)\ell^2\eta_{\alpha\beta} \wedge \stackrel{(\pm)}{\Pi}^{\beta}$ | which $\stackrel{(\pm)}{D}\eta_{\alpha} = \pm(i/2)\ell^2\eta_{\alpha\beta} \wedge \stackrel{(\pm)}{\Pi}^{\beta}$ holds.                        |
| mielke-geometrodynamics_F00598 | 130  | 1.000 | $E_{\alpha}$                                                                                               | $E_{\alpha}$                                                                                               | et al. (2000) that $E_{\alpha}$ is a covariant conserved gravitational energy, i.e., that $DE_{\alpha} =$                                      |
| mielke-geometrodynamics_F00599 | 130  | 1.000 | $D E_{\alpha} =$                                                                                           | $DE_{\alpha} =$                                                                                            | et al. (2000) that $E_{\alpha}$ is a covariant conserved gravitational energy, i.e., that $DE_{\alpha} =$                                      |
| mielke-geometrodynamics_F00600 | 130  | 1.000 | $DDH_{\alpha} \equiv -R_{\alpha}{}^{\beta} \wedge H_{\beta} = 0$                                           | $DDH_{\alpha} \equiv -R_{\alpha}{}^{\beta} \wedge H_{\beta} = 0$                                           | $DDH_{\alpha} \equiv -R_{\alpha}{}^{\beta} \wedge H_{\beta} = 0$ due to the vanishing of the Ricci identity in a teleparallel                  |
| mielke-geometrodynamics_F00602 | 130  | 1.000 | $\Sigma_t$                                                                                                 | $\Sigma_t$                                                                                                 | into a family of spacelike three-dimensional hypersurfaces $\Sigma_t$ that are parametrized                                                    |
| mielke-geometrodynamics_F00603 | 130  | 1.000 | $x^0 = t$                                                                                                  | $x^0 = t$                                                                                                  | by the coordinate time $x^0 = t$ . Then there exists the future-directed <i>timelike</i> vector                                                |
| mielke-geometrodynamics_F00604 | 130  | 0.496 | $u := -N^2 dt, \Longleftrightarrow$                                                                        | $u := -N^2 dt, \Longleftrightarrow$                                                                        | which is uniquely linked to the hypersurface orthogonal 1-form $u := -N^2 dt, \Longleftrightarrow$                                             |
| mielke-geometrodynamics_F00605 | 130  | 0.963 | $du \wedge u = 0$                                                                                          | $du \wedge u = 0$                                                                                          | $du \wedge u = 0$ via $u = g(n, \cdot)$ . The lapse function $N$ and the shift vector $N^A$ , which                                            |
| mielke-geometrodynamics_F00606 | 130  | 0.963 | $u = g(n,$                                                                                                 | $u = g(n,$                                                                                                 | $du \wedge u = 0$ via $u = g(n, \cdot)$ . The lapse function $N$ and the shift vector $N^A$ , which                                            |
| mielke-geometrodynamics_F00607 | 130  | 0.963 | $)$ . The lapse function $N$                                                                               | $)$ . The lapse function $N$                                                                               | $du \wedge u = 0$ via $u = g(n, \cdot)$ . The lapse function $N$ and the shift vector $N^A$ , which                                            |
| mielke-geometrodynamics_F00608 | 130  | 0.963 | $N^A$                                                                                                      | $N^A$                                                                                                      | $du \wedge u = 0$ via $u = g(n, \cdot)$ . The lapse function $N$ and the shift vector $N^A$ , which                                            |
| mielke-geometrodynamics_F00609 | 130  | 0.974 | $p$                                                                                                        | $p$                                                                                                        | The <i>normal</i> part of a $p$ -form $\Psi$ is defined by                                                                                     |
| mielke-geometrodynamics_F00610 | 130  | 0.997 | $(3+1)$                                                                                                    | $(3+1)$                                                                                                    | operators. (For more details on this very concise (3+1) decomposition of exterior                                                              |
| mielke-geometrodynamics_F00611 | 131  | 1.000 | $L = L(\vartheta^{\alpha}, \Psi, D\Psi)$                                                                   | $L = L(\vartheta^{\alpha}, \Psi, D\Psi)$                                                                   | Let us consider a Lagrangian $L = L(\vartheta^{\alpha}, \Psi, D\Psi)$ which is of first differential                                           |
| mielke-geometrodynamics_F00612 | 131  | 0.973 | $\left. \mathscr{K} = L_{\perp} := n \right\rfloor L$                                                      | $\mathscr{K} = L_{\perp} := n \rfloor L$                                                                   | where $\mathscr{K} \equiv L_{\perp} \equiv n \rfloor L$ denotes the normal part of the Lagrangian. Since the “time                             |
| mielke-geometrodynamics_F00613 | 131  | 0.677 | $\{\}^{\bullet}$                                                                                           | $\bullet$                                                                                                  | derivative” $\bullet$ is more precisely given by the Lie derivative $\ell_n := dn \rfloor + n \rfloor d$ along                                 |
| mielke-geometrodynamics_F00614 | 131  | 0.677 | $\left. \ell_n := dn \right\rfloor + n \downarrow d$                                                       | $\ell_n := dn \rfloor + n \downarrow d$                                                                    | derivative” $\bullet$ is more precisely given by the Lie derivative $\ell_n := dn \rfloor + n \rfloor d$ along                                 |
| mielke-geometrodynamics_F00615 | 131  | 1.000 | $n$                                                                                                        | $n$                                                                                                        | the timelike vector field $n$ , we have                                                                                                        |

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| mielke-geometrodynamics_F00616 | 131  | ■ 0.965 | $\left.n^{\alpha}:=n\right\rfloor\vartheta^{\alpha}$                                                                                                                     | $n^{\alpha}:=n\rfloor\vartheta^{\alpha}$                                                                                                                     | and transvect the resulting expression with the four-vector $n^{\alpha}:=n\rfloor\vartheta^{\alpha}$ of <i>lapse and</i>                                                                           |
| mielke-geometrodynamics_F00617 | 131  | ■ 0.605 | $\left.n^{\alpha}e_{\alpha}\right\rfloor\Psi=n\downarrow\Psi$                                                                                                            | $n^{\alpha}e_{\alpha}\rfloor\Psi=n\downarrow\Psi$                                                                                                            | <i>shift</i> , for which the useful relation $n^{\alpha}e_{\alpha}\rfloor\Psi=n\rfloor\Psi$ can be derived. For the projection                                                                     |
| mielke-geometrodynamics_F00618 | 132  | ■ 0.919 | $\mathrm{L}_n=D\ n\downarrow+n\downarrow D$                                                                                                                              | $L_n=Dn\downarrow+n\downarrow D$                                                                                                                             | In introducing the gauge-covariant Lie derivative $L_n=Dn\rfloor+n\rfloor D$ , separating                                                                                                          |
| mielke-geometrodynamics_F00619 | 132  | ■ 0.651 | $\left.n\right\rfloor\Psi=n\downarrow\left(dt\wedge n\right\rfloor\Psi=n\right\rfloor\Psi$                                                                               | $n\rfloor\Psi=n\downarrow(dt\wedge n\rfloor\Psi)=n\rfloor\Psi$                                                                                               | sion is related to the Hamiltonian (6.4.5). Indeed, using $n\rfloor\Psi=n\rfloor(dt\wedge n\rfloor\Psi)=n\rfloor\Psi$                                                                              |
| mielke-geometrodynamics_F00620 | 132  | ■ 0.479 | $\left.\Gamma_{\perp}^{\alpha\beta}:=n\right\rfloor\Gamma^{\alpha\beta}$                                                                                                 | $\Gamma_{\perp}^{\alpha\beta}:=n\rfloor\Gamma^{\alpha\beta}$                                                                                                 | where the normal vector $n^{\alpha}:=n\rfloor\vartheta^{\alpha}$ comprises <i>lapse and shift</i> and $\Gamma_{\perp}^{\alpha\beta}:=n\rfloor\Gamma^{\alpha\beta}$                                 |
| mielke-geometrodynamics_F00621 | 132  | ■ 1.000 | $\mathrm{G}_{\alpha}$                                                                                                                                                    | $\mathcal{G}_{\alpha}$                                                                                                                                       | is the normal part of the Lorentz connection and $\mathcal{G}_{\alpha}$ and $\mathcal{G}_{\alpha\beta}$ represent the tangential                                                                   |
| mielke-geometrodynamics_F00622 | 132  | ■ 1.000 | $\mathrm{G}_{\alpha\beta}$                                                                                                                                               | $\mathcal{G}_{\alpha\beta}$                                                                                                                                  | is the normal part of the Lorentz connection and $\mathcal{G}_{\alpha}$ and $\mathcal{G}_{\alpha\beta}$ represent the tangential                                                                   |
| mielke-geometrodynamics_F00623 | 133  | ■ 0.709 | $D_{\alpha}:=e_{\alpha}\rfloor D$                                                                                                                                        | $D_{\alpha}:=e_{\alpha}\rfloor D$                                                                                                                            | where $D_{\alpha}:=e_{\alpha}\rfloor D$ are the anholonomic components of the exterior covariant                                                                                                   |
| mielke-geometrodynamics_F00624 | 133  | ■ 0.927 | $D\lambda_{\alpha\beta}$                                                                                                                                                 | $D\lambda_{\alpha\beta}$                                                                                                                                     | arise the extra tangential term $D\lambda_{\alpha\beta}$ in the <i>Lorentz</i> generator $\mathcal{G}_{\alpha\beta}$ , where                                                                       |
| mielke-geometrodynamics_F00625 | 133  | ■ 1.000 | $\underline{\Psi}:=\Psi-dt\wedge n\rfloor\Psi$                                                                                                                           | $\underline{\Psi}:=\Psi-dt\wedge n\rfloor\Psi$                                                                                                               | $\underline{\Psi}:=\Psi-dt\wedge n\rfloor\Psi$ in the notation of differential forms. However, as suggested                                                                                        |
| mielke-geometrodynamics_F00626 | 133  | ■ 0.893 | $\underline{\lambda}_{\alpha\beta}=\left(1/2\ell^2\right)*\left(\vartheta_{\alpha}\wedge\vartheta_{\beta}\right)=-\left(1/N\ell^2\right)n_{[\alpha}* \vartheta_{\beta]}$ | $\lambda_{\alpha\beta}=\left(1/2\ell^2\right)*\left(\vartheta_{\alpha}\wedge\vartheta_{\beta}\right)=-\left(1/N\ell^2\right)n_{[\alpha}* \vartheta_{\beta]}$ | by $\lambda_{\alpha\beta}=\left(1/2\ell^2\right)*\left(\vartheta_{\alpha}\wedge\vartheta_{\beta}\right)=-\left(1/N\ell^2\right)n_{[\alpha}* \vartheta_{\beta]}$ . This is another manifestation of |
| mielke-geometrodynamics_F00627 | 133  | ■ 0.925 | $d\stackrel{(\mp)}{C}_{\mathrm{TT}}$                                                                                                                                     | $d\stackrel{(\mp)}{C}_{\mathrm{TT}}$                                                                                                                         | by the boundary term $d\stackrel{(\mp)}{C}_{\mathrm{TT}}$ .                                                                                                                                        |
| mielke-geometrodynamics_F00628 | 133  | ■ 0.989 | $(\alpha=\hat{0})$                                                                                                                                                       | $(\alpha=\hat{0})$                                                                                                                                           | After this trivialization of the Lorentz constraints, the Hamiltonian $(\alpha=\hat{0})$ and                                                                                                       |
| mielke-geometrodynamics_F00629 | 133  | ■ 1.000 | $(\alpha=B)$                                                                                                                                                             | $(\alpha=B)$                                                                                                                                                 | diffeomorphism constraints $(\alpha=B)$ of complexified $\mathrm{GR}_{\parallel}$ , in terms of a complex                                                                                          |
| mielke-geometrodynamics_F00630 | 133  | ■ 0.464 | $\stackrel{(\pm)}{\mathcal{A}}_{\alpha}:=\stackrel{*}{\Pi}_{\alpha}$                                                                                                     | $\stackrel{(\pm)}{\mathcal{A}}_{\alpha}:=\stackrel{*}{\Pi}_{\alpha}$                                                                                         | (1994). Here the tangential Ashtekar-type connection $\stackrel{(\pm)}{\mathcal{A}}_{\alpha}:=\stackrel{*}{\Pi}_{\alpha}$ is three-dual                                                            |
| mielke-geometrodynamics_F00631 | 133  | ■ 0.130 | $\stackrel{*}{-}_{\alpha}$                                                                                                                                               | $\stackrel{*}{-}_{\alpha}$                                                                                                                                   | ties $\stackrel{*}{\vartheta}_{\alpha}$ , where we note that the spatial Hodge dual is involutive, i.e., $\stackrel{*}{*}=+1$ .                                                                    |
| mielke-geometrodynamics_F00632 | 133  | ■ 0.130 | $\stackrel{*}{*}=+1$                                                                                                                                                     | $\stackrel{*}{*}=+1$                                                                                                                                         | ties $\stackrel{*}{\vartheta}_{\alpha}$ , where we note that the spatial Hodge dual is involutive, i.e., $\stackrel{*}{*}=+1$ .                                                                    |
| mielke-geometrodynamics_F00633 | 133  | ■ 1.000 | $\underline{\vartheta}^{\hat{0}}=0$                                                                                                                                      | $\vartheta^{\hat{0}}=0$                                                                                                                                      | Moreover, by adopting tetrads in the temporal gauge $\underline{\vartheta}^{\hat{0}}=0$ of SCHWINGER (1963)                                                                                        |
| mielke-geometrodynamics_F00634 | 133  | ■ 0.858 | $\stackrel{(\pm)}{\mathcal{A}}_{\hat{0}}=d^{(\pm)}$                                                                                                                      | $\stackrel{(\pm)}{\mathcal{A}}_{\hat{0}}=d^{(\pm)}$                                                                                                          | and the gauge $\stackrel{(\pm)}{\mathcal{A}}_{\hat{0}}=d^{(\pm)}\phi$ of NESTER (1989), the Hamiltonian constraint                                                                                 |
| mielke-geometrodynamics_F00635 | 134  | ■ 1.000 | $\alpha=\hat{0}$                                                                                                                                                         | $\alpha=\hat{0}$                                                                                                                                             | i.e., (6.4.19) for $\alpha=\hat{0}$ , vanishes identically.                                                                                                                                        |
| mielke-geometrodynamics_F00636 | 134  | ■ 0.561 | $\stackrel{*}{-}\vartheta_{\alpha}$                                                                                                                                      | $\stackrel{*}{-}\vartheta_{\alpha}$                                                                                                                          | parts of the basic one-forms, i.e., the <i>triad densities</i> $\stackrel{*}{\vartheta}_{\alpha}$ and the tangential part of                                                                       |
| mielke-geometrodynamics_F00637 | 134  | ■ 0.677 | $\stackrel{(\pm)}{\mathcal{A}}_{\alpha}:=\stackrel{*}{\Pi}_{\alpha}$                                                                                                     | $\stackrel{(\pm)}{\mathcal{A}}_{\alpha}:=\stackrel{*}{\Pi}_{\alpha}$                                                                                         | the self-dual or anti-self-dual connection $\stackrel{(\pm)}{\mathcal{A}}_{\alpha}:=\stackrel{*}{\Pi}_{\alpha}$ , the three-dual of the chiral                                                     |
| mielke-geometrodynamics_F00638 | 134  | ■ 0.995 | $\stackrel{\pm}{p}$                                                                                                                                                      | $\stackrel{\pm}{p}$                                                                                                                                          | momenta $\stackrel{(\pm)}{\Pi}_{\alpha}$ , become the generalized coordinates $q$ and momenta $\stackrel{\pm}{p}$ of the bosonic                                                                   |
| mielke-geometrodynamics_F00639 | 134  | ■ 0.337 | $\stackrel{*}{\Pi}_{\alpha}$                                                                                                                                             | $\stackrel{*}{\Pi}_{\alpha}$                                                                                                                                 | prevails (MIELKE 2002), where the complex field “momenta” $\stackrel{(\pm)}{\Pi}_{\alpha}$ become differ-                                                                                          |

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| mielke-geometrodynamics_F00640 | 134  | ■ 0.316 | $\underline{*}^{\{\vartheta^{\beta}\}}$                                                                  | $\underline{*}^{\vartheta^B}$                                                                            | ential operators, whereas the triad densities $\underline{*}^{\vartheta^B}$ remain generalized coordinates $q$ .                                                   |
| mielke-geometrodynamics_F00641 | 134  | ■ 1.000 | $\alpha=B$                                                                                               | $\alpha = B$                                                                                             | For the remaining Gauss constraint (6.4.8) with $\alpha = B$ , let us try the state vector                                                                         |
| mielke-geometrodynamics_F00642 | 134  | ■ 0.731 | $\underline{C}_{\mathrm{TT}}$                                                                            | $\underline{C}_{\mathrm{TT}}$                                                                            | Chern–Simons term $\underline{C}_{\mathrm{TT}}$ in terms of the triads or triad densities and the tangential                                                       |
| mielke-geometrodynamics_F00643 | 134  | ■ 0.862 | $\underline{*}^{\{\underline{\vartheta}^B\}}$                                                            | $\underline{*}^{\underline{\vartheta}^B}$                                                                | rewritten in terms of the triad densities $\underline{*}^{\underline{\vartheta}^B}$ with $B = 1, 2, 3$ and the tangential part                                     |
| mielke-geometrodynamics_F00644 | 134  | ■ 0.862 | $B=1,2,3$                                                                                                | $B = 1, 2, 3$                                                                                            | rewritten in terms of the triad densities $\underline{*}^{\underline{\vartheta}^B}$ with $B = 1, 2, 3$ and the tangential part                                     |
| mielke-geometrodynamics_F00645 | 134  | ■ 0.637 | $\stackrel{(\pm)}{\underline{T}}^B$                                                                      | $\stackrel{(\pm)}{\underline{T}}^B$                                                                      | momentum operator (6.4.22) returns the chiral torsion $\stackrel{(\pm)}{\underline{T}}^B$ as a prefactor:                                                          |
| mielke-geometrodynamics_F00646 | 134  | ■ 1.000 | $\underline{\Gamma}_{\parallel}^{\alpha\beta} = 0$                                                       | $\underline{\Gamma}_{\parallel}^{\alpha\beta} = 0$                                                       | Since the corresponding connection $\underline{\Gamma}_{\parallel}^{\alpha\beta} = 0$ is of pure gauge type, the state vector                                      |
| mielke-geometrodynamics_F00647 | 135  | ■ 0.999 | $\Psi \wedge * \Phi = \Phi \wedge * \Psi$                                                                | $\Psi \wedge * \Phi = \Phi \wedge * \Psi$                                                                | where the rule $\underline{\Psi \wedge * \Phi = \Phi \wedge * \Psi}$ for differential forms of the same degree has                                                 |
| mielke-geometrodynamics_F00648 | 135  | ■ 0.994 | $S \, \mathrm{U}(2)$                                                                                     | $SU(2)$                                                                                                  | borrowed from $SU(2)$ Chern–Simons field theory (WITTEN 1989; GUADAGNINI                                                                                           |
| mielke-geometrodynamics_F00649 | 135  | ■ 1.000 | $\mathcal{H}_{\Lambda} \Psi_{\Lambda}(A) = 0$                                                            | $\mathcal{H}_{\Lambda} \Psi_{\Lambda}(A) = 0$                                                            | straint $\mathcal{H}_{\Lambda} \Psi_{\Lambda}(A) = 0$ of GR with cosmological constant in the nonperturbative                                                      |
| mielke-geometrodynamics_F00650 | 135  | ■ 1.000 | $\ell \rightarrow 0$                                                                                     | $\ell \rightarrow 0$                                                                                     | $\ell$ , it is dimensionless and nonsingular even for $\ell \rightarrow 0$ , due to $\dim \underline{\vartheta}^B = [\ell]$ .                                      |
| mielke-geometrodynamics_F00651 | 135  | ■ 1.000 | $\dim \underline{\vartheta}^B = [\ell]$                                                                  | $\dim \underline{\vartheta}^B = [\ell]$                                                                  | $\ell$ , it is dimensionless and nonsingular even for $\ell \rightarrow 0$ , due to $\dim \underline{\vartheta}^B = [\ell]$ .                                      |
| mielke-geometrodynamics_F00652 | 135  | ■ 0.937 | $\underline{T}^{(\mp)B} = 0$                                                                             | $\underline{T}^{(\mp)B} = 0$                                                                             | configurations with $\underline{T}^{(\mp)B} = 0$ , i.e., <i>self-dual or anti-self-dual</i> tangential torsion. Con-                                               |
| mielke-geometrodynamics_F00653 | 135  | ■ 0.802 | $\stackrel{(\pm)}{\underline{T}} = 0$                                                                    | $\stackrel{(\pm)}{\underline{T}} = 0$                                                                    | straint are dominated by <i>self-dual</i> torsion solutions satisfying $\stackrel{(\pm)}{\underline{T}} = 0$ .                                                     |
| mielke-geometrodynamics_F00654 | 135  | ■ 1.000 | $\mathbb{R}^4 \cup \infty = S^4$                                                                         | $\mathbb{R}^4 \cup \infty = S^4$                                                                         | $\mathbb{R}^4 \cup \infty = S^4$ with the spherically symmetric metric                                                                                             |
| mielke-geometrodynamics_F00655 | 136  | ■ 0.997 | $(r, \psi, \theta, \phi)$                                                                                | $(r, \psi, \theta, \phi)$                                                                                | where $(r, \psi, \theta, \phi)$ are the standard hyperspherical coordinates that parameterize the                                                                  |
| mielke-geometrodynamics_F00656 | 136  | ■ 1.000 | $S^3$                                                                                                    | $S^3$                                                                                                    | unit three-sphere $S^3$ . The function $f$ carries physical dimension [length]. This space                                                                         |
| mielke-geometrodynamics_F00657 | 136  | ■ 1.000 | $f$                                                                                                      | $f$                                                                                                      | unit three-sphere $S^3$ . The function $f$ carries physical dimension [length]. This space                                                                         |
| mielke-geometrodynamics_F00658 | 136  | ■ 1.000 | $df = \pm 2 h dr$                                                                                        | $df = \pm 2 h dr$                                                                                        | which is self-dual or anti-self-dual, provided that $df = \pm 2 h dr$ . Here the self-dual                                                                         |
| mielke-geometrodynamics_F00659 | 136  | ■ 0.998 | $\eta_{\mu\nu}^A := \eta_0^A{}_{\mu\nu} + \delta_{\mu}^A \delta_{\nu}^0 - \delta_{\nu}^A \delta_{\mu}^0$ | $\eta_{\mu\nu}^A := \eta_0^A{}_{\mu\nu} + \delta_{\mu}^A \delta_{\nu}^0 - \delta_{\nu}^A \delta_{\mu}^0$ | generalized Levi-Civita symbol $\eta_{\mu\nu}^A := \eta_0^A{}_{\mu\nu} + \delta_{\mu}^A \delta_{\nu}^0 - \delta_{\nu}^A \delta_{\mu}^0$ of 'T HOOFT (1991),        |
| mielke-geometrodynamics_F00660 | 136  | ■ 1.000 | $r \rightarrow \infty$                                                                                   | $r \rightarrow \infty$                                                                                   | radial infinity $r \rightarrow \infty$ yields                                                                                                                      |
| mielke-geometrodynamics_F00661 | 136  | ■ 0.954 | $\stackrel{(\mp)}{\underline{T}}^B = 0$                                                                  | $\stackrel{(\mp)}{\underline{T}}^B = 0$                                                                  | self-dual, i.e., $\stackrel{(\mp)}{\underline{T}}^B = 0$ , then integration over the chiral CS term $\stackrel{(\pm)}{\underline{C}}_{\mathrm{TT}}$ yields the     |
| mielke-geometrodynamics_F00662 | 136  | ■ 0.954 | $\stackrel{(\pm)}{\underline{C}}_{\mathrm{TT}}$                                                          | $\stackrel{(\pm)}{\underline{C}}_{\mathrm{TT}}$                                                          | self-dual, i.e., $\stackrel{(\mp)}{\underline{T}}^B = 0$ , then integration over the chiral CS term $\stackrel{(\pm)}{\underline{C}}_{\mathrm{TT}}$ yields the     |
| mielke-geometrodynamics_F00663 | 136  | ■ 1.000 | $\Psi_{\parallel}(\vartheta) \propto e^{-C}$                                                             | $\Psi_{\parallel}(\vartheta) \propto e^{-C}$                                                             | Thus one can surmise that the state vector $\Psi_{\parallel}(\vartheta) \propto e^{-C}$ as a solution of the                                                       |
| mielke-geometrodynamics_F00664 | 136  | ■ 1.000 | $\underline{\vartheta}^{\hat{0}} = 3\kappa d\theta_{\mathrm{L}} = h dr = \pm df/2$                       | $\underline{\vartheta}^{\hat{0}} = 3\kappa d\theta_{\mathrm{L}} = h dr = \pm df/2$                       | <sup>4</sup> Interestingly, in the gauge $\underline{\vartheta}^{\hat{0}} = 3\kappa d\theta_{\mathrm{L}} = h dr = \pm df/2$ , such instantons are solutions to the |

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| mielke-geometrodynamics_F00665 | 136  | ■ 0.991 | $T^{\hat{0}}=0$                                                                           | $T^{\hat{0}} = 0$                                                                         | topological Eq. (6.7.12), due to $T^{\hat{0}} = 0$ .                                                                                                                                             |
| mielke-geometrodynamics_F00666 | 137  | ■ 1.000 | $A=A_{\mu}^I \lambda_I$                                                                   | $A = A_{\mu}^I \lambda_I$                                                                 | From the <i>holonomy</i> of a Lie-algebra-valued connection one-form $A = A_{\mu}^I \lambda_I$                                                                                                   |
| mielke-geometrodynamics_F00667 | 137  | ■ 0.967 | $dx^{\mu} \in \mathscr{C}$                                                                | $dx^{\mu} \in \mathcal{C}$                                                                | $dx^{\mu} \in \mathcal{C}$ , one can construct a <i>gauge-invariant</i> operator, or character, along the loop                                                                                   |
| mielke-geometrodynamics_F00668 | 137  | ■ 1.000 | $\gamma^{\mu}(s)$                                                                         | $\gamma^{\mu}(s)$                                                                         | with coordinates $\gamma^{\mu}(s)$ . Then the Wilson loop can be written as                                                                                                                      |
| mielke-geometrodynamics_F00669 | 137  | ■ 0.826 | $\Psi(A)$                                                                                 | $\Psi(A)$                                                                                 | The state vector $\Psi(A)$ for the gauge potential can be transformed via the Wilson                                                                                                             |
| mielke-geometrodynamics_F00670 | 137  | ■ 1.000 | $\mathscr{C} / \mathscr{G}$                                                               | $\mathcal{C} / \mathcal{G}$                                                               | The integration is performed over a coset space $\mathcal{C} / \mathcal{G}$ in order to factor out gauge-                                                                                        |
| mielke-geometrodynamics_F00671 | 137  | ■ 1.000 | $G(\gamma)$                                                                               | $G(\gamma)$                                                                               | of the Wilson loop is the <i>self-linking number</i> $G(\gamma)$ of Gauss. It is interesting to see                                                                                              |
| mielke-geometrodynamics_F00672 | 137  | ■ 1.000 | $V(\gamma)$                                                                               | $V(\gamma)$                                                                               | of the Jones polynomial $V(\gamma)$ . If we could formally transform these results of lattice                                                                                                    |
| mielke-geometrodynamics_F00673 | 138  | ■ 1.000 | $ \Psi\rangle$                                                                            | $ \Psi\rangle$                                                                            | state vector $ \Psi\rangle$ of the universe would be characterized by knot invariants, such as                                                                                                   |
| mielke-geometrodynamics_F00674 | 138  | ■ 1.000 | $\mathbb{R}^4$                                                                            | $\mathbb{R}^4$                                                                            | group $\mathbb{R}^4$ as structure group. Then, instead of Wilson-type loops, we encounter                                                                                                        |
| mielke-geometrodynamics_F00675 | 138  | ■ 0.979 | $n^{\alpha} = n \rfloor \vartheta^{\alpha}$                                               | $n^{\alpha} = n \rfloor \vartheta^{\alpha}$                                               | GUADAGNINI 1993) via one of the orthonormal legs $n^{\alpha} = n \rfloor \vartheta^{\alpha}$ . In view of (6.4.25),                                                                              |
| mielke-geometrodynamics_F00676 | 138  | ■ 1.000 | $(1/16\pi^2)$                                                                             | $(1/16\pi^2)$                                                                             | It is well known from differential geometry that the Euler number $(1/16\pi^2)$                                                                                                                  |
| mielke-geometrodynamics_F00677 | 138  | ■ 1.000 | $\int R^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)}$                                   | $\int R^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)}$                                   | $\int R^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)}$ and the Pontryagin integral $(1/8\pi^2) \int R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}$ bear an intimate                              |
| mielke-geometrodynamics_F00678 | 138  | ■ 1.000 | $(1/8\pi^2) \int R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}$                            | $(1/8\pi^2) \int R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}$                            | $\int R^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)}$ and the Pontryagin integral $(1/8\pi^2) \int R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}$ bear an intimate                              |
| mielke-geometrodynamics_F00679 | 138  | ■ 1.000 | $\epsilon$                                                                                | $\epsilon$                                                                                | to the following argument: For a real constant $\epsilon$ with $0 \leq \epsilon \leq 1$ , the RC connec-                                                                                         |
| mielke-geometrodynamics_F00680 | 138  | ■ 1.000 | $0 \leq \epsilon \leq 1$                                                                  | $0 \leq \epsilon \leq 1$                                                                  | to the following argument: For a real constant $\epsilon$ with $0 \leq \epsilon \leq 1$ , the RC connec-                                                                                         |
| mielke-geometrodynamics_F00681 | 138  | ■ 0.973 | $\Gamma_{\alpha\beta}(\epsilon) = \Gamma_{\alpha\beta}^{\{\}} - \epsilon K_{\alpha\beta}$ | $\Gamma_{\alpha\beta}(\epsilon) = \Gamma_{\alpha\beta}^{\{\}} - \epsilon K_{\alpha\beta}$ | to $\Gamma_{\alpha\beta}(\epsilon) = \Gamma_{\alpha\beta}^{\{\}} - \epsilon K_{\alpha\beta}$ such that $\Gamma_{\alpha\beta}(0) = \Gamma_{\alpha\beta}^{\{\}}$ . Provided a topological quantity |
| mielke-geometrodynamics_F00682 | 138  | ■ 0.973 | $\Gamma_{\alpha\beta}(0) = \Gamma_{\alpha\beta}^{\{\}}$                                   | $\Gamma_{\alpha\beta}(0) = \Gamma_{\alpha\beta}^{\{\}}$                                   | to $\Gamma_{\alpha\beta}(\epsilon) = \Gamma_{\alpha\beta}^{\{\}} - \epsilon K_{\alpha\beta}$ such that $\Gamma_{\alpha\beta}(0) = \Gamma_{\alpha\beta}^{\{\}}$ . Provided a topological quantity |
| mielke-geometrodynamics_F00683 | 138  | ■ 1.000 | $\Gamma_{\alpha\beta}$                                                                    | $\Gamma_{\alpha\beta}$                                                                    | depends on the connection $\Gamma_{\alpha\beta}$ , it must be invariant under this continuous defor-                                                                                             |
| mielke-geometrodynamics_F00684 | 138  | ■ 0.679 | $T := e_{\alpha} \rfloor T^{\alpha} = d\sigma$                                            | $T := e_{\alpha} \rfloor T^{\alpha} = d\sigma$                                            | torsion $T := e_{\alpha} \rfloor T^{\alpha} = d\sigma$ derived from a scalar field via a gradient (GREGORASH &                                                                                   |
| mielke-geometrodynamics_F00685 | 138  | ■ 1.000 | $N$                                                                                       | $N$                                                                                       | manifold, which may enclose the singularity line of the dislocation $N$ times, yields                                                                                                            |
| mielke-geometrodynamics_F00686 | 139  | ■ 1.000 | $\sigma =$                                                                                | $\sigma =$                                                                                | provided the scalar field is not single-valued, as, for instance, in the case of $\sigma =$                                                                                                      |
| mielke-geometrodynamics_F00687 | 139  | ■ 0.997 | $\arcsin( x )$                                                                            | $\arcsin( x )$                                                                            | $\arcsin( x )$ .                                                                                                                                                                                 |
| mielke-geometrodynamics_F00688 | 139  | ■ 1.000 | $\pi_3(SO(4)) =$                                                                          | $\pi_3(SO(4)) =$                                                                          | nontrivial winding number will occur as a result of the homotopy group $\pi_3(SO(4)) \cong$                                                                                                      |
| mielke-geometrodynamics_F00689 | 139  | ■ 1.000 | $\mathbb{Z} \oplus \mathbb{Z}$                                                            | $\mathbb{Z} \oplus \mathbb{Z}$                                                            | $\mathbb{Z} \oplus \mathbb{Z}$ . On the other hand, the 3D hypersurface admits the intriguing mapping $\ell \vartheta^A \rightarrow$                                                             |
| mielke-geometrodynamics_F00690 | 139  | ■ 1.000 | $\ell \underline{\vartheta}^A \rightarrow$                                                | $\ell \underline{\vartheta}^A \rightarrow$                                                | $\mathbb{Z} \oplus \mathbb{Z}$ . On the other hand, the 3D hypersurface admits the intriguing mapping $\ell \underline{\vartheta}^A \rightarrow$                                                 |
| mielke-geometrodynamics_F00691 | 139  | ■ 0.999 | $\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$          | $\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$          | $\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$ of the triads to the corresponding three-dual of the $\mathfrak{su}(2)$ -valued                                 |

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| mielke-geometrodynamics_F00692 | 139  | ■ 0.999 | $\mathfrak{su}(2)$                                                                                | $\mathfrak{su}(2)$                                                                                | $\underline{\Gamma}^{*A} := \frac{1}{2}\eta^{ABC} \underline{\Gamma}_{BC}$ of the triads to the corresponding three-dual of the $\mathfrak{su}(2)$ -valued       |
| mielke-geometrodynamics_F00693 | 139  | ■ 0.718 | $\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}} + \Lambda' \underline{\eta}$ | $\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}} + \Lambda' \underline{\eta}$ | connection (BAEKLER et al. 1992). Since this implies a mapping $\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}} + \Lambda' \underline{\eta}$ |
| mielke-geometrodynamics_F00694 | 139  | ■ 1.000 | $\mathbb{R} \times W^3$                                                                           | $\mathbb{R} \times W^3$                                                                           | small scale of “quantum gravity,” the underlying spacetime manifold $\mathbb{R} \times W^3$ may                                                                  |
| mielke-geometrodynamics_F00695 | 139  | ■ 1.000 | $d j_5$                                                                                           | $dj_5$                                                                                            | In classical EC theory, the net axial current production $dj_5$ seems (CHANG &                                                                                   |
| mielke-geometrodynamics_F00696 | 139  | ■ 1.000 | $d \mathscr{A} \wedge d \mathscr{A}$                                                              | $d\mathscr{A} \wedge d\mathscr{A}$                                                                | $d\mathscr{A} \wedge d\mathscr{A}$ involving the axial torsion. Since torsion instantons are characterized                                                       |
| mielke-geometrodynamics_F00697 | 140  | ■ 0.685 | $\theta_{\mathrm{T}} = 2 / \gamma$                                                                | $\theta_{\mathrm{T}} = 2/\gamma$                                                                  | <sup>5</sup> The translational angle $\theta_{\mathrm{T}} = 2/\gamma$ is at times identified (FREIDEL et al. 2005) with the inverse                              |
| mielke-geometrodynamics_F00698 | 140  | ■ 0.985 | $d C_{\mathrm{TT}}$                                                                               | $dC_{\mathrm{TT}}$                                                                                | lational Chern–Simons term $dC_{\mathrm{TT}}$ were considered earlier by MIELKE (1992).                                                                          |
| mielke-geometrodynamics_F00699 | 141  | ■ 1.000 | $\vartheta_{[\beta} \wedge H_{\alpha]}$                                                           | $\vartheta_{[\beta} \wedge H_{\alpha]}$                                                           | Note that the angular momentum part $\vartheta_{[\beta} \wedge H_{\alpha]}$ of the gauge fields induced by                                                       |
| mielke-geometrodynamics_F00700 | 141  | ■ 1.000 | $D H_{\alpha\beta}$                                                                               | $DH_{\alpha\beta}$                                                                                | the NY term cancels identically against one part of $DH_{\alpha\beta}$ . In the case of Dirac fields                                                             |
| mielke-geometrodynamics_F00701 | 141  | ■ 1.000 | $\mu_{\alpha}$                                                                                    | $\mu_{\alpha}$                                                                                    | (MIELKE 2004), the <i>spin–energy potential</i> $\mu_{\alpha}$ , a two-form, is related to the axial                                                             |
| mielke-geometrodynamics_F00702 | 141  | ■ 1.000 | $j_5 = \bar{\psi}^* \gamma \gamma_5 \psi$                                                         | $j_5 = \bar{\psi}^* \gamma \gamma_5 \psi$                                                         | current three-form $j_5 = \bar{\psi}^* \gamma \gamma_5 \psi$ via                                                                                                 |
| mielke-geometrodynamics_F00703 | 141  | ■ 1.000 | $D \theta_{\mathrm{L}} \neq 0$                                                                    | $D\theta_{\mathrm{L}} \neq 0$                                                                     | that (6.7.6) for $D\theta_{\mathrm{L}} \neq 0$ provides torsion with a dynamical coupling to RC curva-                                                           |
| mielke-geometrodynamics_F00704 | 141  | ■ 1.000 | $\mu_{\alpha} \wedge \vartheta^{\alpha} = 0$                                                      | $\mu_{\alpha} \wedge \vartheta^{\alpha} = 0$                                                      | i.e., $\mu_{\alpha} \wedge \vartheta^{\alpha} = 0$ , in view of (6.7.8).                                                                                         |
| mielke-geometrodynamics_F00705 | 141  | ■ 0.726 | $T = e_{\alpha} \rfloor T^{\alpha}$                                                               | $T = e_{\alpha} \rfloor T^{\alpha}$                                                               | where the vector torsion one-form $T = e_{\alpha} \rfloor T^{\alpha}$ enters as an intermediate source. Due                                                      |
| mielke-geometrodynamics_F00706 | 141  | ■ 0.815 | $DD\theta = dd\theta \equiv 0$                                                                    | $DD\theta = dd\theta \equiv 0$                                                                    | to the Poincaré lemma $\underline{DD\theta} = \underline{dd\theta} = \underline{0}$ for a (pseudo) scalar field, Eq. (6.7.10) has                                |
| mielke-geometrodynamics_F00707 | 141  | ■ 1.000 | $\vartheta^{\beta}$                                                                               | $\vartheta^{\beta}$                                                                               | as a first integral. After a contraction with $\vartheta^{\beta}$ , the dual of the axial torsion one-form                                                       |
| mielke-geometrodynamics_F00708 | 141  | ■ 0.996 | $\mathscr{A} := *(\vartheta^{\alpha} \wedge T_{\alpha})$                                          | $\mathscr{A} := *(\vartheta^{\alpha} \wedge T_{\alpha})$                                          | $\mathscr{A} := *(\vartheta^{\alpha} \wedge T_{\alpha})$ , i.e., the translational CS term, turns out to be related to the volume                                |
| mielke-geometrodynamics_F00709 | 141  | ■ 0.999 | $\theta_{\mathrm{L}}$                                                                             | $\theta_{\mathrm{L}}$                                                                             | a coupling to a kinetic term arising from the pseudoscalar field $\theta_{\mathrm{L}}$ rescaling the                                                             |
| mielke-geometrodynamics_F00710 | 141  | ■ 1.000 | $T^{[\gamma} \wedge T^{\mu]} \eta_{\alpha\beta\gamma\mu} = 0$                                     | $T^{[\gamma} \wedge T^{\mu]} \eta_{\alpha\beta\gamma\mu} = 0$                                     | $T^{[\gamma} \wedge T^{\mu]} \eta_{\alpha\beta\gamma\mu} = 0$ . Again we end up with a first-order equation for torsion, where,                                  |
| mielke-geometrodynamics_F00711 | 142  | ■ 1.000 | $DH$                                                                                              | $DH$                                                                                              | Part of the sources for the Yang–Mills-type divergence terms “ $DH$ ” are the mate-                                                                              |
| mielke-geometrodynamics_F00712 | 143  | ■ 1.000 | $L_g$                                                                                             | $L_g$                                                                                             | aid of the coframe $\vartheta^{\alpha}$ , in general gives us back the gauge Lagrangian $L_g$ amended                                                            |
| mielke-geometrodynamics_F00713 | 143  | ■ 1.000 | ${}^{(i)}T^{\alpha}$                                                                              | ${}^{(i)}T^{\alpha}$                                                                              | Lagrangian, which is at most <i>quadratic</i> in the irreducible pieces ${}^{(i)}T^{\alpha}$ and ${}^{(j)}R^{\alpha\beta}$                                       |
| mielke-geometrodynamics_F00714 | 143  | ■ 1.000 | ${}^{(j)}R^{\alpha\beta}$                                                                         | ${}^{(j)}R^{\alpha\beta}$                                                                         | Lagrangian, which is at most <i>quadratic</i> in the irreducible pieces ${}^{(i)}T^{\alpha}$ and ${}^{(j)}R^{\alpha\beta}$                                       |
| mielke-geometrodynamics_F00715 | 143  | ■ 1.000 | $\ell \simeq 10^{-32} \mathrm{~cm}$                                                               | $\ell \simeq 10^{-32} \mathrm{~cm}$                                                               | The gravitational coupling constant is absorbed in the Planck length $\ell \simeq 10^{-32} \mathrm{~cm}$ ,                                                       |
| mielke-geometrodynamics_F00716 | 143  | ■ 1.000 | $a_0, a_i (i = 1, 2, 3), \kappa$                                                                  | $a_0, a_i (i = 1, 2, 3), \kappa$                                                                  | whereas $a_0, a_i (i = 1, 2, 3), \kappa$ , and $b_j (j = 1, \dots, 6)$ are dimensionless coupling                                                                |
| mielke-geometrodynamics_F00717 | 143  | ■ 1.000 | $b_j (j = 1, \dots, 6)$                                                                           | $b_j (j = 1, \dots, 6)$                                                                           | whereas $a_0, a_i (i = 1, 2, 3), \kappa$ , and $b_j (j = 1, \dots, 6)$ are dimensionless coupling                                                                |
| mielke-geometrodynamics_F00718 | 144  | ■ 1.000 | $C: Q \rightarrow Q' = -Q$                                                                        | $C: Q \rightarrow Q' = -Q$                                                                        | a charge conjugation $\underline{C: Q \rightarrow Q' = -Q}$ , one obtains the same gravitational field.                                                          |

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| mielke-geometrodynamics_F00719 | 144  | ■ 1.000 | C                                                                                                                               | $C$                                                                                                                                    | Maxwell equations are invariant under CPT, charge conjugation $C$ has to be accom-                                                                                                                                  |
| mielke-geometrodynamics_F00720 | 144  | ■ 0.963 | T                                                                                                                               | $T$                                                                                                                                    | panied by an additional time reversal $T$ (particles moving backward in time $\stackrel{\wedge}{=}$ antipar-                                                                                                        |
| mielke-geometrodynamics_F00721 | 144  | ■ 0.963 | $\stackrel{\wedge}{=}$                                                                                                          | $\stackrel{\wedge}{=}$                                                                                                                 | panied by an additional time reversal $T$ (particles moving backward in time $\stackrel{\wedge}{=}$ antipar-                                                                                                        |
| mielke-geometrodynamics_F00722 | 144  | ■ 1.000 | $a_{\{0\}}=0, a_{\{i\}}=1$                                                                                                      | $a_0 = 0, a_i = 1$                                                                                                                     | by the choice $a_0 = 0, a_i = 1$ , and $b_j = 1$ . In order to obtain its complexified version,                                                                                                                     |
| mielke-geometrodynamics_F00723 | 144  | ■ 1.000 | $b_{\{j\}}=1$                                                                                                                   | $b_j = 1$                                                                                                                              | by the choice $a_0 = 0, a_i = 1$ , and $b_j = 1$ . In order to obtain its complexified version,                                                                                                                     |
| mielke-geometrodynamics_F00724 | 144  | ■ 1.000 | $\theta_{\mathrm{T}}$                                                                                                           | $\theta_{\mathrm{T}}$                                                                                                                  | parameters $\theta_{\mathrm{T}}$ and $\theta_{\mathrm{L}}$ :                                                                                                                                                        |
| mielke-geometrodynamics_F00725 | 145  | ■ 1.000 | P: $\vartheta^{\mathrm{A}}=-\vartheta^{\mathrm{A}}$                                                                             | $P : \vartheta'^A = -\vartheta^A$                                                                                                      | are invariant under space reflections $\overline{P \cdot \vartheta'^A} = -\vartheta^A$ , where $A = 1, 2, 3$ , due to the                                                                                           |
| mielke-geometrodynamics_F00726 | 145  | ■ 1.000 | $A=1,2,3$                                                                                                                       | $A = 1, 2, 3$                                                                                                                          | are invariant under space reflections $P : \vartheta'^A = -\vartheta^A$ , where $A = 1, 2, 3$ , due to the                                                                                                          |
| mielke-geometrodynamics_F00727 | 145  | ■ 1.000 | $\eta_{\alpha\beta\gamma\delta}$                                                                                                | $\eta_{\alpha\beta\gamma\delta}$                                                                                                       | occurrence of the determinant of the metric in $\eta_{\alpha\beta\gamma\delta}$ . In contradistinction to these                                                                                                     |
| mielke-geometrodynamics_F00728 | 145  | ■ 1.000 | $R_{\{\alpha\}}{\ }^{\{\beta\}} \wedge R_{\{\beta\}}{\ }^{\{\alpha\}}$                                                          | $R_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha}$                                                                                     | The term $R_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha}$ is known to yield, after integration over spacetime, a topo-                                                                                            |
| mielke-geometrodynamics_F00729 | 145  | ■ 1.000 | $\langle 0   R^{\alpha\beta}   0 \rangle = 0$                                                                                   | $\langle 0   R^{\alpha\beta}   0 \rangle = 0$                                                                                          | Even above the Planck scale, we can at most require $\langle 0   R^{\alpha\beta}   0 \rangle = 0$ , but we could                                                                                                    |
| mielke-geometrodynamics_F00730 | 151  | ■ 0.938 | $\Omega_{\mathrm{C}}$                                                                                                           | $\Omega_{\mathrm{C}}$                                                                                                                  | Let $\Omega_{\mathrm{C}}$ be the 2-form of a generalized curvature tensor (NOMIZU 1972) that is                                                                                                                     |
| mielke-geometrodynamics_F00731 | 151  | ■ 0.929 | $\operatorname{SL}(2, \mathbb{C}) \approx \widetilde{\operatorname{SO}(3, 1)}$                                                  | $\operatorname{SL}(2, \mathbb{C}) \approx \widetilde{\operatorname{SO}(3, 1)}$                                                         | <sup>1</sup> The bispinor representation of the covering group $\operatorname{SL}(2, \mathbb{C}) \approx \widetilde{\operatorname{SO}(3, 1)}$ is inapt for this purpose,                                            |
| mielke-geometrodynamics_F00732 | 151  | ■ 1.000 | $\sigma_{\alpha\beta}=\frac{1}{2}\left[\gamma_{\alpha},\gamma_{\beta}\right]$                                                   | $\sigma_{\alpha\beta} = \frac{1}{2} [\gamma_{\alpha}, \gamma_{\beta}]$                                                                 | since its infinitesimal generators $\sigma_{\alpha\beta} = \frac{1}{2} [\gamma_{\alpha}, \gamma_{\beta}]$ are neither self-dual nor anti-self-dual. As a direct                                                     |
| mielke-geometrodynamics_F00733 | 151  | ■ 1.000 | $\sigma_{\alpha\beta}^{(\star)}=\gamma^5\sigma_{\alpha\beta}$                                                                   | $\sigma_{\alpha\beta}^{(\star)} = \gamma^5 \sigma_{\alpha\beta}$                                                                       | sum of the spinor and its conjugate representation, they rather satisfy the relation $\sigma_{\alpha\beta}^{(\star)} = \gamma^5 \sigma_{\alpha\beta}$ ,                                                             |
| mielke-geometrodynamics_F00734 | 151  | ■ 0.829 | $\gamma^5:=\frac{1}{4!}\varepsilon_{\mathrm{abcd}}\gamma^{\mathrm{a}}\gamma^{\mathrm{b}}\gamma^{\mathrm{c}}\gamma^{\mathrm{d}}$ | $\gamma^5 := \frac{1}{4!} \varepsilon_{\mathrm{abcd}} \gamma^{\mathrm{a}} \gamma^{\mathrm{b}} \gamma^{\mathrm{c}} \gamma^{\mathrm{d}}$ | whereby the matrix $\gamma^5 := \frac{1}{4!} \varepsilon_{\mathrm{abcd}} \gamma^{\mathrm{a}} \gamma^{\mathrm{b}} \gamma^{\mathrm{c}} \gamma^{\mathrm{d}}$ is, within the Clifford algebra, an axial Lorentz scalar. |
| mielke-geometrodynamics_F00735 | 152  | ■ 1.000 | $s=0$                                                                                                                           | $s = 0$                                                                                                                                | In a four-dimensional Riemannian manifold with the Euclidean signature $s = 0$ ,                                                                                                                                    |
| mielke-geometrodynamics_F00736 | 153  | ■ 1.000 | $R_{\alpha\beta}$                                                                                                               | $R_{\alpha\beta}$                                                                                                                      | where $R_{\alpha\beta}$ is the curvature two-form. In WEYL (1919), in the concluding chapter                                                                                                                        |
| mielke-geometrodynamics_F00737 | 153  | ■ 0.813 | $\Gamma^{\alpha\beta}:=\Gamma^{\alpha\beta}_{\mathrm{i}}dx^{\mathrm{i}}$                                                        | $\Gamma^{\alpha\beta} := \Gamma^{\alpha\beta}_{\mathrm{i}} dx^{\mathrm{i}}$                                                            | $\Gamma^{\alpha\beta} := \Gamma^{\alpha\beta}_{\mathrm{i}} dx^{\mathrm{i}}$ . In his purely affine approach, Yang did not consider the canonical                                                                    |
| mielke-geometrodynamics_F00738 | 153  | ■ 0.977 | $\vartheta^{\alpha}=E_{\mathrm{j}}{\ }^{\{\alpha\}}dx^{\mathrm{j}}$                                                             | $\vartheta^{\alpha} = E_j{}^{\alpha} dx^j$                                                                                             | with respect to the coframe $\vartheta^{\alpha} = E_j{}^{\alpha} dx^j$ dual to the local anholonomic frame $e_{\alpha}$ .                                                                                           |
| mielke-geometrodynamics_F00739 | 153  | ■ 0.977 | $e_{\alpha}$                                                                                                                    | $e_{\alpha}$                                                                                                                           | with respect to the coframe $\vartheta^{\alpha} = E_j{}^{\alpha} dx^j$ dual to the local anholonomic frame $e_{\alpha}$ .                                                                                           |
| mielke-geometrodynamics_F00740 | 154  | ■ 1.000 | $G_{\mathrm{i}\mathrm{j}}$                                                                                                      | $G_{ij}$                                                                                                                               | which is dual to the usual Einstein tensor $G_{ij}$ in Riemannian spacetime. In exterior                                                                                                                            |
| mielke-geometrodynamics_F00741 | 154  | ■ 0.494 | $\operatorname{Ric}_{\alpha\beta}:=(-1)^s*\left(R_{\{\alpha\}}{\ }^{\{\delta\}}\wedge\eta_{\delta \beta}\right)$                | $\operatorname{Ric}_{\alpha\beta} := (-1)^s * (R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)})$                                    | $\operatorname{Ric}_{\alpha\beta} := (-1)^s * (R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)})$ .                                                                                                               |
| mielke-geometrodynamics_F00742 | 154  | ■ 1.000 | $\tau_{\alpha\beta}=\vartheta_{[\alpha}\wedge$                                                                                  | $\tau_{\alpha\beta} = \vartheta_{[\alpha} \wedge$                                                                                      | Macroscopically, the matter coupling <sup>2</sup> to the spin current three-form $\tau_{\alpha\beta} = \vartheta_{[\alpha} \wedge$                                                                                  |
| mielke-geometrodynamics_F00743 | 154  | ■ 0.998 | $\mu_{\beta]}$                                                                                                                  | $\mu_{\beta]}$                                                                                                                         | $\mu_{\beta]}$ seems to disfavor such models, a criticism which has already raised by FAIRCHILD                                                                                                                     |
| mielke-geometrodynamics_F00744 | 154  | ■ 0.589 | $\Gamma^{\{\beta\alpha\beta\}}\propto\partial g$                                                                                | $\Gamma^{\{\beta\alpha\beta\}} \propto \partial g$                                                                                     | In Riemannian spacetime, with its Levi–Civita connection $\Gamma^{\{\}}{}^{\alpha\beta} \propto \partial g$ , these                                                                                                 |

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| mielke-geometrodynamics_F00745 | 154  | ■ 1.000 | $\{ \}^{\{\backslashstar\}}$                                                                                                                                                                                                                                                                       | $(\star)$                                                                                                                                                                                     | and the Lie dual $(\star)$ of the curvature $R_{\alpha\beta}$ in a space(time) of signature $s$ .                                                                                                                  |
| mielke-geometrodynamics_F00746 | 154  | ■ 1.000 | $L_{\{\text{SKY}\}^{\{\backslashstar\}}}=2 L_{\{\text{SKY}\}}$                                                                                                                                                                                                                                     | $L_{\text{SKY}}^{(\star)} = 2L_{\text{SKY}}$                                                                                                                                                  | i.e., they lead to $L_{\text{SKY}}^{(\star)} = 2L_{\text{SKY}}$ . Both second-order duality conditions satisfy Yang’s                                                                                              |
| mielke-geometrodynamics_F00747 | 154  | ■ 1.000 | $j_5:=\bar{\psi}^{\{\backslashpsi\}^{\{\star\}}}\gamma\gamma_5\psi$                                                                                                                                                                                                                                | $j_5 := \psi^* \gamma \gamma_5 \psi$                                                                                                                                                          | current $j_5 := \psi^* \gamma \gamma_5 \psi$ .                                                                                                                                                                     |
| mielke-geometrodynamics_F00748 | 155  | ■ 1.000 | $D\eta_{\alpha\beta}=0$                                                                                                                                                                                                                                                                            | $D\eta_{\alpha\beta} = 0$                                                                                                                                                                     | its Lie dual, due to $D\eta_{\alpha\beta} = 0$ in a Riemannian spacetime. More on this “vacuum                                                                                                                     |
| mielke-geometrodynamics_F00749 | 155  | ■ 1.000 | $\zeta=\pm 1$                                                                                                                                                                                                                                                                                      | $\zeta = \pm 1$                                                                                                                                                                               | whereby only $\zeta = \pm 1$ (and zero) are admitted.                                                                                                                                                              |
| mielke-geometrodynamics_F00750 | 155  | ■ 1.000 | $\zeta=+1$                                                                                                                                                                                                                                                                                         | $\zeta = +1$                                                                                                                                                                                  | $\zeta = +1$ : <i>self-double-dual curvature</i>                                                                                                                                                                   |
| mielke-geometrodynamics_F00751 | 155  | ■ 1.000 | $\zeta=-1$                                                                                                                                                                                                                                                                                         | $\zeta = -1$                                                                                                                                                                                  | $\zeta = -1$ : <i>anti-self-double-dual curvature</i>                                                                                                                                                              |
| mielke-geometrodynamics_F00752 | 155  | ■ 0.999 | $\mathscr{A}:=\{ \}^{\{\star\}}\left(\vartheta_{\alpha}\wedge T^{\alpha}\right)$                                                                                                                                                                                                                   | $\mathscr{A} := \star (\vartheta_{\alpha} \wedge T^{\alpha})$                                                                                                                                 | Riemannian curvature $R_{\alpha\beta}^{\{\}}$ and the axial torsion one-form $\mathscr{A} := \star (\vartheta_{\alpha} \wedge \hat{T}^{\alpha})$ is moti-                                                          |
| mielke-geometrodynamics_F00753 | 155  | ■ 0.638 | $\left\langle d j_5 \right\rangle = 2 i^{\star} m \left\langle \bar{\psi} \gamma_5 \psi \right\rangle - \left( R_{\alpha \beta}^{\{\}} \wedge R^{\{j \alpha \beta\}} + \frac{1}{2} d \mathscr{A} \wedge d \mathscr{A} \right) /$                                                                   | $\langle dj_5 \rangle = 2i^* m \langle \bar{\psi} \gamma_5 \psi \rangle - \left( R_{\alpha\beta}^{\{\}} \wedge R^{\{j\alpha\beta\}} + \frac{1}{2} d\mathscr{A} \wedge d\mathscr{A} \right) /$ | vated by the <i>axial anomaly</i> $\langle dj_5 \rangle = 2i^* m \langle \bar{\psi} \gamma_5 \psi \rangle - (R_{\alpha\beta}^{\{\}} \wedge R^{\{\} \alpha\beta} + \frac{1}{2} d\mathscr{A} \wedge d\mathscr{A}) /$ |
| mielke-geometrodynamics_F00754 | 155  | ■ 1.000 | $48 \pi^2$                                                                                                                                                                                                                                                                                         | $48\pi^2$                                                                                                                                                                                     | $48\pi^2$ in the coupling to Dirac fields $\psi$ .                                                                                                                                                                 |
| mielke-geometrodynamics_F00755 | 156  | ■ 1.000 | $d^4 x$                                                                                                                                                                                                                                                                                            | $d^4 x$                                                                                                                                                                                       | arises, where the four-dimensional integration over $d^4 x$ is already implied in our                                                                                                                              |
| mielke-geometrodynamics_F00756 | 156  | ■ 0.996 | $\{ \}^{\{\star\}} R_{\alpha \beta}=\theta_{\mathrm{L}}^{\{\backslashtheta\}\{\backslashstar\}} R_{\alpha \beta}^{\{\backslashalpha\}\{\backslashbeta\}^{\{\backslashstar\}}+\left(\theta_{\mathrm{T}}^{\{\backslashtheta\}\{\backslashstar\}} \backslash \Lambda / 6\right) \eta_{\alpha \beta}}$ | ${}^{\star} R_{\alpha\beta} = \theta_{\text{L}}^{\star} R_{\alpha\beta}^{(\star)} + (\theta_{\text{T}}^{\star} \Lambda / 6) \eta_{\alpha\beta}$                                               | 1994) via a modified duality ${}^{\star} R_{\alpha\beta} = \theta_{\text{L}}^{\star} R_{\alpha\beta}^{(\star)} + (\theta_{\text{T}}^{\star} \Lambda / 6) \eta_{\alpha\beta}$ , which, however, explic-             |
| mielke-geometrodynamics_F00757 | 156  | ■ 0.998 | $\sigma_{\alpha}:=\Sigma_{\alpha}-D^{\{\}\{\}} \mu_{\alpha}$                                                                                                                                                                                                                                       | $\sigma_{\alpha} := \Sigma_{\alpha} - D^{\{\}} \mu_{\alpha}$                                                                                                                                  | & MAGGIOLO 2005) symmetrized energy–momentum current $\sigma_{\alpha} := \Sigma_{\alpha} - D^{\{\}} \mu_{\alpha}.$ )                                                                                               |
| mielke-geometrodynamics_F00758 | 156  | ■ 0.682 | $\left.\left.\left.R:=e_{\alpha}\right\rfloor e_{\beta}\right\rfloor R^{\alpha \beta}\right.$                                                                                                                                                                                                      | $R := e_{\alpha} \rfloor e_{\beta} \rfloor R^{\alpha\beta}$                                                                                                                                   | trace with $R := e_{\alpha} \rfloor e_{\beta} \rfloor R^{\alpha\beta}$ as irreducible pieces. Then the double-dual curvature                                                                                       |
| mielke-geometrodynamics_F00759 | 157  | ■ 1.000 | $o_{ij}^{\text{Mink}}$                                                                                                                                                                                                                                                                             | $o_{ij}^{\text{Mink}}$                                                                                                                                                                        | with $o_{ij}^{\text{Mink}}$ denoting the (pseudo)orthogonal Minkowski metric. Earlier, EINSTEIN &                                                                                                                  |
| mielke-geometrodynamics_F00760 | 157  | ■ 0.529 | $D^{\{\}\{\}} G_{\alpha}^{\{\}\{\}} \equiv 0$                                                                                                                                                                                                                                                      | $D^{\{\}} G_{\alpha}^{\{\}} \equiv 0$                                                                                                                                                         | Due to the contracted Bianchi identity $D^{\{\}} G_{\alpha}^{\{\}} \equiv 0$ in Riemannian spacetime, the                                                                                                          |
| mielke-geometrodynamics_F00761 | 157  | ■ 0.999 | $R=-4 \Lambda$                                                                                                                                                                                                                                                                                     | $R = -4\Lambda$                                                                                                                                                                               | curvature scalar is constant, i.e., more precisely, $R = -4\Lambda$ . Thus, (7.4.6) is equivalent                                                                                                                  |
| mielke-geometrodynamics_F00762 | 157  | ■ 1.000 | $\Lambda \simeq 1 / \kappa$                                                                                                                                                                                                                                                                        | $\Lambda \simeq 1/\kappa$                                                                                                                                                                     | with cosmological constant $\Lambda \simeq 1/\kappa$ . It is crucial to note that the latter carries phys-                                                                                                         |
| mielke-geometrodynamics_F00763 | 157  | ■ 1.000 | $R=0$                                                                                                                                                                                                                                                                                              | $R = 0$                                                                                                                                                                                       | sical Einstein–Fokker background due to $R = 0$ in double-dual Thompson spaces.                                                                                                                                    |
| mielke-geometrodynamics_F00764 | 157  | ■ 1.000 | $h_{ij}:=g_{ij}-g_{ij}^{\mathrm{dS}}$                                                                                                                                                                                                                                                              | $h_{ij} := g_{ij} - g_{ij}^{\text{dS}}$                                                                                                                                                       | On the other hand, the linearized Einstein equations for $h_{ij} := g_{ij} - g_{ij}^{\text{dS}}$ in a de                                                                                                           |
| mielke-geometrodynamics_F00765 | 157  | ■ 1.000 | $g_{ij}^{\mathrm{dS}}$                                                                                                                                                                                                                                                                             | $g_{ij}^{\text{dS}}$                                                                                                                                                                          | Sitter background $g_{ij}^{\text{dS}}$ with a positive cosmological constant are                                                                                                                                   |
| mielke-geometrodynamics_F00766 | 158  | ■ 1.000 | $m / H$                                                                                                                                                                                                                                                                                            | $m/H$                                                                                                                                                                                         | de Sitter space if $m/H$ is kept finite in the massless limit. For gravitons, the mass                                                                                                                             |
| mielke-geometrodynamics_F00767 | 158  | ■ 1.000 | $m \leq 7 \times 10^{-32} \mathrm{eV}$                                                                                                                                                                                                                                                             | $m \leq 7 \times 10^{-32} \text{eV}$                                                                                                                                                          | (GOLDHABER & NIETO 2010) of $m \leq 7 \times 10^{-32} \text{ eV}$ , but larger than the Hubble                                                                                                                     |
| mielke-geometrodynamics_F00768 | 158  | ■ 1.000 | $H \simeq 10^{-33} \mathrm{eV}$                                                                                                                                                                                                                                                                    | $H \simeq 10^{-33} \text{eV}$                                                                                                                                                                 | scale of $H \sim 10^{-33} \text{ eV}$ .                                                                                                                                                                            |
| mielke-geometrodynamics_F00769 | 158  | ■ 0.994 | $\Lambda_{\text{obs}} > 0$                                                                                                                                                                                                                                                                         | $\Lambda_{\text{obs}} > 0$                                                                                                                                                                    | is induced via the tiny cosmological constant $\Lambda_{\text{obs}} > 0$ of our expanding universe.                                                                                                                |

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| mielke-geometrodynamics_F00770 | 158  | ■ 0.994 | $1 / g^2$                                                                                               | $1/g^2$                                                                                                 | constant $1/g^2$ is ‘traded’ for a dimensional one. This “dimensional transmutation” à                                                                                                                                            |
| mielke-geometrodynamics_F00771 | 159  | ■ 0.662 | $\operatorname{SU}(2)$                                                                                  | $SU(2)$                                                                                                 | Originally, Yang and Mills formulated their model as an $SU(2)$ gauge theory with                                                                                                                                                 |
| mielke-geometrodynamics_F00772 | 159  | ■ 0.968 | 1974)                                                                                                   | 1974)                                                                                                   | <a href="#">1974</a> ) is an $SO(1, 3)$ gauge theory, whose covering gauge group, after a Wick rotation                                                                                                                           |
| mielke-geometrodynamics_F00773 | 159  | ■ 1.000 | $\overline{SO}(4) \cong$                                                                                | $\overline{SO}(4) \cong$                                                                                | to Euclidean space in the path integral quantization, is locally isomorphic to $\overline{SO}(4) \cong$                                                                                                                           |
| mielke-geometrodynamics_F00774 | 159  | ■ 1.000 | $SU(2) \otimes SU(2)$                                                                                   | $SU(2) \otimes SU(2)$                                                                                   | $SU(2) \otimes SU(2)$ . This doubling of the (internal) unitary structure group $SU(2)$ was                                                                                                                                       |
| mielke-geometrodynamics_F00775 | 159  | ■ 0.719 | $\kappa \Lambda_{\text{obs}} \simeq 10^{-123}$                                                          | $\kappa \Lambda_{\text{obs}} \simeq 10^{-123}$                                                          | occur at the tiny <i>dimensionless scale</i> of $\kappa \Lambda_{\text{obs}} \simeq 10^{-123}$ , as suggested by observa-                                                                                                         |
| mielke-geometrodynamics_F00776 | 159  | ■ 0.999 | $\Delta L$                                                                                              | $\Delta L$                                                                                              | In perturbative quantum gravity (ALVAREZ <a href="#">1989</a> ), there arise counterterms $\Delta L$ of                                                                                                                           |
| mielke-geometrodynamics_F00777 | 159  | ■ 0.995 | $\operatorname{Ric}_{\alpha\beta} := \left( R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)} \right)$ | $\operatorname{Ric}_{\alpha\beta} := \left( R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)} \right)$ | version of the zero-form $\operatorname{Ric}_{\alpha\beta} := \left( R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)} \right)$ .                                                                                                |
| mielke-geometrodynamics_F00778 | 160  | ■ 1.000 | $H_{\alpha\beta}$                                                                                       | $H_{\alpha\beta}$                                                                                       | Here the field momenta $H_{\alpha}$ and $H_{\alpha\beta}$ are understood as arising from a generating                                                                                                                             |
| mielke-geometrodynamics_F00779 | 160  | ■ 1.000 | $L_{\text{eff}}$                                                                                        | $L_{\text{eff}}$                                                                                        | n-form $G$ as part of some effective gauge Lagrangian $L_{\text{eff}}$ that includes the countert-                                                                                                                                |
| mielke-geometrodynamics_F00780 | 160  | ■ 0.637 | $n-2$                                                                                                   | $n-2$                                                                                                   | In our dynamical approach, the $(n-2)$ -forms $H_{\alpha}$ and $H_{\alpha\beta}$ will be gauge field                                                                                                                              |
| mielke-geometrodynamics_F00781 | 160  | ■ 1.000 | $P := \mathbb{R}^4 \otimes$                                                                             | $P := \mathbb{R}^4 \otimes$                                                                             | tively. Due to the <i>semidirect product</i> structure of the Poincaré group $P := \mathbb{R}^4 \ltimes$                                                                                                                          |
| mielke-geometrodynamics_F00782 | 160  | ■ 0.945 | $H^{\alpha\beta} \rightarrow \vartheta^{\alpha}$                                                        | $H^{\alpha\beta} \rightarrow \vartheta^{\alpha}$                                                        | $SO(1, 3)$ , the gauge field momenta contribute to the gauge potentials via $H^{\alpha\beta} \rightarrow \vartheta^{\alpha}$                                                                                                      |
| mielke-geometrodynamics_F00783 | 160  | ■ 1.000 | $H^{\alpha} \rightarrow \Gamma^{\alpha\beta}$                                                           | $H^{\alpha} \rightarrow \Gamma^{\alpha\beta}$                                                           | and $H^{\alpha} \rightarrow \Gamma^{\alpha\beta}$ in the FR <a href="#">(7.7.2, 7.7.3)</a> just in an <i>intertwined</i> manner.                                                                                                  |
| mielke-geometrodynamics_F00784 | 160  | ■ 0.991 | $*DH_{\alpha}{}^{\beta}$                                                                                | $*DH_{\alpha}{}^{\beta}$                                                                                | similarly as for Yang–Mills fields, a term proportional to $*DH_{\alpha}{}^{\beta}$ . However, “on                                                                                                                                |
| mielke-geometrodynamics_F00785 | 160  | ■ 0.388 | $*\left( H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$                                 | $*\left( H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$                                 | $*\left( H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$ $*H^{\beta}$ due to an algebraic identity. In the FR <a href="#">(7.7.2)</a> of the coframe,                                                              |
| mielke-geometrodynamics_F00786 | 160  | ■ 0.388 | $H^{\beta}$                                                                                             | $H^{\beta}$                                                                                             | $*\left( H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$ $*H^{\beta}$ due to an algebraic identity. In the FR <a href="#">(7.7.2)</a> of the coframe,                                                              |
| mielke-geometrodynamics_F00787 | 160  | ■ 0.644 | $*DH^{\alpha} \cong *E^{\alpha}$                                                                        | $*DH^{\alpha} \cong *E^{\alpha}$                                                                        | the same situation arises, with the modification that the “on shell” term $*DH^{\alpha} \cong *E^{\alpha}$                                                                                                                        |
| mielke-geometrodynamics_F00788 | 160  | ■ 1.000 | $\widetilde{G}$                                                                                         | $\widetilde{G}$                                                                                         | order generation functional $\widetilde{G}$ . When coupling to matter, FRs have to be handled with                                                                                                                                |
| mielke-geometrodynamics_F00789 | 161  | ■ 1.000 | $\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$                     | $\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$                     | momenta $\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$ can be expanded as $\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$ . (For the |
| mielke-geometrodynamics_F00790 | 161  | ■ 1.000 | $\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$ | $\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$ | momenta $\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$ can be expanded as $\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$ . (For the |
| mielke-geometrodynamics_F00791 | 161  | ■ 1.000 | $\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$                               | $\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$                               | time being, the field momentum $\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$ conjugate to the torsion $T^{\alpha} :=$                                                                                 |
| mielke-geometrodynamics_F00792 | 161  | ■ 1.000 | $T^{\alpha} :=$                                                                                         | $T^{\alpha} :=$                                                                                         | time being, the field momentum $\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$ conjugate to the torsion $T^{\alpha} :=$                                                                                 |
| mielke-geometrodynamics_F00793 | 161  | ■ 1.000 | $D\vartheta^{\alpha}$                                                                                   | $D\vartheta^{\alpha}$                                                                                   | $D\vartheta^{\alpha}$ will be set to zero.)                                                                                                                                                                                       |
| mielke-geometrodynamics_F00794 | 161  | ■ 1.000 | $\widetilde{H}_{\alpha\beta\mu\nu}$                                                                     | $\widetilde{H}_{\alpha\beta\mu\nu}$                                                                     | For a <i>fixed point</i> of the transformation <a href="#">(7.7.9)</a> , the Hessian $\widetilde{H}_{\alpha\beta\mu\nu}$ obviously has                                                                                            |
| mielke-geometrodynamics_F00795 | 161  | ■ 1.000 | $\partial \widetilde{H}_{\mu\nu} / \partial R^{\alpha\beta} = 0$                                        | $\partial \widetilde{H}_{\mu\nu} / \partial R^{\alpha\beta} = 0$                                        | to vanish. This condition, i.e., $\partial \widetilde{H}_{\mu\nu} / \partial R^{\alpha\beta} = 0$ , can be readily solved.                                                                                                        |

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| mielke-geometrodynamics_F00796 | 161  | ■ 1.000 | $\mathrm{R}^{\alpha}{}^{\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta}$                                                                                                                                                                                     | $\theta_{\mathrm{T}} R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta}$                                                                                                                                                                               | If parity-violating terms such as $\theta_{\mathrm{T}} R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta}$ arising from the Nieh–Yan                                                                                                                                                                          |
| mielke-geometrodynamics_F00797 | 162  | ■ 1.000 | $\widetilde{L}$                                                                                                                                                                                                                                                        | $\widetilde{L}$                                                                                                                                                                                                                                                        | Lagrangian $\widetilde{L}$ .                                                                                                                                                                                                                                                                                                  |
| mielke-geometrodynamics_F00798 | 162  | ■ 1.000 | $\eta_{\alpha\beta}$                                                                                                                                                                                                                                                   | $\eta_{\alpha\beta}$                                                                                                                                                                                                                                                   | Accordingly, $\eta_{\alpha\beta}$ and $\widetilde{H}_{\alpha\beta}$ interchange their roles as generalized coordinates and                                                                                                                                                                                                    |
| mielke-geometrodynamics_F00799 | 162  | ■ 1.000 | $\widetilde{H}_{\alpha\beta}$                                                                                                                                                                                                                                          | $\widetilde{H}_{\alpha\beta}$                                                                                                                                                                                                                                          | Accordingly, $\eta_{\alpha\beta}$ and $\widetilde{H}_{\alpha\beta}$ interchange their roles as generalized coordinates and                                                                                                                                                                                                    |
| mielke-geometrodynamics_F00800 | 162  | ■ 1.000 | $\widetilde{L} = L_{\mathrm{EC}}$                                                                                                                                                                                                                                      | $\widetilde{L} = L_{\mathrm{EC}}$                                                                                                                                                                                                                                      | momenta, respectively. If we had started from $\widetilde{L} = L_{\mathrm{EC}}$ , then we would be led back                                                                                                                                                                                                                   |
| mielke-geometrodynamics_F00801 | 162  | ■ 1.000 | $L = L_{\mathrm{EC}}$                                                                                                                                                                                                                                                  | $L = L_{\mathrm{EC}}$                                                                                                                                                                                                                                                  | to $L = L_{\mathrm{EC}}$ for the choice $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = 0$ . In the case $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = i$ , this                                                                                                                                              |
| mielke-geometrodynamics_F00802 | 162  | ■ 1.000 | $\theta_{\mathrm{T}}^* = 1$                                                                                                                                                                                                                                            | $\theta_{\mathrm{T}}^* = 1$                                                                                                                                                                                                                                            | to $L = L_{\mathrm{EC}}$ for the choice $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = 0$ . In the case $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = i$ , this                                                                                                                                              |
| mielke-geometrodynamics_F00803 | 162  | ■ 1.000 | $\theta_{\mathrm{T}} = 0$                                                                                                                                                                                                                                              | $\theta_{\mathrm{T}} = 0$                                                                                                                                                                                                                                              | to $L = L_{\mathrm{EC}}$ for the choice $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = 0$ . In the case $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = i$ , this                                                                                                                                              |
| mielke-geometrodynamics_F00804 | 162  | ■ 1.000 | $\theta_{\mathrm{T}} = i$                                                                                                                                                                                                                                              | $\theta_{\mathrm{T}} = i$                                                                                                                                                                                                                                              | to $L = L_{\mathrm{EC}}$ for the choice $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = 0$ . In the case $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = i$ , this                                                                                                                                              |
| mielke-geometrodynamics_F00805 | 162  | ■ 0.436 | $\left. \Gamma_{\alpha}{}^{\beta} \rightarrow \Gamma_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} \mp (i\ell^2/2) e_{\alpha} \right]^{(\mp)\beta}$                                                                                                                  | $\Gamma_{\alpha}{}^{\beta} \rightarrow \Gamma_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} \mp (i\ell^2/2) e_{\alpha} \Big]^{(\mp)\beta}$                                                                                                                           | ition $\Gamma_{\alpha}{}^{\beta} \rightarrow \Gamma_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} \mp (i\ell^2/2) e_{\alpha} \Big]^{(\mp)\beta}$ , induced by the translational Chern–                                                                                                                                      |
| mielke-geometrodynamics_F00806 | 162  | ■ 0.999 | $idC_{\mathrm{TT}}$                                                                                                                                                                                                                                                    | $idC_{\mathrm{TT}}$                                                                                                                                                                                                                                                    | Simons term $idC_{\mathrm{TT}}$ . The resulting Sen-type connection still couples <i>minimally</i> to the                                                                                                                                                                                                                     |
| mielke-geometrodynamics_F00807 | 162  | ■ 0.999 | $\Phi = (-1)^{p(4-p)+s} \Phi$                                                                                                                                                                                                                                          | $**\Phi = (-1)^{p(4-p)+s} \Phi$                                                                                                                                                                                                                                        | holds. Applied to $p$ -forms, it is almost involutive, i.e., $**\Phi = (-1)^{p(4-p)+s} \Phi$ . For                                                                                                                                                                                                                            |
| mielke-geometrodynamics_F00808 | 162  | ■ 0.997 | $s := \operatorname{sig} = 1$                                                                                                                                                                                                                                          | $s := \operatorname{sig} = 1$                                                                                                                                                                                                                                          | spacetimes in which $\boxed{s := \operatorname{sig} = 1}$ holds, it induces an <i>almost complex structure</i> , and                                                                                                                                                                                                          |
| mielke-geometrodynamics_F00809 | 163  | ■ 1.000 | $\{\eta, \eta_{\alpha}, \eta_{\alpha\beta}, \eta_{\alpha\beta\gamma}, \eta_{\alpha\beta\gamma\delta}\}$                                                                                                                                                                | $\{\eta, \eta_{\alpha}, \eta_{\alpha\beta}, \eta_{\alpha\beta\gamma}, \eta_{\alpha\beta\gamma\delta}\}$                                                                                                                                                                | makes it possible to generate the so-called $\eta$ , or dual, basis $\{\eta, \eta_{\alpha}, \eta_{\alpha\beta}, \eta_{\alpha\beta\gamma}, \eta_{\alpha\beta\gamma\delta}\}$                                                                                                                                                   |
| mielke-geometrodynamics_F00810 | 163  | ■ 0.925 | $\eta_{\alpha} := e_{\alpha} \rfloor \eta = *\vartheta_{\alpha}, \eta_{\alpha\beta} :=$                                                                                                                                                                                | $\eta_{\alpha} := e_{\alpha} \rfloor \eta = *\vartheta_{\alpha}, \eta_{\alpha\beta} :=$                                                                                                                                                                                | of exterior forms via consecutive interior products: $\eta_{\alpha} := e_{\alpha} \rfloor \eta = *\vartheta_{\alpha}, \eta_{\alpha\beta} :=$                                                                                                                                                                                  |
| mielke-geometrodynamics_F00811 | 163  | ■ 0.920 | $e_{\beta} \rfloor \eta_{\alpha} = \eta_{\alpha\beta\gamma} \vartheta^{\gamma} = e_{\beta} \rfloor e_{\alpha} \rfloor \eta = *(\vartheta_{\alpha} \wedge \vartheta_{\beta}) = \frac{1}{2} \eta_{\alpha\beta\gamma\delta} \vartheta^{\gamma} \wedge \vartheta^{\delta}$ | $e_{\beta} \rfloor \eta_{\alpha} = \eta_{\alpha\beta\gamma} \vartheta^{\gamma} = e_{\beta} \rfloor e_{\alpha} \rfloor \eta = *(\vartheta_{\alpha} \wedge \vartheta_{\beta}) = \frac{1}{2} \eta_{\alpha\beta\gamma\delta} \vartheta^{\gamma} \wedge \vartheta^{\delta}$ | $\boxed{e_{\beta} \rfloor \eta_{\alpha} = \eta_{\alpha\beta\gamma} \vartheta^{\gamma} = e_{\beta} \rfloor e_{\alpha} \rfloor \eta = *(\vartheta_{\alpha} \wedge \vartheta_{\beta}) = \frac{1}{2} \eta_{\alpha\beta\gamma\delta} \vartheta^{\gamma} \wedge \vartheta^{\delta}}$ , and $\eta_{\alpha\beta\gamma} := e_{\gamma}$ |
| mielke-geometrodynamics_F00812 | 163  | ■ 0.920 | $\eta_{\alpha\beta\gamma} := e_{\gamma}$                                                                                                                                                                                                                               | $\eta_{\alpha\beta\gamma} := e_{\gamma}$                                                                                                                                                                                                                               | $e_{\beta} \rfloor \eta_{\alpha} = \eta_{\alpha\beta\gamma} \vartheta^{\gamma} = e_{\beta} \rfloor e_{\alpha} \rfloor \eta = *(\vartheta_{\alpha} \wedge \vartheta_{\beta}) = \frac{1}{2} \eta_{\alpha\beta\gamma\delta} \vartheta^{\gamma} \wedge \vartheta^{\delta}$ , and $\eta_{\alpha\beta\gamma} := e_{\gamma}$         |
| mielke-geometrodynamics_F00813 | 163  | ■ 0.862 | $\rfloor \eta_{\alpha\beta} = *(\vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge \vartheta_{\gamma})$                                                                                                                                                                | $\rfloor \eta_{\alpha\beta} = *(\vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge \vartheta_{\gamma})$                                                                                                                                                                | $\rfloor \eta_{\alpha\beta} = *(\vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge \vartheta_{\gamma})$ . Anholonomic indices are lowered by $o_{\alpha\beta} = e^i{}_{\alpha} e^j{}_{\beta} g_{ij}$ ,                                                                                                                        |
| mielke-geometrodynamics_F00814 | 163  | ■ 0.862 | $o_{\alpha\beta} = e^i{}_{\alpha} e^j{}_{\beta} g_{ij}$                                                                                                                                                                                                                | $o_{\alpha\beta} = e^i{}_{\alpha} e^j{}_{\beta} g_{ij}$                                                                                                                                                                                                                | $\rfloor \eta_{\alpha\beta} = *(\vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge \vartheta_{\gamma})$ . Anholonomic indices are lowered by $o_{\alpha\beta} = e^i{}_{\alpha} e^j{}_{\beta} g_{ij}$ ,                                                                                                                        |
| mielke-geometrodynamics_F00815 | 163  | ■ 1.000 | $o_{\alpha\beta}$                                                                                                                                                                                                                                                      | $o_{\alpha\beta}$                                                                                                                                                                                                                                                      | where $o_{\alpha\beta}$ denotes the Minkowski metric.                                                                                                                                                                                                                                                                         |
| mielke-geometrodynamics_F00816 | 163  | ■ 1.000 | $\Phi = \frac{1}{p!} \Phi_{i_1 \dots i_p} dx^{i_1} \wedge \dots \wedge dx^{i_p}$                                                                                                                                                                                       | $\Phi = \frac{1}{p!} \Phi_{i_1 \dots i_p} dx^{i_1} \wedge \dots \wedge dx^{i_p}$                                                                                                                                                                                       | dimensional manifold, the Hodge dual of a $p$ -form $\Phi = \frac{1}{p!} \Phi_{i_1 \dots i_p} dx^{i_1} \wedge \dots \wedge dx^{i_p}$                                                                                                                                                                                          |
| mielke-geometrodynamics_F00817 | 163  | ■ 0.769 | $\Phi = (-1)^{p(4-p)+\operatorname{sig}} \Phi$                                                                                                                                                                                                                         | $**\Phi = (-1)^{p(4-p)+\operatorname{sig}} \Phi$                                                                                                                                                                                                                       | so that it is almost involutive, i.e., $**\Phi = (-1)^{p(4-p)+\operatorname{sig}} \Phi$ .                                                                                                                                                                                                                                     |

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| mielke-geometrodynamics_F00818 | 163  | ■ 0.998 | $\left\{\mathbf{1}_4, \gamma_\alpha, \sigma_{\alpha\beta}, \gamma_5, \gamma_5\gamma_\alpha\right\}$                            | $\{\mathbf{1}_4, \gamma_\alpha, \sigma_{\alpha\beta}, \gamma_5, \gamma_5\gamma_\alpha\}$                                       | In the familiar Pauli representation, the 16 matrices $\{\mathbf{1}_4, \gamma_\alpha, \sigma_{\alpha\beta}, \gamma_5, \gamma_5\gamma_\alpha\}$ , where                                                        |
| mielke-geometrodynamics_F00819 | 163  | ■ 0.994 | $\sigma_{\alpha\beta}:=\frac{i}{2}\left(\gamma_\alpha\gamma_\beta-\gamma_\beta\gamma_\alpha\right)$                            | $\sigma_{\alpha\beta}:=\frac{i}{2}(\gamma_\alpha\gamma_\beta-\gamma_\beta\gamma_\alpha)$                                       | $\sigma_{\alpha\beta}:=\frac{i}{2}(\gamma_\alpha\gamma_\beta-\gamma_\beta\gamma_\alpha)$ are the Lorentz generators, and $\gamma_5=-i\gamma_0\gamma_1\gamma_2\gamma_3$ with                                   |
| mielke-geometrodynamics_F00820 | 163  | ■ 0.994 | $\gamma_5=-i\gamma_{\hat{0}}\gamma_{\hat{1}}\gamma_{\hat{2}}\gamma_{\hat{3}}$                                                  | $\gamma_5=-i\gamma_{\hat{0}}\gamma_{\hat{1}}\gamma_{\hat{2}}\gamma_{\hat{3}}$                                                  | $\sigma_{\alpha\beta}:=\frac{i}{2}(\gamma_\alpha\gamma_\beta-\gamma_\beta\gamma_\alpha)$ are the Lorentz generators, and $\gamma_5=-i\gamma_0\gamma_1\gamma_2\gamma_3$ with                                   |
| mielke-geometrodynamics_F00821 | 163  | ■ 0.736 | $\gamma_5^2=(-1)^{\mathrm{sig}+1}\mathbf{1}_4$                                                                                 | $\gamma_5^2=(-1)^{\mathrm{sig}+1}\mathbf{1}_4$                                                                                 | $\gamma_5^2=(-1)^{\mathrm{sig}+1}\mathbf{1}_4$ constitute a basis of the <i>Clifford algebra</i> in four dimensions with                                                                                      |
| mielke-geometrodynamics_F00822 | 163  | ■ 1.000 | $\mathrm{Tr}\left(\gamma_\alpha\gamma_\beta\right)=4\,o_{\alpha\beta}$                                                         | $\mathrm{Tr}\left(\gamma_\alpha\gamma_\beta\right)=4o_{\alpha\beta}$                                                           | They are normalized by the traces $\mathrm{Tr}(\gamma_\alpha\gamma_\beta)=4\,o_{\alpha\beta}$ and $\mathrm{Tr}(\sigma_{\alpha\beta}\sigma^{\gamma\delta})=8\,\delta_{[\alpha}^\gamma\delta_{\beta]}^\delta$ , |
| mielke-geometrodynamics_F00823 | 163  | ■ 1.000 | $\mathrm{Tr}\left(\sigma_{\alpha\beta}\sigma^{\gamma\delta}\right)=8\,\delta_{[\alpha}^\gamma\delta_{\beta]}^\delta$           | $\mathrm{Tr}\left(\sigma_{\alpha\beta}\sigma^{\gamma\delta}\right)=8\delta_{[\alpha}^\gamma\delta_{\beta]}^\delta$             | They are normalized by the traces $\mathrm{Tr}(\gamma_\alpha\gamma_\beta)=4\,o_{\alpha\beta}$ and $\mathrm{Tr}(\sigma_{\alpha\beta}\sigma^{\gamma\delta})=8\,\delta_{[\alpha}^\gamma\delta_{\beta]}^\delta$ , |
| mielke-geometrodynamics_F00824 | 163  | ■ 1.000 | $[\alpha\beta]=\frac{1}{2}(\alpha\beta-\beta\alpha)$                                                                           | $[\alpha\beta]=\frac{1}{2}(\alpha\beta-\beta\alpha)$                                                                           | where $[\alpha\beta]=\frac{1}{2}(\alpha\beta-\beta\alpha)$ denotes the antisymmetrization of indices. For the unit                                                                                            |
| mielke-geometrodynamics_F00825 | 164  | ■ 0.988 | $\mathrm{Lie}$                                                                                                                 | Lie                                                                                                                            | whereas for the <i>Lie</i> (or right) dual $\sigma_{\alpha\beta}^{(\star)}:=\frac{1}{2}\sigma^{\gamma\delta}\eta_{\alpha\beta\gamma\delta}=i\gamma_5\sigma_{\alpha\beta}$ , we obtain the asso-               |
| mielke-geometrodynamics_F00826 | 164  | ■ 0.988 | $\sigma_{\alpha\beta}^{(\star)}:=\frac{1}{2}\sigma^{\gamma\delta}\eta_{\alpha\beta\gamma\delta}=i\gamma_5\sigma_{\alpha\beta}$ | $\sigma_{\alpha\beta}^{(\star)}:=\frac{1}{2}\sigma^{\gamma\delta}\eta_{\alpha\beta\gamma\delta}=i\gamma_5\sigma_{\alpha\beta}$ | whereas for the <i>Lie</i> (or right) dual $\sigma_{\alpha\beta}^{(\star)}:=\frac{1}{2}\sigma^{\gamma\delta}\eta_{\alpha\beta\gamma\delta}=i\gamma_5\sigma_{\alpha\beta}$ , we obtain the asso-               |
| mielke-geometrodynamics_F00827 | 164  | ■ 0.998 | $\sigma^{(\star)}$                                                                                                             | $\sigma^{(\star)}$                                                                                                             | $\sigma^{(\star)}$ solely by multiplication by $\gamma_5$ , which here is regarded as just an anticommuting                                                                                                   |
| mielke-geometrodynamics_F00828 | 164  | ■ 0.998 | $\gamma_5$                                                                                                                     | $\gamma_5$                                                                                                                     | $\sigma^{(\star)}$ solely by multiplication by $\gamma_5$ , which here is regarded as just an anticommuting                                                                                                   |
| mielke-geometrodynamics_F00829 | 164  | ■ 0.995 | $12\,i^*\gamma$                                                                                                                | $12i^*\gamma$                                                                                                                  | relates this to $12i^*\gamma$ , the Hodge dual <sup>3</sup> for the basis of Clifforms has been recovered.                                                                                                    |
| mielke-geometrodynamics_F00830 | 164  | ■ 1.000 | $P_\pm=\frac{1}{2}(1\pm\gamma_5)$                                                                                              | $P_\pm=\frac{1}{2}(1\pm\gamma_5)$                                                                                              | representation involves explicitly the <i>chirality projector</i> $P_\pm=\frac{1}{2}(1\pm\gamma_5)$ , obeying                                                                                                 |
| mielke-geometrodynamics_F00831 | 164  | ■ 1.000 | $P_\pm P_\pm=P_\pm$                                                                                                            | $P_\pm P_\pm=P_\pm$                                                                                                            | $P_\pm P_\pm=P_\pm$ .                                                                                                                                                                                         |
| mielke-geometrodynamics_F00832 | 164  | ■ 0.981 | $\gamma\rightarrow\widetilde{\gamma}=e^{\varphi}\gamma$                                                                        | $\gamma\rightarrow\widetilde{\gamma}=e^\varphi\gamma$                                                                          | (ATIYAH et al. 1978) under the Weyl rescaling $\gamma\rightarrow\widetilde{\gamma}=e^\varphi\gamma$ , where $\varphi$ can be viewed                                                                           |
| mielke-geometrodynamics_F00833 | 164  | ■ 1.000 | $\{ \}^{\#\}$                                                                                                                  | $\#$                                                                                                                           | $\#$ on <i>arbitrary</i> two-forms allows us to <i>reconstruct</i> a spacetime metric $h$ , which is,                                                                                                         |
| mielke-geometrodynamics_F00834 | 164  | ■ 1.000 | $h$                                                                                                                            | $h$                                                                                                                            | $\#$ on <i>arbitrary</i> two-forms allows us to <i>reconstruct</i> a spacetime metric $h$ , which is,                                                                                                         |
| mielke-geometrodynamics_F00835 | 164  | ■ 0.524 | $\{ \}^{* *}=-1$                                                                                                               | $**=-1$                                                                                                                        | <sup>3</sup> For Minkowski signature (sign = 1), the Hodge dual satisfies <span style="border: 1px solid red; padding: 0 2px;"><math>i^*=-1</math></span> , and therefore, $i^*$ is used                      |
| mielke-geometrodynamics_F00836 | 164  | ■ 0.524 | $i^{\ast}$                                                                                                                     | $i^*$                                                                                                                          | <sup>3</sup> For Minkowski signature (sign = 1), the Hodge dual satisfies $**=-1$ , and therefore, $i^*$ is used                                                                                              |
| mielke-geometrodynamics_F00837 | 165  | ■ 1.000 | $\Gamma:=\frac{i}{4}\Gamma^{\alpha\beta}\sigma_{\alpha\beta}$                                                                  | $\Gamma:=\frac{i}{4}\Gamma^{\alpha\beta}\sigma_{\alpha\beta}$                                                                  | of gravity. In terms of the Clifford-algebra-valued <i>connection</i> $\Gamma:=\frac{i}{4}\Gamma^{\alpha\beta}\sigma_{\alpha\beta}$ , the                                                                     |
| mielke-geometrodynamics_F00838 | 165  | ■ 1.000 | $\overline{SO}_0(1,3)\cong SL(2,C)$                                                                                            | $\overline{SO}_0(1,3)\cong SL(2,C)$                                                                                            | $\overline{SO}_0(1,3)\cong SL(2,C)$ -covariant exterior derivative                                                                                                                                            |
| mielke-geometrodynamics_F00839 | 165  | ■ 0.975 | $[\Psi,\Phi]:=\Psi\wedge\Phi-(-1)^{pq}$                                                                                        | $[\Psi,\Phi]:=\Psi\wedge\Phi-(-1)^{pq}$                                                                                        | of p-forms employs the algebra-valued <i>form commutator</i> $[\Psi,\Phi]:=\Psi\wedge\Phi-(-1)^{pq}$                                                                                                          |
| mielke-geometrodynamics_F00840 | 165  | ■ 0.755 | $\Phi\wedge\Psi$                                                                                                               | $\Phi\wedge\Psi$                                                                                                               | $\Phi\wedge\Psi$ .                                                                                                                                                                                            |
| mielke-geometrodynamics_F00841 | 165  | ■ 1.000 | $K:=\frac{i}{4}K^{\alpha\beta}\sigma_{\alpha\beta}$                                                                            | $K:=\frac{i}{4}K^{\alpha\beta}\sigma_{\alpha\beta}$                                                                            | involves $K:=\frac{i}{4}K^{\alpha\beta}\sigma_{\alpha\beta}$ , the contortion one-form.                                                                                                                       |
| mielke-geometrodynamics_F00842 | 166  | ■ 1.000 | $\{ \}^{(2)}\Theta:=$                                                                                                          | $^{(2)}\Theta:=$                                                                                                               | The torsion two-form can be irreducibly decomposed into the trace part $^{(2)}\Theta:=$                                                                                                                       |

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| mielke-geometrodynamics_F00843 | 166  | ■ 1.000 | $\frac{1}{3} \gamma \wedge T$                                                                                               | $\frac{1}{3} \gamma \wedge T$                                                                                  | $\frac{1}{3} \gamma \wedge T$ , the axial torsion ${}^{(3)}\Theta := -\frac{1}{3} * (\gamma \wedge \mathcal{A})$ , and the tensor torsion ${}^{(1)}\Theta :=$                          |
| mielke-geometrodynamics_F00844 | 166  | ■ 1.000 | $\{ \}^{\{3\}} \backslash \Theta := -\frac{1}{3} * (\gamma \wedge \wedge \mathscr{A})$                                      | ${}^{(3)}\Theta := -\frac{1}{3} * (\gamma \wedge \mathcal{A})$                                                 | $\frac{1}{3} \gamma \wedge T$ , the axial torsion ${}^{(3)}\Theta := -\frac{1}{3} * (\gamma \wedge \mathcal{A})$ , and the tensor torsion ${}^{(1)}\Theta :=$                          |
| mielke-geometrodynamics_F00845 | 166  | ■ 1.000 | $\{ \}^{\{1\}} \backslash \Theta :=$                                                                                        | ${}^{(1)}\Theta :=$                                                                                            | $\frac{1}{3} \gamma \wedge T$ , the axial torsion ${}^{(3)}\Theta := -\frac{1}{3} * (\gamma \wedge \mathcal{A})$ , and the tensor torsion ${}^{(1)}\Theta :=$                          |
| mielke-geometrodynamics_F00846 | 166  | ■ 1.000 | $\backslash \Theta - \{ \}^{\{2\}} \backslash \Theta - \{ \}^{\{3\}} \backslash \Theta$                                     | $\Theta - {}^{(2)}\Theta - {}^{(3)}\Theta$                                                                     | $\check{\Theta} - {}^{(2)}\Theta - {}^{(3)}\Theta$ , where the one-forms of the trace and axial vector torsion, respec-                                                                |
| mielke-geometrodynamics_F00847 | 166  | ■ 0.984 | $\widetilde{\mathscr{A}} = \mathscr{A}$                                                                                     | $\widetilde{\mathcal{A}} = \mathcal{A}$                                                                        | Under a Weyl rescaling, the axial torsion remains invariant, i.e., $\widetilde{\mathcal{A}} = \mathcal{A}$ . Both                                                                      |
| mielke-geometrodynamics_F00848 | 167  | ■ 1.000 | $D \Omega^{\mathfrak{g}(\star)} \equiv 0$                                                                                   | $D\Omega^{g(\star)} \equiv 0$                                                                                  | i.e., $D\Omega^{g(\star)} \equiv 0$ , provided that $D\eta_{\alpha\beta\gamma\delta} = 0$ holds as in spaces with vanishing Weyl                                                       |
| mielke-geometrodynamics_F00849 | 167  | ■ 1.000 | $D \eta_{\alpha\beta\gamma\delta} = 0$                                                                                      | $D\eta_{\alpha\beta\gamma\delta} = 0$                                                                          | i.e., $D\Omega^{g(\star)} \equiv 0$ , provided that $D\eta_{\alpha\beta\gamma\delta} = 0$ holds as in spaces with vanishing Weyl                                                       |
| mielke-geometrodynamics_F00850 | 167  | ■ 0.995 | $\mathscr{A} := \{ \}^{\{*\}} \backslash \left( \vartheta_{\alpha} \wedge T^{\alpha} \right) = \mathscr{A}_i dx^i$          | $\mathcal{A} := * (\vartheta_{\alpha} \wedge T^{\alpha}) = \mathcal{A}_i dx^i$                                 | where $\mathcal{A} := * (\vartheta_{\alpha} \wedge T^{\alpha}) = \mathcal{A}_i dx^i$ is the <i>axial torsion</i> one-form. The translational                                           |
| mielke-geometrodynamics_F00851 | 168  | ■ 0.471 | $R^{\{ \} \{ \} } := (-1)^{\text{sig} + 1} * \left( R^{\{j\}\beta} \wedge \eta_{\beta\alpha} \right)$                       | $R^{\{ \} } := (-1)^{\text{sig} + 1} * \left( R^{\{j\}\beta} \wedge \eta_{\beta\alpha} \right)$                | vature scalar $R^{\{ \} } := (-1)^{\text{sig} + 1} * (R^{\{ \} \alpha\beta} \wedge \eta_{\beta\alpha})$ and the axial torsion piece $d\mathcal{A} \wedge d\mathcal{A}$                 |
| mielke-geometrodynamics_F00852 | 168  | ■ 0.995 | $L_{\text{SKY}}$                                                                                                            | $L_{\text{SKY}}$                                                                                               | Lagrangian $L_{\text{SKY}}$ as well as a Ricci-squared term and a scalar-curvature-squared                                                                                             |
| mielke-geometrodynamics_F00853 | 168  | ■ 0.827 | $\operatorname{Ric}_{\alpha\beta} := (-1)^{\text{sig}} * \left( R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)} \right)$ | $\operatorname{Ric}_{\alpha\beta} := (-1)^{\text{sig}} * (R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)})$ | $\operatorname{Ric}_{\alpha\beta} := (-1)^{\text{sig}} * (R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)})$ , is also known as the Gauss–Bonnet term.                               |
| mielke-geometrodynamics_F00854 | 169  | ■ 1.000 | $G_{\alpha} := R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} / 2$                                                         | $G_{\alpha} := R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} / 2$                                            | $G_{\alpha} := R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} / 2$ and the curvature scalar $R$ . The latter is the zero-form                                                         |
| mielke-geometrodynamics_F00855 | 169  | ■ 0.894 | $G^{\{ \} \{ \} } = G_{\alpha}^{\{ \} \beta} \eta_{\beta} \gamma^{\alpha}$                                                  | $G^{\{ \} } = G_{\alpha}^{\{ \} \beta} \eta_{\beta} \gamma^{\alpha}$                                           | decomposes into the Einstein three-form $G^{\{ \} } = G_{\alpha}^{\{ \} \beta} \eta_{\beta} \gamma^{\alpha}$ with respect to the                                                       |
| mielke-geometrodynamics_F00856 | 170  | ■ 0.983 | 2+1                                                                                                                         | 2 + 1                                                                                                          | Feynman RP (1981) The qualitative behavior of Yang-Mills theory in 2+1 dimensions. Nuclear                                                                                             |
| mielke-geometrodynamics_F00857 | 171  | ■ 0.998 | $\mathrm{m} \rightarrow 0$                                                                                                  | $\mathrm{m} \rightarrow 0$                                                                                     | Kogan II, Mouslopoulos S, Papazoglou A (2001) The $\mathrm{m} \rightarrow 0$ limit for massive graviton in dS <sub>4</sub> and                                                         |
| mielke-geometrodynamics_F00858 | 171  | ■ 0.998 | $\mathrm{dS}_4$                                                                                                             | dS <sub>4</sub>                                                                                                | Kogan II, Mouslopoulos S, Papazoglou A (2001) The $\mathrm{m} \rightarrow 0$ limit for massive graviton in dS <sub>4</sub> and                                                         |
| mielke-geometrodynamics_F00859 | 171  | ■ 0.999 | $\mathrm{AdS}_4$                                                                                                            | AdS <sub>4</sub>                                                                                               | AdS <sub>4</sub> . How to circumvent the van Dam-Veltman-Zakharov discontinuity. Phys Lett B 503(1):173–                                                                               |
| mielke-geometrodynamics_F00860 | 171  | ■ 0.878 | 5, $\mathbb{R}$                                                                                                             | 5, $\mathbb{R}$                                                                                                | Mielke EW (2012) Einstein-Weyl gravity from a topological SL(5, $\mathbb{R}$ ) gauge invariant action. Adv                                                                             |
| mielke-geometrodynamics_F00861 | 172  | ■ 1.000 | $R + R^2 + Q^2$                                                                                                             | $R + R^2 + Q^2$                                                                                                | Zhytnikov V (1994) Wavelike exact solutions of $R + R^2 + Q^2$ gravity. J Math Phys 35(11):6001–                                                                                       |
| mielke-geometrodynamics_F00862 | 173  | ■ 0.999 | (2014, 2006)                                                                                                                | (2014, 2006)                                                                                                   | “with flying colors”; see WILL (2014, 2006) for recent reviews. However, Einstein’s                                                                                                    |
| mielke-geometrodynamics_F00863 | 173  | ■ 0.999 | $\lambda_{\mathrm{DE}} = (\hbar c / \rho_{\mathrm{DE}})^{1/4} \simeq 85 \mu \mathrm{m}$                                     | $\lambda_{\mathrm{DE}} = (\hbar c / \rho_{\mathrm{DE}})^{1/4} \simeq 85 \mu \mathrm{m}$                        | cal dark energy (DE) scale of $\lambda_{\mathrm{DE}} = (\hbar c / \rho_{\mathrm{DE}})^{1/4} \simeq 85 \mu \mathrm{m}$ . Thus this “window” of                                          |
| mielke-geometrodynamics_F00864 | 174  | ■ 1.000 | $[\hbar] = [p][q]$                                                                                                          | $[\hbar] = [p][q]$                                                                                             | dimensional transmutation due to the physical dimension $[\hbar] = [p][q]$ of Planck’s                                                                                                 |
| mielke-geometrodynamics_F00865 | 174  | ■ 1.000 | $\hbar$                                                                                                                     | $\hbar$                                                                                                        | (reduced) constant $\hbar$ .                                                                                                                                                           |
| mielke-geometrodynamics_F00866 | 174  | ■ 1.000 | $C := \frac{i}{4} C^{\alpha\beta} \sigma_{\alpha\beta}$                                                                     | $C := \frac{i}{4} C^{\alpha\beta} \sigma_{\alpha\beta}$                                                        | ghosts, into operator transformations. Let $C := \frac{i}{4} C^{\alpha\beta} \sigma_{\alpha\beta}$ denote the zero-form of the                                                         |
| mielke-geometrodynamics_F00867 | 174  | ■ 0.998 | $\Psi := \frac{i}{4} \Psi_i^{\alpha\beta} \sigma_{\alpha\beta} dx^i$                                                        | $\Psi := \frac{i}{4} \Psi_j^{\alpha\beta} \sigma_{\alpha\beta} dx^j$                                           | $\Psi := \frac{i}{4} \Psi_i^{\alpha\beta} \sigma_{\alpha\beta} dx^i$ the topological ghost one-form, and $\Phi := \frac{i}{4} \Phi^{\alpha\beta} \sigma_{\alpha\beta}$ the correspond- |
| mielke-geometrodynamics_F00868 | 174  | ■ 0.998 | $\Phi := \frac{i}{4} \Phi^{\alpha\beta} \sigma_{\alpha\beta}$                                                               | $\Phi := \frac{i}{4} \Phi^{\alpha\beta} \sigma_{\alpha\beta}$                                                  | $\Psi := \frac{i}{4} \Psi_i^{\alpha\beta} \sigma_{\alpha\beta} dx^i$ the topological ghost one-form, and $\Phi := \frac{i}{4} \Phi^{\alpha\beta} \sigma_{\alpha\beta}$ the correspond- |

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| mielke-geometrodynamics_F00869 | 174  | ■ 1.000 | $\sigma_{\alpha\beta}$                                   | $\sigma_{\alpha\beta}$                                   | of the generator $\sigma_{\alpha\beta}$ of the linear (or Lorentz) group.                                                                    |
| mielke-geometrodynamics_F00870 | 175  | ■ 1.000 | $\Gamma$                                                 | $\Gamma$                                                 | This is consistent with the interpretation of $\Gamma$ and $\Omega^g$ as connection one-form and                                             |
| mielke-geometrodynamics_F00871 | 175  | ■ 1.000 | $\Omega^g$                                               | $\Omega^g$                                               | This is consistent with the interpretation of $\Gamma$ and $\Omega^g$ as connection one-form and                                             |
| mielke-geometrodynamics_F00872 | 175  | ■ 1.000 | $C, \Psi$                                                | $C, \Psi$                                                | number of degrees of freedom carried by the ghosts $C, \Psi$ , and $\Phi$ is equal to that of                                                |
| mielke-geometrodynamics_F00873 | 175  | ■ 1.000 | $s^2=0$                                                  | $s^2 = 0$                                                | variables, i.e., $s^2 = 0$ .                                                                                                                 |
| mielke-geometrodynamics_F00874 | 175  | ■ 0.948 | $\wedge^2$                                               | $^2$                                                     | (1988) by introducing the <i>graded</i> <sup>2</sup> connection and curvature forms                                                          |
| mielke-geometrodynamics_F00875 | 175  | ■ 1.000 | $s \ s=0$                                                | $ss = 0$                                                 | Moreover, a straightforward proof of the nilpotency $ss = 0$ of the BRST operator                                                            |
| mielke-geometrodynamics_F00876 | 175  | ■ 1.000 | $(d \oplus s)(d \oplus s) \equiv 0$                      | $(d \oplus s)(d \oplus s) \equiv 0$                      | $s$ follows simply from $\boxed{d \oplus s)(d \oplus s) \equiv 0}$ as a result of the graded Bianchi identity                                |
| mielke-geometrodynamics_F00877 | 175  | ■ 1.000 | $[s, d] := s \ d + d \ s = 0$                            | $[s, d] := sd + ds = 0$                                  | (8.2.3), the anticommutation of the graded commutator $[s, d] := sd + ds = 0$ , and                                                          |
| mielke-geometrodynamics_F00878 | 175  | ■ 0.615 | $\bar{C}, \bar{\chi}$                                    | $\bar{C}, \bar{\chi}$                                    | ism, where the Lorentz-algebra-valued antighosts $\bar{C}, \bar{\chi}$ , and $\bar{\Phi}$ obey the following                                 |
| mielke-geometrodynamics_F00879 | 175  | ■ 0.615 | $\bar{\Phi}$                                             | $\bar{\Phi}$                                             | ism, where the Lorentz-algebra-valued antighosts $\bar{C}, \bar{\chi}$ , and $\bar{\Phi}$ obey the following                                 |
| mielke-geometrodynamics_F00880 | 175  | ■ 1.000 | $[\Psi, \Phi] := \Psi \wedge \Phi -$                     | $[\Psi, \Phi] := \Psi \wedge \Phi -$                     | ghost number $g$ , such that the graded commutator is more generally defined by $[\Psi, \Phi] := \Psi \wedge \Phi -$                         |
| mielke-geometrodynamics_F00881 | 175  | ■ 0.510 | $(-1)^{(p_1+g_1)(p_2+g_2)} \Phi \wedge \Psi$             | $(-1)^{(p_1+g_1)(p_2+g_2)} \Phi \wedge \Psi$             | $\boxed{(-1)^{(p_1+g_1)(p_2+g_2)} \Phi \wedge \Psi}$ , (CHANG & SOO 1992).                                                                   |
| mielke-geometrodynamics_F00882 | 176  | ■ 1.000 | $\bar{\eta}$                                             | $\bar{\eta}$                                             | By construction, $s$ is nilpotent for the antighosts, since the Lagrange multiplier $\bar{\eta}$ ,                                           |
| mielke-geometrodynamics_F00883 | 176  | ■ 1.000 | $\mathbb{B}$                                             | $\mathbb{B}$                                             | the self-dual two-forms $\mathbb{B}$ , and $\beta$ are auxiliary fields introduced as trivial pairs.                                         |
| mielke-geometrodynamics_F00884 | 176  | ■ 0.998 | $\alpha = \alpha_i dx^i$                                 | $\alpha = \alpha_i dx^i$                                 | By introducing a BRST gauge field $\alpha = \alpha_i dx^i$ with ghost number $-1$ and a                                                      |
| mielke-geometrodynamics_F00885 | 176  | ■ 0.999 | $\lambda$                                                | $\lambda$                                                | commuting ghost $\lambda$ of $\alpha$ , one can promote (DE CARVALHO & BAULIEU 1992) the                                                     |
| mielke-geometrodynamics_F00886 | 176  | ■ 0.999 | $\alpha$                                                 | $\alpha$                                                 | commuting ghost $\lambda$ of $\alpha$ , one can promote (DE CARVALHO & BAULIEU 1992) the                                                     |
| mielke-geometrodynamics_F00887 | 176  | ■ 1.000 | $C \rightarrow (\alpha +$                                | $C \rightarrow (\alpha +$                                | promotion, which likewise, can be generated via the field redefinitions $C \rightarrow (\alpha +$                                            |
| mielke-geometrodynamics_F00888 | 176  | ■ 0.712 | $\lambda)C, \Psi \rightarrow (\alpha + \lambda)\Psi$     | $\lambda)C, \Psi \rightarrow (\alpha + \lambda)\Psi$     | $\lambda)C, \Psi \rightarrow (\alpha + \lambda)\Psi$ as well as $\Phi \rightarrow (\alpha + \lambda)^2\Phi$ of the ghosts. Thus, local BRST  |
| mielke-geometrodynamics_F00889 | 176  | ■ 0.712 | $\Phi \rightarrow (\alpha + \lambda)^2\Phi$              | $\Phi \rightarrow (\alpha + \lambda)^2\Phi$              | $\lambda)C, \Psi \rightarrow (\alpha + \lambda)\Psi$ as well as $\Phi \rightarrow (\alpha + \lambda)^2\Phi$ of the ghosts. Thus, local BRST  |
| mielke-geometrodynamics_F00890 | 176  | ■ 1.000 | $\gamma = \vartheta^\alpha \gamma_\alpha$                | $\gamma = \vartheta^\alpha \gamma_\alpha$                | In the affine or (broken) Poincaré gauge theory, the coframe $\gamma = \vartheta^\alpha \gamma_\alpha$ (locally                              |
| mielke-geometrodynamics_F00891 | 176  | ■ 1.000 | $\widetilde{\Theta} := \Theta \oplus \psi$               | $\widetilde{\Theta} := \Theta \oplus \psi$               | respectively. In the graded torsion two-form $\widetilde{\Theta} := \Theta \oplus \psi$ , there occurs the transla-                          |
| mielke-geometrodynamics_F00892 | 176  | ■ 0.999 | $\psi := \psi_j^\alpha \gamma_\alpha dx^j$               | $\psi := \psi_j^\alpha \gamma_\alpha dx^j$               | tional “Clifford” $\psi := \psi_j^\alpha \gamma_\alpha dx^j$ of the corresponding topological ghost.                                         |
| mielke-geometrodynamics_F00893 | 177  | ■ 0.885 | $s \rightarrow \widetilde{s} = s + \mathbb{L}_\zeta$     | $s \rightarrow \widetilde{s} = s + \mathbb{L}_\zeta$     | alizing the BRST transformations $s$ via $s \rightarrow \widetilde{s} = s + \mathbb{L}_\zeta$ involving the covariant Lie                    |
| mielke-geometrodynamics_F00894 | 177  | ■ 0.566 | $\mathbb{L}_\zeta := \zeta \lrcorner D - D\zeta \rfloor$ | $\mathbb{L}_\zeta := \zeta \lrcorner D - D\zeta \rfloor$ | derivative $\mathbb{L}_\zeta := \zeta \lrcorner D - D\zeta \rfloor$ along an anticommuting ghost vector field $\zeta = \zeta^i \partial_i$ . |
| mielke-geometrodynamics_F00895 | 177  | ■ 0.566 | $\zeta = \zeta^i \partial_i$                             | $\zeta = \zeta^i \partial_i$                             | derivative $\mathbb{L}_\zeta := \zeta \lrcorner D - D\zeta \rfloor$ along an anticommuting ghost vector field $\zeta = \zeta^i \partial_i$ . |

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| mielke-geometrodynamics_F00896 | 177  | ■ 0.339 | $\backslash\mathrm{mathfrak{L}}_{\backslash\mathrm{zeta}}$                                                                                                                                                                                                                   | $\mathfrak{L}_\zeta$                                                                          | of $\mathfrak{L}_\zeta$ , since $\zeta$ has ghost number one. Its BRST transformation is $s\zeta = \phi + \mathfrak{L}_\zeta\zeta$ , such                         |
| mielke-geometrodynamics_F00897 | 177  | ■ 0.339 | $\backslash\mathrm{zeta}$                                                                                                                                                                                                                                                    | $\zeta$                                                                                       | of $\mathfrak{L}_\zeta$ , since $\zeta$ has ghost number one. Its BRST transformation is $s\zeta = \phi + \mathfrak{L}_\zeta\zeta$ , such                         |
| mielke-geometrodynamics_F00898 | 177  | ■ 0.339 | $s\ \backslash\mathrm{zeta}=\backslash\mathrm{phi}+\backslash\mathrm{mathfrak{L}}_{\backslash\mathrm{zeta}}\ \backslash\mathrm{zeta}$                                                                                                                                        | $s\zeta = \phi + \mathfrak{L}_\zeta\zeta$                                                     | of $\mathfrak{L}_\zeta$ , since $\zeta$ has ghost number one. Its BRST transformation is $s\zeta = \phi + \mathfrak{L}_\zeta\zeta$ , such                         |
| mielke-geometrodynamics_F00899 | 177  | ■ 1.000 | $p+1$                                                                                                                                                                                                                                                                        | $p + 1$                                                                                       | Its action on a p-form, reduces to a finite series with up to $p + 1$ terms. In effect, the                                                                       |
| mielke-geometrodynamics_F00900 | 177  | ■ 0.497 | $\backslash\mathrm{exp}\ (\backslash\mathrm{zeta}\backslash\mathrm{rfloor})\ \backslash\mathrm{widetilde{\Omega}}^g$                                                                                                                                                         | $\exp(\zeta)]\widetilde{\Omega}^g$                                                            | by $\exp(\zeta)]\widetilde{\Omega}^g$ and $\exp(\zeta)](\widetilde{\Theta} \oplus \phi)$ , respectively, where $\phi:=\phi^\alpha\gamma_\alpha$ are the ghosts of |
| mielke-geometrodynamics_F00901 | 177  | ■ 0.497 | $\backslash\mathrm{left}.\backslash\mathrm{exp}\ (\backslash\mathrm{zeta}\backslash\mathrm{rfloor}\backslash\mathrm{right})(\backslash\mathrm{widetilde{\Theta}}\ \backslash\mathrm{oplus}\ \backslash\mathrm{phi})$                                                         | $\exp(\zeta)](\widetilde{\Theta} \oplus \phi)$                                                | by $\exp(\zeta)]\widetilde{\Omega}^g$ and $\exp(\zeta)](\widetilde{\Theta} \oplus \phi)$ , respectively, where $\phi:=\phi^\alpha\gamma_\alpha$ are the ghosts of |
| mielke-geometrodynamics_F00902 | 177  | ■ 0.497 | $\backslash\mathrm{phi}:=\backslash\mathrm{phi}^{\backslash\mathrm{alpha}}\ \backslash\mathrm{gamma}_{\backslash\mathrm{alpha}}$                                                                                                                                             | $\phi := \phi^\alpha\gamma_\alpha$                                                            | by $\exp(\zeta)]\widetilde{\Omega}^g$ and $\exp(\zeta)](\widetilde{\Theta} \oplus \phi)$ , respectively, where $\phi:=\phi^\alpha\gamma_\alpha$ are the ghosts of |
| mielke-geometrodynamics_F00903 | 177  | ■ 1.000 | $d$                                                                                                                                                                                                                                                                          | $d$                                                                                           | (BAULIEU 1987), the exterior derivative $d$ suffers from the similarity transformation                                                                            |
| mielke-geometrodynamics_F00904 | 177  | ■ 1.000 | $\backslash\mathrm{Gamma}^{\backslash\mathrm{mathrm{T}}}=(\backslash\mathrm{gamma}-D\ \backslash\mathrm{xi})\ /\ \backslash\mathrm{ell}$                                                                                                                                     | $\Gamma^{(\mathrm{T})} = (\gamma - D\xi)/\ell$                                                | <i>translational</i> connection $\Gamma^{(\mathrm{T})} = (\gamma - D\xi)/\ell$ and corresponding curvature $\Omega^{(\mathrm{T})} =$                              |
| mielke-geometrodynamics_F00905 | 177  | ■ 1.000 | $\backslash\mathrm{Omega}^{\backslash\mathrm{mathrm{T}}} =$                                                                                                                                                                                                                  | $\Omega^{(\mathrm{T})} =$                                                                     | <i>translational</i> connection $\Gamma^{(\mathrm{T})} = (\gamma - D\xi)/\ell$ and corresponding curvature $\Omega^{(\mathrm{T})} =$                              |
| mielke-geometrodynamics_F00906 | 177  | ■ 1.000 | $D\ \backslash\mathrm{Gamma}^{\backslash\mathrm{mathrm{T}}}=(\backslash\mathrm{Theta}-D\ D\ \backslash\mathrm{xi})\ /\ \backslash\mathrm{ell}$                                                                                                                               | $D\Gamma^{(\mathrm{T})} = (\Theta - DD\xi)/\ell$                                              | $D\Gamma^{(\mathrm{T})} = (\Theta - DD\xi)/\ell$ . Then the translational connection $\Gamma^{(\mathrm{T})}$ should be graded                                     |
| mielke-geometrodynamics_F00907 | 177  | ■ 1.000 | $\backslash\mathrm{Gamma}^{\backslash\mathrm{mathrm{T}}}$                                                                                                                                                                                                                    | $\Gamma^{(\mathrm{T})}$                                                                       | $D\Gamma^{(\mathrm{T})} = (\Theta - DD\xi)/\ell$ . Then the translational connection $\Gamma^{(\mathrm{T})}$ should be graded                                     |
| mielke-geometrodynamics_F00908 | 177  | ■ 1.000 | $\backslash\mathrm{gamma}\ \backslash\mathrm{rightharpoonrightarrow}\ \backslash\mathrm{widetilde{\gamma}}:=\backslash\mathrm{ell}\backslash\mathrm{left}(\backslash\mathrm{Gamma}^{\backslash\mathrm{mathrm{T}}})\ \backslash\mathrm{oplus}\ c\backslash\mathrm{right})$    | $\gamma \rightarrow \widetilde{\gamma} := \ell \left( \Gamma^{(\mathrm{T})} \oplus c \right)$ | as well, e.g., by the substitution $\gamma \rightarrow \widetilde{\gamma}:=\ell(\Gamma^{(\mathrm{T})} \oplus c)$ in the first structure equation.                 |
| mielke-geometrodynamics_F00909 | 177  | ■ 1.000 | $\backslash\mathrm{xi}:=\backslash\mathrm{xi}^{\backslash\mathrm{alpha}}\ \backslash\mathrm{gamma}_{\backslash\mathrm{alpha}}$                                                                                                                                               | $\xi := \xi^\alpha\gamma_\alpha$                                                              | The “quartet” of scalars $\xi:=\xi^\alpha\gamma_\alpha$ corresponds to the “generalized radius vectors”                                                           |
| mielke-geometrodynamics_F00910 | 177  | ■ 1.000 | $A(4,\ \backslash\mathrm{mathbb{R}})\ /\ G\ L(4,\ \backslash\mathrm{mathbb{R}})\ \backslash\mathrm{approx}\ \backslash\mathrm{mathbb{R}}^4$                                                                                                                                  | $A(4,\mathbb{R})/GL(4,\mathbb{R}) \approx \mathbb{R}^4$                                       | of Cartan and lives on the coset space $A(4,\mathbb{R})/GL(4,\mathbb{R}) \approx \mathbb{R}^4$ of the affine group.                                               |
| mielke-geometrodynamics_F00911 | 178  | ■ 1.000 | $\backslash\mathrm{Gamma}^{\backslash\mathrm{mathrm{T}}}\ \backslash\mathrm{stackrel{*}{=}}\ 0$                                                                                                                                                                              | $\Gamma^{(\mathrm{T})} \stackrel{*}{=} 0$                                                     | naturally arises in the affine gauge theory after, locally, the gauge $\Gamma^{(\mathrm{T})} \stackrel{*}{=} 0$ is imposed                                        |
| mielke-geometrodynamics_F00912 | 178  | ■ 0.995 | $S\ U(N)$                                                                                                                                                                                                                                                                    | $SU(N)$                                                                                       | and replace the internal $SU(N)$ group by the linear group $SL(4,\mathbb{R})$ of the tangent                                                                      |
| mielke-geometrodynamics_F00913 | 178  | ■ 0.995 | $S\ L(4,\ \backslash\mathrm{mathbb{R}})$                                                                                                                                                                                                                                     | $SL(4,\mathbb{R})$                                                                            | and replace the internal $SU(N)$ group by the linear group $SL(4,\mathbb{R})$ of the tangent                                                                      |
| mielke-geometrodynamics_F00914 | 178  | ■ 0.899 | $\backslash\mathrm{mathrm{SO}}(1,3)$                                                                                                                                                                                                                                         | $\mathrm{SO}(1,3)$                                                                            | space embracing the Lorentz group $\mathrm{SO}(1,3)$ (or $\mathrm{SO}(4)$ in Euclidean space with                                                                 |
| mielke-geometrodynamics_F00915 | 178  | ■ 0.467 | $=0$                                                                                                                                                                                                                                                                         | $= 0$                                                                                         | signature $\mathrm{sig} = 0$ ) as subgroup. Then one starts from the gravitational Pontryagin                                                                     |
| mielke-geometrodynamics_F00916 | 178  | ■ 0.999 | $s\backslash\backslash\mathrm{cdots}\backslash$                                                                                                                                                                                                                              | $s\{\cdots\}$                                                                                 | This purely topological action can be amended by any $s$ -exact four-form $s\{\cdots\}$ ,                                                                         |
| mielke-geometrodynamics_F00917 | 178  | ■ 1.000 | $d^{\backslash\mathrm{*}}\backslash\mathrm{Gamma}=0,\ D^{\backslash\mathrm{*}}\backslash\mathrm{Psi}=0$                                                                                                                                                                      | $d^*\Gamma = 0, D^*\Psi = 0$                                                                  | (1988), we may choose the Lorentz-type conditions $d^*\Gamma = 0, D^*\Psi = 0$ on the                                                                             |
| mielke-geometrodynamics_F00918 | 178  | ■ 1.000 | $\backslash\mathrm{Omega}^g(\ \backslash\mathrm{pm})=0$                                                                                                                                                                                                                      | $\Omega^{g(\pm)} = 0$                                                                         | self-duality condition $\Omega^{g(\pm)} = 0$ on the curvature as <i>gauge constraints</i> consistently                                                            |
| mielke-geometrodynamics_F00919 | 178  | ■ 1.000 | $C\ P$                                                                                                                                                                                                                                                                       | $CP$                                                                                          | <sup>3</sup> For the $CP$ -invariant Euler term, a similar result would hold, i.e., $sL_{\mathrm{Euler}} =$                                                       |
| mielke-geometrodynamics_F00920 | 178  | ■ 1.000 | $s\ L_{\backslash\mathrm{text}\ \{\mathrm{Euler}\}} =$                                                                                                                                                                                                                       | $sL_{\mathrm{Euler}} =$                                                                       | <sup>3</sup> For the $CP$ -invariant Euler term, a similar result would hold, i.e., $sL_{\mathrm{Euler}} =$                                                       |
| mielke-geometrodynamics_F00921 | 178  | ■ 0.998 | $(-1)^{\backslash\mathrm{sig}+1}\ d\backslash\mathrm{left}(\backslash\mathrm{Psi}^{\backslash\mathrm{alpha}}\ \backslash\mathrm{beta}\ \backslash\mathrm{wedge}\ R_{\backslash\mathrm{alpha}}\ \backslash\mathrm{beta}}^{\backslash\mathrm{star}})\backslash\mathrm{right})$ | $(-1)^{\mathrm{sig}+1}d\left(\Psi^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)}\right)$          | $(-1)^{\mathrm{sig}+1}d\left(\Psi^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)}\right)$ . However, the latter is only partially metric-free, since it involves       |
| mielke-geometrodynamics_F00922 | 179  | ■ 1.000 | $\backslash\mathrm{bar}\{\backslash\mathrm{chi}\}$                                                                                                                                                                                                                           | $\overline{\chi}$                                                                             | with the four-form character of a Lagrangian, the auxiliary fields $\overline{\chi}$ and $\beta$ have                                                             |

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| mielke-geometrodynamics_F00923 | 179  | ■ 0.993 | $\widetilde{\Psi}_{\mathrm{F}}:=\bar{C}_{\alpha\beta}\left(d^*\Gamma^{\alpha\beta}+\frac{1}{2}*\mathbb{B}^{\alpha\beta}\right)\left(d^*\Gamma^{\alpha\beta}+\frac{1}{2}*\mathbb{B}^{\alpha\beta}\right)$ | $\widetilde{\Psi}_{\mathrm{F}}:=\bar{C}_{\alpha\beta}\left(d^*\Gamma^{\alpha\beta}+\frac{1}{2}*\mathbb{B}^{\alpha\beta}\right)$ |                                                                                                                                                                          |
| mielke-geometrodynamics_F00924 | 179  | ■ 0.835 | $\{\cdots\}$                                                                                                                                                                                             | $\{\cdots\}$                                                                                                                    | four-form $\{\cdots\}$ is able to induce a dependence on the metric that in is not present in                                                                            |
| mielke-geometrodynamics_F00925 | 179  | ■ 0.998 | $d^*\Gamma=\mathbb{B}$                                                                                                                                                                                   | $d^*\Gamma=\mathbb{B}$                                                                                                          | $d^*\Gamma=\mathbb{B}$ on the linear connection. Moreover, for vanishing real <i>gauge parameter</i>                                                                     |
| mielke-geometrodynamics_F00926 | 179  | ■ 1.000 | $\rho=0$                                                                                                                                                                                                 | $\rho=0$                                                                                                                        | $\rho=0$ , the equation of motion for the auxiliary Lagrange-multiplier-type two-form                                                                                    |
| mielke-geometrodynamics_F00927 | 180  | ■ 1.000 | $\rho=1$                                                                                                                                                                                                 | $\rho=1$                                                                                                                        | In the case of the choice $\rho=1$ of the real gauge parameter, the two-form $\beta$ is present                                                                          |
| mielke-geometrodynamics_F00928 | 180  | ■ 1.000 | $D\Omega^g\equiv 0$                                                                                                                                                                                      | $D\Omega^g\equiv 0$                                                                                                             | Due to the second Bianchi identity $D\Omega^g\equiv 0$ , the dual field equation                                                                                         |
| mielke-geometrodynamics_F00929 | 180  | ■ 0.960 | $T_{\mu\nu}:=2^*\left(\partial L/\partial g_{\mu\nu}\right)=0$                                                                                                                                           | $T_{\mu\nu}:=2^*(\partial L/\partial g_{\mu\nu})=0$                                                                             | i.e., $T_{\mu\nu}:=2^*(\partial L/\partial g_{\mu\nu})=0$ . In turn, this implies that the BRST quantization is                                                          |
| mielke-geometrodynamics_F00930 | 181  | ■ 0.999 | $\mathscr{A}:=\left(\vartheta_{\alpha}\wedge\right)$                                                                                                                                                     | $\mathscr{A}:=\left(\vartheta_{\alpha}\wedge\right)$                                                                            | respect to the Riemannian curvature $R_{\alpha\beta}^{\left\  \right\ }$ and the <i>axial torsion</i> one-form $\mathscr{A}:=\left(\vartheta_{\alpha}\wedge\right)$      |
| mielke-geometrodynamics_F00931 | 181  | ■ 0.999 | $T^{\alpha}$                                                                                                                                                                                             | $T^{\alpha}$                                                                                                                    | $T^{\alpha}$ ) is rather well motivated by the <i>axial anomaly</i> in RC spacetime                                                                                      |
| mielke-geometrodynamics_F00932 | 181  | ■ 1.000 | $\gamma:=\vartheta^{\alpha}\gamma_{\alpha}$                                                                                                                                                              | $\gamma:=\vartheta^{\alpha}\gamma_{\alpha}$                                                                                     | In addition, the coframe $\gamma:=\vartheta^{\alpha}\gamma_{\alpha}$ is constrained by the Hilbert–de Donder gauge                                                       |
| mielke-geometrodynamics_F00933 | 181  | ■ 1.000 | $d^*\gamma=0$                                                                                                                                                                                            | $d^*\gamma=0$                                                                                                                   | condition $d^*\gamma=0$ complying with the BRST algebra of translations.                                                                                                 |
| mielke-geometrodynamics_F00934 | 181  | ■ 1.000 | $\widetilde{L}_{\mathrm{FP}}:=L_{\mathrm{FP}}+\Delta L_{\mathrm{FP}}$                                                                                                                                    | $\widetilde{L}_{\mathrm{FP}}:=L_{\mathrm{FP}}+\Delta L_{\mathrm{FP}}$                                                           | Then for $\rho=0$ there arises from $\widetilde{L}_{\mathrm{FP}}:=L_{\mathrm{FP}}+\Delta L_{\mathrm{FP}}$ the <i>modified double duality</i>                             |
| mielke-geometrodynamics_F00935 | 181  | ■ 1.000 | $i$                                                                                                                                                                                                      | $i$                                                                                                                             | on the curvature two-form $\Omega^g$ , which is real due to the extra imaginary unit $i$ .                                                                               |
| mielke-geometrodynamics_F00936 | 181  | ■ 0.692 | $\stackrel{\pm}{\Omega}^g$                                                                                                                                                                               | $\stackrel{\pm}{\Omega}^g$                                                                                                      | only one sign for the dual curvature $\stackrel{\pm}{\Omega}^g$ leads to a self-consistent constraint. This                                                              |
| mielke-geometrodynamics_F00937 | 182  | ■ 0.811 | $\theta=\pi/2$                                                                                                                                                                                           | $\theta=\pi/2$                                                                                                                  | of the curvature, which for $\theta=\pi/2$ specializes to the self-dual curvature $\stackrel{+}{\Omega}^g$ , and for                                                     |
| mielke-geometrodynamics_F00938 | 182  | ■ 0.811 | $\stackrel{+}{\Omega}^g$                                                                                                                                                                                 | $\stackrel{+}{\Omega}^g$                                                                                                        | of the curvature, which for $\theta=\pi/2$ specializes to the self-dual curvature $\stackrel{+}{\Omega}^g$ , and for                                                     |
| mielke-geometrodynamics_F00939 | 182  | ■ 1.000 | $\theta=3\pi/2$                                                                                                                                                                                          | $\theta=3\pi/2$                                                                                                                 | $\theta=3\pi/2$ to the anti-self-dual curvature $\bar{\Omega}^g$ . Since the “unit”-curvature two-form $\sigma$                                                          |
| mielke-geometrodynamics_F00940 | 182  | ■ 1.000 | $\bar{\Omega}^g$                                                                                                                                                                                         | $\bar{\Omega}^g$                                                                                                                | $\theta=3\pi/2$ to the anti-self-dual curvature $\bar{\Omega}^g$ . Since the “unit”-curvature two-form $\sigma$                                                          |
| mielke-geometrodynamics_F00941 | 182  | ■ 1.000 | $\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0$                                                                                                                                                              | $\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0$                                                                                     | a special case of the Ansatz (8.6.4) for the choice $\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0$ and $\theta_{\mathrm{L}}^{\star}=\mp(-1)^{\mathrm{sig}}$ . Interestingly |
| mielke-geometrodynamics_F00942 | 182  | ■ 1.000 | $\theta_{\mathrm{L}}^{\star}=\mp(-1)^{\mathrm{sig}}$                                                                                                                                                     | $\theta_{\mathrm{L}}^{\star}=\mp(-1)^{\mathrm{sig}}$                                                                            | a special case of the Ansatz (8.6.4) for the choice $\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0$ and $\theta_{\mathrm{T}}^{\star}=\mp(-1)^{\mathrm{sig}}$ . Interestingly |
| mielke-geometrodynamics_F00943 | 183  | ■ 0.983 | $L_{\mathrm{qPG}}$                                                                                                                                                                                       | $L_{\mathrm{qPG}}$                                                                                                              | tion parts of the gravitational Lagrangian $L_{\mathrm{qPG}}$ . In the field momenta, each of the                                                                        |
| mielke-geometrodynamics_F00944 | 183  | ■ 1.000 | $a_{(M)}$                                                                                                                                                                                                | $a_{(M)}$                                                                                                                       | the Lagrangian with individual dimensionless weights $a_{(M)}$ and $b_{(N)}$ , respectively.                                                                             |
| mielke-geometrodynamics_F00945 | 183  | ■ 1.000 | $b_{(N)}$                                                                                                                                                                                                | $b_{(N)}$                                                                                                                       | the Lagrangian with individual dimensionless weights $a_{(M)}$ and $b_{(N)}$ , respectively.                                                                             |
| mielke-geometrodynamics_F00946 | 183  | ■ 1.000 | $\theta_{\mathrm{T}},\theta_{\mathrm{T}}^{\star}$                                                                                                                                                        | $\theta_{\mathrm{T}},\theta_{\mathrm{T}}^{\star}$                                                                               | for the rotational field momenta (MIELKE 1992; ZHYTNIKOV 1994), where $\theta_{\mathrm{T}},\theta_{\mathrm{T}}^{\star}$ ,                                                |
| mielke-geometrodynamics_F00947 | 183  | ■ 1.000 | $\theta_{\mathrm{L}}^{\star}$                                                                                                                                                                            | $\theta_{\mathrm{L}}^{\star}$                                                                                                   | $\theta_{\mathrm{L}}$ , and $\theta_{\mathrm{L}}^{\star}$ are dimensionless constants. The constraint (8.5.3) corresponds to the                                         |
| mielke-geometrodynamics_F00948 | 183  | ■ 1.000 | $a_0=0, b_{(N)}=1$                                                                                                                                                                                       | $a_0=0, b_{(N)}=1$                                                                                                              | SKY Lagrangian with $a_0=0, b_{(N)}=1$ , and the choice $\theta_{\mathrm{L}}^{\star}=\mp(-1)^{\mathrm{sig}}$ as well as                                                  |

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| mielke-geometrodynamics_F08949 | 183  | ■ 0.359 | $d\,C_{\mathrm{TT}}\{^*\}:=d\left(\vartheta^{\alpha}\wedge{}^*T_{\alpha}\right)/2\,\ell^2$                                     | $dC_{\mathrm{TT}}^*:=d(\vartheta^{\alpha}\wedge{}^*T_{\alpha})/2\ell^2$                                         | $dC_{\mathrm{TT}}^*:=d(\vartheta^{\alpha}\wedge{}^*T_{\alpha})/2\ell^2$ inducing the teleparallelism equivalence, as well as the                                      |
| mielke-geometrodynamics_F08958 | 183  | ■ 1.000 | $\frac{1}{2}\,s\,\overline{\chi}^{\alpha\beta}\wedge\left(H_{\alpha\beta}+\partial L_{\theta}/\partial R^{\alpha\beta}\right)$ | $\frac{1}{2}s\overline{\chi}^{\alpha\beta}\wedge(H_{\alpha\beta}+\partial L_{\theta}/\partial R^{\alpha\beta})$ | BRST gauge constraint $\frac{1}{2}s\overline{\chi}^{\alpha\beta}\wedge(H_{\alpha\beta}+\partial L_{\theta}/\partial R^{\alpha\beta})$ generalizing the corresponding  |
| mielke-geometrodynamics_F08951 | 183  | ■ 0.680 | $\widetilde{L}_{\mathrm{FP}}$                                                                                                  | $\widetilde{L}_{\mathrm{FP}}$                                                                                   | terms in the Faddeev–Popov Lagrangian $\widetilde{L}_{\mathrm{FP}}$ modified by (8.5.1).                                                                              |
| mielke-geometrodynamics_F08952 | 184  | ■ 1.000 | $\mathsf{D}$                                                                                                                   | $D$                                                                                                             | The covariant derivative $D$ of the “unit”-curvature pieces proportional to $\eta_{\alpha\beta}$ and                                                                  |
| mielke-geometrodynamics_F08953 | 184  | ■ 1.000 | $\vartheta_{\alpha}\wedge\vartheta_{\beta}$                                                                                    | $\vartheta_{\alpha}\wedge\vartheta_{\beta}$                                                                     | $\vartheta_{\alpha}\wedge\vartheta_{\beta}$ in the DD Ansatz is responsible for the two additional torsion terms. For                                                 |
| mielke-geometrodynamics_F08954 | 184  | ■ 0.999 | $\tau_{\alpha\beta}=0$                                                                                                         | $\tau_{\alpha\beta}=0$                                                                                          | spinless matter, i.e., $\tau_{\alpha\beta}=0$ , one can algebraically resolve (8.6.9) using Eq. (A.1.26)                                                              |
| mielke-geometrodynamics_F08955 | 184  | ■ 1.000 | $H_{\alpha}(**)$                                                                                                               | $H_{\alpha}(**)$                                                                                                | of HEHL et al. (1995) for the translational momenta $H_{\alpha}(**)$ constrained by double                                                                            |
| mielke-geometrodynamics_F08956 | 185  | ■ 1.000 | $H_{\alpha\beta}(**)\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}$                                                           | $H_{\alpha\beta}(**) \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}$                                        | The contraction $H_{\alpha\beta}(**) \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}$ of the DD Ansatz (8.6.4) together with the linear                            |
| mielke-geometrodynamics_F08957 | 185  | ■ 1.000 | $\theta_{\mathrm{L}}\neq 0$                                                                                                    | $\theta_{\mathrm{L}}\neq 0$                                                                                     | expansion (8.6.3) implies, for $\theta_{\mathrm{L}}\neq 0$ , that the pseudoscalar curvature vanishes for                                                             |
| mielke-geometrodynamics_F08958 | 185  | ■ 1.000 | $R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}=0$                                                             | $R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}=0$                                              | consistency, i.e., $R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}=0$ . Then the scalar curvature                                                     |
| mielke-geometrodynamics_F08959 | 185  | ■ 1.000 | $\theta_{\mathrm{L}}=0$                                                                                                        | $\theta_{\mathrm{L}}=0$                                                                                         | <sup>5</sup> There occurs an interesting modification in the case that $\theta_{\mathrm{L}}=0$ , since then the pseudoscalar cur-                                     |
| mielke-geometrodynamics_F08960 | 185  | ■ 1.000 | $\theta_{\mathrm{T}}R^{\alpha\beta}\wedge\vartheta_{\alpha}\wedge\vartheta_{\beta}/4\,\ell^2$                                  | $\theta_{\mathrm{T}}R^{\alpha\beta}\wedge\vartheta_{\alpha}\wedge\vartheta_{\beta}/4\ell^2$                     | vature four-form $\theta_{\mathrm{T}}R^{\alpha\beta}\wedge\vartheta_{\alpha}\wedge\vartheta_{\beta}/4\ell^2$ needs to be subtracted from (8.6.12), which would induce |
| mielke-geometrodynamics_F08961 | 186  | ■ 1.000 | $T^{\alpha}\simeq 0$                                                                                                           | $T^{\alpha}\simeq 0$                                                                                            | vanishing torsion $T^{\alpha}\simeq 0$ via Lagrange multipliers in the gravitational gauge field                                                                      |
| mielke-geometrodynamics_F08962 | 186  | ■ 1.000 | $T^{\alpha}=0$                                                                                                                 | $T^{\alpha}=0$                                                                                                  | Then obviously, $T^{\alpha}=0$ will emerge by varying the Lagrange multiplier, and the                                                                                |
| mielke-geometrodynamics_F08963 | 186  | ■ 1.000 | $\lambda_{\alpha}$                                                                                                             | $\lambda_{\alpha}$                                                                                              | second field equation (8.6.7) amounts to an algebraic equation for the two-form $\lambda_{\alpha}$ .                                                                  |
| mielke-geometrodynamics_F08964 | 186  | ■ 0.382 | $\Gamma^{j\alpha\beta}$                                                                                                        | $\Gamma^{j\alpha\beta}$                                                                                         | which, generically, is of third order in the Levi-Civita connection $\Gamma^{(\cdot)\alpha\beta}$ of Rie-                                                             |
| mielke-geometrodynamics_F08965 | 186  | ■ 0.803 | $\mu_{\alpha}=\frac{1}{4}\,\vartheta_{\alpha}\wedge{}^*j_5$                                                                    | $\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge{}^*j_5$                                                       | In the case of Dirac spinors, the <i>spin-energy potential</i> $\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge{}^*j_5$ is dual to                                   |
| mielke-geometrodynamics_F08966 | 186  | ■ 1.000 | $\mu_{\alpha}\wedge\delta T^{\alpha}$                                                                                          | $\mu_{\alpha}\wedge\delta T^{\alpha}$                                                                           | can likewise be obtained from the variation $\mu_{\alpha}\wedge\delta T^{\alpha}$ .                                                                                   |
| mielke-geometrodynamics_F08967 | 186  | ■ 1.000 | $\mathsf{D}\,R_{\alpha\beta}\equiv 0,\,\mathsf{D}\,R_{\alpha\beta}^{\wedge(\star)}\equiv 0$                                    | $DR_{\alpha\beta}\equiv 0, DR_{\alpha\beta}^{(*)}\equiv$                                                        | coupling constants in the $\theta$ -type boundary term (8.6.5): since $DR_{\alpha\beta}\equiv 0$ , $DR_{\alpha\beta}^{(*)}\equiv$                                     |
| mielke-geometrodynamics_F08968 | 186  | ■ 0.420 | $0,\,\mathsf{D}\,\eta_{\alpha\beta}=0$                                                                                         | $0, D\eta_{\alpha\beta}=0$                                                                                      | $0$ , $D\eta_{\alpha\beta}=0$ , and $D(\vartheta_{\alpha}\wedge\vartheta_{\beta})=2T_{[\alpha}\wedge\vartheta_{\beta]}=0$ in a Riemannian spacetime, the              |
| mielke-geometrodynamics_F08969 | 186  | ■ 0.420 | $D\left(\vartheta_{\alpha}\wedge\vartheta_{\beta}\right)=2\,T_{[\alpha}\wedge\vartheta_{\beta]}=0$                             | $D(\vartheta_{\alpha}\wedge\vartheta_{\beta})=2T_{[\alpha}\wedge\vartheta_{\beta]}=0$                           | $0$ , $D\eta_{\alpha\beta}=0$ , and $D(\vartheta_{\alpha}\wedge\vartheta_{\beta})=2T_{[\alpha}\wedge\vartheta_{\beta]}=0$ in a Riemannian spacetime, the              |
| mielke-geometrodynamics_F08970 | 186  | ■ 1.000 | $R_{\alpha\beta}^{\wedge(\star)}$                                                                                              | $R_{\alpha\beta}^{(*)}$                                                                                         | the Lie dual $R_{\alpha\beta}^{(*)}$ of the curvature does not contribute in (8.6.8), due to the Bach–                                                                |
| mielke-geometrodynamics_F08971 | 186  | ■ 0.428 | $G^{\{\cdot\}}:=R^{\{1\}\beta\gamma}\wedge\eta_{\alpha\beta\gamma}\gamma^{\alpha}/2$                                           | $G^{\{\cdot\}}:=R^{\{1\}\beta\gamma}\wedge\eta_{\alpha\beta\gamma}\gamma^{\alpha}/2$                            | Here $G^{\{\cdot\}}:=R^{\{1\}\beta\gamma}\wedge\eta_{\alpha\beta\gamma}\gamma^{\alpha}/2$ is the Einstein three-form dual to the usual Einstein                       |
| mielke-geometrodynamics_F08972 | 186  | ■ 0.997 | $G_{ij}:=\operatorname{Ric}_{ij}^{\{\cdot\}}-\frac{1}{2}g_{ij}$                                                                | $G_{ij}:=\operatorname{Ric}_{ij}^{\{\cdot\}}-\frac{1}{2}g_{ij}$                                                 | tensor $G_{ij}:=\operatorname{Ric}_{ij}^{\{\cdot\}}-\frac{1}{2}g_{ij}$ written in terms of the Riemannian connection $\Gamma^{\{\cdot\}}$ . Matter                    |
| mielke-geometrodynamics_F08973 | 186  | ■ 0.993 | $\kappa_{\mathrm{eff}}=\ell^2/\theta_{\mathrm{T}}^*$                                                                           | $\kappa_{\mathrm{eff}}=\ell^2/\theta_{\mathrm{T}}^*$                                                            | whereas the effective gravitational coupling constant $\kappa_{\mathrm{eff}}=\ell^2/\theta_{\mathrm{T}}^*$ depends on a                                               |
| mielke-geometrodynamics_F08974 | 186  | ■ 1.000 | $\theta_{\mathrm{T}}^*$                                                                                                        | $\theta_{\mathrm{T}}^*$                                                                                         | “bare” length scale $\ell$ “renormalized” by the vacuum angle $\theta_{\mathrm{T}}^*$ .                                                                               |

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| mielke-geometrodynamics_F00975 | 187  | ■ 1.000 | $\widehat{\Sigma}_{\alpha}$                     | $\widehat{\Sigma}_{\alpha}$                     | <i>Einstein equation</i> (8.6.18) to the symmetrized energy–momentum current $\widehat{\Sigma}_{\alpha}$ of             |
| mielke-geometrodynamics_F00976 | 187  | ■ 0.987 | $\kappa_{\text{eff}}=8\pi G_{\mathrm{N}}/c^4$   | $\kappa_{\text{eff}}=8\pi G_{\mathrm{N}}/c^4$   | matter requires, at least locally, the macroscopic Newtonian value $\kappa_{\text{eff}}=8\pi G_{\mathrm{N}}/c^4$        |
| mielke-geometrodynamics_F00977 | 187  | ■ 0.894 | $L_{\text{qPG}}$                                | $L_{\text{qPG}}$                                | most quadratic Lagrangian $L_{\text{qPG}}$ with ten <sup>6</sup> parameters $a_{(M)}$ and $b_{(N)}$ .                   |
| mielke-geometrodynamics_F00978 | 187  | ■ 1.000 | $\varphi\propto 1/\ell$                         | $\langle\varphi\rangle\propto 1/\ell$           | tive to associate this with a (spontaneous) <i>symmetry breaking</i> $\langle\varphi\rangle\propto 1/\ell$ of the scale |
| mielke-geometrodynamics_F00979 | 188  | ■ 1.000 | $a$                                             | $a$                                             | When these covectors are induced by an axion $a$ and dilaton $\varphi$ as respective potentials,                        |
| mielke-geometrodynamics_F00980 | 188  | ■ 1.000 | $\Phi=a+if_{\varphi}\exp(-\varphi/f_{\varphi})$ | $\Phi=a+if_{\varphi}\exp(-\varphi/f_{\varphi})$ | even combine into a single complex scalar, the <i>axidilaton</i> $\Phi=a+if_{\varphi}\exp(-\varphi/f_{\varphi})$ ;      |
| mielke-geometrodynamics_F00981 | 192  | ■ 0.954 | $\Lambda_{\text{eff}}>0$                        | $\Lambda_{\text{eff}}>0$                        | (here generalized to an n-dimensional manifold) and second for $\Lambda_{\text{eff}}>0$ to the Nariai                   |
| mielke-geometrodynamics_F00982 | 193  | ■ 0.989 | $N=N(r\pm t)$                                   | $N=N(r\pm t)$                                   | with $N=N(r\pm t)$ denoting an arbitrary harmonic function of $r$ and $t$ ; cf. Ni (1975).                              |
| mielke-geometrodynamics_F00983 | 193  | ■ 0.989 | $r$                                             | $r$                                             | with $N=N(r\pm t)$ denoting an arbitrary harmonic function of $r$ and $t$ ; cf. Ni (1975).                              |
| mielke-geometrodynamics_F00984 | 193  | ■ 0.998 | $\Lambda_{\text{eff}}=0$                        | $\Lambda_{\text{eff}}=0$                        | spacetime, however, the Schwarzschild solution (9.1.1, 2) with $\Lambda_{\text{eff}}=0$ is of para-                     |
| mielke-geometrodynamics_F00985 | 194  | ■ 0.358 | $\{ \}^{\ast}$                                  | $\ast$                                          | nonsingular one, we first pass to a different coordinate system. Let $r^{\ast}$ be the “tortoise”                       |
| mielke-geometrodynamics_F00986 | 194  | ■ 0.996 | $u^2-v^2=1$                                     | $u^2-v^2=1$                                     | $u^2-v^2=1$ , that is, for $r=r^{\ast}=0$ , this geometry becomes singular in real space-                               |
| mielke-geometrodynamics_F00987 | 194  | ■ 0.996 | $r=r^{\ast}=0$                                  | $r=r^{\ast}=0$                                  | $u^2-v^2=1$ , that is, for $r=r^{\ast}=0$ , this geometry becomes singular in real space-                               |
| mielke-geometrodynamics_F00988 | 194  | ■ 1.000 | $\tilde{v}:=iv$                                 | $\tilde{v}:=iv$                                 | be avoided by making use of a purely “imaginary” coordinate $\tilde{v}:=iv$ . The sphere                                |
| mielke-geometrodynamics_F00989 | 194  | ■ 0.531 | $S^2$                                           | $S^2$                                           | $S^2$ given by $e^v=0$ deteriorates to a rotational axis, while the imaginary time axis                                 |
| mielke-geometrodynamics_F00990 | 194  | ■ 0.531 | $e^v=0$                                         | $e^v=0$                                         | $S^2$ given by $e^v=0$ deteriorates to a rotational axis, while the imaginary time axis                                 |
| mielke-geometrodynamics_F00991 | 194  | ■ 1.000 | $\tau:=$                                        | $\tau:=$                                        | $\tau:=$ it represents an angle variable about this axis in Euclidean spacetime. (Corre-                                |
| mielke-geometrodynamics_F00992 | 194  | ■ 1.000 | $K_3$                                           | $K_3$                                           | $K_3$ -surface represents the only compact manifold that is allowed by a metric with                                    |
| mielke-geometrodynamics_F00993 | 195  | ■ 0.629 | $\omega^g$                                      | $\omega^g$                                      | connection $\omega^g$ into a Christoffel-type connection $\omega^{(l)}$ and the remaining contortion                    |
| mielke-geometrodynamics_F00994 | 195  | ■ 0.975 | $RC$                                            | $RC$                                            | 2-form in an $RC$ spacetime. Its Riemannian part may be decomposed further into                                         |
| mielke-geometrodynamics_F00995 | 195  | ■ 0.917 | $\Omega_C^{\{ \}}$                              | $\Omega_C^{\{ \}}$                              | where $\Omega_C^{\{ \}}$ denotes Weyl’s conformal curvature 2-form in a Riemannian spacetime.                           |
| mielke-geometrodynamics_F00996 | 195  | ■ 1.000 | $f_i=o$                                         | $f_i=o$                                         | choice $f_i=o$ and $\zeta=-1$ , as discussed by BAEKLER et al. (1982), the duality ansatz                               |
| mielke-geometrodynamics_F00997 | 196  | ■ 1.000 | $K\wedge K$                                     | $K\wedge K$                                     | fact that $K\wedge K$ is already self-double-dual.                                                                      |
| mielke-geometrodynamics_F00998 | 196  | ■ 1.000 | $\mu=0$                                         | $\mu=0$                                         | cosmos, vanishes without Schwarzschild’s source term, i.e., for $\mu=0$ . This is in                                    |
| mielke-geometrodynamics_F00999 | 196  | ■ 1.000 | $\mu$                                           | $\mu$                                           | parameter $\mu$ vanishes in the metric ground form. Here the radial functions occurring                                 |
| mielke-geometrodynamics_F01000 | 197  | ■ 1.000 | $\kappa<0$                                      | $\kappa<0$                                      | For a real solution, $\kappa<0$ is necessary. In this case, torsion can exist only in an anti-de                        |
| mielke-geometrodynamics_F01001 | 197  | ■ 1.000 | $t\rightarrow-i\tau$                            | $t\rightarrow-i\tau$                            | Weyl’s trick, i.e., the formal substitution $t\rightarrow-i\tau$ of the timelike variable. It follows                   |

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| mielke-geometrodynamics_F01002 | 197  | 0.687 | $\mathrm{f}=\mathrm{T}_{\hat{0}\hat{1}}^{\hat{0}}$                                                    | $f = T_{01}^{\hat{0}}$                                  | from the matrix representation (9.1.15) that the anholonomic components $f = T_{01}^{\hat{0}}$                      |
| mielke-geometrodynamics_F01003 | 197  | 0.790 | $\mathrm{g}=\mathrm{T}_{\hat{1}\hat{2}}^{\hat{1}\hat{2}}\mathrm{T}_{\hat{1}\hat{3}}^{\hat{2}\hat{3}}$ | $g = T_{12}^{\hat{2}} = T_{13}^{\hat{3}}$               | and $g = T_{12}^{\hat{2}} = T_{13}^{\hat{3}}$ are the only ones that are then allowed to be nonzero. A solution     |
| mielke-geometrodynamics_F01004 | 198  | 1.000 | $\mathrm{n}=4$                                                                                        | $n = 4$                                                 | ourselves to a manifold of dimension $n = 4$ . The previous classification of the source-                           |
| mielke-geometrodynamics_F01005 | 198  | 0.998 | $\mathrm{R}^6$                                                                                        | $R^6$                                                   | space $R^6$ . For then                                                                                              |
| mielke-geometrodynamics_F01006 | 199  | 0.995 | $\gamma_2^g$                                                                                          | $\gamma_2^g$                                            | the form $\gamma_2^g$ over the basis manifold, which provides for a topological invariant. It is                    |
| mielke-geometrodynamics_F01007 | 199  | 1.000 | $\mathrm{H}_{\mathrm{P}}(\mathrm{M},\mathbb{R})$                                                      | $H_P(M, \mathbb{R})$                                    | rank of the pth homology group $H_P(M, \mathbb{R})$ , i.e., from the so-called Betti numbers                        |
| mielke-geometrodynamics_F01008 | 199  | 1.000 | $\mathrm{b}:=\operatorname{dim}\mathrm{H}_{\mathrm{p}}(\mathrm{M},\mathbb{R})$                        | $b := \dim H_p(M, \mathbb{R})$                          | $b := \dim H_p(M, \mathbb{R})$ , which were introduced by Riemann. Similarly as in the case of                      |
| mielke-geometrodynamics_F01009 | 199  | 1.000 | $\chi(\mathrm{M})$                                                                                    | $\chi(M)$                                               | the vanishing of $\chi(M)$ . In particular, it is known that a compact even-dimensional                             |
| mielke-geometrodynamics_F01010 | 199  | 1.000 | $\mathrm{s}=1$                                                                                        | $s = 1$                                                 | manifold admits a pseudo-Riemannian metric of signature $\mathbb{S}=\mathbb{I}$ (“Lorentz metric”)                  |
| mielke-geometrodynamics_F01011 | 199  | 0.996 | $\alpha:\omega^g-\omega^g$                                                                            | $\alpha : \omega^g - \omega^g$                          | $\alpha : \omega^g - \omega^g$ of the embedding of the boundary $\partial M$ in $M$ . Although it is exactly        |
| mielke-geometrodynamics_F01012 | 200  | 1.000 | $\mathrm{B}^n$                                                                                        | $B^n$                                                   | an open ball $B^n$ is to be cut out of $M$ and $N$ in order to identify the then respectively                       |
| mielke-geometrodynamics_F01013 | 200  | 1.000 | $\partial\mathrm{B}^n=\mathrm{S}^{n-1}$                                                               | $\partial B^n = S^{n-1}$                                | existing boundary spheres $\partial B^n = S^{n-1}$ with each other via a diffeomorphism that                        |
| mielke-geometrodynamics_F01014 | 200  | 0.736 | $\mathrm{W}^n:=\mathrm{S}^1\times\mathrm{S}^{n-1}$                                                    | $W^n := S^1 \times S^{n-1}$                             | $W^n := S^1 \times S^{n-1}$ (WHEELER 1955), the manifold $M\#(S^1 \times S^{n-1})$ comes into exis-                 |
| mielke-geometrodynamics_F01015 | 200  | 0.736 | $\left(\mathrm{S}^1\times\mathrm{S}^{n-1}\right)$                                                     | $(S^1 \times S^{n-1})$                                  | $W^n := S^1 \times S^{n-1}$ (WHEELER 1955), the manifold $M\#(S^1 \times S^{n-1})$ comes into exis-                 |
| mielke-geometrodynamics_F01016 | 200  | 0.980 | $\mathbb{C}P^2$                                                                                       | $\mathbb{CP}^2$                                         | number” of a surface (cf. SEIFERT & THRE FALL 1934), and $\mathbb{CP}^2$ denotes the complex                        |
| mielke-geometrodynamics_F01017 | 200  | 1.000 | $\mathrm{E}^2$                                                                                        | $E^2$                                                   | projective two-space in $E^2$ . For spaces with uneven dimensions,                                                  |
| mielke-geometrodynamics_F01018 | 201  | 1.000 | $\mathrm{T}^4=\mathrm{S}^1\times\mathrm{S}^1\times\mathrm{S}^1\times\mathrm{S}^1$                     | $T^4 = S^1 \times S^1 \times S^1 \times S^1$            | by the flat torus $T^4 = S^1 \times S^1 \times S^1 \times S^1$ . Solutions with anti-self-double-dual cur-          |
| mielke-geometrodynamics_F01019 | 202  | 0.982 | $\mathscr{S}_*(\mathrm{M}):=\mathscr{M}/\tilde{\mathscr{D}}$                                          | $\mathcal{S}_*(M) := \mathcal{M} / \tilde{\mathcal{D}}$ | space” $\mathcal{S}_*(M) := \mathcal{M} / \tilde{\mathcal{D}}$ of all Riemannian metrics modulo the diffeomorphisms |
| mielke-geometrodynamics_F01020 | 203  | 1.000 | $S^2$                                                                                                 | $S^2$                                                   | Euler–Poincaré topological invariant (9.3.1). The two-sphere part $S^2$ of these $\mathbb{R} \times$                |
| mielke-geometrodynamics_F01021 | 203  | 1.000 | $\mathbb{R}\times$                                                                                    | $\mathbb{R} \times$                                     | Euler–Poincaré topological invariant (9.3.1). The two-sphere part $S^2$ of these $\mathbb{R} \times$                |
| mielke-geometrodynamics_F01022 | 203  | 1.000 | $S^1\times S^2$                                                                                       | $S^1 \times S^2$                                        | $S^1 \times S^2$ topological configurations may even be knotted (MIELKE 1977a), and within                          |
| mielke-geometrodynamics_F01023 | 203  | 1.000 | $\mathbf{R}^4$                                                                                        | $\mathbf{R}^4$                                          | piecewise linear structures (MILNOR 2011) on a noncompact $\mathbf{R}^4$ . However, mani-                           |
| mielke-geometrodynamics_F01024 | 203  | 1.000 | $U_4$                                                                                                 | $U_4$                                                   | Baekler P (1980) The unique spherically symmetric solution of the $U_4$ -theory of gravity in the                   |
| mielke-geometrodynamics_F01025 | 205  | 0.999 | $\mathrm{R}+\mathrm{R}^2$                                                                             | $R + R^2$                                               | Neville DE (1980) Birkhoff theorems for $R + R^2$ gravity theories with torsion. Phys Rev D                         |
| mielke-geometrodynamics_F01026 | 208  | 0.837 | $C_{\mathrm{TL}}$                                                                                     | $C_{\mathrm{TL}}$                                       | term $C_{\mathrm{TL}}$ , which simulates to some extent Einstein’s theory in 3D, allowing us to                     |
| mielke-geometrodynamics_F01027 | 208  | 0.994 | $\varepsilon\rightarrow 1/\varepsilon$                                                                | $\varepsilon \rightarrow 1/\varepsilon$                 | $\varepsilon \rightarrow 1/\varepsilon$ .                                                                           |
| mielke-geometrodynamics_F01028 | 208  | 0.674 | $\Gamma^{\beta\gamma}=\Gamma_j^{\beta\gamma}dx^j$                                                     | $\Gamma^{\beta\gamma} = \Gamma_j^{\beta\gamma} dx^j$    | connection $\Gamma^{\beta\gamma} = \Gamma_j^{\beta\gamma} dx^j$ , i.e.,                                             |

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| mielke-geometrodynamics_F01029 | 209  | ■ 1.000 | $\Gamma_{\alpha}^{\star}$                                                                    | $\Gamma_{\alpha}^{\star}$                                                                    | Since the coframe $\vartheta_{\alpha}$ has physical dimensions [length] and the connection $\Gamma_{\alpha}^{\star}$ is                                  |
| mielke-geometrodynamics_F01030 | 209  | ■ 1.000 | $\vartheta_{\alpha} / \ell$                                                                  | $\vartheta_{\alpha} / \ell$                                                                  | dimensionless, one can use $\vartheta_{\alpha} / \ell$ and $\Gamma_{\alpha}^{\star}$ as <i>dimensionless</i> gauge potentials of local                   |
| mielke-geometrodynamics_F01031 | 209  | ■ 1.000 | $C = \operatorname{Tr}\{A \wedge F\}$                                                        | $C = \operatorname{Tr}\{A \wedge F\}$                                                        | The corresponding Chern–Simons three-forms of gauge type $C = \operatorname{Tr}\{A \wedge F\}$ are the                                                   |
| mielke-geometrodynamics_F01032 | 209  | ■ 1.000 | $T_{\alpha}$                                                                                 | $T_{\alpha}$                                                                                 | are uniquely related to the torsion $T_{\alpha}$ , the curvature $R_{\alpha}^{\star}$ , and a cosmological term $\eta_{\alpha}$ ,                        |
| mielke-geometrodynamics_F01033 | 209  | ■ 1.000 | $R_{\alpha}^{\star}$                                                                         | $R_{\alpha}^{\star}$                                                                         | are uniquely related to the torsion $T_{\alpha}$ , the curvature $R_{\alpha}^{\star}$ , and a cosmological term $\eta_{\alpha}$ ,                        |
| mielke-geometrodynamics_F01034 | 209  | ■ 1.000 | $\eta_{\alpha}$                                                                              | $\eta_{\alpha}$                                                                              | are uniquely related to the torsion $T_{\alpha}$ , the curvature $R_{\alpha}^{\star}$ , and a cosmological term $\eta_{\alpha}$ ,                        |
| mielke-geometrodynamics_F01041 | 210  | ■ 0.872 | $\mathrm{SO}(1,2)$                                                                           | $SO(1,2)$                                                                                    | dimensional Lorentz group $SO(1,2)$ as a $9 = 5 \otimes 3 \otimes 1$ multiplet after bivectors are                                                       |
| mielke-geometrodynamics_F01043 | 211  | ■ 1.000 | $H = -\partial L / \partial F$                                                               | $H = -\partial L / \partial F$                                                               | are the translational and rotational field momenta (or the excitation $H = -\partial L / \partial F$                                                     |
| mielke-geometrodynamics_F01044 | 211  | ■ 1.000 | $H_{\alpha} = H_{[\alpha\beta]} \vartheta^{\beta}$                                           | $H_{\alpha} = H_{[\alpha\beta]} \vartheta^{\beta}$                                           | (In the particular case of an antisymmetric field $H_{\alpha} = H_{[\alpha\beta]} \vartheta^{\beta}$ , this is dual to the                               |
| mielke-geometrodynamics_F01045 | 211  | ■ 1.000 | $S_{\alpha} = -^* H_{\alpha}$                                                                | $S_{\alpha} = -^* H_{\alpha}$                                                                | translational field strength, i.e., $S_{\alpha} = -^* H_{\alpha}$ .) The sources for the gravitational field                                             |
| mielke-geometrodynamics_F01046 | 211  | ■ 0.957 | $\mathbb{R}^3 \otimes SO(1,2)$                                                               | $\mathbb{R}^3 \otimes SO(1,2)$                                                               | $\mathbb{R}^3 \otimes SO(1,2)$ , we arrive at the MIELKE & BAEKLER (1991) (MB) model, which is                                                           |
| mielke-geometrodynamics_F01047 | 211  | ■ 1.000 | $\theta_{\mathrm{T}}, \theta_{\mathrm{L}}$                                                   | $\theta_{\mathrm{T}}, \theta_{\mathrm{L}}$                                                   | Allowing for arbitrary “vacuum angles” $\theta_{\mathrm{T}}, \theta_{\mathrm{L}}$ , and $\theta_{\mathrm{TL}} = -\chi$ , the most general                |
| mielke-geometrodynamics_F01048 | 211  | ■ 1.000 | $\theta_{\mathrm{TL}} = -\chi$                                                               | $\theta_{\mathrm{TL}} = -\chi$                                                               | Allowing for arbitrary “vacuum angles” $\theta_{\mathrm{T}}, \theta_{\mathrm{L}}$ , and $\theta_{\mathrm{TL}} = -\chi$ , the most general                |
| mielke-geometrodynamics_F01049 | 212  | ■ 0.997 | $C_{\mathrm{T}}, C_{\mathrm{L}}$                                                             | $C_{\mathrm{T}}, C_{\mathrm{L}}$                                                             | where $C_{\mathrm{T}}, C_{\mathrm{L}}$ , and $C_{\mathrm{TL}}$ are respectively the translational, rotational, and “mixed” CS                            |
| mielke-geometrodynamics_F01050 | 212  | ■ 1.000 | $C_{\mathrm{T}}$                                                                             | $C_{\mathrm{T}}$                                                                             | tional term $C_{\mathrm{T}}$ is covariant, it appears that the MB model is only semi-topological,                                                        |
| mielke-geometrodynamics_F01051 | 212  | ■ 1.000 | $\Gamma^{\star\alpha}$                                                                       | $\Gamma^{\star\alpha}$                                                                       | Consequently, varying the Lagrangian (10.2.2) with respect to $\vartheta^{\alpha}$ and $\Gamma^{\star\alpha}$ yields                                     |
| mielke-geometrodynamics_F01052 | 212  | ■ 1.000 | $\theta_{\mathrm{T}} / 2\ell$                                                                | $\theta_{\mathrm{T}} / 2\ell$                                                                | $\theta_{\mathrm{T}} / 2\ell$ , resembling a cosmological constant familiar from 4D gravity.                                                             |
| mielke-geometrodynamics_F01053 | 212  | ■ 1.000 | $\varepsilon = \theta_{\mathrm{TL}} \theta_{\mathrm{T}} / A$                                 | $\varepsilon = \theta_{\mathrm{TL}} \theta_{\mathrm{T}} / A$                                 | where the contortional constants $\varepsilon = \theta_{\mathrm{TL}} \theta_{\mathrm{T}} / A$ and $\rho = -\theta_{\mathrm{T}}^2 / A$ are related to the |
| mielke-geometrodynamics_F01054 | 212  | ■ 1.000 | $\rho = -\theta_{\mathrm{T}}^2 / A$                                                          | $\rho = -\theta_{\mathrm{T}}^2 / A$                                                          | where the contortional constants $\varepsilon = \theta_{\mathrm{TL}} \theta_{\mathrm{T}} / A$ and $\rho = -\theta_{\mathrm{T}}^2 / A$ are related to the |
| mielke-geometrodynamics_F01055 | 212  | ■ 1.000 | $A := -(-1)^s \theta_{\mathrm{TL}}^2 + 2\theta_{\mathrm{T}} \theta_{\mathrm{L}} \neq 0$      | $A := -(-1)^s \theta_{\mathrm{TL}}^2 + 2\theta_{\mathrm{T}} \theta_{\mathrm{L}} \neq 0$      | $A := -(-1)^s \theta_{\mathrm{TL}}^2 + 2\theta_{\mathrm{T}} \theta_{\mathrm{L}} \neq 0$ is nonvanishing.                                                 |
| mielke-geometrodynamics_F01056 | 213  | ■ 0.994 | $K_{\alpha}^{\star} = -^* T_{\alpha} / 2 = -(-1)^s (\varepsilon / 2\ell) \vartheta_{\alpha}$ | $K_{\alpha}^{\star} = -^* T_{\alpha} / 2 = -(-1)^s (\varepsilon / 2\ell) \vartheta_{\alpha}$ | for the RC curvature. From the relation $K_{\alpha}^{\star} = -^* T_{\alpha} / 2 = -(-1)^s (\varepsilon / 2\ell) \vartheta_{\alpha}$ for the             |
| mielke-geometrodynamics_F01057 | 213  | ■ 1.000 | $\varepsilon = 0$                                                                            | $\varepsilon = 0$                                                                            | $\varepsilon = 0$ , i.e., in a purely Riemannian spacetime. Alternatively, for $\rho = 0$ , i.e., in the                                                 |
| mielke-geometrodynamics_F01058 | 213  | ■ 1.000 | $\Lambda_{\mathrm{eff}}$                                                                     | $\Lambda_{\mathrm{eff}}$                                                                     | cosmological term $\Lambda_{\mathrm{eff}}$ has the anti-de Sitter (AdS) metric                                                                           |
| mielke-geometrodynamics_F01059 | 214  | ■ 1.000 | $J$                                                                                          | $J$                                                                                          | respectively. Observe that the shift is proportional to the angular momentum $J$ of                                                                      |
| mielke-geometrodynamics_F01060 | 214  | ■ 1.000 | $J^2 \leq M^2$                                                                               | $J^2 \leq M^2$                                                                               | the solution, which allows for $J^2 \leq M^2$ to interpret this configuration as a <i>rotating</i>                                                       |
| mielke-geometrodynamics_F01061 | 214  | ■ 1.000 | $M = -1$                                                                                     | $M = -1$                                                                                     | normalization $M = -1$ and $J = 0$ , it reduces to the AdS metric.                                                                                       |
| mielke-geometrodynamics_F01062 | 214  | ■ 1.000 | $J = 0$                                                                                      | $J = 0$                                                                                      | normalization $M = -1$ and $J = 0$ , it reduces to the AdS metric.                                                                                       |

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| mielke-geometrodynamics_F01063 [conf 0.080] | 214  | ■ 0.080 | $\backslash\Gamma_{\alpha}^{\{\alpha\}^{\{\}}^{\{\}}\star}\backslashright.\}$<br>$\backslashrightarrow \backslash\Gamma_{\alpha}^{\{\alpha\}^{\{\}}\star}=\backslash\Gamma_{\alpha}^{\{\alpha\}^{\{\}}\star}\backslashleft\{\{\}^{\{\}}\backslash\star\right.\}-K_{\alpha}^{\{\alpha\}^{\{\}}\star}$ | $\Gamma_{\alpha}^{\{\}^{\{\}}^{\{\}}\star} \rightarrow \Gamma_{\alpha}^{\star} = \Gamma_{\alpha}^{\{\}^{\{\}}^{\{\}}\star} - K_{\alpha}^{\star}$ | rest essentially on a “prolongation” $\Gamma_{\alpha}^{\{\}^{\{\}}^{\{\}}\star} \rightarrow \Gamma_{\alpha}^{\star} = \Gamma_{\alpha}^{\{\}^{\{\}}^{\{\}}\star} - K_{\alpha}^{\star}$ of the Riemannian to            |
| mielke-geometrodynamics_F01064              | 214  | ■ 1.000 | $\backslashvarepsilon$                                                                                                                                                                                                                                                                               | $\varepsilon$                                                                                                                                    | effective cosmological constant $\Lambda_{\text{eff}}$ is related to $\varepsilon$ and $\rho$ via (10.2.10), the black hole                                                                                           |
| mielke-geometrodynamics_F01065              | 214  | ■ 1.000 | $\backslash\Gamma_{\alpha}^{\{\alpha\}^{\{\}}\star}=\frac{1}{2}\backslasheta_{\alpha\beta\gamma}\backslash\Gamma^{\beta\gamma}_{\gamma}\backslash\Gamma_{\alpha}^{\{\alpha\}^{\{\}}\star}$                                                                                                           | $\Gamma_{\alpha}^{\star} = \frac{1}{2}\eta_{\alpha\beta\gamma}\Gamma^{\beta\gamma}$                                                              | where $\varepsilon$ is a continuous parameter, involves the Lie dual $\Gamma_{\alpha}^{\star} = \frac{1}{2}\eta_{\alpha\beta\gamma}\Gamma^{\beta\gamma}$ of the                                                       |
| mielke-geometrodynamics_F01066              | 214  | ■ 1.000 | $\backslash\widetilde{\text{T}}_{\alpha}=0$                                                                                                                                                                                                                                                          | $\widetilde{T}_{\alpha} = 0$                                                                                                                     | Riemannian spacetime with $\widetilde{T}_{\alpha} = 0$ and deformed <i>teleparallelism</i> in the local gauge                                                                                                         |
| mielke-geometrodynamics_F01067              | 214  | ■ 0.999 | $\backslash\widetilde{\text{Gamma}}_{\alpha}^{\{\alpha\}^{\{\}}\star}\backslashstackrel{*}{=}\ 0$                                                                                                                                                                                                    | $\widetilde{\Gamma}_{\alpha}^{\star} \stackrel{*}{=} 0$                                                                                          | $\widetilde{\Gamma}_{\alpha}^{\star} \stackrel{*}{=} 0$ , equivalent to the covariant constraint of vanishing modified RC curvature,                                                                                  |
| mielke-geometrodynamics_F01068              | 214  | ■ 1.000 | $\backslash\widetilde{\text{R}}_{\alpha}^{\{\alpha\}^{\{\}}\star}=0$                                                                                                                                                                                                                                 | $\widetilde{R}_{\alpha}^{\star} = 0$                                                                                                             | i.e., $\widetilde{R}_{\alpha}^{\star} = \hat{0}$ . In the latter case, coframe and connection are locally <i>Lie dual</i> to each                                                                                     |
| mielke-geometrodynamics_F01069              | 215  | ■ 1.000 | $1\ /\ \backslashvarepsilon$                                                                                                                                                                                                                                                                         | $1/\varepsilon$                                                                                                                                  | large rotational momentum proportional to $1/\varepsilon$ and conversely, resembling, to some                                                                                                                         |
| mielke-geometrodynamics_F01070              | 215  | ■ 0.915 | $\backslashrho=(-1)^{\{\text{sign}\}}\backslashvarepsilon^{\{2\}}\ /\ 4$                                                                                                                                                                                                                             | $\rho = (-1)^{\text{sign}}\varepsilon^2/4$                                                                                                       | as originally envisioned by CARTAN (1924). Here $\rho = (-1)^{\text{sign}}\varepsilon^2/4$ depends quadrat-                                                                                                           |
| mielke-geometrodynamics_F01071              | 215  | ■ 0.518 | $\text{R}_{\alpha}^{\{\alpha\}^{\{\}}\star}=(-1)^{\{\text{sign}\}}\backslashvarepsilon\ \text{T}_{\alpha}\ /\ 4\ \backslashell$                                                                                                                                                                      | $R_{\alpha}^{\star} = (-1)^{\text{sign}}\varepsilon T_{\alpha}/4\ell$                                                                            | should not ignore that $R_{\alpha}^{\star} = (-1)^{\text{sign}}\varepsilon T_{\alpha}/4\ell$ induces, for $\varepsilon \neq 0$ , a constant-curvature                                                                 |
| mielke-geometrodynamics_F01072              | 215  | ■ 0.518 | $\backslashvarepsilon\ \backslashneq\ 0$                                                                                                                                                                                                                                                             | $\varepsilon \neq 0$                                                                                                                             | should not ignore that $R_{\alpha}^{\star} = (-1)^{\text{sign}}\varepsilon T_{\alpha}/4\ell$ induces, for $\varepsilon \neq 0$ , a constant-curvature                                                                 |
| mielke-geometrodynamics_F01073              | 216  | ■ 1.000 | $\backslashvarepsilon\ \rightarrow\ -\backslashvarepsilon$                                                                                                                                                                                                                                           | $\varepsilon \rightarrow -\varepsilon$                                                                                                           | symmetry” $\varepsilon \rightarrow -\varepsilon$ in (10.3.9) and (10.3.10), we see that the limit $\gamma = (1 - \delta\varepsilon) \rightarrow$                                                                      |
| mielke-geometrodynamics_F01074              | 216  | ■ 1.000 | $\backslash\gamma=(1-\delta\backslashvarepsilon)\ \rightarrow$                                                                                                                                                                                                                                       | $\gamma = (1 - \delta\varepsilon) \rightarrow$                                                                                                   | symmetry” $\varepsilon \rightarrow -\varepsilon$ in (10.3.9) and (10.3.10), we see that the limit <span style="border: 1px solid red; padding: 2px;"><math>\gamma = (1 - \delta\varepsilon) \rightarrow</math></span> |
| mielke-geometrodynamics_F01075              | 216  | ■ 1.000 | $\backslash\gamma\ \backslashneq\ 0$                                                                                                                                                                                                                                                                 | $\gamma \neq 0$                                                                                                                                  | (10.2.1). For $\gamma \neq 0$ , the vacuum solution (10.3.9) yields                                                                                                                                                   |
| mielke-geometrodynamics_F01076              | 216  | ■ 1.000 | $\backslashvarepsilon\ \rightarrow\ 0$                                                                                                                                                                                                                                                               | $\varepsilon \rightarrow 0$                                                                                                                      | in the limit $\varepsilon \rightarrow 0$ of <i>weak</i> axial torsion coupling, the translational field momenta                                                                                                       |
| mielke-geometrodynamics_F01077              | 216  | ■ 1.000 | $\text{H}_{\alpha}^{\{\alpha\}^{\{\}}\star}=\backslash\gamma\ \backslashvartheta_{\alpha}\ /\ \backslashell$                                                                                                                                                                                         | $H_{\alpha}^{\star} = \gamma\vartheta_{\alpha}/\ell$                                                                                             | i.e., $H_{\alpha}^{\star} = \gamma\vartheta_{\alpha}/\ell$ , similar to the case of the EC action. Conversely, in the limit                                                                                           |
| mielke-geometrodynamics_F01078              | 216  | ■ 1.000 | $\backslashvarepsilon\ \rightarrow\ \infty$                                                                                                                                                                                                                                                          | $\varepsilon \rightarrow \infty$                                                                                                                 | $\varepsilon \rightarrow \infty$ of <i>strong</i> axial torsion coupling, the translational momenta (10.3.9) become                                                                                                   |
| mielke-geometrodynamics_F01079              | 216  | ■ 1.000 | $\text{H}_{\alpha}=\backslash\gamma\ \backslashvartheta_{\alpha}\ /\ \delta\backslashell^{\{2\}}$                                                                                                                                                                                                    | $H_{\alpha} = -\gamma\vartheta_{\alpha}/\delta\ell^2$                                                                                            | $H_{\alpha} = -\gamma\vartheta_{\alpha}/\delta\ell^2$ , and the rotational momenta turn out to be $H_{\alpha}^{\star} = 0$ . This implies                                                                             |
| mielke-geometrodynamics_F01080              | 216  | ■ 1.000 | $\text{H}_{\alpha}^{\{\alpha\}^{\{\}}\star}=0$                                                                                                                                                                                                                                                       | $H_{\alpha}^{\star} = 0$                                                                                                                         | $H_{\alpha} = -\gamma\vartheta_{\alpha}/\delta\ell^2$ , and the rotational momenta turn out to be $H_{\alpha}^{\star} = 0$ . This implies                                                                             |
| mielke-geometrodynamics_F01081              | 217  | ■ 1.000 | $\backslash\{\text{H},\ \text{d}\ \text{t},\ \text{d}\ \backslash\mathscr{A},\ \text{d}\ h\}$                                                                                                                                                                                                        | $\{H, dt, d\mathscr{A}, dh\}$                                                                                                                    | a basis $\{H, dt, d\mathscr{A}, dh\}$ of one-forms can be constructed that, for nonzero torsion                                                                                                                       |
| mielke-geometrodynamics_F01082              | 217  | ■ 1.000 | $\text{H}=0$                                                                                                                                                                                                                                                                                         | $H = 0$                                                                                                                                          | For $H = 0$ , the three one-forms on the right-hand side become linearly dependent,                                                                                                                                   |
| mielke-geometrodynamics_F01083              | 217  | ■ 0.998 | $\text{H}_{\alpha}\ \backslashneq\ 0$                                                                                                                                                                                                                                                                | $H_{\alpha} \neq 0$                                                                                                                              | thereby sending us back to the exact solution (10.3.9). However, for $H_{\alpha} \neq 0$ , an exact                                                                                                                   |
| mielke-geometrodynamics_F01084              | 217  | ■ 1.000 | $\text{d}\ \backslash\mathscr{A}$                                                                                                                                                                                                                                                                    | $d\mathscr{A}$                                                                                                                                   | can be proposed in which one coordinate “leg,” for instance $d\mathscr{A}$ , needs to be con-                                                                                                                         |
| mielke-geometrodynamics_F01085              | 218  | ■ 1.000 | $\text{d}^{\{*\}}\ \backslashvartheta_{\alpha}=0$                                                                                                                                                                                                                                                    | $d^*\vartheta_{\alpha} = 0$                                                                                                                      | generalizes the Hilbert–de Donder or <i>transversality condition</i> $d^*\vartheta_{\alpha} = 0$ for the                                                                                                              |
| mielke-geometrodynamics_F01086              | 218  | ■ 1.000 | $\backslasheta_{\alpha}=\{\}^{\{*\}}\ \backslashvartheta_{\alpha}$                                                                                                                                                                                                                                   | $\eta_{\alpha} = ^*\vartheta_{\alpha}$                                                                                                           | To this end, we will employ the relation $\eta_{\alpha} = ^*\vartheta_{\alpha}$ for the $\eta_{\alpha}$ basis and iterate the                                                                                         |
| mielke-geometrodynamics_F01087              | 219  | ■ 1.000 | $\text{m}=\backslashvarepsilon\ /\ \backslashell$                                                                                                                                                                                                                                                    | $m = \varepsilon/\ell$                                                                                                                           | where $m = \varepsilon/\ell$ , results in vacuum and the gauge-covariant Hilbert–de Donder or                                                                                                                         |
| mielke-geometrodynamics_F01088              | 219  | ■ 0.782 | $\backslashvartheta^{\{\alpha\}}=\text{E}_{\{i\}}\{\}^{\{\alpha\}}\ \text{d}\ x^{\{i\}}$                                                                                                                                                                                                             | $\vartheta^{\alpha} = E_i^{\alpha}dx^i$                                                                                                          | is as follows: In 3D, the coframe $\vartheta^{\alpha} = E_i^{\alpha}dx^i$ has $3 \times 3 = 9$ components, of which                                                                                                   |

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| mielke-geometrodynamics_F01089 | 219  | 0.782 | $3 \times 3 = 9$                                                                      | $3 \times 3 = 9$                                                                   | is as follows: In 3D, the coframe $\vartheta^\alpha = E_i^\alpha dx^i$ has <del><math>3 \times 3 = 9</math></del> components, of which                                              |
| mielke-geometrodynamics_F01090 | 219  | 0.924 | $S O(1,2)$                                                                            | $SO(1,2)$                                                                          | 3 are pure gauge due to the $SO(1,2)$ Lorentz transformation $\vartheta'^\alpha = A_\beta^\alpha(x)\vartheta^\beta$ . The                                                           |
| mielke-geometrodynamics_F01091 | 219  | 0.924 | $\vartheta^{\prime\alpha}=\Lambda_\beta^\alpha(x)\vartheta^\beta$                     | $\vartheta'^\alpha = \Lambda_\beta^\alpha(x)\vartheta^\beta$                       | 3 are pure gauge due to the $SO(1,2)$ Lorentz transformation $\vartheta'^\alpha = A_\beta^\alpha(x)\vartheta^\beta$ . The                                                           |
| mielke-geometrodynamics_F01092 | 220  | 0.992 | $2/\theta_{\mathrm{L}}$                                                               | $2/\theta_{\mathrm{L}}$                                                            | contributes to the mass with an “anyon”-type g-factor of $2/\theta_{\mathrm{L}}$ ; cf. JACKIW & NAIR                                                                                |
| mielke-geometrodynamics_F01093 | 220  | 1.000 | $\theta_{\mathrm{T}}\theta_{\mathrm{L}}/2\chi=1$                                      | $\theta_{\mathrm{T}}\theta_{\mathrm{L}}/2\chi = 1$                                 | to be constrained to $\theta_{\mathrm{T}}\theta_{\mathrm{L}}/2\chi = 1$ , or at least to be positive in order to avoid nega-                                                        |
| mielke-geometrodynamics_F01094 | 220  | 1.000 | $\theta_{\mathrm{T}}\neq 0$                                                           | $\theta_{\mathrm{T}} \neq 0$                                                       | $\theta_{\mathrm{T}} \neq 0$ , the vacuum angle of the translational Chern–Simons term $C_{\mathrm{T}}$ . In any case,                                                              |
| mielke-geometrodynamics_F01095 | 221  | 1.000 | $\varepsilon\neq\tilde{\varepsilon}$                                                  | $\varepsilon \neq \tilde{\varepsilon}$                                             | Then, for <i>different</i> continuous parameters $\varepsilon \neq \tilde{\varepsilon}$ , we obtain                                                                                 |
| mielke-geometrodynamics_F01096 | 221  | 1.000 | $\tilde{\theta}_{\mathrm{L}}$                                                         | $\tilde{\theta}_{\mathrm{L}}$                                                      | parent. Consequently, the constants $\theta_{\mathrm{L}}$ and $\tilde{\theta}_{\mathrm{L}}$ give rise to the two central charges                                                    |
| mielke-geometrodynamics_F01097 | 222  | 1.000 | $C^{kl}=C^{lk}$                                                                       | $C^{kl} = C^{lk}$                                                                  | is the one-form associated with the symmetric Cotton tensor $C^{kl} = C^{lk}$ . In general,                                                                                         |
| mielke-geometrodynamics_F01098 | 222  | 0.940 | $\Gamma^{\{\lambda\alpha\beta\}}$                                                     | $\Gamma^{\{\lambda\alpha\beta\}}$                                                  | this is a third-order equation in the Levi-Civita connection $\Gamma^{\{\} \alpha\beta}$ , i.e., of fourth order                                                                    |
| mielke-geometrodynamics_F01099 | 222  | 1.000 | $T_{\alpha}=D\vartheta_{\alpha}=0$                                                    | $T_\alpha = D\vartheta_\alpha = 0$                                                 | Due to $T_\alpha = D\vartheta_\alpha = 0$ in Riemannian spacetime, implying $H_\alpha = 0$ in 3D, the mod-                                                                          |
| mielke-geometrodynamics_F01100 | 222  | 1.000 | $H_{\alpha}=0$                                                                        | $H_\alpha = 0$                                                                     | Due to $T_\alpha = D\vartheta_\alpha = 0$ in Riemannian spacetime, implying $H_\alpha = 0$ in 3D, the mod-                                                                          |
| mielke-geometrodynamics_F01101 | 222  | 0.891 | $DH_{\alpha}^{\star}=\delta\ell DH_{\alpha}+(\gamma/\ell)D\vartheta_{\alpha}\equiv 0$ | $DH_\alpha^\star = \delta\ell DH_\alpha + (\gamma/\ell)D\vartheta_\alpha \equiv 0$ | ified $S$ -duality leads to <del><math>DH_\alpha^\star = \delta\ell DH_\alpha + (\gamma/\ell)D\vartheta_\alpha \equiv 0</math></del> , and the two-form $D^{[\!]}H_{[\alpha\beta]}$ |
| mielke-geometrodynamics_F01102 | 222  | 0.891 | $D^{[4]}H_{[\alpha\beta]}$                                                            | $D^{[4]}H_{[\alpha\beta]}$                                                         | ified $S$ -duality leads to $DH_\alpha^\star = \delta\ell DH_\alpha + (\gamma/\ell)D\vartheta_\alpha \equiv 0$ , and the two-form $D^{[\!]}H_{[\alpha\beta]}$                       |
| mielke-geometrodynamics_F01103 | 222  | 0.539 | $G_{\alpha}=R_{\alpha}^{\{\left\{\right\}\}\star}$                                    | $G_\alpha = R_\alpha^{\{\}^\star}$                                                 | in Riemannian spacetime. Since $G_\alpha = R_\alpha^{[\!]\star}$ is the Einstein current two-form in 3D,                                                                            |
| mielke-geometrodynamics_F01104 | 222  | 0.990 | $\widetilde{\Sigma}_{\alpha}:=\Sigma_{\alpha}-D^{\{\}\}\mu_{\alpha}$                  | $\tilde{\Sigma}_\alpha := \Sigma_\alpha - D^{\{\} \} \mu_\alpha$                   | <i>symmetric</i> Belinfante–Rosenfeld two-form $\tilde{\Sigma}_\alpha := \Sigma_\alpha - D^{[\!]} \mu_\alpha$ as source.                                                            |
| mielke-geometrodynamics_F01105 | 223  | 1.000 | $E=p\,c_{\mathrm{F}}$                                                                 | $E = pc_{\mathrm{F}}$                                                              | spectrum is $E = pc_{\mathrm{F}}$ , where $c_{\mathrm{F}}$ is the effective speed of fermions in the 2D carbon                                                                      |
| mielke-geometrodynamics_F01106 | 223  | 1.000 | $c_{\mathrm{F}}$                                                                      | $c_{\mathrm{F}}$                                                                   | spectrum is $E = pc_{\mathrm{F}}$ , where $c_{\mathrm{F}}$ is the effective speed of fermions in the 2D carbon                                                                      |
| mielke-geometrodynamics_F01107 | 223  | 0.825 | $\mathrm{SO}(3)$                                                                      | $SO(3)$                                                                            | space(time): in the Euclidean case, the covering group of the rotation group $SO(3)$                                                                                                |
| mielke-geometrodynamics_F01108 | 224  | 0.902 | $s=1$                                                                                 | $s = 1$                                                                            | On the other hand, for Lorentzian signature $s = 1$ , the covering group of $SO(1,2)$ is                                                                                            |
| mielke-geometrodynamics_F01109 | 224  | 0.942 | $S L(2,\,\mathbb{R})$                                                                 | $SL(2, \mathbb{R})$                                                                | isomorphic to the real group $SL(2, \mathbb{R})$ , or as well to the symplectic algebra $\mathfrak{sp}(2, \mathbb{R})$ .                                                            |
| mielke-geometrodynamics_F01110 | 224  | 0.942 | $\mathfrak{sp}(2,\,\mathbb{R})$                                                       | $\mathfrak{sp}(2, \mathbb{R})$                                                     | isomorphic to the real group $SL(2, \mathbb{R})$ , or as well to the symplectic algebra $\mathfrak{sp}(2, \mathbb{R})$ .                                                            |
| mielke-geometrodynamics_F01111 | 224  | 1.000 | $\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2g_{\alpha\beta}$        | $\gamma_\alpha\gamma_\beta + \gamma_\beta\gamma_\alpha = 2g_{\alpha\beta}$         | thus providing a realization of the Clifford algebra $\gamma_\alpha\gamma_\beta + \gamma_\beta\gamma_\alpha = 2g_{\alpha\beta}$ in 3D.                                              |
| mielke-geometrodynamics_F01112 | 224  | 1.000 | $D\psi:=d\psi-\frac{1}{2}\gamma_{\alpha}\wedge\Gamma^{\star\alpha}\psi$               | $D\psi := d\psi - \frac{1}{2}\gamma_\alpha \wedge \Gamma^{\star\alpha}\psi$        | where $D\psi := d\psi - \frac{1}{2}\gamma_\alpha \wedge \Gamma^{\star\alpha}\psi$ denotes the covariant spinor derivative in 3D.                                                    |
| mielke-geometrodynamics_F01113 | 224  | 1.000 | $L_{\mathrm{D}}$                                                                      | $L_{\mathrm{D}}$                                                                   | is obtained by varying $L_{\mathrm{D}}$ with respect to $\overline{\Psi}$ . Since                                                                                                   |
| mielke-geometrodynamics_F01114 | 224  | 1.000 | $\bar{\Psi}$                                                                          | $\bar{\Psi}$                                                                       | is obtained by varying $L_{\mathrm{D}}$ with respect to $\overline{\Psi}$ . Since                                                                                                   |
| mielke-geometrodynamics_F01115 | 224  | 1.000 | $\hat{D}$                                                                             | $\hat{D}$                                                                          | The curvature associated with the deformed covariant derivative $\hat{D}$ is                                                                                                        |

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| mielke-geometrodynamics_F01116 | 225  | ■ 0.999 | F:=d A                                                                              | $F := dA$                                                                                   | field strength reads $F := dA$ , i.e., it remains an exact two-form. In 3D, the Maxwell                                                                              |
| mielke-geometrodynamics_F01117 | 225  | ■ 0.539 | d\left(A \wedge^{\ast} F\right) / 2                                                 | $d(A \wedge^{\ast} F) / 2$                                                                  | However, this three-form is gauge-invariant only modulo an exact form $d(A \wedge^{\ast} F) / 2$ .                                                                   |
| mielke-geometrodynamics_F01118 | 225  | ■ 0.528 | \square:=d^{\ast}\{*\} d^{\ast}\{*\}-\{*\}^{\ast}\{*\} d^{\ast}\{*\} d              | $\square := d^{\ast} d^{\ast} - {}^{\ast} d^{\ast} d$                                       | where $\square := d^{\ast} d^{\ast} - {}^{\ast} d^{\ast} d$ is the d'Alembertian in 2 + 1 dimensions. Since (10.6.3) is a                                            |
| mielke-geometrodynamics_F01119 | 225  | ■ 1.000 | \lambda=\hbar / c m_{\text {photon }}                                               | $\lambda = \hbar / c m_{\text{photon}}$                                                     | reasons, via the Compton wavelength $\lambda = \hbar / c m_{\text{photon}}$ for massive photons.                                                                     |
| mielke-geometrodynamics_F01120 | 226  | ■ 0.987 | x^{\alpha}                                                                          | $x^{\alpha}$                                                                                | vention that $x^{\alpha}$ and $y^{\alpha}$ are spacelike orthogonal vectors that span the $(x, y)$ -plane                                                            |
| mielke-geometrodynamics_F01121 | 226  | ■ 0.987 | y^{\alpha}                                                                          | $y^{\alpha}$                                                                                | vention that $x^{\alpha}$ and $y^{\alpha}$ are spacelike orthogonal vectors that span the $(x, y)$ -plane                                                            |
| mielke-geometrodynamics_F01122 | 226  | ■ 0.987 | (x, y)                                                                              | $(x, y)$                                                                                    | vention that $x^{\alpha}$ and $y^{\alpha}$ are spacelike orthogonal vectors that span the $(x, y)$ -plane                                                            |
| mielke-geometrodynamics_F01123 | 226  | ■ 1.000 | n^{\alpha}                                                                          | $n^{\alpha}$                                                                                | Moreover, the vector $n^{\alpha}$ is a timelike unit vector normal to the hypersurface with                                                                          |
| mielke-geometrodynamics_F01124 | 226  | ■ 1.000 | n^{\alpha} n_{\alpha}=s                                                             | $n^{\alpha} n_{\alpha} = s$                                                                 | $n^{\alpha} n_{\alpha} = s$ , the signature $s$ of our 3D spacetime. Following SOLENG (1992), ANANDAN                                                                |
| mielke-geometrodynamics_F01125 | 226  | ■ 1.000 | \tau_{\alpha}^{\star}                                                               | $\tau_{\alpha}^{\star}$                                                                     | (1994), and BAKKE et al. (2009), we assume that the two-forms $\Sigma_{\alpha}$ and $\tau_{\alpha}^{\star}$ of the                                                   |
| mielke-geometrodynamics_F01126 | 227  | ■ 1.000 | X                                                                                   | $X$                                                                                         | of the defect; cf. Fig. 10.2. From the specification (10.7.1) of the one-forms $X$ and                                                                               |
| mielke-geometrodynamics_F01127 | 227  | ■ 1.000 | Y                                                                                   | $Y$                                                                                         | $Y$ , it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}} \neq 0$ and $\tau_{\hat{1}\hat{2}} =$                                         |
| mielke-geometrodynamics_F01128 | 227  | ■ 1.000 | \Sigma_{\hat{0}} \neq 0                                                             | $\Sigma_{\hat{0}} \neq 0$                                                                   | $Y$ , it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}} \neq 0$ and $\tau_{\hat{1}\hat{2}} =$                                         |
| mielke-geometrodynamics_F01129 | 227  | ■ 1.000 | \tau_{\hat{1}\hat{2}}=                                                              | $\tau_{\hat{1}\hat{2}} =$                                                                   | $Y$ , it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}} \neq 0$ and $\tau_{\hat{1}\hat{2}} =$                                         |
| mielke-geometrodynamics_F01130 | 227  | ■ 0.975 | -\tau_{\hat{2}\hat{1}} \neq 0                                                       | $-\tau_{\hat{2}\hat{1}} \neq 0$                                                             | $-\tau_{\hat{2}\hat{1}} \neq 0$ .                                                                                                                                    |
| mielke-geometrodynamics_F01131 | 227  | ■ 1.000 | x^{\alpha} y^{\beta} R_{\alpha \beta}=R_{\hat{1}\hat{2}}=-R_{\hat{2}\hat{1}} \neq 0 | $x^{[\alpha} y^{\beta]} R_{\alpha \beta} = R_{\hat{1}\hat{2}} = -R_{\hat{2}\hat{1}} \neq 0$ | with $x^{\alpha}$ and $y^{\alpha}$ reveal that $x^{[\alpha} y^{\beta]} R_{\alpha \beta} = R_{\hat{1}\hat{2}} = -R_{\hat{2}\hat{1}} \neq 0$ are the only nonvanishing |
| mielke-geometrodynamics_F01132 | 227  | ■ 0.511 | \left.N^{\alpha}=n\right\rfloor \vartheta^{\alpha}                                  | $N^{\alpha} = n \rfloor \vartheta^{\alpha}$                                                 | We recall that $N^{\alpha} = n \rfloor \vartheta^{\alpha}$ is the lapse and shift vector in the (2+1) decomposition à                                                |
| mielke-geometrodynamics_F01133 | 228  | ■ 1.000 | d(\mathscr{A} \wedge d \mathscr{A})                                                 | $d(\mathscr{A} \wedge d \mathscr{A})$                                                       | Pointryagin-type term $d(\mathscr{A} \wedge d \mathscr{A})$ from the axial torsion. Moreover, the Nieh–Yan                                                           |
| mielke-geometrodynamics_F01134 | 228  | ■ 1.000 | d C_{\mathrm{T}}                                                                    | $d C_{\text{T}}$                                                                            | term $d C_{\text{T}}$ proportional to $d^{\ast} \mathscr{A}$ vanishes identically for this example of a spinning                                                     |
| mielke-geometrodynamics_F01135 | 228  | ■ 1.000 | d^{\ast} \mathscr{A}                                                                | $d^{\ast} \mathscr{A}$                                                                      | term $d C_{\text{T}}$ proportional to $d^{\ast} \mathscr{A}$ vanishes identically for this example of a spinning                                                     |
| mielke-geometrodynamics_F01136 | 228  | ■ 1.000 | \mathscr{N}=1                                                                       | $\mathcal{N} = 1$                                                                           | try generator, i.e., $\mathcal{N} = 1$ , represents the simplest consistent coupling of a Rarita–                                                                    |
| mielke-geometrodynamics_F01137 | 228  | ■ 0.961 | \left.\Psi_{\alpha}:=e_{\alpha}\right\rfloor \Psi                                   | $\Psi_{\alpha} := e_{\alpha} \rfloor \Psi$                                                  | as $\Psi_{\alpha} := e_{\alpha} \rfloor \Psi$ involves the tetrad that is inverse to the frame in the anholonomic                                                    |
| mielke-geometrodynamics_F01138 | 228  | ■ 1.000 | i, j, k, \ldots=0,1,2                                                               | $i, j, k, \dots = 0, 1, 2$                                                                  | run over $i, j, k, \dots = 0, 1, 2$ , whereas $\alpha, \beta, \dots = \hat{0}, \hat{1}, \hat{2}$ for the anholonomic indices.                                        |
| mielke-geometrodynamics_F01139 | 228  | ■ 1.000 | \alpha, \beta, \ldots=\hat{0}, \hat{1}, \hat{2}                                     | $\alpha, \beta, \dots = \hat{0}, \hat{1}, \hat{2}$                                          | run over $i, j, k, \dots = 0, 1, 2$ , whereas $\alpha, \beta, \dots = \hat{0}, \hat{1}, \hat{2}$ for the anholonomic indices.                                        |
| mielke-geometrodynamics_F01140 | 228  | ■ 1.000 | \bar{\Psi}:=\Psi^{\dagger} \gamma^0                                                 | $\bar{\Psi} := \Psi^{\dagger} \gamma^0$                                                     | $\bar{\Psi} := \Psi^{\dagger} \gamma^0$ .                                                                                                                            |
| mielke-geometrodynamics_F01141 | 228  | ■ 1.000 | \Psi:=\Psi_{\alpha} \vartheta^{\alpha}                                              | $\Psi := \Psi_{\alpha} \vartheta^{\alpha}$                                                  | <sup>3</sup> In four dimensions (4D), the RS field $\Psi := \Psi_{\alpha} \vartheta^{\alpha}$ entering (10.8.1) is a <i>Majorana-spinor</i> -valued                  |
| mielke-geometrodynamics_F01142 | 228  | ■ 0.700 | \Psi=C \bar{\Psi}^{\mathrm{T}}                                                      | $\Psi = C \bar{\Psi}^{\text{T}}$                                                            | i.e., $\Psi = C \bar{\Psi}^{\text{T}}$ , where $C$ is the charge conjugation matrix given by $C = -i \gamma_0$ satisfying $C^{\dagger} =$                            |

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| mielke-geometrodynamics_F01143 | 228  | 0.700 | $C=-i\ \gamma_{\{0\}}$                                                                                                            | $C = -i\gamma_0$                                                                                              | i.e., $\Psi = C\bar{\Psi}^T$ , where $C$ is the charge conjugation matrix given by $C = -i\gamma_0$ satisfying $C^\dagger =$                             |
| mielke-geometrodynamics_F01144 | 228  | 0.700 | $C^{\dagger}=$                                                                                                                    | $C^\dagger =$                                                                                                 | i.e., $\Psi = C\bar{\Psi}^T$ , where $C$ is the charge conjugation matrix given by $C = -i\gamma_0$ satisfying $C^\dagger =$                             |
| mielke-geometrodynamics_F01145 | 228  | 1.000 | $C^{-1},\ \quad C^T=-C$                                                                                                           | $C^{-1},\ \quad C^T = -C$                                                                                     | $C^{-1},\ \quad C^T = -C$ and $C^{-1}\gamma^\alpha C = -(\gamma^\alpha)^T$ . Consequently,                                                               |
| mielke-geometrodynamics_F01146 | 228  | 1.000 | $C^{-1}\ \gamma^\alpha\ C=-\left(\gamma^\alpha\right)^T$                                                                          | $C^{-1}\gamma^\alpha C = -(\gamma^\alpha)^T$                                                                  | $C^{-1},\ \quad C^T = -C$ and $C^{-1}\gamma^\alpha C = -(\gamma^\alpha)^T$ . Consequently,                                                               |
| mielke-geometrodynamics_F01147 | 228  | 1.000 | $\gamma^\alpha$                                                                                                                   | $\gamma^\alpha$                                                                                               | For the real Majorana representation, all $\gamma^\alpha$ are purely imaginary, and the components of the                                                |
| mielke-geometrodynamics_F01148 | 229  | 0.803 | $L_{\mathrm{RS}}$                                                                                                                 | $L_{\mathrm{RS}}$                                                                                             | The generalization of $L_{\mathrm{RS}}$ analyzed by MIELKE & MAGGIOLO (2012) exists only                                                                 |
| mielke-geometrodynamics_F01149 | 230  | 0.995 | $\vartheta^\alpha,\Gamma_\alpha^\star,\Psi$                                                                                       | $\vartheta^\alpha,\Gamma_\alpha^\star,\Psi$                                                                   | independent variables $(\vartheta^\alpha,\Gamma_\alpha^\star,\Psi)$ yields                                                                               |
| mielke-geometrodynamics_F01150 | 230  | 1.000 | $c$                                                                                                                               | $c$                                                                                                           | where $\sigma$ stands in for a spinor-valued zero-form and $c$ is a real constant. Let us probe                                                          |
| mielke-geometrodynamics_F01151 | 230  | 1.000 | $\overline{c\ \gamma\ \sigma}=c\ \overline{\sigma}\ \gamma$                                                                       | $\overline{c\gamma\sigma} = c\overline{\sigma}\gamma$                                                         | where $\overline{c\gamma\sigma} = c\overline{\sigma}\gamma$ holds for the Dirac adjoint.                                                                 |
| mielke-geometrodynamics_F01152 | 230  | 1.000 | $\delta L\ /\ \delta\Gamma_\alpha^\star\ \cong\ 0$                                                                                | $\delta L/\delta\Gamma_\alpha^\star \cong 0$                                                                  | In the following, let us assume that the second field equation $\delta L/\delta\Gamma_\alpha^\star \cong 0$ satisfies,                                   |
| mielke-geometrodynamics_F01153 | 231  | 1.000 | $c=-m$                                                                                                                            | $c = -m$                                                                                                      | $\tau_\alpha^\star$ , vanish. Moreover, we require $c = -m$ , in order to eliminate kinetic terms like                                                   |
| mielke-geometrodynamics_F01154 | 231  | 1.000 | $\gamma\ \wedge\ D\ \Psi$                                                                                                         | $\gamma \wedge D\Psi$                                                                                         | $\gamma \wedge D\Psi$ . Then, using the 3D formula                                                                                                       |
| mielke-geometrodynamics_F01155 | 232  | 0.999 | $L_{\infty}$                                                                                                                      | $L_\infty$                                                                                                    | for the coupling constants of the bosonic part of $L_\infty$ . Consequently, massless RS                                                                 |
| mielke-geometrodynamics_F01156 | 232  | 1.000 | $\mathscr{N}=2$                                                                                                                   | $\mathcal{N} = 2$                                                                                             | More recently, ALVAREZ et al. (2015) considered an $\mathcal{N} = 2$ supersymmetry with-                                                                 |
| mielke-geometrodynamics_F01157 | 233  | 0.988 | $\mathrm{D}=3$                                                                                                                    | $D = 3$                                                                                                       | Deser S, McCarthy J (1990) Self-dual formulations of $D = 3$ gravity theories. Phys Lett B                                                               |
| mielke-geometrodynamics_F01158 | 233  | 0.666 | $(2+1)$                                                                                                                           | $(2 + 1)$                                                                                                     | Garcia-Compean H, Obregon O, Ramirez C, Sabido M (2001) On S duality in (2+1) Chern–Simons                                                               |
| mielke-geometrodynamics_F01159 | 237  | 0.772 | $1\ /\ 2$                                                                                                                         | $1/2$                                                                                                         | $1/2$ are described by bispinor representations $D^{(-\frac{1}{2},\frac{3}{2})} \otimes D^{(\frac{1}{2},\frac{3}{2})}(\widetilde{SO}_\circ(1,3))$ of the |
| mielke-geometrodynamics_F01160 | 237  | 0.772 | $D^{\left(-\frac{1}{2},\frac{3}{2}\right)}\otimes D^{\left(\frac{1}{2},\frac{3}{2}\right)}\left(\widetilde{SO}_\circ(1,3)\right)$ | $D^{(-\frac{1}{2},\frac{3}{2})} \otimes D^{(\frac{1}{2},\frac{3}{2})} \left(\widetilde{SO}_\circ(1,3)\right)$ | $1/2$ are described by bispinor representations $D^{(-\frac{1}{2},\frac{3}{2})} \otimes D^{(\frac{1}{2},\frac{3}{2})}(\widetilde{SO}_\circ(1,3))$ of the |
| mielke-geometrodynamics_F01161 | 237  | 1.000 | $S\ O_\circ(1,3)$                                                                                                                 | $SO_\circ(1,3)$                                                                                               | proper Lorentz group $SO_\circ(1,3)$ (GEL’FAND et al. 1963). <sup>2</sup> These representations                                                          |
| mielke-geometrodynamics_F01162 | 237  | 0.983 | $P_\circ:=\mathbb{R}^4\otimes S\ O_\circ(1,3)$                                                                                    | $P_\circ := \mathbb{R}^4 \otimes SO_\circ(1,3)$                                                               | resentations of the (restricted) Poincaré group $P_\circ := \mathbb{R}^4 \otimes SO_\circ(1,3)$ of flat                                                  |
| mielke-geometrodynamics_F01163 | 237  | 1.000 | $\psi_\infty:=\psi_{1\ /\ 2}\otimes\psi_{5\ /\ 2}\otimes\psi_{9\ /\ 2}\otimes\ldots$                                              | $\psi_\infty := \psi_{1/2} \otimes \psi_{5/2} \otimes \psi_{9/2} \otimes \dots$                               | $\psi_\infty := \psi_{1/2} \otimes \psi_{5/2} \otimes \psi_{9/2} \otimes \dots$ .                                                                        |
| mielke-geometrodynamics_F01164 | 237  | 0.706 | $\mathrm{SO}(s,\ n-s)$                                                                                                            | $SO(s, n - s)$                                                                                                | (pseudo-orthogonal) structure group $SO(s, n - s)$ of the tangent bundle of an                                                                           |
| mielke-geometrodynamics_F01165 | 237  | 1.000 | $s\ \neq\ 0$                                                                                                                      | $s \neq 0$                                                                                                    | n-dimensional manifold <sup>3</sup> of signature $s$ . Provided $s \neq 0$ , this topological space                                                      |
| mielke-geometrodynamics_F01166 | 237  | 1.000 | $S\ O_\circ(s,\ n-s)$                                                                                                             | $SO_\circ(s, n - s)$                                                                                          | ing to homotopy theory. Concerning the connected component $SO_\circ(s, n - s)$ of the                                                                   |
| mielke-geometrodynamics_F01167 | 237  | 0.929 | $\widetilde{S\ O}_\circ(s,\ n-s)(=:\operatorname{Spin}(n))$                                                                       | $\widetilde{SO}_\circ(s, n - s)(=: \operatorname{Spin}(n))$                                                   | $\widetilde{SO}_\circ(s, n - s) (=:\operatorname{Spin}(n)$ for $s = 0$ ), whose irreducible representations provide                                      |
| mielke-geometrodynamics_F01168 | 237  | 0.929 | $s=0)$                                                                                                                            | $s = 0)$                                                                                                      | $\widetilde{SO}_\circ(s, n - s) (=:\operatorname{Spin}(n)$ for $s = 0$ ), whose irreducible representations provide                                      |
| mielke-geometrodynamics_F01169 | 238  | 1.000 | $\Lambda:\ \widetilde{\mathrm{G}}\rightarrow\mathrm{G}$                                                                           | $\Lambda : \tilde{G} \rightarrow G$                                                                           | This transition is achieved by a group homomorphism $\Lambda : \tilde{G} \rightarrow G$ , which is                                                       |

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| mielke-geometrodynamics_F01170 | 238  | ■ 1.000 | $C(s, n-s)$                                               | $C(s, n-s)$                                       | based on the following explicit construction: Let $C(s, n-s)$ denote the so-called                           |
| mielke-geometrodynamics_F01171 | 238  | ■ 1.000 | $\gamma_{\alpha}$                                         | $\gamma_{\alpha}$                                 | Clifford algebra, which is spanned by <i>generalized</i> complex Dirac matrices $\gamma_{\alpha}$ . These    |
| mielke-geometrodynamics_F01172 | 238  | ■ 1.000 | $N \times N$                                              | $N \times N$                                      | and admit realizations by $N \times N$ matrices for which $N = 2^{\lfloor n/2 \rfloor}$ corresponds to the   |
| mielke-geometrodynamics_F01173 | 238  | ■ 1.000 | $N=2^{\lfloor n/2 \rfloor}$                               | $N = 2^{\lfloor n/2 \rfloor}$                     | and admit realizations by $N \times N$ matrices for which $N = 2^{\lfloor n/2 \rfloor}$ corresponds to the   |
| mielke-geometrodynamics_F01174 | 238  | ■ 1.000 | $\mathbf{E}^n$                                            | $\mathbf{E}^n$                                    | Let the transformations of the Euclidean vector space $\mathbf{E}^n$ with coordinates $x^{\alpha}$ be        |
| mielke-geometrodynamics_F01175 | 238  | ■ 0.963 | $\mathrm{g} \in SO(s, n-s)$                               | $g \in SO(s, n-s)$                                | determined by the (Lorentz) rotations $g \in SO(s, n-s)$ on behalf of the group action                       |
| mielke-geometrodynamics_F01176 | 238  | ■ 0.983 | $2^N$                                                     | $2^N$                                             | to a $2^N$ -dimensional bundle, the so-called Clifford bundle CV. For details, see ATIYAH                    |
| mielke-geometrodynamics_F01177 | 239  | ■ 0.976 | $\widetilde{SO}(1,3) \approx SL(2, \mathbb{C})$           | $\widetilde{SO}_o(1,3) \approx SL(2, \mathbb{C})$ | orthochronous Lorentz group $SO_o(1,3)$ is represented by $\widetilde{SO}_o(1,3) \approx SL(2, \mathbb{C})$  |
| mielke-geometrodynamics_F01178 | 239  | ■ 1.000 | $\widetilde{SO}(s, n-s)$                                  | $\widetilde{SO}_o(s, n-s)$                        | infinitesimal generators of the group $\widetilde{SO}_o(s, n-s)$ are                                         |
| mielke-geometrodynamics_F01179 | 239  | ■ 0.588 | $\mathrm{P}_0 := \mathbb{R}^n \otimes \widetilde{SO}$     | $P_0 := \mathbb{R}^n \otimes \widetilde{SO}$      | as induced representations of the “covering” Poincaré group $P_0 := \mathbb{R}^n \otimes \widetilde{SO}_o$ . |
| mielke-geometrodynamics_F01180 | 239  | ■ 1.000 | $(s, n-s)$                                                | $(s, n-s)$                                        | $(s, n-s)$ in a curved spacetime. It is necessary to obtain from a more general                              |
| mielke-geometrodynamics_F01181 | 239  | ■ 1.000 | $L^g(\mathrm{g})(M)$                                      | $L^g(M)$                                          | “geometric arena” the bundle $L^g(M)$ of orthogonal frames concerning the foundation                         |
| mielke-geometrodynamics_F01182 | 239  | ■ 0.974 | $L_{\circ}^g(M)$                                          | $L_o^g(M)$                                        | a <i>spin structure</i> is imposed on a base space $M$ as follows: Besides $L_o^g(M)$ , the                  |
| mielke-geometrodynamics_F01183 | 239  | ■ 0.998 | $\mathrm{f}: \widetilde{L}(M) \rightarrow L_{\circ}^g(M)$ | $f: \widetilde{L}(M) \rightarrow L_o^g(M)$        | is introduced (MILNOR 1963) together with the bundle map $f: \widetilde{L}(M) \rightarrow L_o^g(M)$ ,        |
| mielke-geometrodynamics_F01184 | 239  | ■ 0.950 | $L_o^g(M)$                                                | $L_o^g(M)$                                        | A spin structure over $M$ is equivalent to a double covering of $L_o^g(M)$ . This spin                       |
| mielke-geometrodynamics_F01185 | 239  | ■ 0.949 | $\widetilde{L}^{\prime}(M), f^{\prime}$                   | $\widetilde{L}'(M), f'$                           | structure is identical to a second one that is represented by $(\widetilde{L}'(M), f')$ if the iso-          |
| mielke-geometrodynamics_F01186 | 239  | ■ 1.000 | $f^{\prime}$                                              | $f'$                                              | bundle mapping $f'$ . Additionally, a <i>spin manifold</i> is understood as a Riemannian                     |
| mielke-geometrodynamics_F01187 | 239  | ■ 1.000 | $\mathrm{w}_2 \in H^2(M, \mathbb{Z}_2)$                   | $w_2 \in H^2(M, \mathbb{Z}_2)$                    | it is necessary and sufficient that the Stiefel–Whitney class $w_2 \in H^2(M, \mathbb{Z}_2)$ of              |
| mielke-geometrodynamics_F01188 | 239  | ■ 1.000 | $\mathrm{g}^{\text{th}}$                                  | $g^{\text{th}}$                                   | the base space $M$ be zero. Here the $g^{\text{th}}$ Stiefel–Whitney class is considered as the              |
| mielke-geometrodynamics_F01189 | 240  | ■ 1.000 | $H^g(M, \mathbb{Z}_2)$                                    | $H^g(M, \mathbb{Z}_2)$                            | characteristic class of the $g^{\text{th}}$ relative cohomology group $H^g(M, \mathbb{Z}_2)$ of the manifold |
| mielke-geometrodynamics_F01190 | 240  | ■ 1.000 | $\widetilde{s} = \dim H^1(M, \mathbb{Z}_2)$               | $\tilde{s} = \dim H^1(M, \mathbb{Z}_2)$           | defined by the dimension $\tilde{s} = \dim H^1(M, \mathbb{Z}_2)$ of the first relative cohomology group      |
| mielke-geometrodynamics_F01191 | 240  | ■ 0.813 | $\widetilde{s} = 2^v$                                     | $\tilde{s} = 2^v$                                 | (MILNOR 1963). Accordingly, there are exactly $\tilde{s} = 2^v$ spin structures possible                     |
| mielke-geometrodynamics_F01192 | 240  | ■ 0.997 | $\nu$                                                     | $\nu$                                             | hypersurface consists of the connected sum of $\nu$ handlebodies (“wormholes”) $W^3 :=$                      |
| mielke-geometrodynamics_F01193 | 240  | ■ 0.997 | $W^3 :=$                                                  | $W^3 :=$                                          | hypersurface consists of the connected sum of $\nu$ handlebodies (“wormholes”) $W^3 :=$                      |
| mielke-geometrodynamics_F01194 | 240  | ■ 0.999 | $\mathrm{w}_1=0$                                          | $w_1 = 0$                                         | $s$ ) bundles. The last bundle can be oriented if $w_1 = 0$ is valid, while the number of                    |
| mielke-geometrodynamics_F01195 | 240  | ■ 0.993 | $H^{\circ}(M, \mathbb{Z}_2)$                              | $H^{\circ}(M, \mathbb{Z}_2)$                      | distinguishable orientations has to be processed out of the dimension of $H^{\circ}(M, \mathbb{Z}_2)$        |
| mielke-geometrodynamics_F01196 | 240  | ■ 1.000 | $S^7$                                                     | $S^7$                                             | dimensional sphere $S^7$ (MILNOR 1965). Attention can be drawn to other topologically                        |

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| mielke-geometrodynamics_F01197 | 240  | ■ 0.998 | (1968,1970)                                                                           | (1968,1970)                                                                           | However, it was GEROCH (1968, 1970) who pointed out an intuitive and in the                                                                             |
| mielke-geometrodynamics_F01198 | 240  | ■ 0.969 | $T(M)$                                                                                | $T(M)$                                                                                | structures on $T(M)$ is dependent only on the mapping $f$ (ISHAM 1978).                                                                                 |
| mielke-geometrodynamics_F01199 | 240  | ■ 0.925 | $S\ U(3)$                                                                             | $SU(3)$                                                                               | invariance or Gell-Mann’s $SU(3)$ -group as well into these global constructions, the                                                                   |
| mielke-geometrodynamics_F01200 | 241  | ■ 1.000 | $\tilde{\mathrm{L}}(\mathrm{M})$                                                      | $\tilde{L}(M)$                                                                        | <i>associated</i> to the spin bundle $\tilde{L}(M)$ . For this reason, it is also called an <i>associi-</i>                                             |
| mielke-geometrodynamics_F01201 | 241  | ■ 0.614 | $D^{\mathrm{F}}$                                                                      | $D^F$                                                                                 | plex) representation $D^F$ of dimension $N = 2^{[n/2]}$ of the structure group $\widetilde{SO}_\circ$ .                                                 |
| mielke-geometrodynamics_F01202 | 241  | ■ 0.614 | $\widetilde{S\ 0}_{\circ}$                                                            | $\widetilde{SO}_\circ$                                                                | plex) representation $D^F$ of dimension $N = 2^{[n/2]}$ of the structure group $\widetilde{SO}_\circ$ .                                                 |
| mielke-geometrodynamics_F01203 | 241  | ■ 0.986 | $\widetilde{S\ 0}_{\circ}(1,3)$                                                       | $\widetilde{SO}_\circ(1,3)$                                                           | $(s, n - s)$ , which contains the physical Lorentz group $\widetilde{SO}_\circ(1,3)$ as a subgroup.                                                     |
| mielke-geometrodynamics_F01204 | 241  | ■ 0.996 | $S\ L(2, \mathbb{C}) \approx \widetilde{S\ 0}_{\circ}(1,3)$                           | $SL(2, \mathbb{C}) \approx \widetilde{SO}_\circ(1,3)$                                 | denotes the bispinor representation of the covering group $\boxed{SL(2, \mathbb{C}) \approx \widetilde{SO}_\circ(1,3)}$ of                              |
| mielke-geometrodynamics_F01205 | 241  | ■ 1.000 | $\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus \left(\frac{1}{2}, \frac{3}{2}\right)$ | $\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus \left(\frac{1}{2}, \frac{3}{2}\right)$ | the proper Lorentz group, which itself is the direct sum $\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus \left(\frac{1}{2}, \frac{3}{2}\right)$ of spin- |
| mielke-geometrodynamics_F01206 | 241  | ■ 0.999 | $\frac{1}{2}$                                                                         | $\frac{1}{2}$                                                                         | $\frac{1}{2}$ representations labeled according to GEL’FAND et al. (1963). This formulation                                                             |
| mielke-geometrodynamics_F01207 | 241  | ■ 1.000 | $\tilde{0}(1,3)$                                                                      | $\tilde{O}(1,3)$                                                                      | group $\tilde{O}(1,3)$ if required. Concerning the spinor basis $\tilde{b}_{(1)}, i = 1, \dots, N$ , their cross                                        |
| mielke-geometrodynamics_F01208 | 241  | ■ 1.000 | $\tilde{b}_{(1)}, i=1, \ldots, N$                                                     | $\tilde{b}_{(1)}, i = 1, \dots, N$                                                    | group $\tilde{O}(1,3)$ if required. Concerning the spinor basis $\boxed{\tilde{b}_{(1)}, i = 1, \dots, N}$ , their cross                                |
| mielke-geometrodynamics_F01209 | 242  | ■ 0.999 | $\mathscr{G}_S$                                                                       | $\mathscr{G}_S$                                                                       | of the group $\mathscr{G}_S$ of (local) spin gauge transformations. This group results from $\mathscr{G}_p$ , a                                         |
| mielke-geometrodynamics_F01210 | 242  | ■ 0.999 | $\mathscr{G}_p$                                                                       | $\mathscr{G}_p$                                                                       | of the group $\mathscr{G}_S$ of (local) spin gauge transformations. This group results from $\mathscr{G}_p$ , a                                         |
| mielke-geometrodynamics_F01211 | 242  | ■ 1.000 | $\tilde{L}(M)$                                                                        | $\tilde{L}(M)$                                                                        | The spin bundle $\tilde{L}(M)$ can, analogously to the bundle $L^g(M)$ of the orthonormal                                                               |
| mielke-geometrodynamics_F01212 | 242  | ■ 1.000 | $\gamma^{n+1}$                                                                        | $\gamma^{n+1}$                                                                        | The matrix $\gamma^{n+1}$ that is thereby implicitly defined is an axial Lorentz scalar and can                                                         |
| mielke-geometrodynamics_F01213 | 243  | ■ 1.000 | $\gamma^{n+1}$                                                                        | $\gamma^{n+1}$                                                                        | If one realizes the Dirac matrices in the “Cartan basis,” then $\gamma^{n+1}$ can be regarded as                                                        |
| mielke-geometrodynamics_F01214 | 243  | ■ 0.999 | $\tilde{\omega}$                                                                      | $\tilde{\omega}$                                                                      | The introduction of a $\widetilde{SO}_\circ(s, n - s)$ -valued 1-form $\tilde{\omega}$ of the connection in the asso-                                   |
| mielke-geometrodynamics_F01215 | 243  | ■ 0.790 | $\Gamma_i^{\alpha\beta}$                                                              | $\Gamma_i^{\alpha\beta}$                                                              | The local components of the spin connection, i.e., Ricci’s rotation coefficients $\Gamma_i^{-\alpha\beta}$ ,                                            |
| mielke-geometrodynamics_F01216 | 243  | ■ 0.958 | $\Gamma_{ij}^{\cdot k}$                                                               | $\Gamma_{ij}^{\cdot k}$                                                               | can be generated from the holonomic connection coefficients $\Gamma_{ij}^{\cdot k}$ by means of                                                         |
| mielke-geometrodynamics_F01217 | 244  | ■ 0.995 | $\mathbb{R}^n \otimes S\ 0$                                                           | $\mathbb{R}^n \otimes SO$                                                             | If we had started from a spinor representation of the Poincaré group $\mathbb{R}^n \ltimes SO_\circ$ .                                                  |
| mielke-geometrodynamics_F01218 | 244  | ■ 0.996 | $e_{\alpha}^{-i}(m)$                                                                  | $e_{\alpha}^{-i}(m)$                                                                  | strived for an extension by inserting the tetrad fields $e_{\alpha}^i(m)$ , i.e., the local bundle coor-                                                |
| mielke-geometrodynamics_F01219 | 245  | ■ 1.000 | $\mathcal{D}(M)$                                                                      | $\mathcal{D}(M)$                                                                      | with respect to the group $\mathcal{D}(M)$ of differentiable coordinate transformations is again                                                        |
| mielke-geometrodynamics_F01220 | 245  | ■ 1.000 | $\delta L_{\mathrm{D}} / \delta \bar{\psi}$                                           | $\delta L_D / \delta \bar{\psi}$                                                      | By variation of (11.2.1) for $\delta L_D / \delta \bar{\psi}$ or for $\delta L_D / \delta \psi$ , respectively, the generally                           |
| mielke-geometrodynamics_F01221 | 245  | ■ 1.000 | $\delta L_{\mathrm{D}} / \delta \psi$                                                 | $\delta L_D / \delta \psi$                                                            | By variation of (11.2.1) for $\delta L_D / \delta \bar{\psi}$ or for $\delta L_D / \delta \psi$ , respectively, the generally                           |
| mielke-geometrodynamics_F01222 | 246  | ■ 1.000 | $\mathrm{D}\gamma = 0$                                                                | $D\gamma = 0$                                                                         | for a covariant constant $\gamma$ , i.e., for $\boxed{D\gamma = 0}$ . However, the latter condition is globally                                         |

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| mielke-geometrodynamics_F01223 | 246  | ■ 1.000 | $\tau:=\delta\mathrm{L} / \delta\mathrm{\omega}$                              | $\tau := \delta L / \delta \omega$                       | using the rules concerning exterior forms. According to the definitions $\tau := \delta L / \delta \omega$                                |
| mielke-geometrodynamics_F01224 | 246  | ■ 1.000 | $\tau=i\,\overline{\psi}\,\gamma\,\psi$                                       | $\tau = i\bar{\psi}\gamma\psi$                           | $\tau = i\bar{\psi}\gamma\psi$ of its charge current, but also on the (Lorentz) rotational gauge fields of                                |
| mielke-geometrodynamics_F01225 | 246  | ■ 0.992 | $\tau_{\mathrm{s}}$                                                           | $\tau_s$                                                 | respect to the current $\tau$ or $\tau_s$ , and this according to whether the “internal” Yang–                                            |
| mielke-geometrodynamics_F01226 | 246  | ■ 0.999 | $j_{\mathrm{n+1}}:=i\,\overline{\psi}^*\,\gamma\,\gamma^{\mathrm{n+1}}\,\psi$ | $j_{n+1} := i\bar{\psi}^*\gamma\gamma^{n+1}\psi$         | <sup>6</sup> A corresponding conservation law is also valid for the axial vector current $j_{n+1} := i\bar{\psi}^*\gamma\gamma^{n+1}\psi$ |
| mielke-geometrodynamics_F01227 | 247  | ■ 1.000 | $\mathrm{H}^{\mathrm{n}-1}$                                                   | $\mathrm{H}^{n-1}$                                       | hypersurfaces $\mathrm{H}^{n-1}$ , this is not only independent of time, but even <i>positive definite</i> ,                              |
| mielke-geometrodynamics_F01228 | 247  | ■ 1.000 | $\tau_{\mathrm{s}}^0$                                                         | $\tau_s^0$                                               | Nevertheless, one could not interpret $\tau_s^0$ as the probability density in a non quantized                                            |
| mielke-geometrodynamics_F01229 | 247  | ■ 0.999 | $\tilde{\omega}^{\{\}}$                                                       | $\tilde{\omega}^{\{}}$                                   | into the Christoffel-like connection $\tilde{\omega}^{\{}}$ and the spin contortion that is projected by                                  |
| mielke-geometrodynamics_F01230 | 247  | ■ 0.826 | $\mathrm{D}^{\{\}}$                                                           | $\mathrm{D}^{\{}}$                                       | Here $\mathrm{D}^{\{}}$ denotes—similarly as in HEHL et al. (1976)—the covariant derivative with                                          |
| mielke-geometrodynamics_F01231 | 247  | ■ 0.993 | $\Theta=[\mathrm{K},\,\vartheta]$                                             | $\Theta = [K, \vartheta]$                                | occurs in RC spacetime. Considering the resolution of the relation $\Theta = [K, \vartheta]$ for                                          |
| mielke-geometrodynamics_F01232 | 248  | ■ 1.000 | $[\Theta,\vartheta]$                                                          | $[\Theta, \vartheta]$                                    | it follows that the modified torsion $[\Theta, \vartheta]$ couples <i>algebraically</i> to the canonical                                  |
| mielke-geometrodynamics_F01233 | 248  | ■ 1.000 | $\delta\mathrm{L}_{\mathrm{D}} / \delta\overline{\psi}$                       | $\delta L_D / \delta \bar{\psi}$                         | $\delta L_D / \delta \bar{\psi}$ . This equation will be referred to as the <i>nonlinear Heisenberg–Pauli–Weyl</i>                        |
| mielke-geometrodynamics_F01234 | 249  | ■ 1.000 | $\mathrm{D}=\vartheta^{\alpha}\nabla_{\alpha}$                                | $D = \vartheta^\alpha \nabla_\alpha$                     | with $D = \vartheta^\alpha \nabla_\alpha$ taken into considerations, as follows:                                                          |
| mielke-geometrodynamics_F01235 | 249  | ■ 1.000 | (1938,1957)                                                                   | (1938, 1957)                                             | was IVANENKO (1938, 1957) who, for quantum-theoretic reasons, pointed out the                                                             |
| mielke-geometrodynamics_F01236 | 249  | ■ 1.000 | P C                                                                           | $PC$                                                     | reflection and charge conjugation $PC$ , and the time reflection $T$ . To be added is the                                                 |
| mielke-geometrodynamics_F01237 | 249  | ■ 0.520 | $\psi:=\mathrm{binom}\{\psi_{\mathrm{p}}\}\{\psi_{\mathrm{N}}\}$              | $\psi := \begin{pmatrix} \psi_p \\ \psi_N \end{pmatrix}$ | fundamental equation for the isotopic doublet $\psi := \begin{pmatrix} \psi_p \\ \psi_N \end{pmatrix}$ , this equation had to             |
| mielke-geometrodynamics_F01238 | 249  | ■ 1.000 | $\mathrm{S}\,\mathrm{L}(2,\,\mathbb{C})$                                      | $SL(2, \mathbb{C})$                                      | group $SL(2, \mathbb{C})$ of Weyl to a gauge theory with $\overline{\mathrm{G}} = SL(2\mathrm{f}, \mathbb{C})$ as a structure group,      |
| mielke-geometrodynamics_F01239 | 249  | ■ 1.000 | $\overline{\mathrm{G}}=\mathrm{S}\,\mathrm{L}(2\,\mathrm{f},\,\mathbb{C})$    | $\overline{\mathrm{G}} = SL(2\mathrm{f}, \mathbb{C})$    | group $SL(2, \mathbb{C})$ of Weyl to a gauge theory with $\overline{\mathrm{G}} = SL(2\mathrm{f}, \mathbb{C})$ as a structure group,      |
| mielke-geometrodynamics_F01240 | 249  | ■ 0.994 | $\overline{\mathrm{G}}$                                                       | $\overline{\mathrm{G}}$                                  | results in a $\overline{\mathrm{G}}$ -invariant nonlinear Heisenberg–Pauli–Weyl spinor equation (MIELKE                                   |
| mielke-geometrodynamics_F01241 | 251  | ■ 1.000 | $1 / r^2$                                                                     | $1/r^2$                                                  | and thus replaces Coulomb’s $1/r^2$ -dependence on the distance $r :=  \vec{\mathbf{x}} $ from the cen-                                   |
| mielke-geometrodynamics_F01242 | 251  | ■ 1.000 | $r:= \overrightarrow{\mathbf{x}} $                                            | $r :=  \vec{\mathbf{x}} $                                | and thus replaces Coulomb’s $1/r^2$ -dependence on the distance $r :=  \vec{\mathbf{x}} $ from the cen-                                   |
| mielke-geometrodynamics_F01243 | 251  | ■ 1.000 | (1934,1935)                                                                   | (1934, 1935)                                             | context. Theoretically, it was coined by BORN & INFELD (1934, 1935) and, as far as a quantum-                                             |
| mielke-geometrodynamics_F01244 | 252  | ■ 0.979 | n=4                                                                           | $n = 4$                                                  | that is given by (11.3.7) exclusively in the physically(!) relevant case of $n = 4$ dimen-                                                |
| mielke-geometrodynamics_F01245 | 252  | ■ 1.000 | $\varepsilon=\pm 1,\,\ell$                                                    | $\varepsilon = \pm 1, \ell$                              | the following analysis. On the other hand, the constants $\varepsilon = \pm 1, \ell$ , and $a$ are to be                                  |
| mielke-geometrodynamics_F01246 | 253  | ■ 0.999 | $v=v(\rho)$                                                                   | $v = v(\rho)$                                            | Concerning <i>static</i> spinor fields, the functions $v = v(\rho)$ and $\mu = \mu(\rho)$ depend only                                     |
| mielke-geometrodynamics_F01247 | 253  | ■ 0.999 | $\mu=\mu(\rho)$                                                               | $\mu = \mu(\rho)$                                        | Concerning <i>static</i> spinor fields, the functions $v = v(\rho)$ and $\mu = \mu(\rho)$ depend only                                     |
| mielke-geometrodynamics_F01248 | 253  | ■ 1.000 | $\mathrm{r}:= \overrightarrow{\mathbf{x}} ,\,\vartheta$                       | $\mathbf{r} :=  \vec{\mathbf{x}} , \vartheta$            | and the metric becomes static itself. Thereby the spherical coordinates $\mathbf{r} :=  \vec{\mathbf{x}} , \vartheta$ ,                   |

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| mielke-geometrodynamics_F01249 | 253  | ■ 1.000 | $\overrightarrow{\mathrm{x}}$                                | $\vec{x}$                                           | and $\phi$ are related to the Cartesian coordinates $\vec{x}$ in the familiar manner. The search for                          |
| mielke-geometrodynamics_F01250 | 253  | ■ 1.000 | $g_{kl}$                                                     | $g_{kl}$                                            | can be simplified by conformally relating the components $g_{kl}$ of the metric with those                                    |
| mielke-geometrodynamics_F01251 | 253  | ■ 0.919 | $\Gamma_{\kappa}^{\{f \cdot \alpha \beta\}}$                 | $\Gamma_{\kappa}^{\{f \cdot \alpha \beta\}}$        | between the Christoffel symbols and the torsion-free rotation coefficients $\Gamma_{\kappa}^{\{ \cdot \} \alpha \beta}$ , the |
| mielke-geometrodynamics_F01252 | 254  | ■ 0.742 | $\mathrm{R}_{\circ}$                                         | $R_{\circ}$                                         | $R_{\circ}$ . For this purpose, the de Sitter space is to be covered with the so-called horospher-                            |
| mielke-geometrodynamics_F01253 | 254  | ■ 0.990 | $\lambda, \mathbf{y}$                                        | $\lambda, \mathbf{y}$                               | ical coordinate system $(\lambda, \mathbf{y})$ . Concerning these coordinates, the line element of the                        |
| mielke-geometrodynamics_F01254 | 254  | ■ 0.984 | $S_{O_{\circ}}(1, 4)$                                        | $SO_{\circ}(1, 4)$                                  | de Sitter group $SO_{\circ}(1, 4)$ (see MIELKE (1977c), Eqs. (1.23) and (4.5) in the case of                                  |
| mielke-geometrodynamics_F01255 | 254  | ■ 1.000 | $\bar{\Gamma}_k$                                             | $\bar{\Gamma}_k$                                    | For the sake of completeness, the conformally related component $\bar{\Gamma}_k$ of the spinor                                |
| mielke-geometrodynamics_F01256 | 254  | ■ 1.000 | $i \vec{\gamma} \cdot \vec{\partial}$                        | $i \vec{\gamma} \cdot \vec{\partial}$               | spacelike Dirac operator $i \vec{\gamma} \cdot \vec{\partial}$ as being written again in Cartesian coordinates.               |
| mielke-geometrodynamics_F01257 | 255  | ■ 0.999 | $\exp(-\mu/2 - \nu/4)$                                       | $\exp(-\mu/2 - \nu/4)$                              | $\exp(-\mu/2 - \nu/4)$ , as stated, into the separation ansatz                                                                |
| mielke-geometrodynamics_F01258 | 255  | ■ 0.643 | $\chi_k^m$                                                   | $\chi_k^m$                                          | Following the notation of ROSE (1961), the spin-weighted spherical harmonics $\chi_k^m$                                       |
| mielke-geometrodynamics_F01259 | 255  | ■ 1.000 | $\mathrm{P} = (-1)^l$                                        | $P = (-1)^l$                                        | of parity $P = (-1)^l$ are given by                                                                                           |
| mielke-geometrodynamics_F01260 | 255  | ■ 1.000 | $l$                                                          | $l$                                                 | Here the parameter $\kappa$ is related to the quantum numbers $j$ and $l$ of the total angular                                |
| mielke-geometrodynamics_F01261 | 255  | ■ 1.000 | $\mathrm{C}\left(l\frac{1}{2}j; m - \bar{m}, \bar{m}\right)$ | $C\left(l\frac{1}{2}j; m - \bar{m}, \bar{m}\right)$ | Here $C(l\frac{1}{2}j; m - \bar{m}, \bar{m})$ denote the Clebsch–Gordan coefficients of the rotation group                    |
| mielke-geometrodynamics_F01262 | 255  | ■ 0.996 | $\mathrm{Y}_l^m(\vartheta, \phi)$                            | $Y_l^m(\vartheta, \phi)$                            | SU(2), and $Y_l^m(\vartheta, \phi)$ are the spherical harmonics. Concerning the operators $\vec{J}$ and                       |
| mielke-geometrodynamics_F01263 | 255  | ■ 0.996 | $\vec{J}$                                                    | $\vec{J}$                                           | SU(2), and $Y_l^m(\vartheta, \phi)$ are the spherical harmonics. Concerning the operators $\vec{J}$ and                       |
| mielke-geometrodynamics_F01264 | 255  | ■ 1.000 | $\vec{L}$                                                    | $\vec{L}$                                           | $\vec{L}$ of the total spin and the orbital angular momentum, these spinors satisfy the                                       |
| mielke-geometrodynamics_F01265 | 256  | ■ 1.000 | $ \kappa =1$                                                 | $ \kappa  = 1$                                      | follows from the properties of the spherical harmonics that this is the case for $ \kappa  = 1$                               |
| mielke-geometrodynamics_F01266 | 256  | ■ 1.000 | $j=\frac{1}{2}, l=0, 1$                                      | $j = \frac{1}{2}, l = 0, 1$                         | to the values $j = \frac{1}{2}, l = 0, 1$ , and $m = \pm \frac{1}{2}$ , respectively. States of higher quantum                |
| mielke-geometrodynamics_F01267 | 256  | ■ 1.000 | $m = \pm \frac{1}{2}$                                        | $m = \pm \frac{1}{2}$                               | to the values $j = \frac{1}{2}, l = 0, 1$ , and $m = \pm \frac{1}{2}$ , respectively. States of higher quantum                |
| mielke-geometrodynamics_F01268 | 256  | ■ 1.000 | $\int L_{D'}$                                                | $\int L_{D'}$                                       | functional $\int L_{D'}$ belonging to (11.4.3). After integrating over the angular variables                                  |
| mielke-geometrodynamics_F01269 | 256  | ■ 1.000 | $\rho^*$                                                     | $\rho^*$                                            | schild solution, an utmost slow tending of $\rho^*$ to the “coordinate singularity” is                                        |
| mielke-geometrodynamics_F01270 | 257  | ■ 1.000 | $M^*$                                                        | $M^*$                                               | mass $M^*$ of the (possibly “strong”) gravitational field is absorbed in the coefficients                                     |
| mielke-geometrodynamics_F01271 | 257  | ■ 0.999 | $j=\frac{1}{2}$                                              | $j = \frac{1}{2}$                                   | localized and regular solutions. For a soliton with total spin $j = \frac{1}{2}$ , i.e., for $\kappa = -1$ ,                  |
| mielke-geometrodynamics_F01272 | 257  | ■ 0.999 | $\kappa=-1$                                                  | $\kappa = -1$                                       | localized and regular solutions. For a soliton with total spin $j = \frac{1}{2}$ , i.e., for $\kappa = -1$ ,                  |
| mielke-geometrodynamics_F01273 | 257  | ■ 1.000 | $H(\rho)$                                                    | $H(\rho)$                                           | For a soliton with total spin $j = \frac{1}{2}$ and $\kappa = -1$ , the radial functions $H(\rho)$ in the                     |
| mielke-geometrodynamics_F01274 | 257  | ■ 1.000 | $l=0$                                                        | $l = 0$                                             | ansatz (11.4.15) account for a spherically symmetric distribution with $l = 0$ , while                                        |

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| mielke-geometrodynamics_F01275 | 257  | ■ 1.000 | $l=1$                                                                          | $l = 1$                                                                        | according to (11.4.17), the quantum number $l = 1$ of angular momentum can be                                                                      |
| mielke-geometrodynamics_F01276 | 257  | ■ 1.000 | $F(\rho)$                                                                      | $F(\rho)$                                                                      | ascribed to $F(\rho)$ . Let us compare this with the situation in a nonlocal version of the                                                        |
| mielke-geometrodynamics_F01277 | 258  | ■ 0.943 | $ \varphi ^6$                                                                  | $ \varphi ^6$                                                                  | $ \varphi ^6$ -model of ROSEN (1965). With reference to an expansion in terms of spherical                                                         |
| mielke-geometrodynamics_F01278 | 258  | ■ 1.000 | $\mathrm{Y}_{\ell}^m(\vartheta, \phi)$                                         | $Y_{\ell}^m(\vartheta, \phi)$                                                  | harmonics $Y_{\ell}^m(\vartheta, \phi)$ , the occurring <i>exact</i> radial solutions                                                              |
| mielke-geometrodynamics_F01279 | 259  | ■ 0.726 | $ \omega  < 1$                                                                 | $ \omega  < 1$                                                                 | ansatz (11.4.15) satisfies $ \omega  < 1$ . On the other hand, it is known (VÁZQUEZ 1977)                                                          |
| mielke-geometrodynamics_F01280 | 259  | ■ 1.000 | $ \omega  > 1$                                                                 | $ \omega  > 1$                                                                 | that bounded and square-integrable radial solutions <i>do not</i> exist for $ \omega  > 1$ .                                                       |
| mielke-geometrodynamics_F01281 | 259  | ■ 1.000 | $ \psi  \sim e^{-m_{\pi} r}$                                                   | $ \psi  \sim e^{-m_{\pi} r}$                                                   | $ \psi  \sim e^{-m_{\pi} r}$ between the centers of charge of the solitons. This is the case if the                                                |
| mielke-geometrodynamics_F01282 | 259  | ■ 1.000 | $m_s$                                                                          | $m_s$                                                                          | The total mass $m_s$ of a spinor soliton is obtained by the integration                                                                            |
| mielke-geometrodynamics_F01283 | 259  | ■ 1.000 | $\Sigma_D$                                                                     | $\Sigma_D$                                                                     | of the canonical energy momentum current $\Sigma_D$ over a spacelike hypersurface $H^3$ .                                                          |
| mielke-geometrodynamics_F01284 | 259  | ■ 1.000 | $m_s$                                                                          | $m_s$                                                                          | (11.4.40) is inversely proportional to the “fundamental” length $\ell^*$ and that $m_s$ would                                                      |
| mielke-geometrodynamics_F01285 | 260  | ■ 0.979 | (1951, 1956)                                                                   | (1951, 1956)                                                                   | Following the preceding studies of FINKELSTEIN et al. (1951, 1956) and SOLER                                                                       |
| mielke-geometrodynamics_F01286 | 260  | ■ 0.356 | $a=0$                                                                          | $a = 0$                                                                        | (1970), which were mainly concerned with a model (11.4.3) with $a = 0$ , RAÑADA                                                                    |
| mielke-geometrodynamics_F01287 | 260  | ■ 0.988 | $a=1$                                                                          | $a = 1$                                                                        | equation (11.3.7), i.e., on the Eq.(11.4.3) with $a = 1$ but with the opposite                                                                     |
| mielke-geometrodynamics_F01288 | 260  | ■ 1.000 | $\varepsilon=+1$                                                               | $\varepsilon = +1$                                                             | sign $\varepsilon = +1$ . For solitons corresponding to an $S_{1/2}$ -bound state with normalized                                                  |
| mielke-geometrodynamics_F01289 | 260  | ■ 1.000 | $S_{1/2}$                                                                      | $S_{1/2}$                                                                      | sign $\varepsilon = +1$ . For solitons corresponding to an $S_{1/2}$ -bound state with normalized                                                  |
| mielke-geometrodynamics_F01290 | 260  | ■ 1.000 | $Q=e$                                                                          | $Q = e$                                                                        | charge $Q = e$ , only nodeless radial solutions whose total rest mass shows an                                                                     |
| mielke-geometrodynamics_F01291 | 263  | ■ 1.000 | $D_F(x)$                                                                       | $D_F(x)$                                                                       | tudes are not yet subjected to field (“second”) quantization. Here $D_F(x)$ denotes                                                                |
| mielke-geometrodynamics_F01292 | 263  | ■ 1.000 | (1957, 1958)                                                                   | (1957, 1958)                                                                   | A further reason for IVANENKO (1957, 1958) to consider the necessity of nonlin-                                                                    |
| mielke-geometrodynamics_F01293 | 263  | ■ 1.000 | $\ell^*=8.1 \times 10^{-33} \text{ cm}$                                        | $\ell^* = 8.1 \times 10^{-33} \text{ cm}$                                      | $\ell^* = 8.1 \times 10^{-33} \text{ cm}$ to the far larger amount of the Compton wavelength $\ell = 10^{-13}$                                     |
| mielke-geometrodynamics_F01294 | 263  | ■ 1.000 | $\ell=10^{-13}$                                                                | $\ell = 10^{-13}$                                                              | $\ell^* = 8.1 \times 10^{-33} \text{ cm}$ to the far larger amount of the Compton wavelength $\ell = 10^{-13}$                                     |
| mielke-geometrodynamics_F01295 | 263  | ■ 1.000 | $\ell=10^{-13} \text{ cm}$                                                     | $\ell = 10^{-13} \text{ cm}$                                                   | field theory (1967), which starts from $\ell = 10^{-13} \text{ cm}$ , can refrain from using a “bare”                                              |
| mielke-geometrodynamics_F01296 | 269  | ■ 1.000 | $\langle dj \rangle = 0$                                                       | $\langle dj \rangle = 0$                                                       | charge current $j$ can be conserved, i.e., $\langle dj \rangle = 0$ , whereas the conservation of the                                              |
| mielke-geometrodynamics_F01297 | 269  | ■ 1.000 | $\langle dj_5 \rangle \neq 0$                                                  | $\langle dj_5 \rangle \neq 0$                                                  | axial current $j_5$ is broken, $\langle dj_5 \rangle \neq 0$ , in the most common convention.                                                      |
| mielke-geometrodynamics_F01298 | 269  | ■ 0.999 | $j_5(x; \varepsilon) := \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon)$ | $j_5(x; \varepsilon) := \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon)$ | sidering the point-split current $j_5(x; \varepsilon) := \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon)$ , where $\varepsilon$ is an infin- |
| mielke-geometrodynamics_F01299 | 269  | ■ 1.000 | $\delta / \delta A$                                                            | $\delta / \delta A$                                                            | The variation $\delta / \delta A$ of the current $j_5(x; \varepsilon)$ is compensated by the variation of the                                      |
| mielke-geometrodynamics_F01300 | 269  | ■ 1.000 | $j_5(x; \varepsilon)$                                                          | $j_5(x; \varepsilon)$                                                          | The variation $\delta / \delta A$ of the current $j_5(x; \varepsilon)$ is compensated by the variation of the                                      |
| mielke-geometrodynamics_F01301 | 269  | ■ 0.999 | $x^i \rightarrow x^i + \varepsilon^i$                                          | $x^i \rightarrow x^i + \varepsilon^i$                                          | exponential. Since the parallel transport from $x' \rightarrow x' + \varepsilon$ along the infinitesimal                                           |

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|--------------------------------|------|---------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| mielke-geometrodynamics_F01302 | 270  | ■ 1.000 | $P=\bar{\psi}\gamma_5\psi$                                                                                          | $P = \bar{\psi}\gamma_5\psi$                                                                      | for massive fermions, where $P = \overline{\psi}\gamma_5\psi$ is the pseudoscalar current and $F := dA$                                                  |
| mielke-geometrodynamics_F01303 | 270  | ■ 1.000 | $C:=A\wedge dA$                                                                                                     | $C := A \wedge dA$                                                                                | Simons (CS) term $C := A \wedge dA$ corresponds to the spin or helicity of the photon,                                                                   |
| mielke-geometrodynamics_F01304 | 270  | ■ 0.760 | $\mathbf{A}\cdot\mathbf{B}$                                                                                         | $\mathbf{A}\cdot\mathbf{B}$                                                                       | with its spacelike part $\mathbf{A}\cdot\mathbf{B}$ known as magnetic helicity (JACKIW & PI 2000). Since                                                 |
| mielke-geometrodynamics_F01305 | 270  | ■ 1.000 | $\mathscr{D}:=D+iA\wedge$                                                                                           | $\mathscr{D} := D + iA\wedge$                                                                     | potential $A = A_idx^i$ is accounted for via $\mathscr{D} := D + iA\wedge$ , where $D\psi := d\psi + \Gamma\wedge$                                       |
| mielke-geometrodynamics_F01306 | 270  | ■ 1.000 | $D\psi:=d\psi+\Gamma\wedge$                                                                                         | $D\psi := d\psi + \Gamma\wedge$                                                                   | potential $A = A_idx^i$ is accounted for via $\mathscr{D} := D + iA\wedge$ , where $D\psi := d\psi + \Gamma\wedge$                                       |
| mielke-geometrodynamics_F01307 | 270  | ■ 0.566 | $\Gamma^{\alpha\beta}=\Gamma_i^{\alpha\beta}dx^i$                                                                   | $\Gamma^{\alpha\beta} = \Gamma_i^{\alpha\beta}dx^i$                                               | nection one-form $\Gamma^{\alpha\beta} = \Gamma_i^{\alpha\beta}dx^i$ .                                                                                   |
| mielke-geometrodynamics_F01308 | 270  | ■ 1.000 | $\Theta:=D\gamma$                                                                                                   | $\Theta := D\gamma$                                                                               | with respect to $\overline{\psi}$ and $\psi$ . Making use of the torsion $\Theta := D\gamma$ and of the properties                                       |
| mielke-geometrodynamics_F01309 | 271  | ■ 0.652 | $\left.\left.T:=\frac{1}{4}\operatorname{Tr}(\check{\gamma}\lfloor\Theta\right)=e_{\alpha}\right\rfloor T^{\alpha}$ | $T := \frac{1}{4}\operatorname{Tr}(\check{\gamma}\rfloor\Theta) = e_{\alpha}\lrcorner T^{\alpha}$ | where $T := \frac{1}{4}Tr(\check{\gamma}\rfloor\Theta) = e_{\alpha}\lrcorner T^{\alpha}$ is the one-form of the trace (or vector) torsion.               |
| mielke-geometrodynamics_F01310 | 271  | ■ 1.000 | $K=\frac{i}{4}K^{\alpha\beta}\sigma_{\alpha\beta}$                                                                  | $K = \frac{i}{4}K^{\alpha\beta}\sigma_{\alpha\beta}$                                              | (or Christoffel) connection $\Gamma^{(\rfloor}$ and the <i>contortion</i> one-form $K = \frac{i}{4}K^{\alpha\beta}\sigma_{\alpha\beta}$ , sat-           |
| mielke-geometrodynamics_F01311 | 271  | ■ 1.000 | $D\gamma=[\gamma,K]=\gamma_{\alpha}T^{\alpha}$                                                                      | $D\gamma = [\gamma, K] = \gamma_{\alpha}T^{\alpha}$                                               | isfying $D\gamma = [\gamma, K] = \gamma_{\alpha}T^{\alpha}$ . Accordingly, the Dirac Lagrangian (12.2.1) splits                                          |
| mielke-geometrodynamics_F01312 | 271  | ■ 0.509 | $D^{(t)}\gamma=0$                                                                                                   | $D^{(t)}\gamma = 0$                                                                               | $D^{(\rfloor)}\gamma = 0$ . Hence, in an RC spacetime, a Dirac spinor feels only the <i>axial torsion</i> ,                                              |
| mielke-geometrodynamics_F01313 | 271  | ■ 1.000 | $\gamma\rightarrow\gamma^{\beta}=$                                                                                  | $\gamma \rightarrow \gamma^{\beta} =$                                                             | which is invariant under Weyl rescalings and <i>chiral transformations</i> $\gamma \rightarrow \gamma^{\beta} =$                                         |
| mielke-geometrodynamics_F01314 | 271  | ■ 1.000 | $e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$                                                                      | $e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$                                                    | $e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P:\vartheta^{\vec{B}}\rightarrow-\vartheta^{\vec{B}}$ , where $B =$ |
| mielke-geometrodynamics_F01315 | 271  | ■ 1.000 | $P:\vartheta^B\rightarrow-\vartheta^B$                                                                              | $P:\vartheta^B\rightarrow-\vartheta^B$                                                            | $e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P:\vartheta^{\vec{B}}\rightarrow-\vartheta^{\vec{B}}$ , where $B =$ |
| mielke-geometrodynamics_F01316 | 271  | ■ 1.000 | $B=$                                                                                                                | $B =$                                                                                             | $e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P:\vartheta^{\vec{B}}\rightarrow-\vartheta^{\vec{B}}$ , where $B =$ |
| mielke-geometrodynamics_F01317 | 271  | ■ 0.992 | 1,2,3                                                                                                               | 1,2,3                                                                                             | 1,2,3.                                                                                                                                                   |
| mielke-geometrodynamics_F01318 | 271  | ■ 1.000 | $dj\simeq 0$                                                                                                        | $dj \simeq 0$                                                                                     | From the Dirac equation (12.2.2) and its adjoint one can readily deduce that $dj \simeq 0$                                                               |
| mielke-geometrodynamics_F01319 | 271  | ■ 1.000 | $m\rightarrow 0$                                                                                                    | $m \rightarrow 0$                                                                                 | spacetime. If we restore chiral symmetry in the limit $m \rightarrow 0$ , this leads to the classical                                                    |
| mielke-geometrodynamics_F01320 | 272  | ■ 1.000 | $dj_5=0$                                                                                                            | $dj_5 = 0$                                                                                        | conservation law $dj_5 = 0$ of the axial current for massless Weyl spinors, or since                                                                     |
| mielke-geometrodynamics_F01321 | 272  | ■ 1.000 | $\mathscr{A}=2d\theta$                                                                                              | $\mathscr{A} = 2d\theta$                                                                          | torsion via $\mathscr{A} = 2d\theta$ . Then there arises in (12.2.3) a derivative coupling of the                                                        |
| mielke-geometrodynamics_F01322 | 272  | ■ 1.000 | $a=\theta f_a$                                                                                                      | $a = \theta f_a$                                                                                  | would-be axion $a = \theta f_a$ to two fermions via the CPT-invariant term                                                                               |
| mielke-geometrodynamics_F01323 | 272  | ■ 0.972 | $U(1)_{\mathrm{PQ}}$                                                                                                | $U(1)_{\mathrm{PQ}}$                                                                              | associated with a spontaneously broken Peccei–Quinn symmetry $U(1)_{\mathrm{PQ}}$ ; cf. KIM &                                                            |
| mielke-geometrodynamics_F01324 | 273  | ■ 1.000 | $\eta^{\alpha\beta}:=*(\vartheta^{\alpha}\wedge\vartheta^{\beta})$                                                  | $\eta^{\alpha\beta} := *(\vartheta^{\alpha} \wedge \vartheta^{\beta})$                            | where $\eta^{\alpha\beta} := *(\vartheta^{\alpha} \wedge \vartheta^{\beta})$ is dual to the unit two-form.                                               |
| mielke-geometrodynamics_F01325 | 273  | ■ 1.000 | $\langle dj_5\rangle$                                                                                               | $\langle dj_5 \rangle$                                                                            | quantization, where the vacuum expectation value $\langle dj_5 \rangle$ of the axial current picks up                                                    |
| mielke-geometrodynamics_F01326 | 273  | ■ 0.999 | $dC_{\mathrm{TT}}\cong(1/4)dj_5\rightarrow 0$                                                                       | $dC_{\mathrm{TT}} \cong (1/4)dj_5 \rightarrow 0$                                                  | NY four-form tends to zero “on shell,” i.e., $dC_{\mathrm{TT}} \cong (1/4)dj_5 \rightarrow 0$ . This is consistent                                       |
| mielke-geometrodynamics_F01327 | 273  | ■ 0.997 | $*j_5$                                                                                                              | $*j_5$                                                                                            | Here we implicitly assume that the light-cone structure of the axial covector $*j_5$                                                                     |



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| mielke-geometrodynamics_F01354 | 276  | ■ 0.960 | $d^{*}\!\mathscr{A}=2\,\ell^2\,dC_{\mathrm{TT}}$                                                                 | $d^{*}\mathscr{A}=2\ell^2dC_{\mathrm{TT}}$                                                                       | $d^{*}\mathscr{A}=2\ell^2dC_{\mathrm{TT}}$ carries dimensions. Since a <i>massive</i> Dirac spinor has canonical                                                       |
| mielke-geometrodynamics_F01355 | 276  | ■ 0.986 | $[\mathrm{length}]^{-3/2}$                                                                                       | $[\mathrm{length}]^{-3/2}$                                                                                       | dimension $[length]^{-3/2}$ , it scales as $\psi\sim m^{3/2}$ . Moreover, the term $t=1/M^2$ is                                                                        |
| mielke-geometrodynamics_F01356 | 276  | ■ 0.986 | $\psi\sim m^{3/2}$                                                                                               | $\psi\sim m^{3/2}$                                                                                               | dimension $[length]^{-3/2}$ , it scales as $\psi\sim m^{3/2}$ . Moreover, the term $t=1/M^2$ is                                                                        |
| mielke-geometrodynamics_F01357 | 276  | ■ 0.986 | $t=1/M^2$                                                                                                        | $t=1/M^2$                                                                                                        | dimension $[length]^{-3/2}$ , it scales as $\psi\sim m^{3/2}$ . Moreover, the term $t=1/M^2$ is                                                                        |
| mielke-geometrodynamics_F01358 | 276  | ■ 1.000 | $M\rightarrow\infty$                                                                                             | $M\rightarrow\infty$                                                                                             | related to the regulator mass $M\rightarrow\infty$ in the FUJIKAWA (1979) method. Then the                                                                             |
| mielke-geometrodynamics_F01359 | 276  | ■ 1.000 | $m\neq 0$                                                                                                        | $m\neq 0$                                                                                                        | heat kernel expansion will tend to zero in the chiral limit $m\rightarrow 0$ . In the case $m\neq 0$ ,                                                                 |
| mielke-geometrodynamics_F01360 | 276  | ■ 1.000 | $\vartheta^{\alpha}\rightarrow\widetilde{\vartheta}^{\alpha}=M\vartheta^{\alpha}$                                | $\vartheta^{\alpha}\rightarrow\widetilde{\vartheta}^{\alpha}=M\vartheta^{\alpha}$                                | To rescale the coframe by $\vartheta^{\alpha}\rightarrow\widetilde{\vartheta}^{\alpha}=M\vartheta^{\alpha}$ does not help, since that would also                       |
| mielke-geometrodynamics_F01361 | 276  | ■ 1.000 | $[\hbar]$                                                                                                        | $[\hbar]$                                                                                                        | change the dimension of the Dirac field in order to retain the physical dimension $[\hbar]$                                                                            |
| mielke-geometrodynamics_F01362 | 276  | ■ 1.000 | $(n=2)$                                                                                                          | $(n=2)$                                                                                                          | sions, neither classically nor in QFT. One would surmise that in $(n=2)$ -dimensional                                                                                  |
| mielke-geometrodynamics_F01363 | 276  | ■ 1.000 | $\sigma_{\alpha}:=\Sigma_{\alpha}-D\mu_{\alpha}$                                                                 | $\sigma_{\alpha}:=\Sigma_{\alpha}-D\mu_{\alpha}$                                                                 | where $\sigma_{\alpha}:=\Sigma_{\alpha}-D\mu_{\alpha}$ is the energy–momentum current Belinfante symmetrized                                                           |
| mielke-geometrodynamics_F01364 | 277  | ■ 0.996 | $\stackrel{(\pm)}{\Gamma}$                                                                                       | $\stackrel{(\pm)}{\Gamma}$                                                                                       | anti-self-dual variables $\stackrel{(\pm)}{\Gamma}$ in EC theory as well as in simple supergravity (MIELKE                                                             |
| mielke-geometrodynamics_F01365 | 277  | ■ 0.839 | $dC_{\mathrm{RR}}^{\{\}}$                                                                                        | $dC_{\mathrm{RR}}^{\{\}}$                                                                                        | The appearance of the Riemannian Pontryagin term $dC_{\mathrm{RR}}^{\{\}}$ in the anomaly (12.3.1)                                                                     |
| mielke-geometrodynamics_F01366 | 277  | ■ 1.000 | $\Lambda\neq 0$                                                                                                  | $\Lambda\neq 0$                                                                                                  | essary condition $\Lambda\neq 0$ could even be the dominating configurations, one gets a net                                                                           |
| mielke-geometrodynamics_F01367 | 277  | ■ 0.570 | $\underline{C}_{\mathrm{RR}}^{\{\}}$                                                                             | $\underline{C}_{\mathrm{RR}}^{\{\}}$                                                                             | to gravity, the tangential complexified CS term $\underline{C}_{\mathrm{RR}}^{\{\}}$ is known (KODAMA 1990) to                                                         |
| mielke-geometrodynamics_F01368 | 277  | ■ 0.936 | $\stackrel{(\pm)}{\underline{\Gamma}}$                                                                           | $\stackrel{(\pm)}{\underline{\Gamma}}$                                                                           | of gravity with cosmological constant, where the complex Ashtekar variable $\stackrel{(\pm)}{\underline{\Gamma}}$                                                      |
| mielke-geometrodynamics_F01369 | 277  | ■ 1.000 | $\mathscr{C}:=\Gamma/\mathscr{G}$                                                                                | $\mathscr{C}:=\Gamma/\mathscr{G}$                                                                                | theory on a three-dimensional hypersurface, with $\mathscr{C}:=\Gamma/\mathscr{G}$ of inequivalent gauge                                                               |
| mielke-geometrodynamics_F01370 | 278  | ■ 0.423 | $\mathbb{L}_n:=n\downarrow D+Dn$                                                                                 | $\mathbb{L}_n:=n\downarrow D+Dn$                                                                                 | where $\mathbb{L}_n:=n\downarrow D+Dn$ is the gauge-covariant Lie derivative along the normal                                                                          |
| mielke-geometrodynamics_F01371 | 278  | ■ 0.523 | $\tau^A$                                                                                                         | $\tau^A$                                                                                                         | for the matter current $\tau^A$ on the spacelike hypersurface (JIANG 1991). Only when                                                                                  |
| mielke-geometrodynamics_F01372 | 278  | ■ 1.000 | $\tau^A:=(1/2)\eta^{0A\beta\gamma}\tau_{\beta\gamma}$                                                            | $\tau^A:=(1/2)\eta^{0A\beta\gamma}\tau_{\beta\gamma}$                                                            | field, it is the canonical spin $\tau^A:=(1/2)\eta^{0A\beta\gamma}\tau_{\beta\gamma}$ that appears on the right-hand                                                   |
| mielke-geometrodynamics_F01373 | 278  | ■ 1.000 | $\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2\mathrm{o}_{\alpha\beta}$                          | $\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2\mathrm{o}_{\alpha\beta}$                          | When the constant Dirac matrices $\gamma_{\alpha}$ satisfying $\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2\mathrm{o}_{\alpha\beta}$ saturate the     |
| mielke-geometrodynamics_F01374 | 278  | ■ 1.000 | $\eta^{\alpha}:=$                                                                                                | $\eta^{\alpha}:=$                                                                                                | index of the orthonormal coframe one-form $\vartheta^{\alpha}=E_j{}^{\alpha}dx^j$ and its Hodge dual $\eta^{\alpha}:=$                                                 |
| mielke-geometrodynamics_F01375 | 278  | ■ 0.996 | $*\vartheta^{\alpha}$                                                                                            | $*\vartheta^{\alpha}$                                                                                            | $*\vartheta^{\alpha}$ , we obtain a basis of Clifford-algebra-valued exterior forms via                                                                                |
| mielke-geometrodynamics_F01376 | 278  | ■ 0.500 | $\Gamma:=\frac{i}{4}\Gamma_i^{\alpha\beta}\sigma_{\alpha\beta}dx^i$                                              | $\Gamma:=\frac{i}{4}\Gamma_i^{\alpha\beta}\sigma_{\alpha\beta}dx^i$                                              | In terms of the Clifford-algebra-valued <i>connection</i> $\Gamma:=\frac{i}{4}\Gamma_i^{\alpha\beta}\sigma_{\alpha\beta}dx^i$ , the $SL(2,\mathbb{C})$ -               |
| mielke-geometrodynamics_F01377 | 278  | ■ 1.000 | $D=d+\Gamma\wedge$                                                                                               | $D=d+\Gamma\wedge$                                                                                               | covariant exterior derivative is given by $D=d+\Gamma\wedge$ , where $\sigma_{\alpha\beta}=\frac{i}{2}(\gamma_{\alpha}\gamma_{\beta}-\gamma_{\beta}\gamma_{\alpha})$   |
| mielke-geometrodynamics_F01378 | 278  | ■ 1.000 | $\sigma_{\alpha\beta}=\frac{i}{2}(\gamma_{\alpha}\gamma_{\beta}-\gamma_{\beta}\gamma_{\alpha})$                  | $\sigma_{\alpha\beta}=\frac{i}{2}(\gamma_{\alpha}\gamma_{\beta}-\gamma_{\beta}\gamma_{\alpha})$                  | covariant exterior derivative is given by $D=d+\Gamma\wedge$ , where $\sigma_{\alpha\beta}=\frac{i}{2}(\gamma_{\alpha}\gamma_{\beta}-\gamma_{\beta}\gamma_{\alpha})$   |
| mielke-geometrodynamics_F01379 | 278  | ■ 1.000 | $\sigma:=\frac{i}{2}\gamma\wedge\gamma=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}$ | $\sigma:=\frac{i}{2}\gamma\wedge\gamma=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}$ | are the Lorentz generators entering in the two-form $\sigma:=\frac{i}{2}\gamma\wedge\gamma=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}$ . |

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| mielke-geometrodynamics_F01380 | 279  | ■ 1.000 | $R:=\{ \ }^*\!\!\left(R^{\alpha}\!_{\beta}\wedge\! \eta_{\beta\alpha}\!_{\alpha}\right)$ | $R:=\!^*(R^{\alpha\beta}\wedge\eta_{\beta\alpha})$                                       | to the curvature scalar $R:=\!^*(R^{\alpha\beta}\wedge\eta_{\beta\alpha})$ and the axial torsion piece $d\mathscr{A}\wedge d\mathscr{A}$ of                           |
| mielke-geometrodynamics_F01381 | 279  | ■ 1.000 | $\mathfrak{sl}(5,\mathbb{R})$                                                            | $\mathfrak{sl}(5,\mathbb{R})$                                                            | the $\mathfrak{sl}(5,\mathbb{R})$ -valued connection $\hat{F}=\Gamma+(\vartheta^{\alpha}L^4_{\alpha}+\vartheta_{\beta}\tilde{L}^{\beta}_4)/\ell$ is expanded into the |
| mielke-geometrodynamics_F01382 | 279  | ■ 1.000 | $\hat{\Gamma}=\Gamma+(\vartheta^{\alpha}L^4_{\alpha}+\vartheta_{\beta}L^{\beta}_4)/\ell$ | $\hat{\Gamma}=\Gamma+(\vartheta^{\alpha}L^4_{\alpha}+\vartheta_{\beta}L^{\beta}_4)/\ell$ | the $\mathfrak{sl}(5,\mathbb{R})$ -valued connection $\hat{F}=\Gamma+(\vartheta^{\alpha}L^4_{\alpha}+\vartheta_{\beta}\tilde{L}^{\beta}_4)/\ell$ is expanded into the |
| mielke-geometrodynamics_F01383 | 279  | ■ 0.763 | $\hat{C}_{\mathrm{RR}}$                                                                  | $\hat{C}_{\mathrm{RR}}$                                                                  | canonical dimension $[length]$ . The corresponding Pontryagin term $\hat{C}_{\mathrm{RR}}$ splits via                                                                 |
| mielke-geometrodynamics_F01384 | 280  | ■ 1.000 | $\pi^0\rightarrow 2\gamma$                                                               | $\pi^0\rightarrow 2\gamma$                                                               | Bell JS, Jackiw R (1969) A PCAC puzzle: $\pi^0\rightarrow 2\gamma$ in the $\sigma$ model. Il Nuovo Cimento A 60(1):                                                   |
| mielke-geometrodynamics_F01385 | 282  | ■ 0.775 | $\mathbb{R}^4\otimes\mathrm{SO}(1,3)$                                                    | $\mathbb{R}^4\otimes\mathrm{SO}(1,3)$                                                    | Lorentz or Poincaré group $\mathbb{R}^4\ltimes\mathrm{SO}(1,3)$ to a metalinear group $\mathrm{SL}(5,\mathbb{R})$ as a gauge                                          |
| mielke-geometrodynamics_F01386 | 283  | ■ 1.000 | $dB=0$                                                                                   | $dB=0$                                                                                   | Independent variations with respect to $A$ and $B$ lead to $dB=0$ and to the <i>constraint</i>                                                                        |
| mielke-geometrodynamics_F01387 | 284  | ■ 0.999 | $\sigma=\sigma_id x^i$                                                                   | $\sigma=\sigma_id x^i$                                                                   | where $\sigma=\sigma_id x^i$ is a one-form or a covector; cf. LUCCHESI et al. (1993). Since Bianchi                                                                   |
| mielke-geometrodynamics_F01388 | 284  | ■ 0.672 | $\mathrm{SL}(5,\mathbb{R})$                                                              | $\mathrm{SL}(5,\mathbb{R})$                                                              | “semi-topological” BF scheme based on the metric-free structure group $\mathrm{SL}(5,\mathbb{R})$                                                                     |
| mielke-geometrodynamics_F01389 | 284  | ■ 1.000 | $\mathrm{SL}(n+1,\mathbb{R})$                                                            | $\mathrm{SL}(n+1,\mathbb{R})$                                                            | In general, the metalinear group $\mathrm{SL}(n+1,\mathbb{R})$ with $\det g=1$ is generated by                                                                        |
| mielke-geometrodynamics_F01390 | 284  | ■ 1.000 | $\det g=1$                                                                               | $\det g=1$                                                                               | In general, the metalinear group $\mathrm{SL}(n+1,\mathbb{R})$ with $\det g=1$ is generated by                                                                        |
| mielke-geometrodynamics_F01391 | 284  | ■ 1.000 | $n(n+2)$                                                                                 | $n(n+2)$                                                                                 | $n(n+2)$ trace-free generators                                                                                                                                        |
| mielke-geometrodynamics_F01392 | 284  | ■ 1.000 | $\mathfrak{gl}(n,\mathbb{R})$                                                            | $\mathfrak{gl}(n,\mathbb{R})$                                                            | comprise those of a $\mathfrak{gl}(n,\mathbb{R})$ subalgebra                                                                                                          |
| mielke-geometrodynamics_F01393 | 284  | ■ 1.000 | $\alpha,\ldots,\delta=0,\ldots,n-1$                                                      | $\alpha,\ldots,\delta=0,\ldots,n-1$                                                      | where $\alpha,\ldots,\delta=0,\ldots,n-1$ .                                                                                                                           |
| mielke-geometrodynamics_F01394 | 284  | ■ 0.367 | $\ell P_{\alpha}:=\mathbb{L}^{n+1}_{\alpha}=L^{n+1}_{\alpha}$                            | $\ell P_{\alpha}:=\mathbb{L}^{n+1}_{\alpha}=L^{n+1}_{\alpha}$                            | The generators $\ell P_{\alpha}:=\mathbb{L}^{n+1}_{\alpha}=L^{n+1}_{\alpha}$ and $\ell P^{\beta}_{*}:=\mathbb{L}^{\beta}_{n+1}=L^{\beta}_{n+1}$ transform             |
| mielke-geometrodynamics_F01395 | 284  | ■ 0.367 | $\ell P^{\beta}_{*}:=\mathbb{L}^{\beta}_{n+1}=L^{\beta}_{n+1}$                           | $\ell P^{\beta}_{*}:=\mathbb{L}^{\beta}_{n+1}=L^{\beta}_{n+1}$                           | The generators $\ell P_{\alpha}:=\mathbb{L}^{n+1}_{\alpha}=L^{n+1}_{\alpha}$ and $\ell P^{\beta}_{*}:=\mathbb{L}^{\beta}_{n+1}=L^{\beta}_{n+1}$ transform             |
| mielke-geometrodynamics_F01396 | 285  | ■ 0.998 | $P^{\alpha}_{*}$                                                                         | $P^{\alpha}_{*}$                                                                         | This reveals that $P^{\alpha}_{*}$ is a vector and $P_{\beta}$ is a covector with respect to the $\mathrm{GL}(n,\mathbb{R})$                                          |
| mielke-geometrodynamics_F01397 | 285  | ■ 0.998 | $P_{\beta}$                                                                              | $P_{\beta}$                                                                              | This reveals that $P^{\alpha}_{*}$ is a vector and $P_{\beta}$ is a covector with respect to the $\mathrm{GL}(n,\mathbb{R})$                                          |
| mielke-geometrodynamics_F01398 | 285  | ■ 0.998 | $\mathrm{GL}(n,\mathbb{R})$                                                              | $\mathrm{GL}(n,\mathbb{R})$                                                              | This reveals that $P^{\alpha}_{*}$ is a vector and $P_{\beta}$ is a covector with respect to the $\mathrm{GL}(n,\mathbb{R})$                                          |
| mielke-geometrodynamics_F01399 | 285  | ■ 0.996 | $\mathrm{A}_{*}(n,\mathbb{R})$                                                           | $\mathrm{A}_{*}(n,\mathbb{R})$                                                           | to the semisimple <i>graded affine</i> group $\mathrm{A}_{*}(n,\mathbb{R})$ of the same rank:                                                                         |
| mielke-geometrodynamics_F01400 | 285  | ■ 0.998 | $\mathrm{SO}(n+1)$                                                                       | $\mathrm{SO}(n+1)$                                                                       | (Pseudo)orthogonal groups like $\mathrm{SO}(n+1)$ involve a Cartan–Killing metric with                                                                                |
| mielke-geometrodynamics_F01401 | 285  | ■ 1.000 | $\hat{g}_{AB}$                                                                           | $\hat{g}_{AB}$                                                                           | components $\hat{g}_{AB}$ . Then a basis of the Lie generators                                                                                                        |
| mielke-geometrodynamics_F01402 | 285  | ■ 0.998 | $n(n+1)/2$                                                                               | $n(n+1)/2$                                                                               | is only $n(n+1)/2$ -dimensional. The same would apply to the respective connections                                                                                   |
| mielke-geometrodynamics_F01403 | 285  | ■ 0.999 | $A(x)\in\mathrm{A}_{*}(n,R)$                                                             | $A(x)\in\mathrm{A}_{*}(n,R)$                                                             | where $A(x)\in\mathrm{A}_{*}(n,R)$ and the $\Lambda(x)\in\mathrm{GL}(n,\mathbb{R})$ are linear gauge transformations,                                                 |
| mielke-geometrodynamics_F01404 | 285  | ■ 0.999 | $\Lambda(x)\in\mathrm{GL}(n,\mathbb{R})$                                                 | $\Lambda(x)\in\mathrm{GL}(n,\mathbb{R})$                                                 | where $A(x)\in\mathrm{A}_{*}(n,R)$ and the $\Lambda(x)\in\mathrm{GL}(n,\mathbb{R})$ are linear gauge transformations,                                                 |
| mielke-geometrodynamics_F01405 | 285  | ■ 1.000 | $\tau(x):=\exp\left[\xi^{\alpha}P_{\alpha}\right]\in\mathrm{T}(n,R)$                     | $\tau(x):=\exp\left[\xi^{\alpha}P_{\alpha}\right]\in\mathcal{T}(n,R)$                    | while $\tau(x):=\exp[\xi^{\alpha}P_{\alpha}]\in\mathcal{T}(n,R)$ and $\tau_{*}(x):=\exp[\xi^{\beta}_{*}P^{\beta}_{*}]\in\mathcal{T}_{*}(n,R)$ represent               |
| mielke-geometrodynamics_F01406 | 285  | ■ 1.000 | $\tau_{*}(x):=\exp\left[\xi^{\beta}_{*}P^{\beta}_{*}\right]\in\mathcal{T}_{*}(n,R)$      | $\tau_{*}(x):=\exp\left[\xi^{\beta}_{*}P^{\beta}_{*}\right]\in\mathcal{T}_{*}(n,R)$      | while $\tau(x):=\exp[\xi^{\alpha}P_{\alpha}]\in\mathcal{T}(n,R)$ and $\tau_{*}(x):=\exp[\xi^{\beta}_{*}P^{\beta}_{*}]\in\mathcal{T}_{*}(n,R)$ represent               |

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| mielke-geometrodynamics_F01407 | 286  | ■ 1.000 | <code>\ell \rightarrow \infty</code>                                          | $\ell \rightarrow \infty$                            | For $\ell \rightarrow \infty$ , this holds automatically due to the degeneracy of the commutation                                                    |
| mielke-geometrodynamics_F01408 | 286  | ■ 1.000 | <code>\operatorname{Ad} A(B)=[A, B]</code>                                    | $\operatorname{Ad} A(B) = [A, B]$                    | the adjoint representation $\operatorname{Ad} A(B) = [A, B]$ . The curvature                                                                         |
| mielke-geometrodynamics_F01409 | 286  | ■ 1.000 | <code>\hat{\Gamma}</code>                                                     | $\hat{\Gamma}$                                       | The inhomogeneous transformation law (13.4.3) for the graded affine connection $\hat{\Gamma}$                                                        |
| mielke-geometrodynamics_F01410 | 286  | ■ 1.000 | <code>\Gamma^{\{L\}}</code>                                                   | $\Gamma^{(L)}$                                       | can be split into that of the linear connection $\Gamma^{(L)}$ , which acquires the conventional                                                     |
| mielke-geometrodynamics_F01411 | 286  | ■ 0.990 | <code>\mathrm{GL}(n, \mathbb{R})</code>                                       | $\mathrm{GL}(n, \mathbb{R})$                         | of a Yang–Mills-type connection for the gauge group $\mathrm{GL}(n, \mathbb{R})$ . The remaining trans-                                              |
| mielke-geometrodynamics_F01412 | 287  | ■ 0.998 | <code>D \tau(x):=d \tau(x)+\Gamma^{\{L\}} \tau(x)</code>                      | $D\tau(x) := d\tau(x) + \Gamma^{(L)} \tau(x)$        | derivative $D\tau(x) := d\tau(x) + \Gamma^{(L)} \tau(x)$ , the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not                        |
| mielke-geometrodynamics_F01413 | 287  | ■ 0.998 | <code>\Gamma^{\{T\}}</code>                                                   | $\Gamma^{(T)}$                                       | derivative $D\tau(x) := d\tau(x) + \Gamma^{(L)} \tau(x)$ , the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not                        |
| mielke-geometrodynamics_F01414 | 287  | ■ 0.998 | <code>\Gamma_*^{\{T\}}</code>                                                 | $\Gamma_*^{(T)}$                                     | derivative $D\tau(x) := d\tau(x) + \Gamma^{(L)} \tau(x)$ , the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not                        |
| mielke-geometrodynamics_F01415 | 287  | ■ 1.000 | <code>\xi=\xi^{\{\alpha\}} P_{\{\alpha\}}</code>                              | $\xi = \xi^\alpha P_\alpha$                          | Since $\xi = \xi^\alpha P_\alpha$ and $\xi_* = \xi_\beta^* P_\beta^*$ acquire their values in the coset space                                        |
| mielke-geometrodynamics_F01416 | 287  | ■ 1.000 | <code>\xi_{\{*\}}=\xi_{\{\beta\}}^{\{*\}} P_{\{*\}}^{\{\beta\}}</code>        | $\xi_* = \xi_\beta^* P_\beta^*$                      | Since $\xi = \xi^\alpha P_\alpha$ and $\xi_* = \xi_\beta^* P_\beta^*$ acquire their values in the coset space                                        |
| mielke-geometrodynamics_F01417 | 287  | ■ 1.000 | <code>\mathscr{T}(n, R) \otimes \mathscr{T}_*(n, R)</code>                    | $\mathcal{T}(n, R) \otimes \mathcal{T}_*(n, R)$      | “hide” the action of the local pseudotranslations $\mathcal{T}(n, R) \otimes \mathcal{T}_*(n, R)$ on $\vartheta$ and $\theta$ .                      |
| mielke-geometrodynamics_F01418 | 287  | ■ 1.000 | <code>\xi=\xi^{\{*\}}=0</code>                                                | $\xi = \xi^* = 0$                                    | If we postulate even stronger constraints $\xi = \xi^* = 0$ of “zero section” vector                                                                 |
| mielke-geometrodynamics_F01419 | 287  | ■ 1.000 | <code>\theta_{\{\alpha\}}=g_{\{\alpha \beta\}} \vartheta^{\{\beta\}}</code>   | $\theta_\alpha = g_{\alpha\beta} \vartheta^\beta$    | mutually transposed one-forms, i.e., $\theta_\alpha = g_{\alpha\beta} \vartheta^\beta$ .                                                             |
| mielke-geometrodynamics_F01420 | 287  | ■ 1.000 | <code>D \xi^{\{\alpha\}} \stackrel{*}{=} \vartheta^{\alpha}</code>            | $D\xi^\alpha \stackrel{*}{=} \vartheta^\alpha$       | When $D\xi^\alpha \stackrel{*}{=} \vartheta^\alpha$ , a Cartan transport of a radius vector $\xi^\alpha$ occurs via “rolling                         |
| mielke-geometrodynamics_F01421 | 287  | ■ 1.000 | <code>\xi^{\{\alpha\}}</code>                                                 | $\xi^\alpha$                                         | When $D\xi^\alpha \stackrel{*}{=} \vartheta^\alpha$ , a Cartan transport of a radius vector $\xi^\alpha$ occurs via “rolling                         |
| mielke-geometrodynamics_F01422 | 288  | ■ 0.593 | <code>1 / [</code>                                                            | $1/[$                                                | In view of the canonical dimension 1/[length] of the generators $P_\alpha$ and $P_\beta^*$ of pseudo-                                                |
| mielke-geometrodynamics_F01423 | 288  | ■ 0.593 | <code>]</code>                                                                | $]$                                                  | In view of the canonical dimension 1/[length] of the generators $P_\alpha$ and $P_\beta^*$ of pseudo-                                                |
| mielke-geometrodynamics_F01424 | 288  | ■ 0.593 | <code>P_{\{*\}}^{\{\beta\}}</code>                                            | $P_\beta^*$                                          | In view of the canonical dimension 1/[length] of the generators $P_\alpha$ and $P_\beta^*$ of pseudo-                                                |
| mielke-geometrodynamics_F01425 | 288  | ■ 0.647 | <code>\vartheta^{\{\alpha\}}=\ell \Gamma_{\{\hat{4}\}}^{\alpha}</code>        | $\vartheta^\alpha = \ell \Gamma_4^\alpha$            | i.e., it becomes a Cartan connection, where $\vartheta^\alpha = \ell \Gamma_4^\alpha$ and $\theta_\beta = \ell \Gamma_\beta^{\hat{4}}$ are two inde- |
| mielke-geometrodynamics_F01426 | 288  | ■ 0.647 | <code>\theta_{\{\beta\}}=\ell \Gamma_{\{\beta\}}^{\{\hat{4}\}}</code>         | $\theta_\beta = \ell \Gamma_\beta^{\hat{4}}$         | i.e., it becomes a Cartan connection, where $\vartheta^\alpha = \ell \Gamma_4^\alpha$ and $\theta_\beta = \ell \Gamma_\beta^{\hat{4}}$ are two inde- |
| mielke-geometrodynamics_F01427 | 288  | ■ 1.000 | <code>\Lambda=3 / \ell^2</code>                                               | $\Lambda = 3/\ell^2$                                 | where $\Lambda = 3/\ell^2$ will later be identified with the cosmological constant.                                                                  |
| mielke-geometrodynamics_F01428 | 289  | ■ 0.998 | <code>d \hat{C}</code>                                                        | $d\hat{C}$                                           | $d\hat{C}$ provides us, after variation with respect to $\Gamma_A^B$ , with the Bianchi identity                                                     |
| mielke-geometrodynamics_F01429 | 289  | ■ 0.998 | <code>\Gamma_{\{A\}}^{\{B\}}</code>                                           | $\Gamma_A^B$                                         | $d\hat{C}$ provides us, after variation with respect to $\Gamma_A^B$ , with the Bianchi identity                                                     |
| mielke-geometrodynamics_F01430 | 289  | ■ 1.000 | <code>\hat{B}</code>                                                          | $\hat{B}$                                            | The variation with respect to $\hat{B}$ would lead to $\hat{R} = 0$ , a physically too strong constraint.                                            |
| mielke-geometrodynamics_F01431 | 289  | ■ 1.000 | <code>\hat{R}=0</code>                                                        | $\hat{R} = 0$                                        | The variation with respect to $\hat{B}$ would lead to $\hat{R} = 0$ , a physically too strong constraint.                                            |
| mielke-geometrodynamics_F01432 | 289  | ■ 0.994 | <code>\eta_{\{A B C D E\}}:=\sqrt{\hat{g}} \varepsilon_{\{A B C D E\}}</code> | $\eta_{ABCDE} := \sqrt{\hat{g}} \varepsilon_{ABCDE}$ | Here, the antisymmetric unit tensor $\eta_{ABCDE} := \sqrt{\hat{g}} \varepsilon_{ABCDE}$ is constructed from the                                     |

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| mielke-geometrodynamics_F01433 | 289  | ■ 1.000 | <code>\varepsilon_{A B C D E}</code>                                                                       | $\varepsilon_{ABCDE}$                                                                          | <i>metric-free</i> Levi-Civita symbol $\varepsilon_{ABCDE}$ , being invariant under the five-dimensional                                                                                |
| mielke-geometrodynamics_F01434 | 289  | ■ 0.913 | <code>\hat{g}</code>                                                                                       | $\hat{g}$                                                                                      | structure group $\mathrm{SL}(5, \mathbb{R})$ , together with the determinant $\hat{g}$ of the invertible group                                                                          |
| mielke-geometrodynamics_F01435 | 289  | ■ 1.000 | <code>B^{A B}:=\hat{g}^{A C} B_{C}\{ \}^{B}</code>                                                         | $B^{AB} := \hat{g}^{AC} B_C{}^B$                                                               | $B^{AB} := \hat{g}^{AC} B_C{}^B$ . Thus, the additional quadratic term provides a <i>dynamical symmetry</i> -                                                                           |
| mielke-geometrodynamics_F01436 | 289  | ■ 0.853 | <code>\operatorname{SO}(2,3)</code>                                                                        | $\mathrm{SO}(2, 3)$                                                                            | or $\mathrm{SO}(2, 3)$ , respectively. Moreover, the five-index structure of $\eta_{ABCDE}$ has forced us                                                                               |
| mielke-geometrodynamics_F01437 | 289  | ■ 0.853 | <code>\eta_{A B C D E}</code>                                                                              | $\eta_{ABCDE}$                                                                                 | or $\mathrm{SO}(2, 3)$ , respectively. Moreover, the five-index structure of $\eta_{ABCDE}$ has forced us                                                                               |
| mielke-geometrodynamics_F01438 | 289  | ■ 0.997 | <code>\Phi^A=\left\{\phi^{\alpha}, \phi^{\hat{4}}\right\}</code>                                           | $\Phi^A = \left\{\phi^\alpha, \phi^{\hat{4}}\right\}$                                          | to complement the construction by a quintet of pseudoscalars $\Phi^A = \{\phi^\alpha, \phi^{\hat{4}}\}$ , our                                                                           |
| mielke-geometrodynamics_F01439 | 289  | ■ 0.988 | <code>U\left(\Phi^B \Phi_B\right)</code>                                                                   | $U\left(\Phi^B \Phi_B\right)$                                                                  | arise from a minimum of a Higgs-type potential $U(\Phi^B \Phi_B)$ ; cf. PAGELS (1984) and                                                                                               |
| mielke-geometrodynamics_F01440 | 290  | ■ 1.000 | <code>\Phi=\Re \Phi+i \Im \Phi</code>                                                                      | $\Phi = \Re \Phi + i \Im \Phi$                                                                 | $\Phi = \Re \Phi + i \Im \Phi$ is invariant                                                                                                                                             |
| mielke-geometrodynamics_F01441 | 290  | ■ 0.995 | <code>d C_{\mathrm{RR}}</code>                                                                             | $dC_{\mathrm{RR}}$                                                                             | where the resulting Pontryagin term $dC_{\mathrm{RR}}$ serves merely to reproduce the Bianchi                                                                                           |
| mielke-geometrodynamics_F01442 | 290  | ■ 0.989 | <code>\stackrel{\circ}{\Gamma}^{\alpha \beta} =</code>                                                     | $\overset{\circ}{\Gamma}^{\alpha \beta} =$                                                     | identity (13.5.5) for the curvature, now in terms of the Lorentz connection $\overset{\circ}{\Gamma}^{\alpha \beta} =$                                                                  |
| mielke-geometrodynamics_F01443 | 290  | ■ 1.000 | <code>-\stackrel{\circ}{\Gamma}^{\beta \alpha}</code>                                                      | $-\overset{\circ}{\Gamma}^{\beta \alpha}$                                                      | $-\overset{\circ}{\Gamma}^{\beta \alpha}$ . The variation with respect to the Lorentz-valued two-form $B^{\alpha \beta}$ leads to                                                       |
| mielke-geometrodynamics_F01444 | 290  | ■ 1.000 | <code>B^{\alpha \beta}</code>                                                                              | $B^{\alpha \beta}$                                                                             | $-\overset{\circ}{\Gamma}^{\beta \alpha}$ . The variation with respect to the Lorentz-valued two-form $B^{\alpha \beta}$ leads to                                                       |
| mielke-geometrodynamics_F01445 | 290  | ■ 1.000 | <code>\mu \neq 0</code>                                                                                    | $\mu \neq 0$                                                                                   | as an equation of motion. Thus, for $\mu \neq 0$ , our Lagrangian is “on shell” equivalent                                                                                              |
| mielke-geometrodynamics_F01446 | 291  | ■ 1.000 | <code>\vartheta^{\alpha}=g^{\alpha \beta} \theta_{\beta}</code>                                            | $\vartheta^\alpha = g^{\alpha \beta} \theta_\beta$                                             | For a nondegenerate coframe $\vartheta^\alpha = g^{\alpha \beta} \theta_\beta$ and vanishing torsion, the covariant                                                                     |
| mielke-geometrodynamics_F01447 | 291  | ■ 1.000 | <code>\kappa =</code>                                                                                      | $\kappa =$                                                                                     | Pontryagin four-forms (here suppressed). Einstein’s gravitational constant $\kappa =$                                                                                                   |
| mielke-geometrodynamics_F01448 | 291  | ■ 1.000 | <code>8 \pi G_{\mathrm{N}}=8 \pi \ell_{\text {Planck }}^2</code>                                           | $8 \pi G_{\mathrm{N}} = 8 \pi \ell_{\mathrm{Planck}}^2$                                        | $8 \pi G_{\mathrm{N}} = 8 \pi \ell_{\mathrm{Planck}}^2$ and the density $\rho_\Lambda = \Lambda / \kappa = \Omega_\Lambda \rho_{\mathrm{crit}} < \rho_{\mathrm{crit}}$ corresponding to |
| mielke-geometrodynamics_F01449 | 291  | ■ 1.000 | <code>\rho_{\Lambda}=\Lambda / \kappa=\Omega_{\Lambda} \rho_{\text {crit }}&lt;\rho_{\text {crit }}</code> | $\rho_\Lambda = \Lambda / \kappa = \Omega_\Lambda \rho_{\mathrm{crit}} < \rho_{\mathrm{crit}}$ | $8 \pi G_{\mathrm{N}} = 8 \pi \ell_{\mathrm{Planck}}^2$ and the density $\rho_\Lambda = \Lambda / \kappa = \Omega_\Lambda \rho_{\mathrm{crit}} < \rho_{\mathrm{crit}}$ corresponding to |
| mielke-geometrodynamics_F01450 | 291  | ■ 1.000 | <code>\kappa=\mu \ell^2 / 4</code>                                                                         | $\kappa = \mu \ell^2 / 4$                                                                      | implies that these constants are related via $\kappa = \mu \ell^2 / 4$ and $\Lambda = 12 \kappa / \mu \ell^4 = 3 / \ell^2$ to                                                           |
| mielke-geometrodynamics_F01451 | 291  | ■ 1.000 | <code>\Lambda=12 \kappa / \mu \ell^4=3 / \ell^2</code>                                                     | $\Lambda = 12 \kappa / \mu \ell^4 = 3 / \ell^2$                                                | implies that these constants are related via $\kappa = \mu \ell^2 / 4$ and $\Lambda = 12 \kappa / \mu \ell^4 = 3 / \ell^2$ to                                                           |
| mielke-geometrodynamics_F01452 | 291  | ■ 1.000 | <code>\mu \simeq 10^{-123}</code>                                                                          | $\mu \simeq 10^{-123}$                                                                         | The symmetry-breaking parameter $\mu \simeq 10^{-123}$ is observationally an extremely small                                                                                            |
| mielke-geometrodynamics_F01453 | 291  | ■ 1.000 | <code>\hat{B} \wedge \hat{B}</code>                                                                        | $\hat{B} \wedge \hat{B}$                                                                       | breaking $\hat{B} \wedge \hat{B}$ term, as anticipated by SMOLIN (2000) in the restricted case of                                                                                       |
| mielke-geometrodynamics_F01454 | 291  | ■ 1.000 | <code>G_{\mathrm{N}}</code>                                                                                | $G_{\mathrm{N}}$                                                                               | gravitational constant $G_{\mathrm{N}}$ has its origin in SSB, as suggested already by ENGLERT et                                                                                       |
| mielke-geometrodynamics_F01455 | 292  | ■ 0.997 | <code>O(5)</code>                                                                                          | $O(5)$                                                                                         | Since our $\mathrm{SL}(5, \mathbb{R})$ gauge model is broken into two steps from $O(5)$ down to the                                                                                     |
| mielke-geometrodynamics_F01456 | 292  | ■ 0.466 | <code>\operatorname{SO}(1,3)</code>                                                                        | $\mathrm{SO}(1, 3)$                                                                            | Lorentz group $\mathrm{SO}(1, 3)$ as an exact subgroup, a local holonomic spacetime metric                                                                                              |
| mielke-geometrodynamics_F01457 | 292  | ■ 1.000 | <code>g_{i j}</code>                                                                                       | $g_{ij}$                                                                                       | $g_{ij}$ is then induced. In the last step, we assumed that the translational connections                                                                                               |
| mielke-geometrodynamics_F01458 | 292  | ■ 0.999 | <code>\Gamma_4^{\alpha} \otimes_{\mathrm{s}} \Gamma_{\alpha}^4</code>                                      | $\Gamma_4{}^\alpha \otimes_{\mathrm{s}} \Gamma_\alpha{}^4$                                     | $\Gamma_4{}^\alpha \otimes_{\mathrm{s}} \Gamma_\alpha{}^4$ is dimensionless, we had to multiply the vacuum expectation (13.6.1) of                                                      |
| mielke-geometrodynamics_F01459 | 292  | ■ 1.000 | <code>\ell^2 / \mu^2=\left(\kappa / 4 \Lambda_{\mathrm{H}}\right)</code>                                   | $\ell^2 / \mu^2 = (\kappa / 4 \Lambda_{\mathrm{H}})$                                           | the derivative of the Higgs field by a <i>huge</i> dimensional factor $\ell^2 / \mu^2 = (\kappa / 4 \Lambda_{\mathrm{H}})$ . This                                                       |

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| mielke-geometrodynamics_F01466 | 294  | ■ 1.000 | $W^{\{+\}}$                                                       | $W^+$                                                     | $W^+$ and $W^-$ gauge bosons or even “heavy photons” $Z$ , again via pair creation. These                                                                       |
| mielke-geometrodynamics_F01461 | 294  | ■ 1.000 | $W^{\{-\}}$                                                       | $W^-$                                                     | $W^+$ and $W^-$ gauge bosons or even “heavy photons” $Z$ , again via pair creation. These                                                                       |
| mielke-geometrodynamics_F01462 | 294  | ■ 0.633 | $\mu \rightarrow 0$                                               | $\mu \rightarrow 0$                                       | In the limit $\mu \rightarrow 0$ of $\mathrm{SL}(5, \mathbb{R})$ symmetry restoration, concepts like macroscopic                                                |
| mielke-geometrodynamics_F01463 | 294  | ■ 1.000 | $d\,s$                                                            | $ds$                                                      | logical field theory does a line element $ds$ emerge as a necessary means for measuring                                                                         |
| mielke-geometrodynamics_F01464 | 294  | ■ 1.000 | $1/\alpha :=$                                                     | $1/\alpha :=$                                             | slowly decay (ZEE 2004). The inverse <i>dimensionless</i> coupling constants $1/\alpha :=$                                                                      |
| mielke-geometrodynamics_F01465 | 295  | ■ 1.000 | $\hbar c/e^2 \simeq 137$                                          | $\hbar c/e^2 \simeq 137$                                  | <span style="border: 1px solid red;"><math>\hbar c/e^2 \simeq 137</math></span> of Sommerfeld and $1/\mu \simeq 10^{123}$ of gravity seem to be on a par except |
| mielke-geometrodynamics_F01466 | 295  | ■ 1.000 | $1/\mu \simeq 10^{123}$                                           | $1/\mu \simeq 10^{123}$                                   | $\hbar c/e^2 \simeq 137$ of Sommerfeld and $1/\mu \simeq 10^{123}$ of gravity seem to be on a par except                                                        |
| mielke-geometrodynamics_F01467 | 295  | ■ 0.991 | $C_{\mathrm{TT}}$                                                 | $C_{\mathrm{TT}}$                                         | contains a translational CS three-form $C_{\mathrm{TT}}$ that gives rise to the <i>parity-violating</i>                                                         |
| mielke-geometrodynamics_F01468 | 295  | ■ 1.000 | $\mathrm{gl}(4, \mathbb{R})$                                      | $\mathrm{gl}(4, \mathbb{R})$                              | of Nieh and Yan (NY) (NIEH 2007). On the other hand, the $\mathrm{gl}(4, \mathbb{R})$ -valued Pontryagin                                                        |
| mielke-geometrodynamics_F01469 | 296  | ■ 0.865 | $\operatorname{Ric}_{\alpha\beta} := (-1)^{\text{sig}} *$         | $\operatorname{Ric}_{\alpha\beta} := (-1)^{\text{sig}} *$ | sion involving the symmetric Ricci tensor, i.e., the zero-form $\operatorname{Ric}_{\alpha\beta} := (-1)^{\text{sig}} *$                                        |
| mielke-geometrodynamics_F01470 | 296  | ■ 0.998 | $\left(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)}\right)$ | $(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)})$    | $(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)})$ , is known as the <i>Gauss-Bonnet identity</i> .                                                         |
| mielke-geometrodynamics_F01471 | 300  | ■ 1.000 | $\mathscr{S} := \mathscr{M} / \mathscr{D}$                        | $\mathscr{S} := \mathscr{M} / \mathscr{D}$                | <sup>2</sup> The quotient space $\mathscr{S} := \mathscr{M} / \mathscr{D}$ of all Riemannian metrics on a 3-dimensional manifold modulo the                     |
| mielke-geometrodynamics_F01472 | 300  | ■ 1.000 | $\mathscr{D}$                                                     | $\mathscr{D}$                                             | group $\mathscr{D}$ of the diffeomorphisms of $M^3$ is called the “superspace”. Accordingly, it has a completely                                                |
| mielke-geometrodynamics_F01473 | 300  | ■ 1.000 | $M^3$                                                             | $M^3$                                                     | group $\mathscr{D}$ of the diffeomorphisms of $M^3$ is called the “superspace”. Accordingly, it has a completely                                                |
| mielke-geometrodynamics_F01474 | 301  | ■ 1.000 | $F_{ij}$                                                          | $F_{ij}$                                                  | ponents $F_{ij}$ of the electromagnetic field strengths. Such a theory was developed not                                                                        |
| mielke-geometrodynamics_F01475 | 301  | ■ 1.000 | $g_{[ij]} g^{[ij]}$                                               | $g_{[ij]} g^{[ij]}$                                       | term proportional to $g_{[ij]} g^{[ij]}$ occurs side by side with Hilbert’s Lagrangian 4-form,                                                                  |
| mielke-geometrodynamics_F01476 | 301  | ■ 1.000 | $\operatorname{g}_{[ij]} := \ell^{*2} F_{ij}$                     | $\operatorname{g}_{[ij]} := \ell^{*2} F_{ij}$             | FAT (1979a), in a more recent study, has given up the identification $\operatorname{g}_{[ij]} := \ell^{*2} F_{ij}$ .                                            |
| mielke-geometrodynamics_F01477 | 301  | ■ 1.000 | $m \in M$                                                         | $m \in M$                                                 | be dependent not only on the point $m \in M$ but also on locally distinguished direc-                                                                           |
| mielke-geometrodynamics_F01478 | 303  | ■ 1.000 | $\Sigma_{\omega}$                                                 | $\Sigma_{\omega}$                                         | mathematical structure of the canonical energy–momentum current $\Sigma_{\omega}$ of the Yang–                                                                  |
| mielke-geometrodynamics_F01479 | 304  | ■ 0.995 | $\Sigma_{\text{mat}} = 0$                                         | $\Sigma_{\text{mat}} = 0$                                 | $\Sigma_{\text{mat}} = 0$ .                                                                                                                                     |
| mielke-geometrodynamics_F01480 | 304  | ■ 1.000 | $\delta = \pi/2$                                                  | $\delta = \pi/2$                                          | $\delta = \pi/2$ is exactly equal to the operation of taking the Hodge dual <sup>3</sup>                                                                        |
| mielke-geometrodynamics_F01481 | 305  | ■ 1.000 | $\Sigma_{\mathrm{YM}}$                                            | $\Sigma_{\mathrm{YM}}$                                    | transformation). The <i>invariance</i> of $\Sigma_{\mathrm{YM}}$ with respect to duality rotations can easily                                                   |
| mielke-geometrodynamics_F01482 | 305  | ■ 1.000 | $\cos^2 \delta +$                                                 | $\cos^2 \delta +$                                         | be proved by writing out and applying the elementary functional relation $\cos^2 \delta +$                                                                      |
| mielke-geometrodynamics_F01483 | 305  | ■ 1.000 | $\sin^2 \delta = 1$                                               | $\sin^2 \delta = 1$                                       | $\sin^2 \delta = 1$ for trigonometric functions.                                                                                                                |
| mielke-geometrodynamics_F01484 | 307  | ■ 0.999 | $R_{ij}^{\{ \}}$                                                  | $R_{ij}^{\{ \}}$                                          | for the holonomically written components $R_{ij}^{\{ \}}$ or the Ricci tensor. Combined with                                                                    |
| mielke-geometrodynamics_F01485 | 307  | ■ 0.993 | $\bar{\tau}_e$                                                    | $\bar{\tau}_e$                                            | out of the matter currents $\bar{\tau}_e$ and $\bar{\tau}_p$ of a theory (SCHWINGER 1968; JULIA &                                                               |
| mielke-geometrodynamics_F01486 | 307  | ■ 0.993 | $\bar{\tau}_p$                                                    | $\bar{\tau}_p$                                            | out of the matter currents $\bar{\tau}_e$ and $\bar{\tau}_p$ of a theory (SCHWINGER 1968; JULIA &                                                               |

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| mielke-geometrodynamics_F01487 | 308  | ■ 1.000 | $\delta=0$                                                  | $\delta = 0$                                                | traditional electrodynamics, which is defined by the choice $\delta = 0$ of a “dual gauge”                                 |
| mielke-geometrodynamics_F01488 | 308  | ■ 0.562 | $\xi \wedge \vartheta$                                      | $\xi \wedge \vartheta$                                      | After the exterior multiplication of (14.1.32) by $\xi \wedge \vartheta$ and (14.1.33) by $\ast \xi \wedge \vartheta$ and  |
| mielke-geometrodynamics_F01489 | 308  | ■ 0.562 | $\ast \xi \wedge \vartheta$                                 | $\ast \xi \wedge \vartheta$                                 | After the exterior multiplication of (14.1.32) by $\xi \wedge \vartheta$ and (14.1.33) by $\ast \xi \wedge \vartheta$ and  |
| mielke-geometrodynamics_F01490 | 308  | ■ 0.966 | $d \delta$                                                  | $d\delta$                                                   | $d\delta$ , Eq. (14.1.37) has to be contracted by $\ast \Sigma_{\mathrm{YM}}(\xi)$ . On account of the invariance          |
| mielke-geometrodynamics_F01491 | 308  | ■ 0.966 | $\ast \Sigma_{\mathrm{YM}}(\xi)$                            | $\ast \Sigma_{\mathrm{YM}}(\xi)$                            | $d\delta$ , Eq. (14.1.37) has to be contracted by $\ast \Sigma_{\mathrm{YM}}(\xi)$ . On account of the invariance          |
| mielke-geometrodynamics_F01492 | 309  | ■ 1.000 | $dd\delta \equiv 0$                                         | $dd\delta \equiv 0$                                         | Due to the identity $dd\delta \equiv 0$ , this follows in an unequivocal way from the exactness                            |
| mielke-geometrodynamics_F01493 | 309  | ■ 0.928 | $b_1 = \dim H^1(M) = \nu$                                   | $b_1 = \dim H^1(M) = \nu$                                   | of the form $d\delta$ . For manifolds with Betti number $b_1 = \dim H^1(M) = \nu$ , e.g., for                              |
| mielke-geometrodynamics_F01494 | 309  | ■ 1.000 | $\nu$                                                       | $\nu$                                                       | those that exhibit $\nu$ -fold global connectivity, however,                                                               |
| mielke-geometrodynamics_F01495 | 310  | ■ 1.000 | (1962, 1969)                                                | (1962, 1969)                                                | Encouraged by this remarkable result, WHEELER (1962, 1969) was inclined not                                                |
| mielke-geometrodynamics_F01496 | 311  | ■ 1.000 | $n = 4 + K$                                                 | $n = 4 + K$                                                 | (or supergravity) in $n = 4 + K$ dimensions, which is then shown or assumed to                                             |
| mielke-geometrodynamics_F01497 | 311  | ■ 0.844 | (1926, 1928)                                                | (1926, 1928)                                                | (1926, 1928) proved that this is possible after proceeding to 5-dimensional <sup>5</sup> man-                              |
| mielke-geometrodynamics_F01498 | 312  | ■ 1.000 | $M^{4+K}$                                                   | $M^{4+K}$                                                   | $M^{4+K}$ that is endowed with a bundle $\hat{L}(M^{4+K})$ of linear frames and is furthermore                             |
| mielke-geometrodynamics_F01499 | 312  | ■ 1.000 | $\hat{L}(M^{4+K})$                                          | $\hat{L}(M^{4+K})$                                          | $M^{4+K}$ that is endowed with a bundle $\hat{L}(M^{4+K})$ of linear frames and is furthermore                             |
| mielke-geometrodynamics_F01500 | 312  | ■ 0.998 | $\hat{\vartheta}$                                           | $\hat{\vartheta}$                                           | equipped with a canonical 1-form $\hat{\vartheta}$ associated with $\hat{L}(M^{4+K})$ . Only later does one                |
| mielke-geometrodynamics_F01501 | 312  | ■ 1.000 | $\hat{E} \approx M^4 \times B$                              | $\hat{E} \approx M^4 \times B$                              | such that $M^{4+K}$ takes on the form $\hat{E} \approx M^4 \times B$ , where B is a (non)compact K-                        |
| mielke-geometrodynamics_F01502 | 312  | ■ 1.000 | $G/H$                                                       | $G/H$                                                       | dimensional homogeneous space, $G/H$ . Here G denotes a semisimple Lie group of                                            |
| mielke-geometrodynamics_F01503 | 312  | ■ 0.951 | $K = \dim G$                                                | $K = \dim G$                                                | dimension $K = \dim G$ , and H an isotropic subgroup. As shown by COQUEREAUX &                                             |
| mielke-geometrodynamics_F01504 | 312  | ■ 0.600 | $P(M^4, G, \pi, \delta)$                                    | $P(M^4, G, \pi, \delta)$                                    | JADCZYK (1983), the “geometric arena” is not the fiber bundle $P(M^4, G, \pi, \delta)$ but                                 |
| mielke-geometrodynamics_F01505 | 312  | ■ 0.983 | $P(M^4, N/H, \pi, \delta)$                                  | $P(M^4, N/H, \pi, \delta)$                                  | $P(M^4, N/H, \pi, \delta)$ , where N denotes the normalizer of H in G. The extended space                                  |
| mielke-geometrodynamics_F01506 | 312  | ■ 1.000 | $\hat{E}$                                                   | $\hat{E}$                                                   | $\hat{E}$ , which is locally isomorphic to $M^4 \times G/H$ , is globally the associated bundle                            |
| mielke-geometrodynamics_F01507 | 312  | ■ 1.000 | $M^4 \times G/H$                                            | $M^4 \times G/H$                                            | $\hat{E}$ , which is locally isomorphic to $M^4 \times G/H$ , is globally the associated bundle                            |
| mielke-geometrodynamics_F01508 | 312  | ■ 0.996 | $\hat{E}(M^4, F = G/H, N/H, P)$                             | $\hat{E}(M^4, F = G/H, N/H, P)$                             | $\hat{E}(M^4, F = G/H, N/H, P)$ with $N/H$ as structure group.                                                             |
| mielke-geometrodynamics_F01509 | 312  | ■ 0.996 | $N/H$                                                       | $N/H$                                                       | $\hat{E}(M^4, F = G/H, N/H, P)$ with $N/H$ as structure group.                                                             |
| mielke-geometrodynamics_F01510 | 312  | ■ 0.858 | $\omega = \omega^j_{\alpha} L_j \otimes \vartheta^{\alpha}$ | $\omega = \omega^j_{\alpha} L_j \otimes \vartheta^{\alpha}$ | and $\omega = \omega^j_{\alpha} L_j \otimes \vartheta^{\alpha}$ is a 1-form with values in the Lie algebra of G or $N/H$ , |
| mielke-geometrodynamics_F01511 | 312  | ■ 0.760 | $\overset{\circ}{\vartheta}$                                | $\overset{\circ}{\vartheta}$                                | Mills-type gauge fields. The 1-form $\overset{\circ}{\vartheta}$ is defined on the cotangent bundle $T_e^*(G)$ of          |
| mielke-geometrodynamics_F01512 | 312  | ■ 0.760 | $T_e^*(G)$                                                  | $T_e^*(G)$                                                  | Mills-type gauge fields. The 1-form $\overset{\circ}{\vartheta}$ is defined on the cotangent bundle $T_e^*(G)$ of          |

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| mielke-geometrodynamics_F01513 | 312  | ■ 1.000 | $\xi^j$                                                                                                        | $\xi^j$                                                                            | the group G, being parametrized by the coordinates $\xi^j$ . In its expansion                                                                                                                                                                           |
| mielke-geometrodynamics_F01514 | 312  | ■ 1.000 | $\varphi_j$                                                                                                    | $\varphi_j$                                                                        | the coefficients $\varphi_j$ may be regarded as a multiplet of (Higgs-type) scalar fields trans-                                                                                                                                                        |
| mielke-geometrodynamics_F01515 | 312  | ■ 1.000 | $d \xi^j$                                                                                                      | $d\xi^j$                                                                           | (holonomic) basis $d\xi^j$ is “dual” to the infinitesimal generators $I_j$ of G. Moreover,                                                                                                                                                              |
| mielke-geometrodynamics_F01516 | 312  | ■ 1.000 | $I_{\{j\}}$                                                                                                    | $I_j$                                                                              | (holonomic) basis $d\xi^j$ is “dual” to the infinitesimal generators $I_j$ of G. Moreover,                                                                                                                                                              |
| mielke-geometrodynamics_F01517 | 312  | ■ 0.881 | $V=V^{\{j\}} I_{\{j\}}, \stackrel{\circ}{\relax}\circ{\vartheta}$                                              | $V = V^j I_j, \overset{\circ}{\vartheta}$                                          | as a frame “dual” to the Lie vector field $V = V^j I_j, \overset{\circ}{\vartheta}$ satisfies the Maurer–Cartan                                                                                                                                         |
| mielke-geometrodynamics_F01518 | 312  | ■ 1.000 | $\stackrel{\circ}{\relax}\circ{\omega}$                                                                        | $\overset{\circ}{\omega}$                                                          | A connection $\overset{\circ}{\omega}$ on G arises naturally via the Ansatz                                                                                                                                                                             |
| mielke-geometrodynamics_F01519 | 313  | ■ 1.000 | $\stackrel{\circ}{\relax}\circ{\Theta}$                                                                        | $\overset{\circ}{\Theta}$                                                          | The <i>internal</i> torsion $\overset{\circ}{\Theta}$ and curvature $\overset{\circ}{\Omega}$ , respectively, are obtained from Cartan’s                                                                                                                |
| mielke-geometrodynamics_F01520 | 313  | ■ 1.000 | $\stackrel{\circ}{\relax}\circ{\Omega}$                                                                        | $\overset{\circ}{\Omega}$                                                          | The <i>internal</i> torsion $\overset{\circ}{\Theta}$ and curvature $\overset{\circ}{\Omega}$ , respectively, are obtained from Cartan’s                                                                                                                |
| mielke-geometrodynamics_F01521 | 313  | ■ 0.811 | $b=1 / 2$                                                                                                      | $b = 1/2$                                                                          | $b = 1/2$ : torsionless connection in a symmetric coset space B of constant                                                                                                                                                                             |
| mielke-geometrodynamics_F01522 | 313  | ■ 1.000 | $\stackrel{\circ}{\relax}\circ{R}=1 / 4$                                                                       | $\overset{\circ}{R}= 1/4$                                                          | curvature $\overset{\circ}{R} = 1/4$ .                                                                                                                                                                                                                  |
| mielke-geometrodynamics_F01523 | 313  | ■ 1.000 | $b=0, 1$                                                                                                       | $b = 0, 1$                                                                         | $b = 0, 1$ : Flat connection <i>with</i> parallelizing <i>torsion</i> .                                                                                                                                                                                 |
| mielke-geometrodynamics_F01524 | 313  | ■ 1.000 | $\operatorname{Tr}(\stackrel{\circ}{\relax}\circ{\vartheta} \otimes \stackrel{\circ}{\relax}\circ{\vartheta})$ | $\operatorname{Tr}(\overset{\circ}{\vartheta} \otimes \overset{\circ}{\vartheta})$ | G. The term $\operatorname{Tr}(\overset{\circ}{\vartheta} \otimes \overset{\circ}{\vartheta})$ just gives the internal line element                                                                                                                     |
| mielke-geometrodynamics_F01525 | 313  | ■ 1.000 | $\stackrel{\circ}{\relax}\circ{g}_{ij}$                                                                        | $\overset{\circ}{g}_{ij}$                                                          | where $\overset{\circ}{g}_{ij}$ denotes the Cartan–Killing metric of G. As far as <i>simple</i> Lie groups are                                                                                                                                          |
| mielke-geometrodynamics_F01526 | 313  | ■ 0.997 | $\operatorname{Tr}(\omega \otimes \stackrel{\circ}{\relax}\circ{\vartheta})$                                   | $\operatorname{Tr}(\omega \otimes \overset{\circ}{\vartheta})$                     | the terms $\operatorname{Tr}(\overset{\circ}{\vartheta} \otimes \overset{\circ}{\vartheta})$ and $\operatorname{Tr}(\omega \otimes \overset{\circ}{\vartheta})$ but also in $\operatorname{Tr}(\omega \otimes \omega)$ , since $\omega$ takes values in |
| mielke-geometrodynamics_F01527 | 313  | ■ 0.997 | $\operatorname{Tr}(\omega \otimes \omega)$                                                                     | $\operatorname{Tr}(\omega \otimes \omega)$                                         | the terms $\operatorname{Tr}(\overset{\circ}{\vartheta} \otimes \overset{\circ}{\vartheta})$ and $\operatorname{Tr}(\omega \otimes \overset{\circ}{\vartheta})$ but also in $\operatorname{Tr}(\omega \otimes \omega)$ , since $\omega$ takes values in |
| mielke-geometrodynamics_F01528 | 313  | ■ 1.000 | $d \hat{s}^2$                                                                                                  | $d\hat{s}^2$                                                                       | classical, i.e., nonquantized models, the total signature of the line element $d\hat{s}^2$ has                                                                                                                                                          |
| mielke-geometrodynamics_F01529 | 313  | ■ 0.983 | $\mathrm{G} \times \mathrm{G} / \mathrm{G}_{\mathrm{D}}$                                                       | $G \times G/G_D$                                                                   | <sup>6</sup> More precisely, the group manifold of G is the symmetric space $G \times G/G_D$ , where $G_D$ is the                                                                                                                                       |
| mielke-geometrodynamics_F01530 | 313  | ■ 0.983 | $\mathrm{G}_{\mathrm{D}}$                                                                                      | $G_D$                                                                              | <sup>6</sup> More precisely, the group manifold of G is the symmetric space $G \times G/G_D$ , where $G_D$ is the                                                                                                                                       |
| mielke-geometrodynamics_F01531 | 314  | ■ 1.000 | $\hat{L}(\hat{E})$                                                                                             | $\hat{L}(\hat{E})$                                                                 | may be imprinted on the bundle $\hat{L}(\hat{E})$ of the linear frames. Here $\omega^g$ denotes the                                                                                                                                                     |
| mielke-geometrodynamics_F01532 | 314  | ■ 0.998 | $L\left(M^4\right), \stackrel{\circ}{\relax}\circ{\omega}$                                                     | $L\left(M^4\right), \overset{\circ}{\omega}$                                       | gravitational connection in $L(M^4)$ , $\overset{\circ}{\omega}$ a connection in the group manifold, and $\hat{\alpha}$                                                                                                                                 |
| mielke-geometrodynamics_F01533 | 314  | ■ 0.998 | $\hat{\alpha}$                                                                                                 | $\hat{\alpha}$                                                                     | gravitational connection in $L(M^4)$ , $\overset{\circ}{\omega}$ a connection in the group manifold, and $\hat{\alpha}$                                                                                                                                 |
| mielke-geometrodynamics_F01534 | 314  | ■ 0.514 | $4+\mathrm{K}$                                                                                                 | $4 + K$                                                                            | is valid. Connections possessing a nonvanishing $(\textcolor{red}{4}+\textcolor{red}{K})$ -dimensional torsion and                                                                                                                                      |
| mielke-geometrodynamics_F01535 | 314  | ■ 1.000 | $\hat{\Omega}$                                                                                                 | $\hat{\Omega}$                                                                     | Accordingly, the higher-dimensional Riemann–Cartan curvature 2-form $\hat{\Omega}$ , given                                                                                                                                                              |
| mielke-geometrodynamics_F01536 | 314  | ■ 0.917 | $(4+\mathrm{K})$                                                                                               | $(4 + K)$                                                                          | In our matrix notation, the $\textcolor{red}{[4+K]}$ -dimensional exterior derivative $\hat{d}$ is given by                                                                                                                                             |
| mielke-geometrodynamics_F01537 | 314  | ■ 0.917 | $\hat{\mathrm{d}}$                                                                                             | $\hat{d}$                                                                          | In our matrix notation, the $(4 + K)$ -dimensional exterior derivative $\hat{d}$ is given by                                                                                                                                                            |
| mielke-geometrodynamics_F01538 | 315  | ■ 0.980 | $M^4$                                                                                                          | $M^4$                                                                              | internal space B relative to the four-dimensional spacetime $M^4$ in the bundle $\hat{E}$ .                                                                                                                                                             |

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| mielke-geometrodynamics_F01539 | 315  | ■ 0.999 | $\Omega^{\prime}$                               | $\Omega'$                              | $\hat{\alpha}$ and the 2-form $\Omega'$ of the Yang–Mills field strengths, we have obtained the decisive                      |
| mielke-geometrodynamics_F01540 | 316  | ■ 0.988 | $\alpha_g:=g^2/\hbar c$                         | $\alpha_g := g^2/\hbar c$              | modification by the gauge-coupling constant $\alpha_g := g^2/\hbar c$ . The term $\tilde{L}_W^G$ depends                      |
| mielke-geometrodynamics_F01541 | 316  | ■ 0.988 | $\hat{L}_{\mathrm{W}}^G$                        | $\hat{L}_W^G$                          | modification by the gauge-coupling constant $\alpha_g := g^2/\hbar c$ . The term $\tilde{L}_W^G$ depends                      |
| mielke-geometrodynamics_F01542 | 316  | ■ 0.928 | $\stackrel{\circ}{R}$                           | $\overset{\circ}{R}$                   | whose value is determined by the scalar curvature $\overset{\circ}{R}$ and by the density $\sqrt{\det \overset{\circ}{g}}$ of |
| mielke-geometrodynamics_F01543 | 316  | ■ 0.928 | $\sqrt{\operatorname{det} \stackrel{\circ}{g}}$ | $\sqrt{\det \overset{\circ}{g}}$       | whose value is determined by the scalar curvature $\overset{\circ}{R}$ and by the density $\sqrt{\det \overset{\circ}{g}}$ of |
| mielke-geometrodynamics_F01544 | 316  | ■ 1.000 | $M^{4+K} \approx M^4 \times B$                  | $M^{4+K} \approx M^4 \times B$         | (i) $M^{4+K} \approx M^4 \times B$ locally and                                                                                |
| mielke-geometrodynamics_F01545 | 316  | ■ 1.000 | $R_{\{4\}} \gg R_{\{K\}}$                       | $R_4 \gg R_K$                          | (ii) $R_4 \gg R_K$ for the corresponding characteristic radii.                                                                |
| mielke-geometrodynamics_F01546 | 316  | ■ 0.611 | $\mathscr{D}\left(M^{4+K}\right)$               | $\mathscr{D}\left(M^{4+K}\right)$      | “spontaneous” symmetry breaking of the general $(4 + K)$ -dimensional covariance group $\mathscr{D}(M^{4+K})$                 |
| mielke-geometrodynamics_F01547 | 316  | ■ 0.686 | $R_{\circ}$                                     | $R_{\circ}$                            | which tends to diminish the characteristic radius $R_{\circ}$ of B to the order of the Planck length $\ell^*$                 |
| mielke-geometrodynamics_F01548 | 317  | ■ 0.998 | $Y_{\{n\}}(B)$                                  | $Y_n(B)$                               | into a complete set of harmonic functions $Y_n(B)$ on $B = G/H$ ; cf. SALAM &                                                 |
| mielke-geometrodynamics_F01549 | 317  | ■ 0.998 | $B=G/H$                                         | $B = G/H$                              | into a complete set of harmonic functions $Y_n(B)$ on $B = G/H$ ; cf. SALAM &                                                 |
| mielke-geometrodynamics_F01550 | 317  | ■ 1.000 | $M^4 \times S^1$                                | $M^4 \times S^1$                       | closing up of the five-dimensional world to a cylinder $M^4 \times S^1$ of an exceedingly                                     |
| mielke-geometrodynamics_F01551 | 317  | ■ 0.725 | $R_{\mathrm{o}}$                                | $R_{\circ}$                            | small radius $R_{\circ}$ is taken for granted, Eq. (14.2.21) degenerates to a Fourier expansion                               |
| mielke-geometrodynamics_F01552 | 317  | ■ 1.000 | $S^1$                                           | $S^1$                                  | on the circle $S^1$ . The coefficients in the expansion (14.2.21) are fields of short range                                   |
| mielke-geometrodynamics_F01553 | 318  | ■ 1.000 | $S\,U(2) \times U(1)$                           | $SU(2) \times U(1)$                    | of strong interactions and the $SU(2) \times U(1)$ gauge group of the Weinberg–Salam                                          |
| mielke-geometrodynamics_F01554 | 318  | ■ 0.948 | $\mathbb{C}P^2=S\,U(3)/U(2)$                    | $\mathbb{C}P^2 = SU(3)/U(2)$           | where $\mathbb{C}P^2 = SU(3)/U(2)$ denotes the complex projective two-space and $S^n$ the                                     |
| mielke-geometrodynamics_F01555 | 318  | ■ 0.948 | $S^n$                                           | $S^n$                                  | where $\mathbb{C}P^2 = SU(3)/U(2)$ denotes the complex projective two-space and $S^n$ the                                     |
| mielke-geometrodynamics_F01556 | 318  | ■ 0.994 | $M^4 \times B$                                  | $M^4 \times B$                         | difficulty arises due to the fact that the Cartesian product $M^4 \times B$ of the Minkowski                                  |
| mielke-geometrodynamics_F01557 | 318  | ■ 1.000 | $\mathrm{n}=11$                                 | $n = 11$                               | Einstein space. In order to remedy this deficiency in, e.g., $\bar{n}=11$ dimensions, it is                                   |
| mielke-geometrodynamics_F01558 | 318  | ■ 0.874 | $S^7=S\,O(8)/S\,O(7)$                           | $S^7 = SO(8)/SO(7)$                    | necessary to “squash” the (round) sphere $S^7 = SO(8)/SO(7)$ by partial rescalings                                            |
| mielke-geometrodynamics_F01559 | 318  | ■ 0.999 | $\mathrm{K}=4+2+1=7$                            | $K = 4 + 2 + 1 = 7$                    | As observed by WITTEN (1981), $\mathbb{K} = 4 + 2 + 1 = 7$ of (14.2.25) represents the min-                                   |
| mielke-geometrodynamics_F01560 | 319  | ■ 1.000 | $\mathrm{n}=4+7=11$                             | $n = 4 + 7 = 11$                       | metric, belong to a single representation. Most remarkably, $\bar{n} = 4 + 7 = 11$ is the                                     |
| mielke-geometrodynamics_F01561 | 319  | ■ 1.000 | $(\mathscr{N}=1)$                               | $(\mathscr{N} = 1)$                    | <i>maximal</i> dimension of a theory of simple ( $\mathscr{N} = 1$ ) supergravity, cf. DEWITT &                               |
| mielke-geometrodynamics_F01562 | 319  | ■ 1.000 | $\overline{\mathrm{s}} \leq 3/2$                | $\bar{s} \leq 3/2$                     | STORA (1984), that contains the necessary fermion fields of spin $\bar{s} \leq 3/2$ without                                   |
| mielke-geometrodynamics_F01563 | 319  | ■ 1.000 | $\overline{\mathrm{s}}=2$                       | $\bar{s} = 2$                          | the bosonic fields exceeding the spin or helicity $\bar{s} = 2$ of the graviton. The bosonic                                  |
| mielke-geometrodynamics_F01564 | 319  | ■ 0.928 | $A_{\mathrm{IJK}}$                              | $A_{\mathrm{IJK}}$                     | metric tensor potential $A_{\mathrm{IJK}}$ , a remnant of the internal torsion, which may induce a                            |
| mielke-geometrodynamics_F01565 | 319  | ■ 0.835 | $M^{11}=(\mathrm{AdS})^4 \times S^7$            | $M^{11} = (\mathrm{AdS})^4 \times S^7$ | tions, the underlying vacuum geometry factors into the product $M^{11} = (\mathrm{AdS})^4 \times S^7$                         |

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| mielke-geometrodynamics_F01566 | 319  | ■ 1.000 | $\mathrm{S}^7$                                     | $\tilde{S}^7$                                      | ical model of the real world. For the “squashed” seven-sphere $\tilde{S}^7$ , there exists a                                      |
| mielke-geometrodynamics_F01567 | 319  | ■ 1.000 | $M^5=\mathbb{R}^4\times S^1$                       | $M^5=\mathbb{R}^4\times S^1$                       | BERGMANN (1938), who utilized the topology $M^5=\mathbb{R}^4\times S^1$ of a cylinder for the                                     |
| mielke-geometrodynamics_F01568 | 320  | ■ 0.994 | $\mathrm{E}_{2,4}$                                 | $E_{2,4}$                                          | pseudo-Euclidean space $E_{2,4}$ of conformal symmetries was pointed out by BUDINI                                                |
| mielke-geometrodynamics_F01569 | 320  | ■ 0.998 | $\hat{\omega}$                                     | $\hat{\omega}$                                     | point. Employing the dissociation (14.2.9) of the connection 1-form $\hat{\omega}$ , one obtains                                  |
| mielke-geometrodynamics_F01570 | 320  | ■ 1.000 | $\hat{\psi}$                                       | $\hat{\psi}$                                       | field $\hat{\psi}$ with $2^{[2+K/2]}$ components is adopted:                                                                      |
| mielke-geometrodynamics_F01571 | 320  | ■ 1.000 | $2^{\{2+\mathrm{K}/2\}}$                           | $2^{[2+K/2]}$                                      | field $\hat{\psi}$ with $2^{[2+K/2]}$ components is adopted:                                                                      |
| mielke-geometrodynamics_F01572 | 320  | ■ 1.000 | $\chi_{\rho F}^{\pm}(\xi)$                         | $\chi_{\rho F}^{\pm}(\xi)$                         | Here the spinorial harmonics $\chi_{\rho F}^{\pm}(\xi)$ furnish a representation $\rho$ of the symmetry                           |
| mielke-geometrodynamics_F01573 | 320  | ■ 0.989 | $\stackrel{\circ}{\gamma}^{K+1}$                   | $\stackrel{\circ}{\gamma}^{K+1}$                   | of the internal helicity operator $\stackrel{\circ}{\gamma}^{K+1}$ . After projection $\pi$ to four-dimensional compo-            |
| mielke-geometrodynamics_F01574 | 321  | ■ 1.000 | $\tilde{\omega}^g$                                 | $\tilde{\omega}^g$                                 | the (spinor) connections $\tilde{\omega}^g$ and $\omega$ , respectively, but that in addition, inherits a gauge                   |
| mielke-geometrodynamics_F01575 | 321  | ■ 1.000 | $\gamma\wedge\gamma\wedge^*\Omega\psi$             | $\gamma\wedge\gamma\wedge^*\Omega\psi$             | covariant term $\gamma\wedge\gamma\wedge^*\Omega\psi$ of electric dipole type. This term in the Lagrangian                        |
| mielke-geometrodynamics_F01576 | 321  | ■ 0.527 | $\hat{m}=0$                                        | $\hat{m}=0$                                        | which follows from the Lagrangian (4 + K)-form in the case $\hat{m}=0$ . This equation                                            |
| mielke-geometrodynamics_F01577 | 322  | ■ 1.000 | $\stackrel{\circ}{D}$                              | $\stackrel{\circ}{D}$                              | depends via the covariant derivative $\stackrel{\circ}{D}$ on the spinor connection $\stackrel{\circ}{\omega}$ and therefore acts |
| mielke-geometrodynamics_F01578 | 322  | ■ 1.000 | $\stackrel{\circ}{\omega}$                         | $\stackrel{\circ}{\omega}$                         | depends via the covariant derivative $\stackrel{\circ}{D}$ on the spinor connection $\stackrel{\circ}{\omega}$ and therefore acts |
| mielke-geometrodynamics_F01579 | 322  | ■ 0.978 | $i\not{D}^{\{\text{(int)}\}}$                      | $i\mathcal{D}^{(\text{int})}$                      | only on the internal space B. Provided B is compact, $i\mathcal{D}^{(\text{int})}$ has a discrete spectrum                        |
| mielke-geometrodynamics_F01580 | 322  | ■ 1.000 | $\gamma^5\psi=(\mp)\psi$                           | $\gamma^5\psi=(\mp)\psi$                           | obeying $\gamma^5\psi=(\mp)\psi$ , transform according to inequivalent representations of the                                     |
| mielke-geometrodynamics_F01581 | 322  | ■ 1.000 | $n=4+K$                                            | $n=4+K$                                            | dimension of the total space. For $n=4+K$ , the spinor fields transform accord-                                                   |
| mielke-geometrodynamics_F01582 | 322  | ■ 0.999 | $\widetilde{SO}(1,3+K)$                            | $\widetilde{SO}(1,3+K)$                            | ing to the fundamental representation of the covering tangent group $\widetilde{SO}(1,3+K)$ .                                     |
| mielke-geometrodynamics_F01583 | 322  | ■ 0.995 | $\hat{\gamma}^{n+1}$                               | $\hat{\gamma}^{n+1}$                               | eigenvalues of the n-dimensional chirality operator $\hat{\gamma}^{n+1}$ . Under the subgroup                                     |
| mielke-geometrodynamics_F01584 | 322  | ■ 0.615 | $\overparen{SO}(1,3)\otimes\widetilde{SO}(K)$      | $\widehat{SO}(1,3)\otimes\widetilde{SO}(K)$        | $\widetilde{SO}(1,3)\otimes\widetilde{SO}(K)$ , such a representation branches according to                                       |
| mielke-geometrodynamics_F01585 | 322  | ■ 1.000 | $n=2p+1$                                           | $n=2p+1$                                           | (i) $n=2p+1$ : The transformation $\hat{\gamma}^A\rightarrow-\hat{\gamma}^A$ does not affect the infinitesimal                    |
| mielke-geometrodynamics_F01586 | 322  | ■ 1.000 | $\hat{\gamma}^A\rightarrow-\hat{\gamma}^A$         | $\hat{\gamma}^A\rightarrow-\hat{\gamma}^A$         | (i) $n=2p+1$ : The transformation $\hat{\gamma}^A\rightarrow-\hat{\gamma}^A$ does not affect the infinitesimal                    |
| mielke-geometrodynamics_F01587 | 322  | ■ 1.000 | $\hat{\sigma}^{AB}$                                | $\hat{\sigma}^{AB}$                                | generators $\hat{\sigma}^{AB}$ but $\hat{\gamma}^{n+1}\rightarrow-\hat{\gamma}^{n+1}$ . Since G-transformations are               |
| mielke-geometrodynamics_F01588 | 322  | ■ 1.000 | $\hat{\gamma}^{n+1}\rightarrow-\hat{\gamma}^{n+1}$ | $\hat{\gamma}^{n+1}\rightarrow-\hat{\gamma}^{n+1}$ | generators $\hat{\sigma}^{AB}$ but $\hat{\gamma}^{n+1}\rightarrow-\hat{\gamma}^{n+1}$ . Since G-transformations are               |
| mielke-geometrodynamics_F01589 | 322  | ■ 1.000 | $n=4p$                                             | $n=4p$                                             | (ii) $n=4p$ : This implies that                                                                                                   |
| mielke-geometrodynamics_F01590 | 322  | ■ 1.000 | $n=4p+2$                                           | $n=4p+2$                                           | (iii) $n=4p+2$ : Then it follows that                                                                                             |
| mielke-geometrodynamics_F01591 | 323  | ■ 0.543 | $i\not{D}^{\{\text{(int)}\}}$                      | $i\mathcal{D}^{(\text{int})}$                      | Then it remains to reveal the conditions under which $i\mathcal{D}^{(\text{int})}$ has zero eigenmodes.                           |

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| mielke-geometrodynamics_F01592 | 323  | ■ 1.000 | $\mathrm{D}^{\wedge\{*\}}=\{ \ }^{\wedge\{*\}} \mathrm{D}$ | $D^* = {}^*D$                   | Since $\overline{D^*} = {}^*D$ holds for a metric-compatible (spinor) connection,                           |
| mielke-geometrodynamics_F01593 | 323  | ■ 0.922 | $\stackrel{\circ}{\Theta}=\Theta=0$                        | $\overset{\circ}{\Theta}=0$     | for vanishing torsion, i.e., $\overset{\circ}{\Theta}=0$ , Eq. (14.3.18) is the sum of a nonnegative and a  |
| mielke-geometrodynamics_F01594 | 324  | ■ 1.000 | $n>2$                                                      | $n>2$                           | izable in $n>2$ dimensions according to perturbation theory. For supergravity with                          |
| mielke-geometrodynamics_F01595 | 324  | ■ 0.999 | $n=11$                                                     | $n=11$                          | turbative expansion. Moreover, simple ( $\mathcal{N}=1$ ) supergravity in $n=11$ dimensions                 |
| mielke-geometrodynamics_F01596 | 324  | ■ 0.999 | $G=\mathrm{SO}(32)$                                        | $G=\mathrm{SO}(32)$             | ical strings. These <i>superstring theories</i> with $G=\mathrm{SO}(32)$ or the exceptional group           |
| mielke-geometrodynamics_F01597 | 324  | ■ 1.000 | $\mathrm{E}_8\times\mathrm{E}_8$                           | $E_8\times E_8$                 | $E_8\times E_8$ as structure group seem to provide a unification of all physical interactions               |
| mielke-geometrodynamics_F01598 | 324  | ■ 1.000 | $n=10+\operatorname{dim} G=506$                            | $n=10+\dim G=506$               | in $\overline{n=10+\dim G=506}$ dimensions (DUFF et al. 1985) and a consistent and <i>finite</i>            |
| mielke-geometrodynamics_F01599 | 324  | ■ 0.993 | $\{ \ }^{\wedge\{10\}} J_3 \hbar$                          | ${}^{10}J_3\hbar$               | by the mass M, the z-component <sup>10</sup> $J_3\hbar$ of the angular momentum, and the total angu-        |
| mielke-geometrodynamics_F01600 | 324  | ■ 1.000 | $\sqrt{J(J+1)} \hbar$                                      | $\sqrt{J(J+1)}\hbar$            | lar momentum $\sqrt{J(J+1)}\hbar$ . Furthermore, it may carry the generalized “internal”                    |
| mielke-geometrodynamics_F01601 | 324  | ■ 1.000 | $Q^{\{i\}}$                                                | $Q^{(i)}$                       | Here $Q^{(i)}$ and $P^{(i)}$ denote the “electric”- and “magnetic”-type charges of a “dually”               |
| mielke-geometrodynamics_F01602 | 324  | ■ 1.000 | $P^{\{i\}}$                                                | $P^{(i)}$                       | Here $Q^{(i)}$ and $P^{(i)}$ denote the “electric”- and “magnetic”-type charges of a “dually”               |
| mielke-geometrodynamics_F01603 | 324  | ■ 1.000 | $J_3\hat{=}J$                                              | $J_3\hat{=}J$                   | rotation group. As for our still purely <i>classical</i> considerations, $J_3\hat{=}J$ remains valid.       |
| mielke-geometrodynamics_F01604 | 325  | ■ 0.798 | $r^2e^v$                                                   | $r^2e^v$                        | substitutes the function $r^2e^v$ in the Schwarzschild metric. The generalization of                        |
| mielke-geometrodynamics_F01605 | 326  | ■ 0.976 | $\bar{Q}^{\{o\}}$                                          | $\bar{Q}^{(o)}$                 | mass M, generalized charge $\bar{Q}^{(o)}$ , and (classical) angular momentum component $J_3\hbar$          |
| mielke-geometrodynamics_F01606 | 326  | ■ 0.976 | $J_3 \hbar$                                                | $J_3\hbar$                      | mass M, generalized charge $\bar{Q}^{(o)}$ , and (classical) angular momentum component $J_3\hbar$          |
| mielke-geometrodynamics_F01607 | 326  | ■ 1.000 | $\mu_m=\frac{Q_e}{M}J$                                     | $\mu_m=\frac{Q_e}{M}J$          | with respect to the axis of rotation, but it also has the magnetic moment $\mu_m=\frac{Q_e}{M}J$ ,          |
| mielke-geometrodynamics_F01608 | 326  | ■ 0.823 | $M_{\mathrm{ir}}$                                          | $M_{\mathrm{ir}}$               | The total mass M of a black hole depends on its geometric remnant mass $M_{\mathrm{ir}}$ ,                  |
| mielke-geometrodynamics_F01609 | 326  | ■ 0.938 | $M_{\mathrm{ir}},\bar{Q}^{\{o\}}$                          | $M_{\mathrm{ir}},\bar{Q}^{(o)}$ | dependence of M on the characteristic values $M_{\mathrm{ir}}$ , $\bar{Q}^{(o)}$ , and J is nonlinear. More |
| mielke-geometrodynamics_F01610 | 326  | ■ 0.811 | $J\hat{=}J_3=0$                                            | $J\hat{=}J_3=0$                 | formulated by REISSNER (1916) and NORDSTRÖM (1918) if $J\hat{=}J_3=0$ holds.                                |
| mielke-geometrodynamics_F01611 | 327  | ■ 1.000 | $\lambda_3$                                                | $\lambda_3$                     | The generalized Gell-Mann matrix $\lambda_3$ , which is identical to the Pauli matrix $\sigma_3$ for        |
| mielke-geometrodynamics_F01612 | 327  | ■ 1.000 | $\sigma_3$                                                 | $\sigma_3$                      | The generalized Gell-Mann matrix $\lambda_3$ , which is identical to the Pauli matrix $\sigma_3$ for        |
| mielke-geometrodynamics_F01613 | 327  | ■ 1.000 | $\Delta$                                                   | $\Delta$                        | Here $\Delta$ is to be substituted by the term (14.4.5) with $J=0$ . Apart from the horizon                 |
| mielke-geometrodynamics_F01614 | 327  | ■ 0.993 | $r=r_{\mathrm{o}}$                                         | $r=r_{\mathrm{o}}$              | given by (14.4.6), this spacetime geometry has obviously still a further one at $r=r_{\mathrm{o}}$ ,        |
| mielke-geometrodynamics_F01615 | 327  | ■ 1.000 | $r=0$                                                      | $r=0$                           | away from the essential singularity at $r=0$ .                                                              |
| mielke-geometrodynamics_F01616 | 328  | ■ 0.672 | $Q=-\bar{Q}\cos\delta$                                     | $Q=-\bar{Q}\cos\delta$          | a particle of mass M with an electric charge $\overline{Q}=-Q\cos\delta$ and a magnetic monopole            |
| mielke-geometrodynamics_F01617 | 328  | ■ 1.000 | $P=-\bar{Q}\sin\delta$                                     | $P=-\bar{Q}\sin\delta$          | strength $P=-\bar{Q}\sin\delta$ . If the surface area $S_A$ of the event horizon, which is defined          |
| mielke-geometrodynamics_F01618 | 328  | ■ 1.000 | $S_{\mathrm{A}}$                                           | $S_A$                           | strength $P=-\bar{Q}\sin\delta$ . If the surface area $S_A$ of the event horizon, which is defined          |

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| mielke-geometrodynamics_F01619 | 328  | ■ 0.546 | $\backslash\mathrm{bar}\{Q\}$           | $\overline{Q}$                          | and a large dual charge $\overline{Q}$ , the charge–mass relation and the total mass M of the                                |
| mielke-geometrodynamics_F01620 | 328  | ■ 0.990 | $1 \ / \ \kappa$                        | $1/\kappa$                              | necessarily fixed by the coupling constant $1/\kappa$ of the Lorentz “rotons”.                                               |
| mielke-geometrodynamics_F01621 | 330  | ■ 1.000 | $\mathrm{d}=10$                         | $d = 10$                                | Green MB, Schwarz JH (1984) Anomaly cancellations in supersymmetric d = 10 gauge theory and                                  |
| mielke-geometrodynamics_F01622 | 335  | ■ 1.000 | $1 \ / \ \alpha_e$                      | $1/\alpha_e$                            | than electromagnetic forces by the factor $1/\alpha_e$ . On the other hand, all experiments                                  |
| mielke-geometrodynamics_F01623 | 335  | ■ 0.967 | $m_{\pi}=2 \ m_e \ / \ \alpha_e$        | $m_{\pi} = 2m_e/\alpha_e$               | (1964), is explained by a virtual exchange of scalar $\pi$ -mesons of mass $m_{\pi} = 2 \ m_e/\alpha_e$ .                    |
| mielke-geometrodynamics_F01624 | 335  | ■ 0.829 | $S \ U(3)^c$                            | $SU(3)^c$                               | ory with an exact $SU(3)^c$ structure group. The introduction of these additional color                                      |
| mielke-geometrodynamics_F01625 | 336  | ■ 0.924 | $f_{i \ j}$                             | $f_{ij}$                                | tensor fields $f_{ij}$ and $g_{ij}$ or via the canonical 1-forms $\vartheta^f$ and $\vartheta^g$ as follows:                 |
| mielke-geometrodynamics_F01626 | 336  | ■ 0.924 | $\vartheta^f$                           | $\vartheta^f$                           | tensor fields $f_{ij}$ and $g_{ij}$ or via the canonical 1-forms $\vartheta^f$ and $\vartheta^g$ as follows:                 |
| mielke-geometrodynamics_F01627 | 336  | ■ 0.924 | $\vartheta^g$                           | $\vartheta^g$                           | tensor fields $f_{ij}$ and $g_{ij}$ or via the canonical 1-forms $\vartheta^f$ and $\vartheta^g$ as follows:                 |
| mielke-geometrodynamics_F01628 | 336  | ■ 1.000 | $L_{f \ g}$                             | $L_{fg}$                                | necessitates the introduction of an additional mixing term $L_{fg}$ . Were it not for this                                   |
| mielke-geometrodynamics_F01629 | 336  | ■ 1.000 | $\vartheta^f$                           | $\vartheta^f$                           | mixing term is understood to have only a polynomial dependence on $\vartheta^f$ and $\vartheta^g$ of                         |
| mielke-geometrodynamics_F01630 | 336  | ■ 1.000 | $b-4$                                   | $b - 4$                                 | degree $a$ or $b - 4$ ,                                                                                                      |
| mielke-geometrodynamics_F01631 | 337  | ■ 1.000 | $\sum^f$                                | $\sum^f$                                | Let $\sum^f$ and $\sum^g$ be the canonical energy–momentum currents that are generated                                       |
| mielke-geometrodynamics_F01632 | 337  | ■ 1.000 | $\sum^g$                                | $\sum^g$                                | Let $\sum^f$ and $\sum^g$ be the canonical energy–momentum currents that are generated                                       |
| mielke-geometrodynamics_F01633 | 338  | ■ 0.921 | $\mathrm{G}_{\mathrm{N}}$               | $G_N$                                   | of the (weak) Newtonian constant $G_N$ of gravity to the coupling constant $G_S$ of strong                                   |
| mielke-geometrodynamics_F01634 | 338  | ■ 0.921 | $\mathrm{G}_{\mathrm{S}}$               | $G_S$                                   | of the (weak) Newtonian constant $G_N$ of gravity to the coupling constant $G_S$ of strong                                   |
| mielke-geometrodynamics_F01635 | 338  | ■ 1.000 | $=10^{-13} \ \mathrm{cm}$               | $= 10^{-13} \ \mathrm{cm}$              | spacetime structure beyond hadronic distances of the size of one fermi = $10^{-13} \ \mathrm{cm}$ .                          |
| mielke-geometrodynamics_F01636 | 338  | ■ 1.000 | $f^0$                                   | $f^0$                                   | $f^0$ -meson and the massless graviton in a first approximation.                                                             |
| mielke-geometrodynamics_F01637 | 339  | ■ 1.000 | $\mu^g, \mu^f$                          | $\mu^g, \mu^f$                          | is the result for the f-metric. The integration constants are $\mu^g, \mu^f$ , and $\gamma$ . For u = 1/2,                   |
| mielke-geometrodynamics_F01638 | 339  | ■ 1.000 | $u=1 \ / \ 2$                           | $u = 1/2$                               | is the result for the f-metric. The integration constants are $\mu^g, \mu^f$ , and $\gamma$ . For u = 1/2,                   |
| mielke-geometrodynamics_F01639 | 339  | ■ 0.992 | $\mu^g=\Lambda=0$                       | $\mu^g = \Lambda = 0$                   | <u><math>\mu^g = \Lambda = 0</math></u> , this solution reduces to the configuration that was given by SALAM &               |
| mielke-geometrodynamics_F01640 | 339  | ■ 1.000 | $m_f=1-2 \ \mathrm{GeV} \ / \ c^2$      | $m_f = 1 - 2\mathrm{GeV}/c^2$           | f-mesons of mass $m_f = 1 - 2 \ \mathrm{GeV}/c^2$ . This additional property of the tensor dom-                              |
| mielke-geometrodynamics_F01641 | 340  | ■ 1.000 | $E_j^{\beta}$                           | $E_j^{\beta}$                           | the Poincaré gauge theory. These are given by the tetrads $E_j^{\beta}$ and Ricci’s rotation                                 |
| mielke-geometrodynamics_F01642 | 340  | ■ 1.000 | $\Gamma_i^{\alpha \beta}$               | $\Gamma_i^{\alpha \beta}$               | coefficients $\Gamma_i^{\alpha \beta}$ , respectively. With respect to anholonomic frame, these potentials                   |
| mielke-geometrodynamics_F01643 | 340  | ■ 1.000 | $\left(\frac{1}{2}, \frac{1}{2}\right)$ | $\left(\frac{1}{2}, \frac{1}{2}\right)$ | transform under the Lorentz group according to the representation $\left(\frac{1}{2}, \frac{1}{2}\right)$ or $(1, 0) \oplus$ |
| mielke-geometrodynamics_F01644 | 340  | ■ 1.000 | $(1,0) \ \oplus$                        | $(1,0) \oplus$                          | transform under the Lorentz group according to the representation $\left(\frac{1}{2}, \frac{1}{2}\right)$ or $(1, 0) \oplus$ |

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| mielke-geometrodynamics_F01645 | 340  | 0.609 | $(0,1)$                                                       | $(0,1)$                                                      | $(0,1)$ , respectively, and are therefore have an holonomic spin $\bar{s}_a = 1$ . After the tran-                          |
| mielke-geometrodynamics_F01646 | 340  | 0.609 | $\operatorname{spin} \bar{s}_a=1$                             | $\operatorname{spin} \bar{s}_a = 1$                          | $(0,1)$ , respectively, and are therefore have an holonomic spin $\bar{s}_a = 1$ . After the tran-                          |
| mielke-geometrodynamics_F01647 | 340  | 1.000 | $\bar{s}=2$                                                   | $\bar{s} = 2$                                                | induces gauge potentials of maximal (reducible) spin or helicity $\bar{s} = 2$ (HEHL et al.                                 |
| mielke-geometrodynamics_F01648 | 340  | 1.000 | $\Lambda_{\text{eff}} \neq 0$                                 | $\Lambda_{\text{eff}} \neq 0$                                | nonlinear theory with $\Lambda_{\text{eff}} \neq 0$ . Other quantization procedures are needed in order to isolate the      |
| mielke-geometrodynamics_F01649 | 341  | 1.000 | $\ell=\ell^{*} / \sqrt{\kappa}$                               | $\ell = \ell^{*} / \sqrt{\kappa}$                            | proportional to $\ell = \ell^{*} / \sqrt{\kappa}$ is superposed in the metric background of the solutions.                  |
| mielke-geometrodynamics_F01650 | 341  | 1.000 | $1 / \sqrt{\kappa}$                                           | $1 / \sqrt{\kappa}$                                          | $\ell^{*}$ scaled by $1 / \sqrt{\kappa}$ is identical to the Compton wavelength of a proton according to                    |
| mielke-geometrodynamics_F01651 | 341  | 1.000 | $10^{13} \mathrm{~cm}$                                        | $10^{13} \text{ cm}$                                         | minimum radius would be increased to about $10^{13} \text{ cm}$ as a result of the hypothesis of                            |
| mielke-geometrodynamics_F01652 | 341  | 0.999 | $\ell^2$                                                      | $\ell^2$                                                     | “strong” gravity. The relative strength of the coupling constants $\ell^2$ and $\ell^{*2}$ , which                          |
| mielke-geometrodynamics_F01653 | 341  | 0.999 | $\ell^{*2}$                                                   | $\ell^{*2}$                                                  | “strong” gravity. The relative strength of the coupling constants $\ell^2$ and $\ell^{*2}$ , which                          |
| mielke-geometrodynamics_F01654 | 341  | 1.000 | $\kappa=0,38 \times 10^{-38}$                                 | $\kappa = 0,38 \times 10^{-38}$                              | small number of order of magnitude $\kappa = 0,38 \times 10^{-38}$ . It was again EINSTEIN in                               |
| mielke-geometrodynamics_F01655 | 342  | 1.000 | $10^{20}$                                                     | $10^{20}$                                                    | there would be a fundamental length $\ell$ being enlarged by about $10^{20}$ . This hypothe-                                |
| mielke-geometrodynamics_F01656 | 342  | 1.000 | $\mathrm{M}_{\mathrm{U}}$                                     | $\mathrm{M}_{\mathrm{U}}$                                    | between the total mass $\mathrm{M}_{\mathrm{U}}$ of the universe and the mass $\mathrm{M}_{\mathrm{P}}$ of the proton while |
| mielke-geometrodynamics_F01657 | 342  | 1.000 | $\mathrm{M}_{\mathrm{P}}$                                     | $\mathrm{M}_{\mathrm{P}}$                                    | between the total mass $\mathrm{M}_{\mathrm{U}}$ of the universe and the mass $\mathrm{M}_{\mathrm{P}}$ of the proton while |
| mielke-geometrodynamics_F01658 | 342  | 1.000 | $\ell_{\mathrm{H}}$                                           | $\ell_{\mathrm{H}}$                                          | relating it to the product of the relative Hubble radius $\ell_{\mathrm{H}}$ and the relative Compton                       |
| mielke-geometrodynamics_F01659 | 342  | 1.000 | $S U(f)$                                                      | $SU(f)$                                                      | resentations of the unitary groups $SU(f)$ . Concerning this aspect, particular success                                     |
| mielke-geometrodynamics_F01660 | 342  | 0.995 | $S U(4)$                                                      | $SU(4)$                                                      | to a far larger extent for the enlarged group $SU(4)$ that is postulated in the context of the charm                        |
| mielke-geometrodynamics_F01661 | 342  | 0.997 | $S U(5)$                                                      | $SU(5)$                                                      | hypothesis, and also for the grand unified theories (GUTs) that embarks upon $SU(5)$ as internal                            |
| mielke-geometrodynamics_F01662 | 343  | 0.999 | $S O(3) \approx S U(2)$                                       | $SO(3) \approx SU(2)$                                        | transform according to spinor representations of the group $SO(3) \approx SU(2)$ of the                                     |
| mielke-geometrodynamics_F01663 | 343  | 1.000 | $S U(4) \approx S O(6) \supset$                               | $SU(4) \approx SO(6) \supset$                                | nal spin within a common group-theoretic scheme, the group $SU(4) \approx SO(6) \supset$                                    |
| mielke-geometrodynamics_F01664 | 343  | 1.000 | $S O(4) \approx S U(2) \times S U(2)$                         | $SO(4) \approx SU(2) \times SU(2)$                           | $SO(4) \approx SU(2) \times SU(2)$ was proposed by Wigner. Later on, the group $SU(6)$ was                                  |
| mielke-geometrodynamics_F01665 | 343  | 1.000 | $S U(6)$                                                      | $SU(6)$                                                      | $SO(4) \approx SU(2) \times SU(2)$ was proposed by Wigner. Later on, the group $SU(6)$ was                                  |
| mielke-geometrodynamics_F01666 | 343  | 1.000 | $J=1 / 2$                                                     | $J = 1/2$                                                    | $J = 1/2$ and $J = 3/2$ of total angular momentum are thought to fill out a 56-fold                                         |
| mielke-geometrodynamics_F01667 | 343  | 1.000 | $J=3 / 2$                                                     | $J = 3/2$                                                    | $J = 1/2$ and $J = 3/2$ of total angular momentum are thought to fill out a 56-fold                                         |
| mielke-geometrodynamics_F01668 | 343  | 1.000 | $S L(6, \mathbb{C}) \supset S L(2, \mathbb{C}) \times S U(3)$ | $SL(6, \mathbb{C}) \supset SL(2, \mathbb{C}) \times SU(3)$   | this problem is to extend $SU(6)$ to the linear group $SL(6, \mathbb{C}) \supset SL(2, \mathbb{C}) \times SU(3)$ .          |
| mielke-geometrodynamics_F01669 | 343  | 0.962 | $\mathrm{SU}(3)$                                              | $SU(3)$                                                      | Thereby a relativistic union of the internal symmetry group $SU(3)$ with the (simply                                        |
| mielke-geometrodynamics_F01670 | 343  | 1.000 | $S L(6, \mathbb{C})$                                          | $SL(6, \mathbb{C})$                                          | spacetime, is achieved. If the symmetry is regarded only as a local one, an $SL(6, \mathbb{C})$                             |
| mielke-geometrodynamics_F01671 | 343  | 0.933 | $\mathrm{P}(\mathrm{M}, \overline{\mathrm{G}}, \pi, \delta)$  | $\mathrm{P}(\mathrm{M}, \overline{\mathrm{G}}, \pi, \delta)$ | fiber bundle $\mathrm{P}(\mathrm{M}, \overline{\mathrm{G}}, \pi, \delta)$ as a geometric arena. But this time, it is        |
| mielke-geometrodynamics_F01672 | 343  | 1.000 | $[S U(4) \otimes]^4$                                          | $[SU(4) \otimes]^4$                                          | chiral $[SU(4) \otimes]^4$ model of ELIAS et al. (1978) ready to hand for the case $f = c = 4$ .                            |

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| mielke-geometrodynamics_F01673 | 343  | ■ 1.000 | f=c=4                                                                                              | $f = c = 4$                                            | chiral $[SU(4)\otimes]^4$ model of ELIAS et al. (1978) ready to hand for the case $f = c = 4$ .               |
| mielke-geometrodynamics_F01674 | 344  | ■ 0.993 | $\overline{\mathrm{L}}(\mathrm{M})$                                                                | $\overline{\mathrm{L}}(\mathrm{M})$                    | with $\overline{\mathrm{L}}(\mathrm{M})$ . This 1-form is supposed to have holonomic components in the local  |
| mielke-geometrodynamics_F01675 | 344  | ■ 0.993 | $\mathrm{A}_2$                                                                                     | $A_2$                                                  | fields—as is, for instance, realized in nature via the $SU(3)$ -nonet $f, f', A_2$ , and $K^*$                |
| mielke-geometrodynamics_F01676 | 344  | ■ 0.995 | $S\mathrm{L}(2\mathrm{N},\mathbb{C})\supset S\mathrm{L}(2,\mathbb{C})\times\mathrm{U}(\mathrm{N})$ | $SL(2N,\mathbb{C})\supset SL(2,\mathbb{C})\times U(N)$ | erators of $SL(2N,\mathbb{C})\supset SL(2,\mathbb{C})\times U(N)$ can be realized by the matrices             |
| mielke-geometrodynamics_F01677 | 344  | ■ 1.000 | $\lambda_{\mathrm{i}}$                                                                             | $\lambda_i$                                            | where the generalized Gell-Mann matrices $\lambda_i$ are the $N^2-1$ vector operators of the                  |
| mielke-geometrodynamics_F01678 | 344  | ■ 1.000 | $\mathrm{N}^2-1$                                                                                   | $N^2 - 1$                                              | where the generalized Gell-Mann matrices $\lambda_i$ are the $N^2-1$ vector operators of the                  |
| mielke-geometrodynamics_F01679 | 344  | ■ 0.983 | $\mathrm{SU}(\mathrm{N})$                                                                          | $SU(N)$                                                | $SU(N)$ subgroup. If the gauge potentials of the unitary groups—which are contained                           |
| mielke-geometrodynamics_F01680 | 345  | ■ 1.000 | $\mathrm{A}_{\mathrm{i}}$                                                                          | $A_i$                                                  | The Yang–Mills-like gauge potentials $A_i$ aside, this total connection 1-form com-                           |
| mielke-geometrodynamics_F01681 | 345  | ■ 0.998 | $\overline{\gamma}$                                                                                | $\overline{\gamma}$                                    | field strengths that belong to the 1-forms $\overline{\gamma}$ and $\overline{\omega}$ are the 2-form         |
| mielke-geometrodynamics_F01682 | 345  | ■ 1.000 | $L_{\mathrm{W}}(\mathrm{f})$                                                                       | $L_{\mathrm{W}}(f)$                                    | freedom, this is achieved by replacing the Weyl term $L_{\mathrm{W}}(f)$ in the total Lagrangian              |
| mielke-geometrodynamics_F01683 | 345  | ■ 1.000 | f=c=0                                                                                              | $f = c = 0$                                            | If the internal degrees of freedom are suppressed, i.e., for the case $f = c = 0$ , the                       |
| mielke-geometrodynamics_F01684 | 345  | ■ 1.000 | f=3                                                                                                | $f = 3$                                                | as a structure group. In the framework of an $SL(6,\mathbb{C})$ gauge theory, i.e., for $f = 3$               |
| mielke-geometrodynamics_F01685 | 345  | ■ 0.987 | c=0                                                                                                | $c = 0$                                                | and $c = 0$ , they also proposed a mixing term $\overline{L}_{\mathrm{fg}}$ that is responsible for the medi- |
| mielke-geometrodynamics_F01686 | 345  | ■ 0.987 | $\overline{L}_{\mathrm{fg}}$                                                                       | $\overline{L}_{\mathrm{fg}}$                           | and $c = 0$ , they also proposed a mixing term $\overline{L}_{\mathrm{fg}}$ that is responsible for the medi- |
| mielke-geometrodynamics_F01687 | 346  | ■ 1.000 | G=S U(2)                                                                                           | $G = SU(2)$                                            | made allowance only for the isospin group, i.e., $G = SU(2)$ . If we followed that                            |
| mielke-geometrodynamics_F01688 | 346  | ■ 0.989 | $S\mathrm{U}(4)^{\mathrm{f}}\otimes S\mathrm{U}(4)^{\mathrm{c}}$                                   | $SU(4)^f\otimes SU(4)^c$                               | theories. Within the scheme A of this $SU(4)^f\otimes SU(4)^c$ -gauge theory, the quarklike                   |
| mielke-geometrodynamics_F01689 | 346  | ■ 1.000 | $S\mathrm{L}(8,\mathbb{C})^{\mathrm{f}}\otimes S\mathrm{U}(8,\mathbb{C})^{\mathrm{c}}$             | $SL(8,\mathbb{C})^f\otimes SU(8,\mathbb{C})^c$         | $SL(8,\mathbb{C})^f\otimes SU(8,\mathbb{C})^c$ -gauge theory, however, is rather more related—following       |
| mielke-geometrodynamics_F01690 | 347  | ■ 1.000 | $M=\mathbb{R}\times S^1\times S^2$                                                                 | $M = \mathbb{R} \times S^1 \times S^2$                 | fields within the nontrivial topology of the space $M=\mathbb{R}\times S^1\times S^2$ . Its apparent          |
| mielke-geometrodynamics_F01691 | 348  | ■ 0.994 | $G\mathrm{L}(8,\mathbb{C})^{\mathrm{f}}\otimes$                                                    | $GL(8,\mathbb{C})^f\otimes$                            | 1980) as indicated previously; see also CASTRO (2012). Thereby, a $GL(8,\mathbb{C})^f\otimes$                 |
| mielke-geometrodynamics_F01692 | 348  | ■ 1.000 | $G\mathrm{L}(8,\mathbb{C})^{\mathrm{c}}$                                                           | $GL(8,\mathbb{C})^c$                                   | $GL(8,\mathbb{C})^c$ gauge unification of all basic particle forces emerged into a rather spec-               |
| mielke-geometrodynamics_F01693 | 348  | ■ 1.000 | U(5)                                                                                               | $U(5)$                                                 | Since $SU(4)$ and $U(5)$ together with the local Lorentz group can also be embedded                           |
| mielke-geometrodynamics_F01694 | 348  | ■ 0.987 | S O(1,9)                                                                                           | $SO(1,9)$                                              | (FADDEEV 2011) such as $SO(1,9)$ and the embedding of the $SO(10)$ GUT into the                               |
| mielke-geometrodynamics_F01695 | 348  | ■ 0.987 | S O(10)                                                                                            | $SO(10)$                                               | (FADDEEV 2011) such as $SO(1,9)$ and the embedding of the $SO(10)$ GUT into the                               |
| mielke-geometrodynamics_F01696 | 348  | ■ 0.994 | S O(1,13)                                                                                          | $SO(1,13)$                                             | group $SO(1,13)$ have been proposed (PERCACCI 1991). Even “graviweak” groups                                  |
| mielke-geometrodynamics_F01697 | 348  | ■ 0.998 | S O(4,mathbb{C})                                                                                   | $SO(4,\mathbb{C})$                                     | such as $SO(4,\mathbb{C})$ have been considered by NESTI & PERCACCI (2010). Again, in                         |
| mielke-geometrodynamics_F01698 | 348  | ■ 0.995 | E(8)                                                                                               | $E(8)$                                                 | such as $E(8)$ and hyperbolic Kac–Moody algebras (KLEINSCHMIDT & NICOLAI                                      |
| mielke-geometrodynamics_F01699 | 348  | ■ 0.995 | E(10)                                                                                              | $E(10)$                                                | 2010) like $E(10)$ have been suggested to achieve this. However, these approaches                             |
| mielke-geometrodynamics_F01700 | 349  | ■ 0.984 | (5,mathbb{C})                                                                                      | $(5,\mathbb{C})$                                       | Castro C (2012) A Clifford Cl $(5,\mathbb{C})$ unified gauge field theory of conformal gravity, Maxwell and   |

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|--------------------------------|------|---------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| mielke-geometrodynamics_F01701 | 349  | ■ 0.992 | $\operatorname{SL}(2\,\mathrm{n},\,\mathrm{mathbb{C}})\,\otimes\,\operatorname{SL}(2,\,\mathrm{mathbb{C}})$ | $SL(2n, \mathbb{C}) \otimes SL(2, \mathbb{C})$   | Dennis PW, Huang JC (1977) Spontaneous symmetry breakdown in an <del><math>SL(2n, \mathbb{C}) \otimes SL(2, \mathbb{C})</math></del> . |
| mielke-geometrodynamics_F01702 | 349  | ■ 0.998 | $1000\,\mathrm{m}_W$                                                                                        | $1000\,m_W$                                      | $1000\,m_W$ . Phys Rev Lett 40(14):920                                                                                                 |
| mielke-geometrodynamics_F01703 | 350  | ■ 0.473 | $2\,\mathrm{n},\,\mathrm{mathbb{C}}$                                                                        | $2n, \mathbb{C}$                                 | Huang JC, Dennis PW (1981a) Spin-1 and spin-2 gauge fields in unified $SL(2n, \mathbb{C})$ model. Phys                                 |
| mielke-geometrodynamics_F01704 | 350  | ■ 0.805 | $2^{+}$                                                                                                     | $2^{+}$                                          | Isham C, Salam A, Strathdee J (1973a) $2^{+}$ nonet as gauge particles for $SL(6, \mathbb{C})$ symmetry. Phys                          |
| mielke-geometrodynamics_F01705 | 354  | ■ 0.952 | $\mathrm{R}-\mathrm{R}_{\mathrm{o}}$                                                                        | $R - R_o$                                        | $R - R_o$ ) whose gradient balances the electrodynamic force. <sup>1</sup>                                                             |
| mielke-geometrodynamics_F01706 | 354  | ■ 1.000 | (1966,1967)                                                                                                 | (1966,1967)                                      | for instance, PRASAD (1966, 1967) and ROMAN & KOH (1966). These notions have                                                           |
| mielke-geometrodynamics_F01707 | 354  | ■ 1.000 | $\kappa>0$                                                                                                  | $\kappa > 0$                                     | responsible for the negative (for $\kappa > 0$ ) pressure term                                                                         |
| mielke-geometrodynamics_F01708 | 354  | ■ 0.983 | $2\,\mu\,/\,\mathrm{r}$                                                                                     | $2\mu/r$                                         | sented by a Schwarzschild–de Sitter microuniverse, whereby the $2\mu/r$ -singularity                                                   |
| mielke-geometrodynamics_F01709 | 355  | ■ 1.000 | $\ell_{\mathrm{m}}$                                                                                         | $\ell_m$                                         | coordinate. The (dimensionless) ratio of the Compton wavelength $\ell_m$ of the test                                                   |
| mielke-geometrodynamics_F01710 | 355  | ■ 0.998 | $\mathrm{F}=\mathrm{F}_{\mathrm{nm}}^{\iota}\left(\rho^{*}\right)$                                          | $F = F_{\mathrm{nm}}^{\iota}(\rho^{*})$          | with reference to the still to be determined functions $F = F_{\mathrm{nm}}^{\iota}(\rho^{*})$ .                                       |
| mielke-geometrodynamics_F01711 | 356  | ■ 0.877 | $\iota=0$                                                                                                   | $\iota = 0$                                      | For the sake of illustration (Fig. 16.1), the simplest case $\iota = 0$ for $\kappa > 0$ , i.e., the                                   |
| mielke-geometrodynamics_F01712 | 357  | ■ 1.000 | $\rho^{*}=\frac{\pi}{2}$                                                                                    | $\rho^{*} = \frac{\pi}{2}$                       | At $\rho^{*} = \frac{\pi}{2}$ there arises for $\beta^2 < 2$ a steeply rising potential wall that can never be                         |
| mielke-geometrodynamics_F01713 | 357  | ■ 1.000 | $\beta^2<2$                                                                                                 | $\beta^2 < 2$                                    | At $\rho^{*} = \frac{\pi}{2}$ there arises for $\beta^2 < 2$ a steeply rising potential wall that can never be                         |
| mielke-geometrodynamics_F01714 | 357  | ■ 1.000 | $\mathrm{V}(\mathrm{r})=2\,\mu\,/\,\mathrm{r}+\mathrm{cr}$                                                  | $V(r) = 2\mu/r + cr$                             | In the latter, it is usually the potential $V(r) = 2\mu/r + cr$ that is postulated for the                                             |
| mielke-geometrodynamics_F01715 | 358  | ■ 0.997 | $\overline{\mathrm{Q}}$                                                                                     | $\bar{Q}$                                        | quantum number J of total angular momentum, and generalized charge $\bar{Q}$ is (almost)                                               |
| mielke-geometrodynamics_F01716 | 358  | ■ 1.000 | $\mathrm{Q}_{\mathrm{e}}$                                                                                   | $Q_e$                                            | while the electric charge $Q_e$ is determined via the generalized Gell-Mann–Nishijima                                                  |
| mielke-geometrodynamics_F01717 | 358  | ■ 0.522 | $\mathrm{M}_{\mathrm{ir}}$                                                                                  | $M_{\mathrm{ir}}$                                | irreducible mass $M_{\mathrm{ir}}$ ; concerning the tensor dominance model, it is thus suggestive                                      |
| mielke-geometrodynamics_F01718 | 358  | ■ 0.630 | $\mathrm{M}^{*}:=\hbar\sqrt{8\,\pi}\,/\,\ell^{*}\mathrm{c}$                                                 | $M^{*} := \hbar\sqrt{8\pi}/\ell^{*}c$            | to propose that $M_{\mathrm{ir}}$ as well as the modified Planck mass $M^{*} := \hbar\sqrt{8\pi}/\ell^{*}c$ are                        |
| mielke-geometrodynamics_F01719 | 358  | ■ 0.966 | $\mathrm{M}_{\mathrm{ir}}=\mathrm{M}^{*}=1\,\mathrm{GeV}\,/\,\mathrm{c}^2$                                  | $M_{\mathrm{ir}} = M^{*} = 1\mathrm{GeV}/c^2$    | approximately identical to the mass of a proton, i.e., $M_{\mathrm{ir}} = M^{*} = 1\mathrm{GeV}/c^2$ . For                             |
| mielke-geometrodynamics_F01720 | 359  | ■ 0.999 | $M,\,J$                                                                                                     | $M, J$                                           | $M, J$ , and $I$ are the characteristics of particles that are here defined <i>canonically</i> in                                      |
| mielke-geometrodynamics_F01721 | 359  | ■ 0.999 | $I$                                                                                                         | $I$                                              | $M, J$ , and $I$ are the characteristics of particles that are here defined <i>canonically</i> in                                      |
| mielke-geometrodynamics_F01722 | 359  | ■ 1.000 | $\mathrm{J}=0$                                                                                              | $J = 0$                                          | (However, if $J=0$ , this leads precisely back to the linear Gell-Mann–Okubo formula!)                                                 |
| mielke-geometrodynamics_F01723 | 359  | ■ 0.999 | $\mathrm{b}=\frac{1}{5}$                                                                                    | $b = \frac{1}{5}$                                | surprisingly well represented by (16.3.3). (In Fig. 16.2, $b = \frac{1}{5}$ has been chosen, while                                     |
| mielke-geometrodynamics_F01724 | 359  | ■ 0.998 | $\Lambda(1116)$                                                                                             | $\Lambda(1116)$                                  | $M^{*}$ has been accommodated to the state $\Lambda(1116)$ of excited baryons.) A detailed                                             |
| mielke-geometrodynamics_F01725 | 360  | ■ 1.000 | $k_{\mathrm{B}}=8.6\,\times\,10^{-5}\,\mathrm{eV}\,/\,\mathrm{K}$                                           | $k_B = 8.6 \times 10^{-5}\mathrm{eV}/\mathrm{K}$ | Here $k_B = 8.6 \times 10^{-5}\mathrm{eV}/\mathrm{K}$ denotes the Boltzmann constant. Under the assump-                                |
| mielke-geometrodynamics_F01726 | 362  | ■ 0.877 | $S\,\mathrm{U}(8)$                                                                                          | $SU(8)$                                          | Okubo S (1975) $SU(4)$ and $SU(8)$ mass formulas and weak interactions. Phys Rev D 11(11):3261                                         |
| mielke-geometrodynamics_F01727 | 362  | ■ 0.824 | $(6,\,\mathrm{mathbb{C}})$                                                                                  | $(6, \mathbb{C})$                                | Salam A (1973). On $SL(6, \mathbb{C})$ gauge invariance, In: Fundamental Interactions in Physics, 1973 Coral                           |
| mielke-geometrodynamics_F01738 | 365  | ■ 0.998 | $2\,\times\,2$                                                                                              | $2 \times 2$                                     | $2 \times 2$ Pauli matrices                                                                                                            |

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| mielke-geometrodynamics_F01739 | 365  | ■ 0.984 | <code>\mathrm{C}(0,2)=\mathbb{H}</code>                                                                              | $C(0,2) = \mathbb{H}$                                         | of the Clifford algebra $\mathbb{C}(0,2) = \mathbb{H}$ (field of quaternions).                                                               |
| mielke-geometrodynamics_F01740 | 365  | ■ 0.989 | <code>4 \times 4</code>                                                                                              | $4 \times 4$                                                  | $4 \times 4$ Dirac matrices in the so-called Pauli realization:                                                                              |
| mielke-geometrodynamics_F01741 | 366  | ■ 1.000 | <code>4&lt;2 \mathrm{p}+1&lt;2 \mathrm{p}+2 \leq n=2 l+2</code>                                                      | $4 < 2p + 1 < 2p + 2 \leq n = 2l + 2$                         | where $4 < 2p + 1 < 2p + 2 \leq n = 2l + 2$ ; cf. KALINOWSKI (1984).                                                                         |
| mielke-geometrodynamics_F01742 | 366  | ■ 0.997 | <code>\emptyset</code>                                                                                               | $\emptyset$                                                   | (iii) Both $X$ itself and the empty set $\emptyset$ are open.                                                                                |
| mielke-geometrodynamics_F01743 | 366  | ■ 1.000 | <code>\mathrm{T}_{1}</code>                                                                                          | $T_1$                                                         | $T_1$ : For each two distinct points $m_1, m_2 \in X$ , there exist neighborhoods $U_i \ni m_i$ of                                           |
| mielke-geometrodynamics_F01744 | 366  | ■ 1.000 | <code>\mathrm{m}_{1}, \mathrm{~m}_{2} \in \mathrm{X}</code>                                                          | $m_1, m_2 \in X$                                              | $T_1$ : For each two distinct points $m_1, m_2 \in X$ , there exist neighborhoods $U_i \ni m_i$ of                                           |
| mielke-geometrodynamics_F01745 | 366  | ■ 1.000 | <code>\mathrm{U}_{\{\mathrm{i}\}} \ni \mathrm{m}_{\{\mathrm{i}\}}</code>                                             | $U_i \ni m_i$                                                 | $T_1$ : For each two distinct points $m_1, m_2 \in X$ , there exist neighborhoods $U_i \ni m_i$ of                                           |
| mielke-geometrodynamics_F01746 | 366  | ■ 1.000 | <code>\mathrm{T}_{2}</code>                                                                                          | $T_2$                                                         | $T_2$ : For any two distinct points $m_1, m_2 \in X$ there exist neighborhoods $U_i \ni m_i$ that                                            |
| mielke-geometrodynamics_F01747 | 366  | ■ 0.566 | <code>\mathrm{U}_{1} \bigcap \mathrm{U}_{2}=\varnothing</code>                                                       | $U_1 \bigcap U_2 = \emptyset$                                 | are disjoint, namely $U_1 \bigcap U_2 = \emptyset$ .                                                                                         |
| mielke-geometrodynamics_F01748 | 367  | ■ 1.000 | <code>\sigma: \mathrm{M} \rightarrow \mathrm{N}</code>                                                               | $\sigma : M \rightarrow N$                                    | A diffeomorphism $\sigma : M \rightarrow N$ between manifolds of the same dimension (e.g., $M \approx$                                       |
| mielke-geometrodynamics_F01749 | 367  | ■ 1.000 | <code>\mathrm{M} \approx</code>                                                                                      | $M \approx$                                                   | A diffeomorphism $\sigma : M \rightarrow N$ between manifolds of the same dimension (e.g., $M \approx$                                       |
| mielke-geometrodynamics_F01750 | 367  | ■ 1.000 | <code>\mathrm{N} \approx \mathbb{R}^n</code>                                                                         | $N \approx \mathbb{R}^n$                                      | $N \approx \mathbb{R}^n$ ) is an injective mapping for which $\sigma$ and the corresponding inverse mapping                                  |
| mielke-geometrodynamics_F01751 | 367  | ■ 0.980 | <code>{ }^{-1}</code>                                                                                                | $^{-1}$                                                       | $\sigma^{-1}$ are $C^\infty$ functions, i.e., every partial derivative of $\sigma$ exists and is continuous in                               |
| mielke-geometrodynamics_F01752 | 367  | ■ 0.990 | <code>\sigma_{*}: \mathrm{T}_{\mathrm{m}}(\mathrm{M}) \rightarrow \mathrm{T}_{\sigma(\mathrm{m})}(\mathrm{N})</code> | $\sigma_* : T_m(M) \rightarrow T_{\sigma(m)}(N)$              | $m$ gives rise to a mapping $\sigma_* : T_m(M) \rightarrow T_{\sigma(m)}(N)$ within the space of tangent                                     |
| mielke-geometrodynamics_F01753 | 367  | ■ 0.999 | <code>\sigma(\mathrm{s}(\mathrm{t}))</code>                                                                          | $\sigma(s(t))$                                                | vectors along the curve $s(t)$ or $\sigma(s(t))$ if constructed according to the directional                                                 |
| mielke-geometrodynamics_F01754 | 367  | ■ 1.000 | <code>e^f</code>                                                                                                     | $e^f$                                                         | derivative $e^f$ . On the other hand, the transposed mapping of $\sigma_*$ defines a linear                                                  |
| mielke-geometrodynamics_F01755 | 367  | ■ 1.000 | <code>\sigma_{*}</code>                                                                                              | $\sigma_*$                                                    | derivative $e^f$ . On the other hand, the transposed mapping of $\sigma_*$ defines a linear                                                  |
| mielke-geometrodynamics_F01756 | 367  | ■ 1.000 | <code>\sigma_{*}^T: T_{\sigma(\mathrm{m})}^{*}(\mathrm{N}) \rightarrow T_{\mathrm{M}}^{*}(\mathrm{M})</code>         | $\sigma_*^T : T_{\sigma(m)}^*(N) \rightarrow T_M^*(M)$        | mapping $\sigma_*^T : T_{\sigma(m)}^*(N) \rightarrow T_M^*(M)$ between the “dual” cotangent spaces. As such,                                 |
| mielke-geometrodynamics_F01757 | 367  | ■ 1.000 | <code>{ }^{*} \alpha</code>                                                                                          | $^*\alpha$                                                    | Consequently, $^*\alpha$ denotes the inverse image of the differential form $\alpha$ . For cross                                             |
| mielke-geometrodynamics_F01758 | 368  | ■ 0.858 | <code>(p, 0)</code>                                                                                                  | $(p, 0)$                                                      | Completely <i>antisymmetric</i> covariant tensor fields of degree $(p, 0)$ are of prime impor-                                               |
| mielke-geometrodynamics_F01759 | 369  | ■ 0.982 | <code>{ }^1 \mathrm{d} f=\vartheta^{\alpha} \partial_{\alpha} f</code>                                               | $^1 \mathrm{d} f = \vartheta^\alpha \partial_\alpha f$        | $f$ , it is nothing but the total differential <sup>1</sup> $\mathrm{d} f = \vartheta^\alpha \partial_\alpha f$ . This action is extended to |
| mielke-geometrodynamics_F01760 | 369  | ■ 0.937 | <code>\sigma_{*} \alpha^{(0)}:=\alpha^{(0)} \circ \sigma</code>                                                      | $\sigma_* \alpha^{(0)} := \alpha^{(0)} \circ \sigma$          | that for a 0-form is obtained by composition the $\sigma_* \alpha^{(0)} := \alpha^{(0)} \circ \sigma$ .                                      |
| mielke-geometrodynamics_F01761 | 369  | ■ 1.000 | <code>\varepsilon_{\alpha_1 \ldots \alpha_n}</code>                                                                  | $\varepsilon_{\alpha_1 \dots \alpha_n}$                       | Let $\varepsilon_{\alpha_1 \dots \alpha_n}$ be the totally antisymmetric unit tensor (Levi-Civita tensor; see MTW,                           |
| mielke-geometrodynamics_F01762 | 369  | ■ 0.999 | <code>\varepsilon_{\hat{1} \ldots \tilde{n}}=+1</code>                                                               | $\varepsilon_{\hat{1} \dots \tilde{n}} = +1$                  | p. 87) with the conventional choice of signs $\varepsilon_{\hat{1} \dots \tilde{n}} = +1$ . In a pseudo-Riemannian                           |
| mielke-geometrodynamics_F01763 | 369  | ■ 1.000 | <code>g_{\{\mathrm{i} j\}}</code>                                                                                    | $g_{ij}$                                                      | manifold endowed with the metric tensor $g_{ij}$ , this unit tensor allows one to define the                                                 |
| mielke-geometrodynamics_F01764 | 369  | ■ 0.997 | <code>\mathrm{d} f=\left(\partial f / \partial x^i\right) d x^i</code>                                               | $\mathrm{d} f = \left(\partial f / \partial x^i\right) d x^i$ | to a coordinate basis, it would read $\mathrm{d} f = (\partial f / \partial x^i) d x^i$ .                                                    |
| mielke-geometrodynamics_F01765 | 371  | ■ 1.000 | <code>\delta \alpha=0</code>                                                                                         | $\delta \alpha = 0$                                           | if and only if it is closed as well as <i>coclosed</i> , i.e., $\delta \alpha = 0$ .                                                         |

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| mielke-geometrodynamics_F01766 | 371  | ■ 0.901 | $\mathrm{B}(\mathrm{M}) := \mathrm{d} \mathrm{D}(\mathrm{M})$                                                                                                                    | $B(M) := dD(M)$                                                                                  | <u><math>B(M) := d D(M)</math></u> of <i>exact</i> forms                                                                                           |
| mielke-geometrodynamics_F01767 | 371  | ■ 1.000 | $\mathrm{B}^{\wedge\{0\}}(\mathrm{M}) = 0$                                                                                                                                       | $B^0(M) = 0$                                                                                     | On the other hand, we have $B^0(M) = 0$ . Consequently,                                                                                            |
| mielke-geometrodynamics_F01768 | 372  | ■ 0.995 | $\mathrm{H}^{\wedge\{\mathrm{p}\}}(\mathrm{M})$                                                                                                                                  | $H^p(M)$                                                                                         | manifolds), the dimensions of the spaces $H^p(M)$ are called the <i>Betti numbers</i>                                                              |
| mielke-geometrodynamics_F01769 | 372  | ■ 1.000 | $\mathrm{t} = -1$                                                                                                                                                                | $t = -1$                                                                                         | with the Betti numbers as coefficients, gives for <u><math>t = -1</math></u> the alternating sum                                                   |
| mielke-geometrodynamics_F01770 | 372  | ■ 0.998 | $\eta_{\mathrm{i}}$                                                                                                                                                              | $\eta_i$                                                                                         | with a countable basis $\eta_i$ , a p-dimensional standard simplex is defined by the set                                                           |
| mielke-geometrodynamics_F01771 | 373  | ■ 0.843 | $\mathrm{c} : \Delta_{\mathrm{p}} \rightarrow \mathrm{M}$                                                                                                                        | $c : \Delta_p \rightarrow M$                                                                     | the $C^\infty$ -mapping $c : \Delta_p \rightarrow M$ . These simplices form a free $\mathbb{R}$ -module $C_p(M, \mathbb{R})$ ,                     |
| mielke-geometrodynamics_F01772 | 373  | ■ 0.843 | $\mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$                                                                                                                                | $C_p(M, \mathbb{R})$                                                                             | the $C^\infty$ -mapping $c : \Delta_p \rightarrow M$ . These simplices form a free $\mathbb{R}$ -module $C_p(M, \mathbb{R})$ ,                     |
| mielke-geometrodynamics_F01773 | 373  | ■ 1.000 | $\partial \mathrm{c}$                                                                                                                                                            | $\partial c$                                                                                     | whose elements are called p-chains. The boundary $\partial c$ of a chain will be generated                                                         |
| mielke-geometrodynamics_F01774 | 373  | ■ 0.997 | $\mathrm{Z}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}) = (\mathrm{Ker} \partial)_{\mathrm{p}}, \mathrm{B}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}) := (\mathrm{Im} \partial)_{\mathrm{p}}$ | $Z_p(M, \mathbb{R}) = (\mathrm{Ker} \partial)_p, B_p(M, \mathbb{R}) := (\mathrm{Im} \partial)_p$ | tically.” The spaces <u><math>Z_p(M, \mathbb{R}) = (\mathrm{Ker} \partial)_p, B_p(M, \mathbb{R}) := (\mathrm{Im} \partial)_p</math></u> as well as |
| mielke-geometrodynamics_F01775 | 373  | ■ 1.000 | $\mathrm{N}_{(\mathrm{p})} \subset \mathrm{M}$                                                                                                                                   | $N_{(p)} \subset M$                                                                              | dimensional (compact) submanifold $N_{(p)} \subset M$ of M. Consequently, they can be                                                              |
| mielke-geometrodynamics_F01776 | 373  | ■ 1.000 | $\mathrm{N}_{(\mathrm{p})}$                                                                                                                                                      | $N_{(p)}$                                                                                        | integrated, provided $N_{(p)}$ admits a simplicial covering by the chain $c_p \in C_p(M, \mathbb{R})$ .                                            |
| mielke-geometrodynamics_F01777 | 373  | ■ 1.000 | $c_{\mathrm{p}} \in C_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$                                                                                                                      | $c_p \in C_p(M, \mathbb{R})$                                                                     | integrated, provided $N_{(p)}$ admits a simplicial covering by the chain <u><math>c_p \in C_p(M, \mathbb{R})</math></u> .                          |
| mielke-geometrodynamics_F01778 | 374  | ■ 1.000 | $\mathrm{H}^{\mathrm{P}}(\mathrm{M}) \times \mathrm{H}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}) \rightarrow \mathbb{R}$                                                              | $H^P(M) \times H_p(M, \mathbb{R}) \rightarrow \mathbb{R}$                                        | ping $H^P(M) \times H_p(M, \mathbb{R}) \rightarrow \mathbb{R}$ from the Cartesian product of the cohomology and                                    |
| mielke-geometrodynamics_F01779 | 375  | ■ 1.000 | $\mathrm{g} \in \mathrm{G}$                                                                                                                                                      | $g \in G$                                                                                        | (i) the elements $g \in G$ form an (analytic) manifold,                                                                                            |
| mielke-geometrodynamics_F01780 | 375  | ■ 1.000 | $\mathrm{G} \times \mathrm{G} \rightarrow \mathrm{G}$                                                                                                                            | $G \times G \rightarrow G$                                                                       | (ii) the mapping $G \times G \rightarrow G$ originating from the pointwise multiplication                                                          |
| mielke-geometrodynamics_F01781 | 375  | ■ 0.999 | $(\mathrm{g}_1, \mathrm{g}_2) \rightarrow \mathrm{g}_1 \mathrm{g}_2^{-1}$                                                                                                        | $(g_1, g_2) \rightarrow g_1 g_2^{-1}$                                                            | $(g_1, g_2) \rightarrow g_1 g_2^{-1}$ of (group) elements is analytic as well.                                                                     |
| mielke-geometrodynamics_F01782 | 375  | ■ 1.000 | $\mathrm{G} = \mathrm{GL}(\mathrm{N}, \mathbb{C})$                                                                                                                               | $G = GL(N, \mathbb{C})$                                                                          | The most general example is given by $G = GL(N, \mathbb{C})$ , i.e., by the general linear                                                         |
| mielke-geometrodynamics_F01783 | 375  | ■ 0.999 | $\mathrm{T}_{\mathrm{e}}(\mathrm{G})$                                                                                                                                            | $T_e(G)$                                                                                         | The tangent space $T_e(G)$ attached to the unit element $e \in G$ of the group possesses                                                           |
| mielke-geometrodynamics_F01784 | 375  | ■ 0.999 | $\mathrm{e} \in \mathrm{G}$                                                                                                                                                      | $e \in G$                                                                                        | The tangent space $T_e(G)$ attached to the unit element $e \in G$ of the group possesses                                                           |
| mielke-geometrodynamics_F01785 | 375  | ■ 0.952 | $\xi = \{\xi^j, j = 1, \dots, K\}$                                                                                                                                               | $\xi = \{\xi^j, j = 1, \dots, K\}$                                                               | the <i>Lie algebra</i> of $\mathfrak{g}$ of G (KN I, p. 38). Let $\xi = \{\xi^j, j = 1, \dots, K\}$ be an appropriate                              |
| mielke-geometrodynamics_F01786 | 375  | ■ 0.943 | $c_{ij}^{\mathrm{k}}$                                                                                                                                                            | $c_{ij}^k$                                                                                       | Here the real <i>structure constants</i> $c_{ij}^k$ are characteristic of the group G. They are always                                             |
| mielke-geometrodynamics_F01787 | 376  | ■ 1.000 | $\mathrm{A} = \mathrm{A}^{\mathrm{i}} \mathrm{I}_{\mathrm{i}}$                                                                                                                   | $A = A^i I_i$                                                                                    | defined for generic Lie-algebra-valued vectors $A = A^i I_i$ and $B = B^j I_j$ is linear and                                                       |
| mielke-geometrodynamics_F01788 | 376  | ■ 1.000 | $\mathrm{B} = \mathrm{B}^{\mathrm{j}} \mathrm{I}_{\mathrm{j}}$                                                                                                                   | $B = B^j I_j$                                                                                    | defined for generic Lie-algebra-valued vectors $A = A^i I_i$ and $B = B^j I_j$ is linear and                                                       |
| mielke-geometrodynamics_F01789 | 376  | ■ 1.000 | $\stackrel{\circ}{g}_{ls}$                                                                                                                                                       | $\overset{\circ}{g}_{ls}$                                                                        | may be introduced that is determined by the <i>Cartan–Killing metric</i> $\overset{\circ}{g}_{ls}$ of the Lie                                      |
| mielke-geometrodynamics_F01790 | 376  | ■ 0.863 | $\mathfrak{gl}(\mathrm{N}, \mathbb{C})$                                                                                                                                          | $\mathfrak{gl}(N, \mathbb{C})$                                                                   | of $\mathfrak{g}$ . In the case of the Lie algebra $\mathfrak{gl}(N, \mathbb{C})$ of the general linear group, this product                        |

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| mielke-geometrodynamics_F01791 | 376  | ■ 0.737 | $\mathrm{G}=\exp ^{\mathrm{Trg}}$ | $G = \exp^{\mathrm{Trg}}$ | $\det G = \exp^{\mathrm{Trg}}$ . The scalar product (C.12) is nondegenerate if and only if $G$ is |
| mielke-geometrodynamics_F01792 | 377  | ■ 0.995 | $\mathrm{\xi}^{\mathrm{j}}$       | $\xi^j$                   | algebra. In terms of a real parametrization $\xi^j$ , an arbitrary group element is generated     |
| mielke-geometrodynamics_F01793 | 377  | ■ 1.000 | $\lambda_{\mathrm{j}}$            | $\lambda_j$               | Here the $\lambda_j$ denote the generalized Gell-Mann matrices, which are Hermitian and           |